

Section 1

Geometry recap

Euclidean geometry

- After Euclid of Alexandria

Euclid of Alexandria (325-265 BCE)



School of Athens (Raphael)

Marie-Lan Nguyen

Euclidean **geometry**

- After Euclid of Alexandria

Around 300 BCE he wrote “The Elements” (13 books)



Scroll from “The Elements”
Euclid | Public domain, Wikimedia Commons

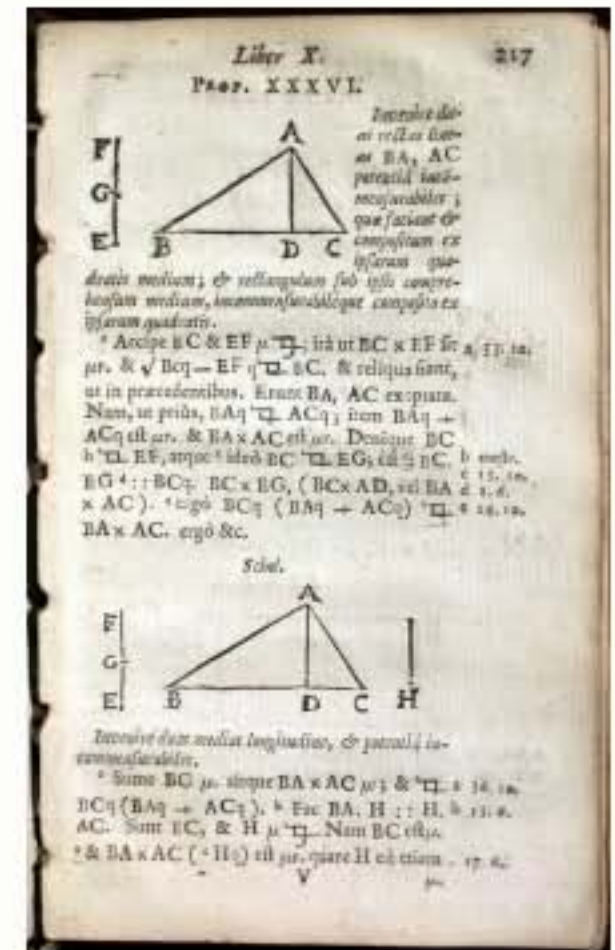
Euclid of Alexandria (325-265 BCE)



School of Athens (Raphael)
Marie-Lan Nguyen

Euclidean geometry

- The geometry you learnt at school



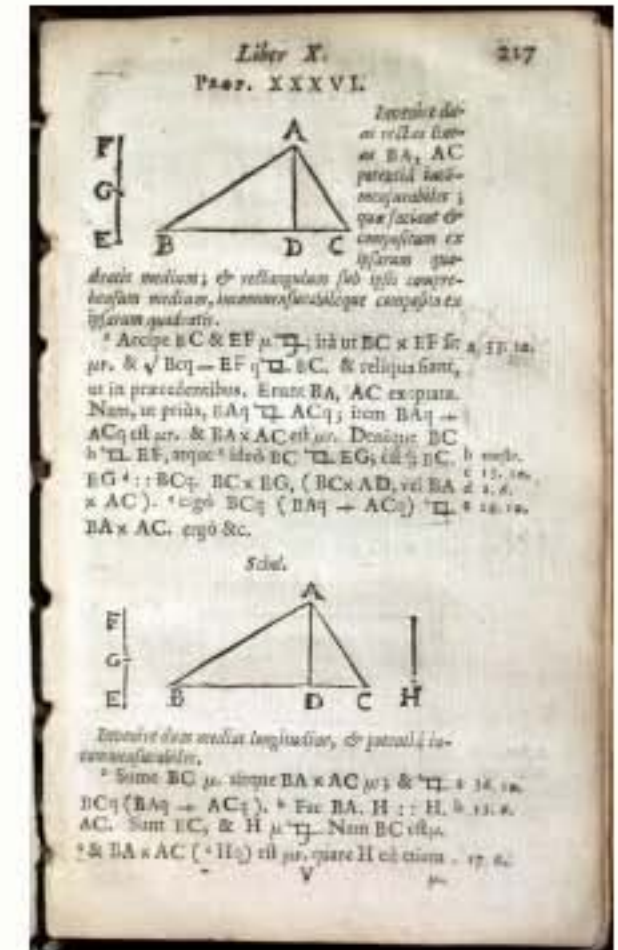
Euclid, Elementorum Libri XV: Cantabrigiae, 1655 2005

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Euclidean geometry

- The geometry you learnt at school
- The geometry of a **plane** and **3D space**
 - ➔ lines, triangles, circles etc.



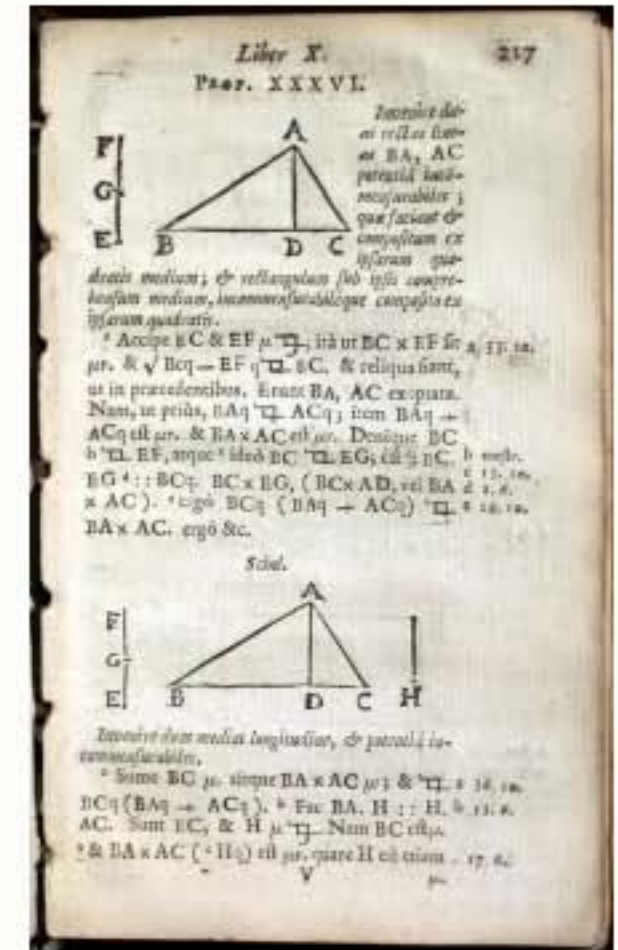
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Euclidean geometry

- The geometry you learnt at school
- The geometry of a **plane** and **3D space**
 - ➔ lines, triangles, circles etc.
- Based on a few axioms



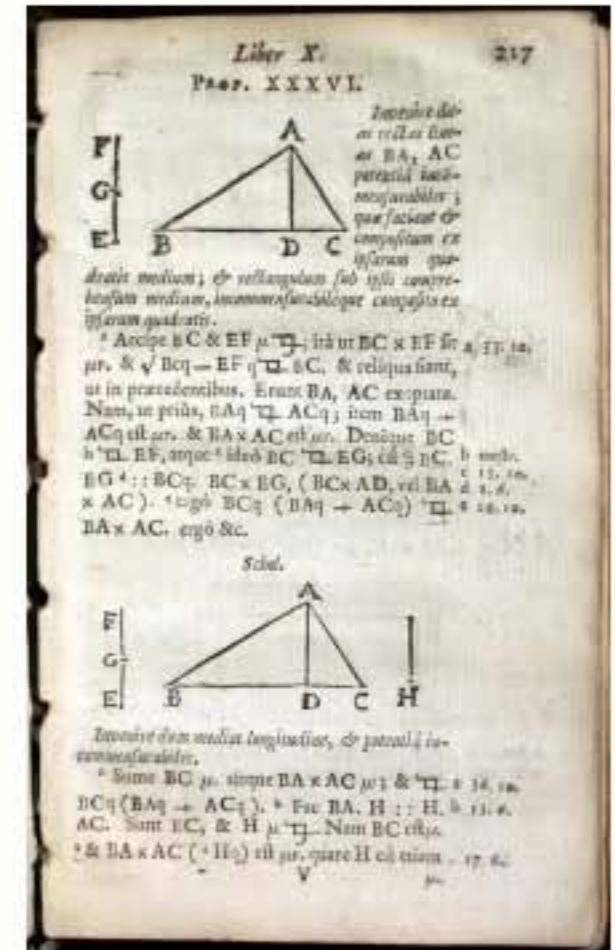
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Euclidean geometry

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 - ➔ lines, triangles, circles etc.
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- No numbers

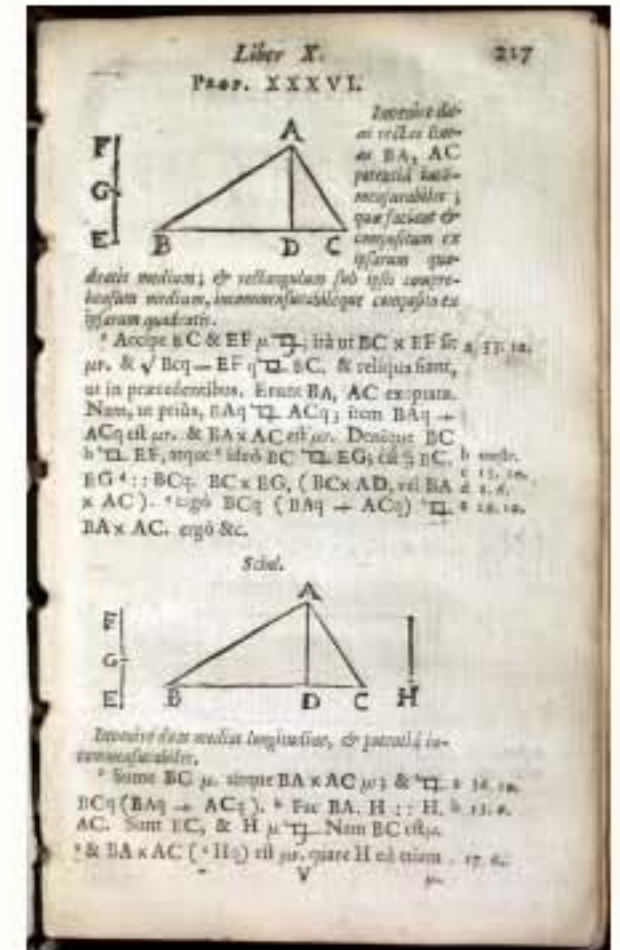


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Euclidean geometry

- The geometry you learnt at school
- The geometry of a **plane** and **3D space**
 - ➔ lines, triangles, circles etc.
- Based on a few axioms
- No numbers
- No coordinate frames



Euclid, Elementorum Libri XV: Cantabrigiae, 1655 2005
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cogito,
ergo sum



René Descartes (1596-1650)

- Philosopher
- Mathematician
- Part-time mercenary

**cogito,
ergo sum**



René Descartes (1596-1650)

- **Philosopher**
- **Mathematician**
- **Part-time mercenary**
- **The father of modern philosophy**

cogito,
ergo sum



I think, therefore I am



Rene Descartes (1596-1650)

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Eiffel Tower
Waithamai (Own work) | CC BY-SA 3.0. via Wikimedia Commons
Panoramic view of the French countryside 2013
Mary Harrsch | CC BY-SA 2.0.

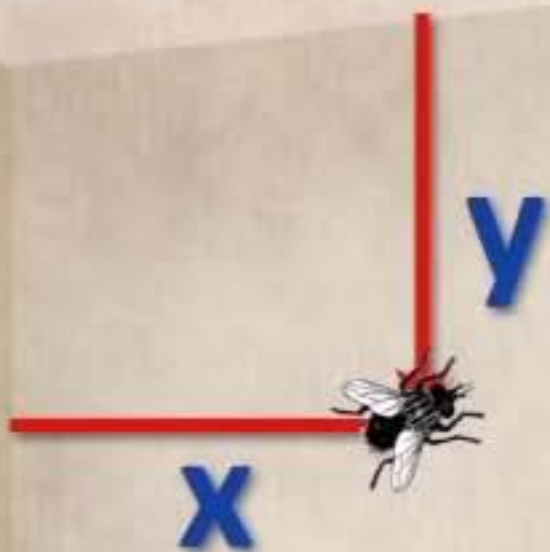


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Waithamai (Own work) | CC BY-SA 3.0. via Wikimedia Commons

Panoramic view of the French countryside 2013

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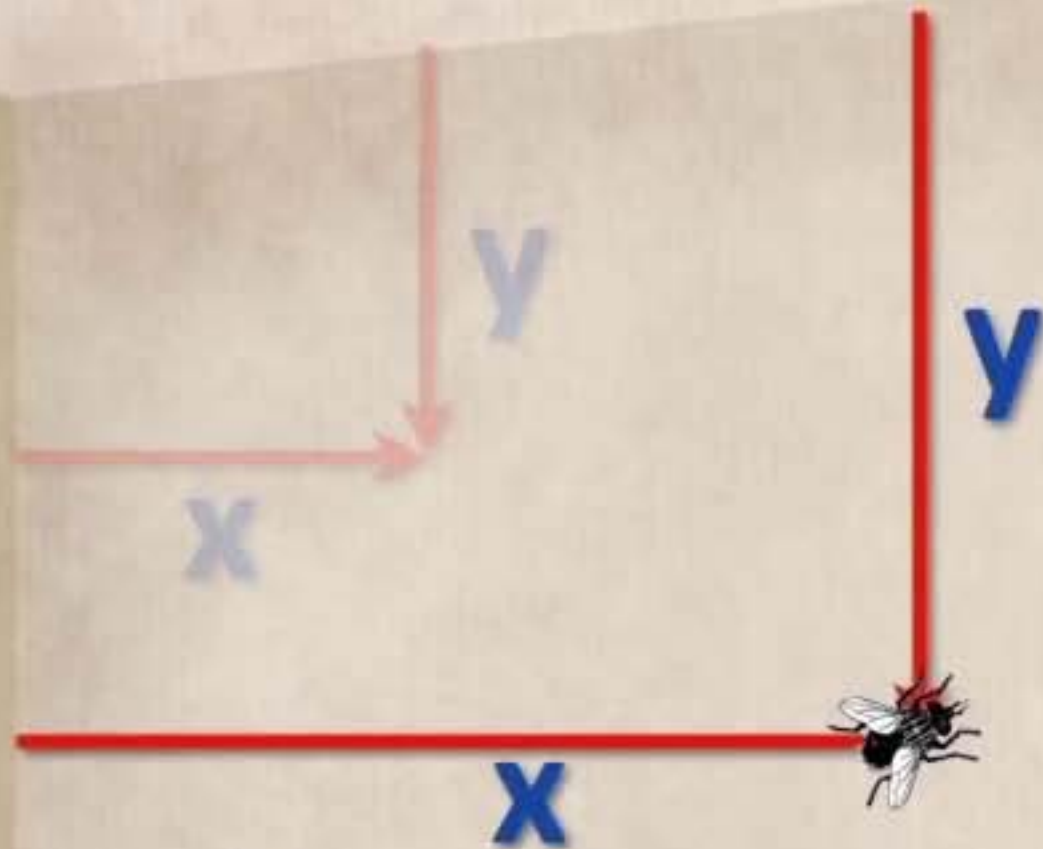


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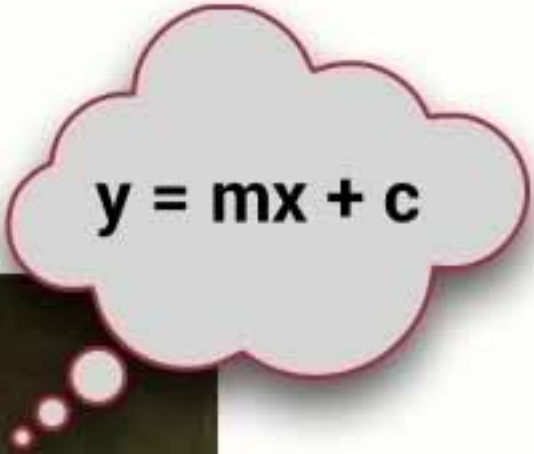
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Cartesian **geometry**

- Added algebra to Euclidean geometry



Cartesian **geometry**


$$y = mx + c$$



- Added algebra to Euclidean geometry
 - ➔ Shapes could be expressed by equations

Cartesian geometry


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- Added algebra to Euclidean geometry
 - ➔ Shapes could be expressed by equations
 - ➔ Cartesian (or analytic) geometry



His name in Latin was
Renatus Cartesius

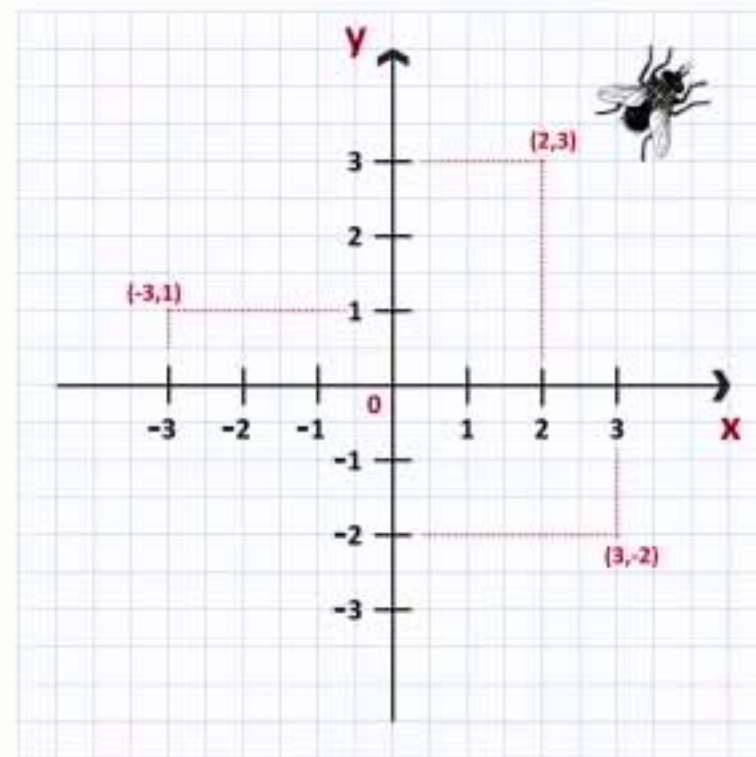
Cartesian geometry

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Descartes

After Frans Hals | Public domain, Wikimedia Commons

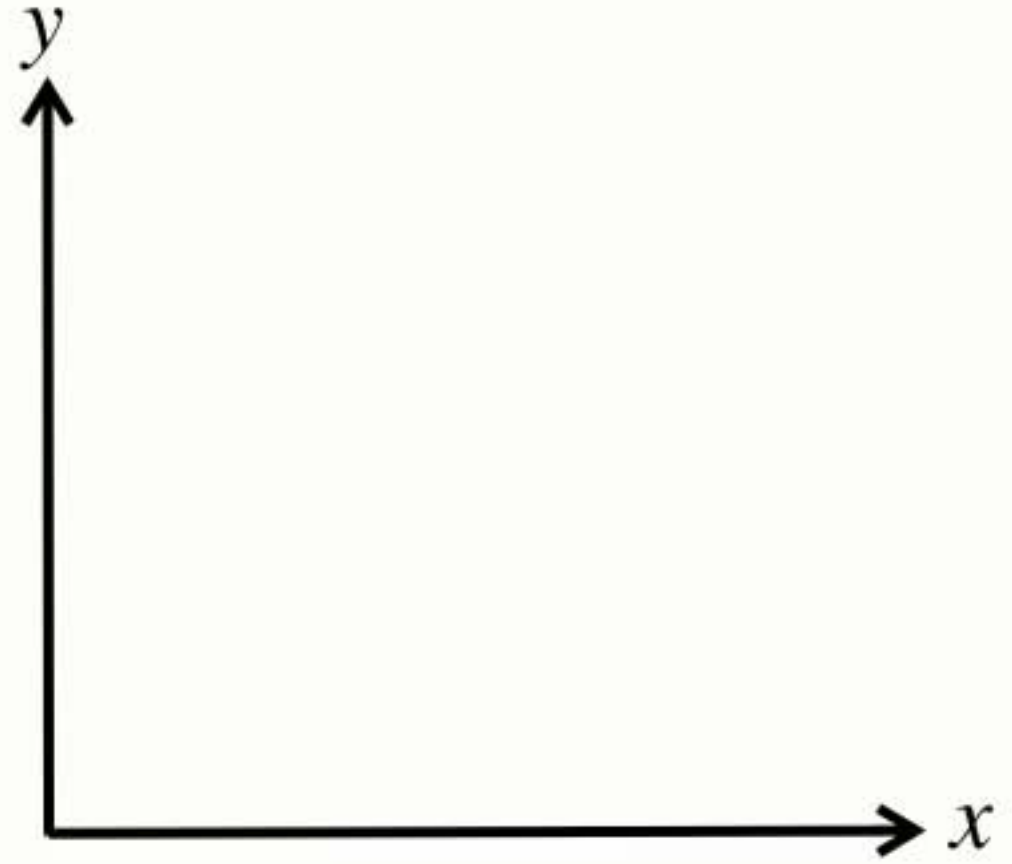


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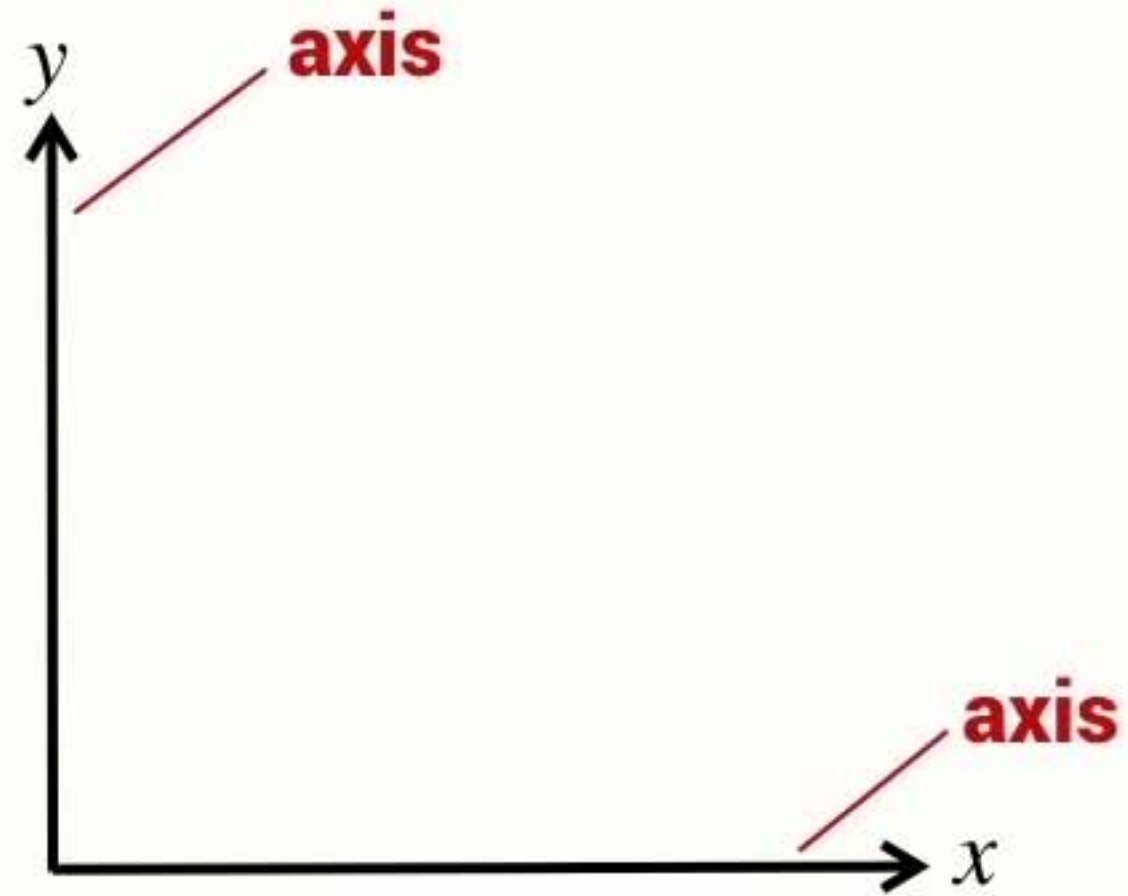
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- Developed the coordinate system

2D coordinate frame

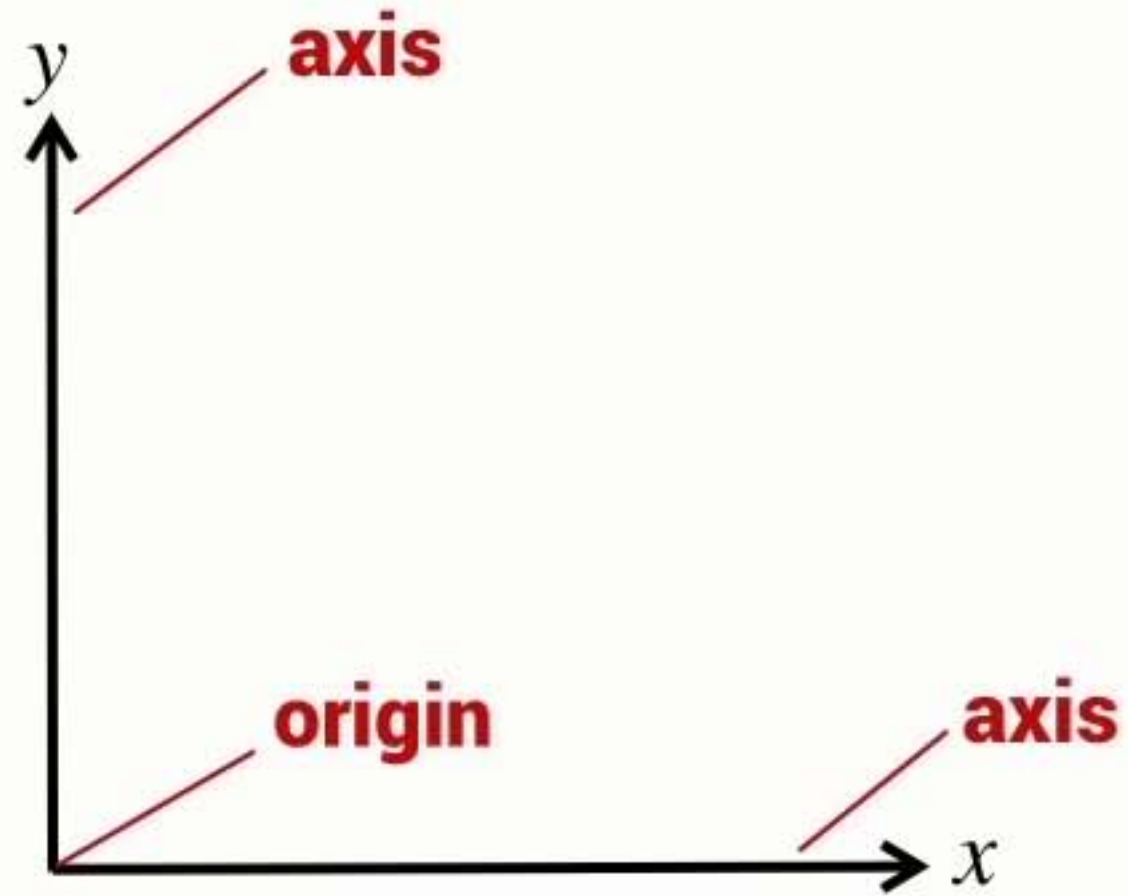


2D coordinate frame



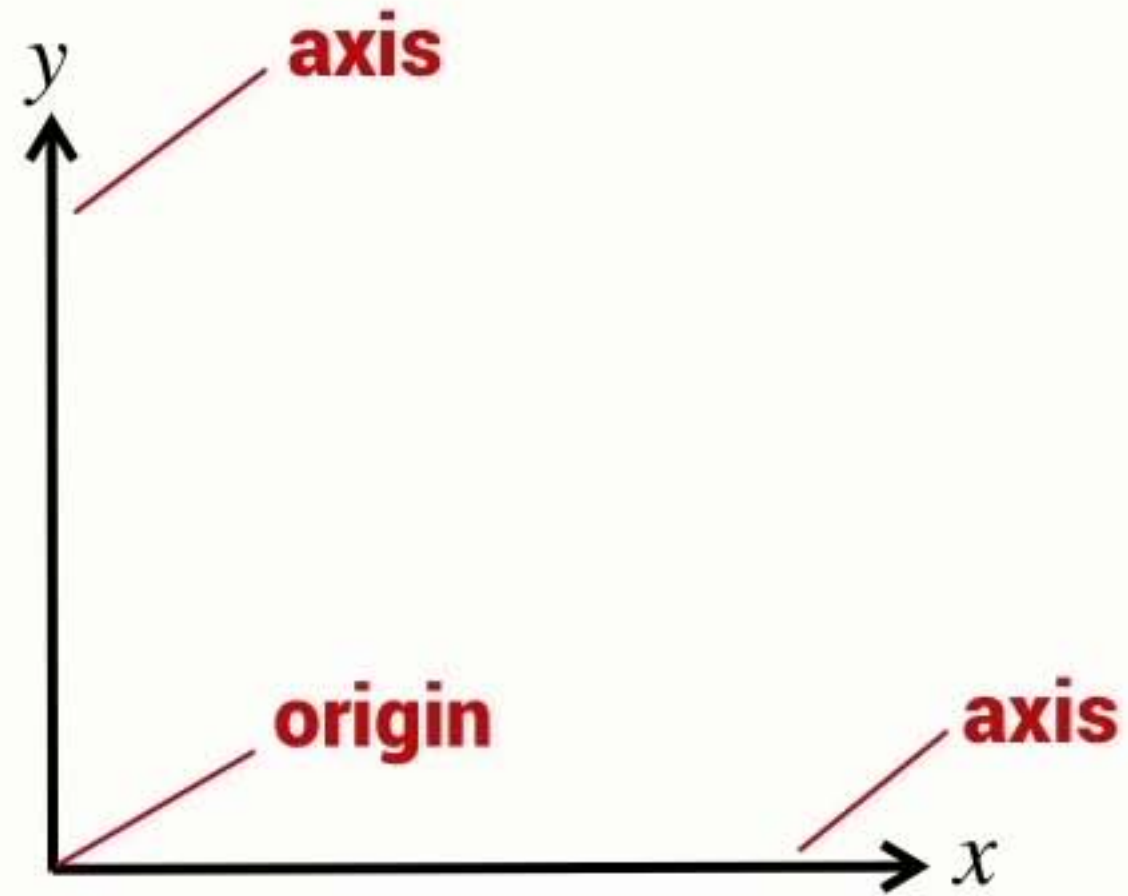
- Two axes form a 2D coordinate frame

2D coordinate frame



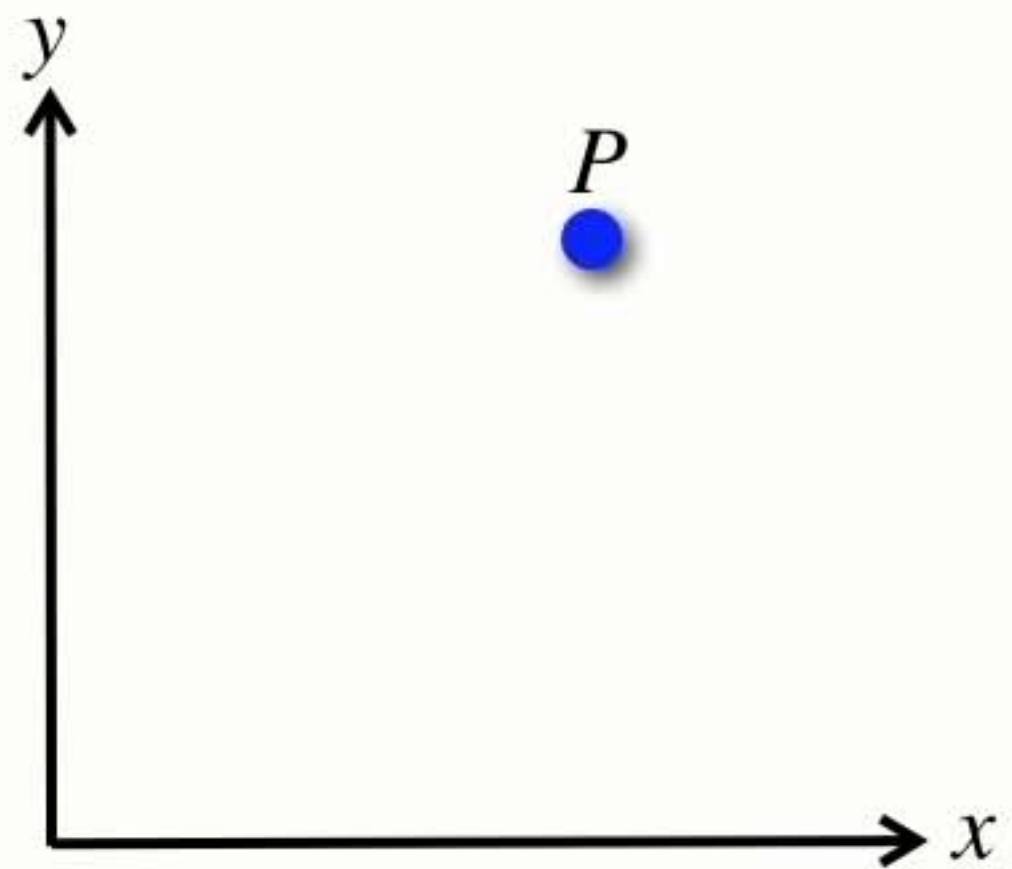
- Two axes form a 2D **coordinate frame**

2D coordinate frame

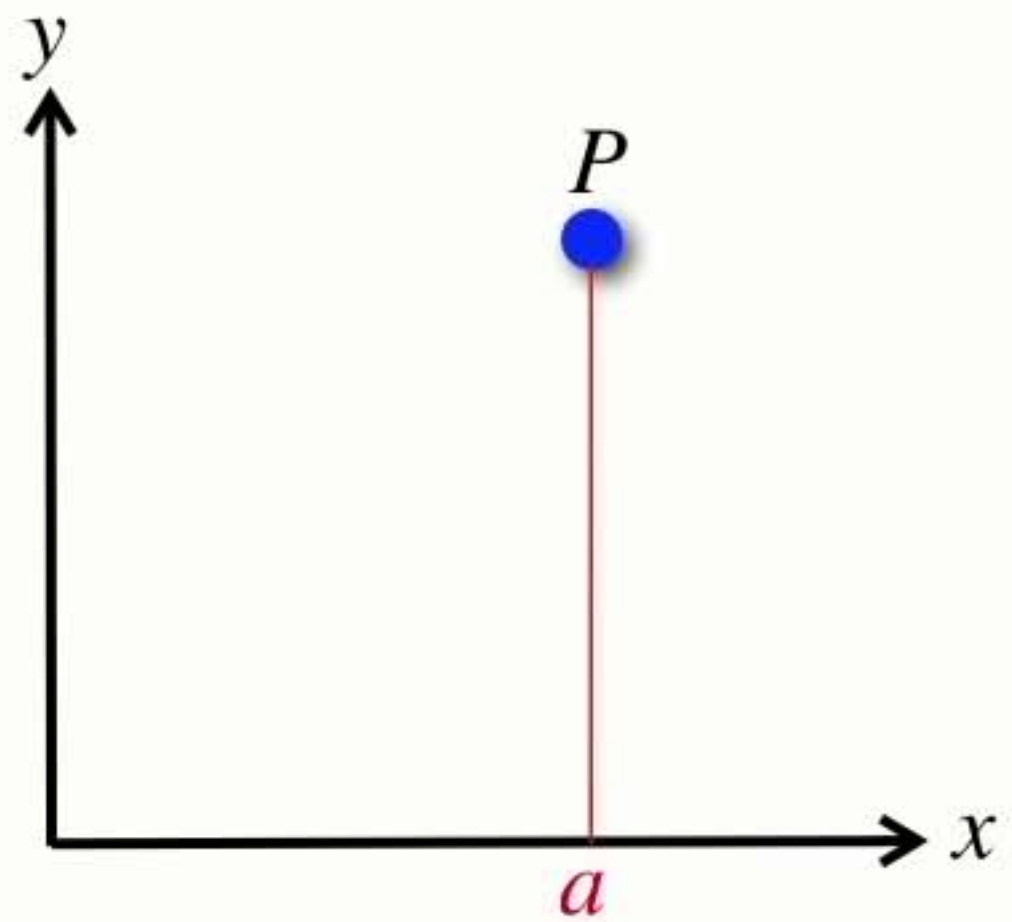


- Two axes form a 2D **coordinate frame**

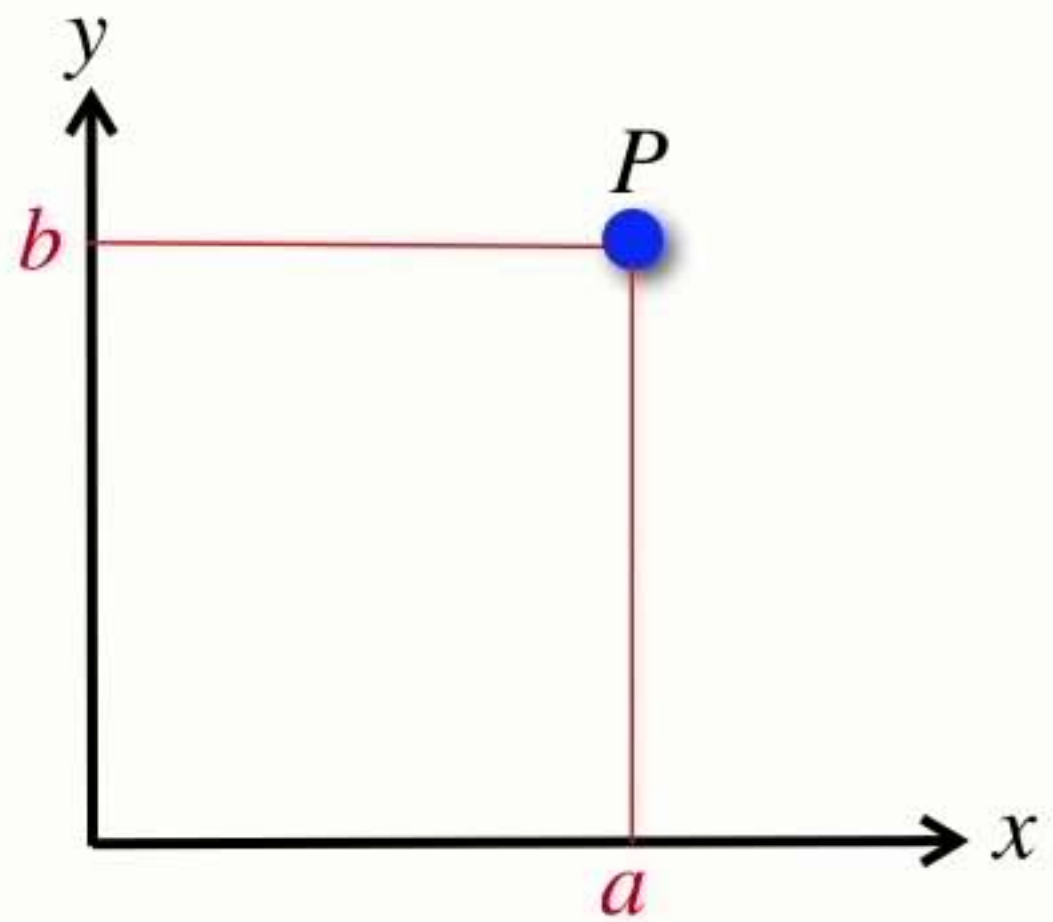
Points



Points

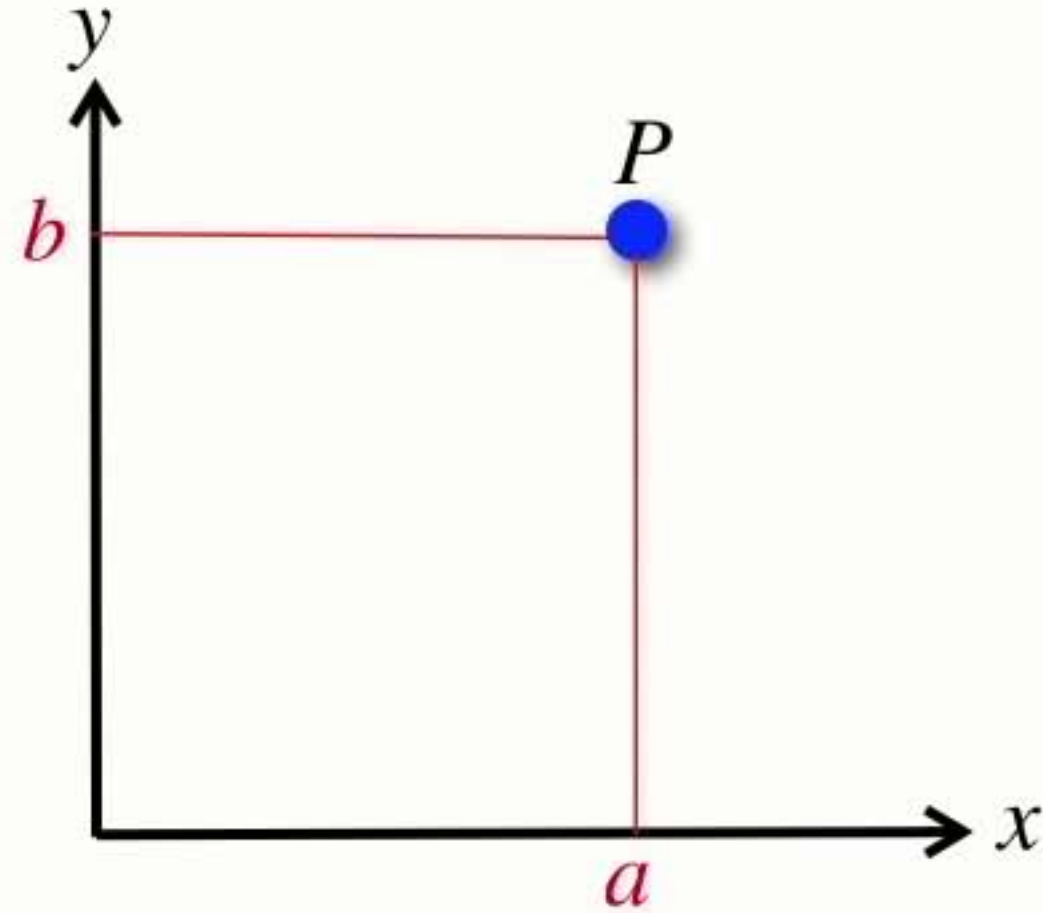


Points



$$P = (a, b)$$

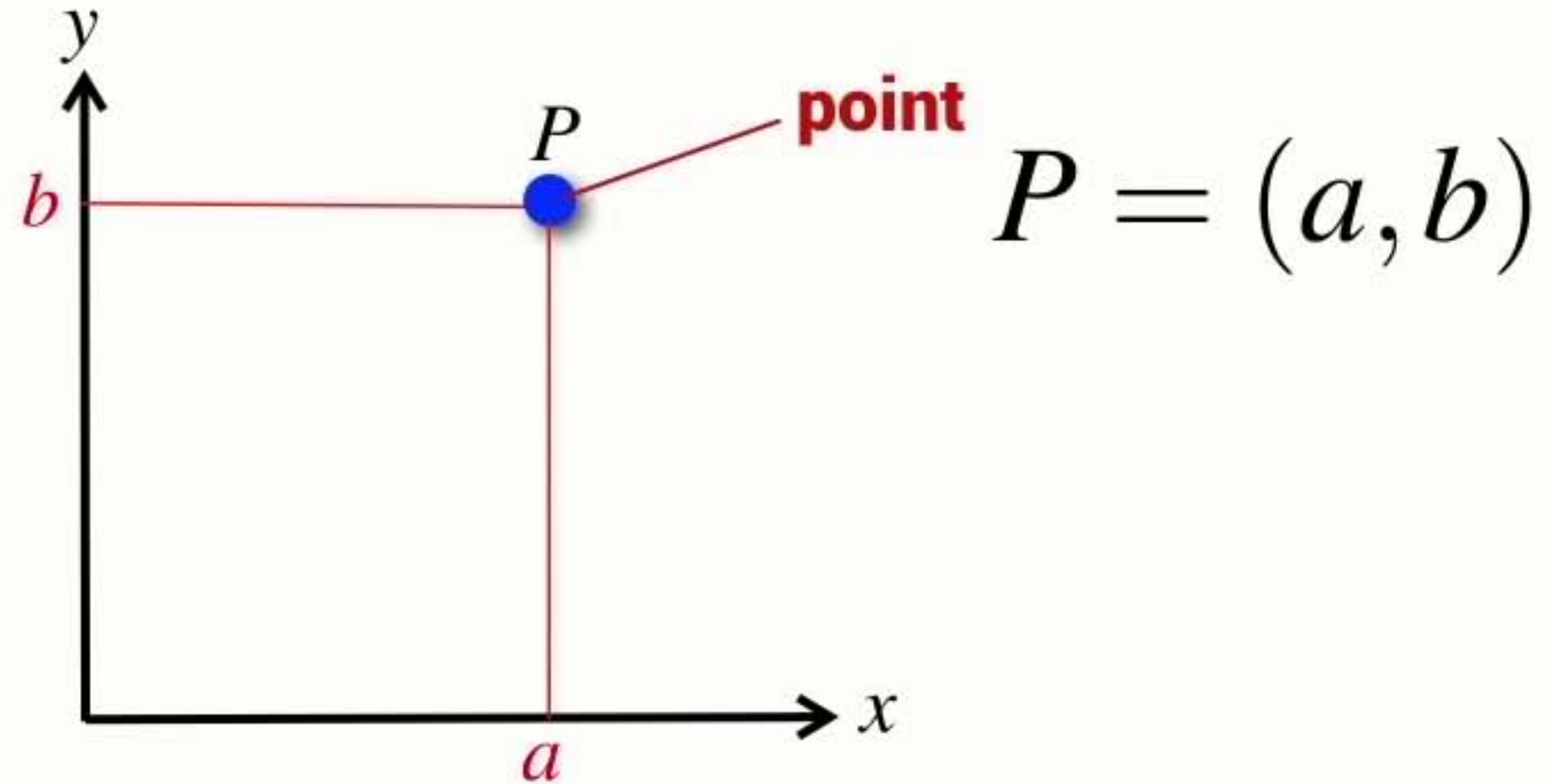
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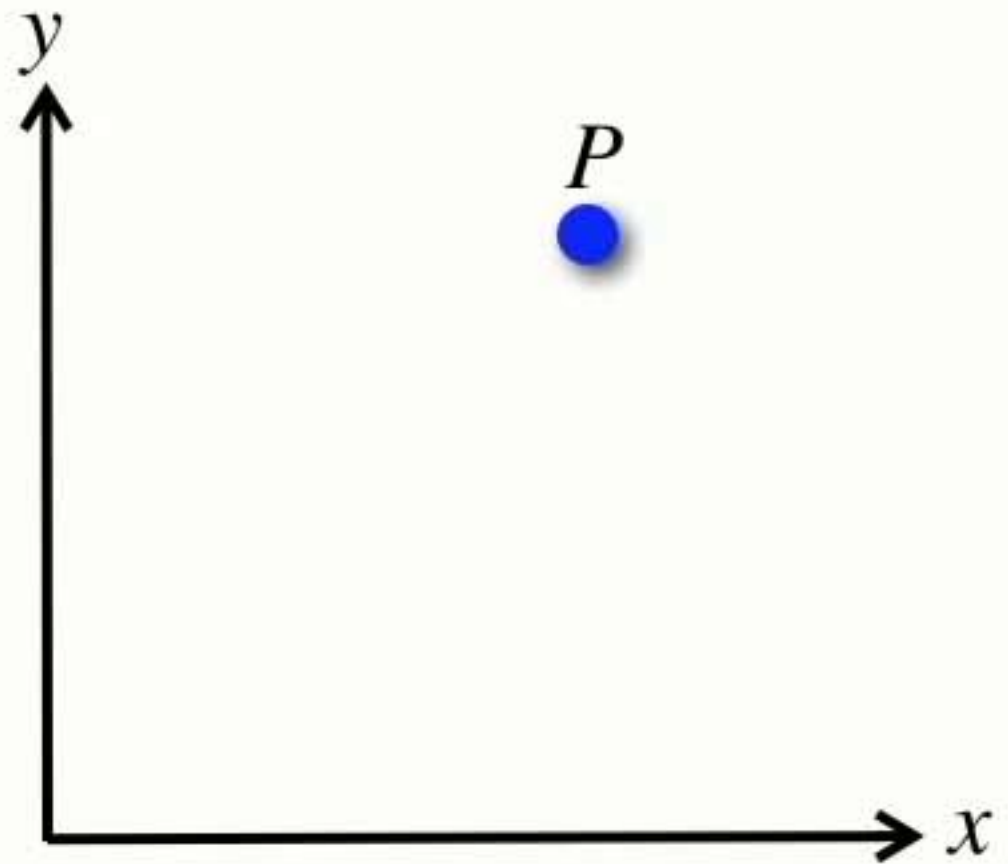
- A point is a location in n-dimensional space $P \in \mathbb{R}^n$

Points

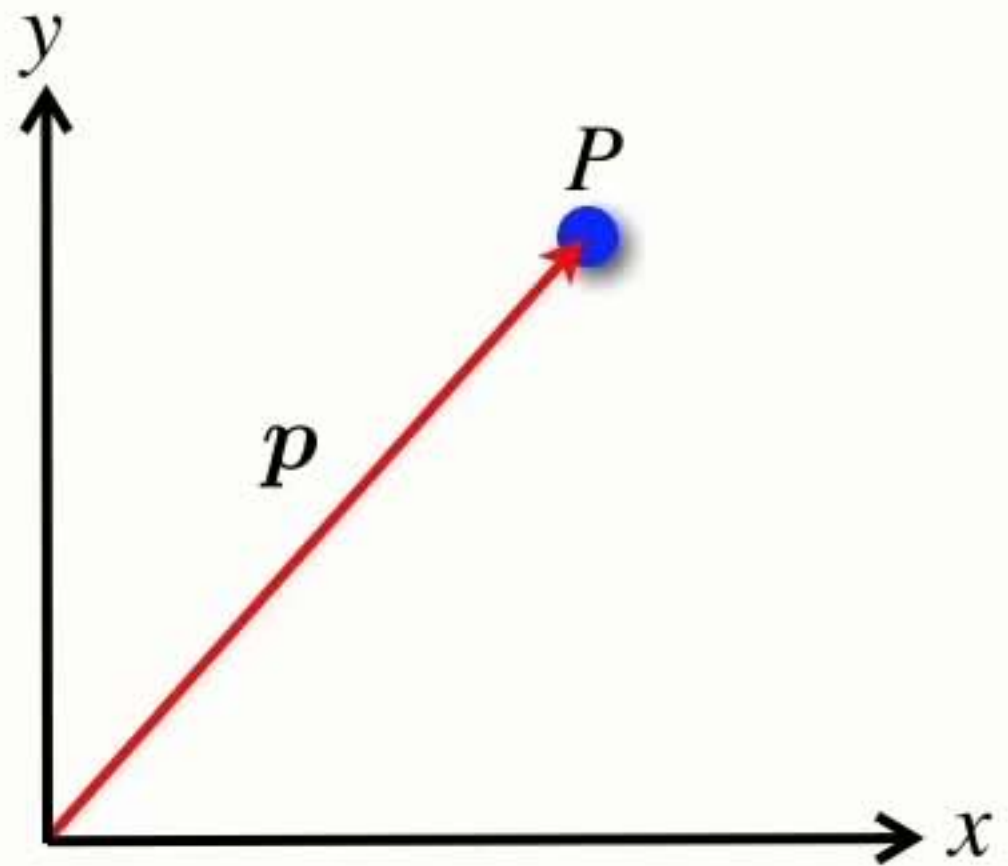


- A point is a location in n-dimensional space $P \in \mathbb{R}^n$
- Represented by a Cartesian coordinate or an n-tuple $(x_1, x_2 \cdots x_n)$

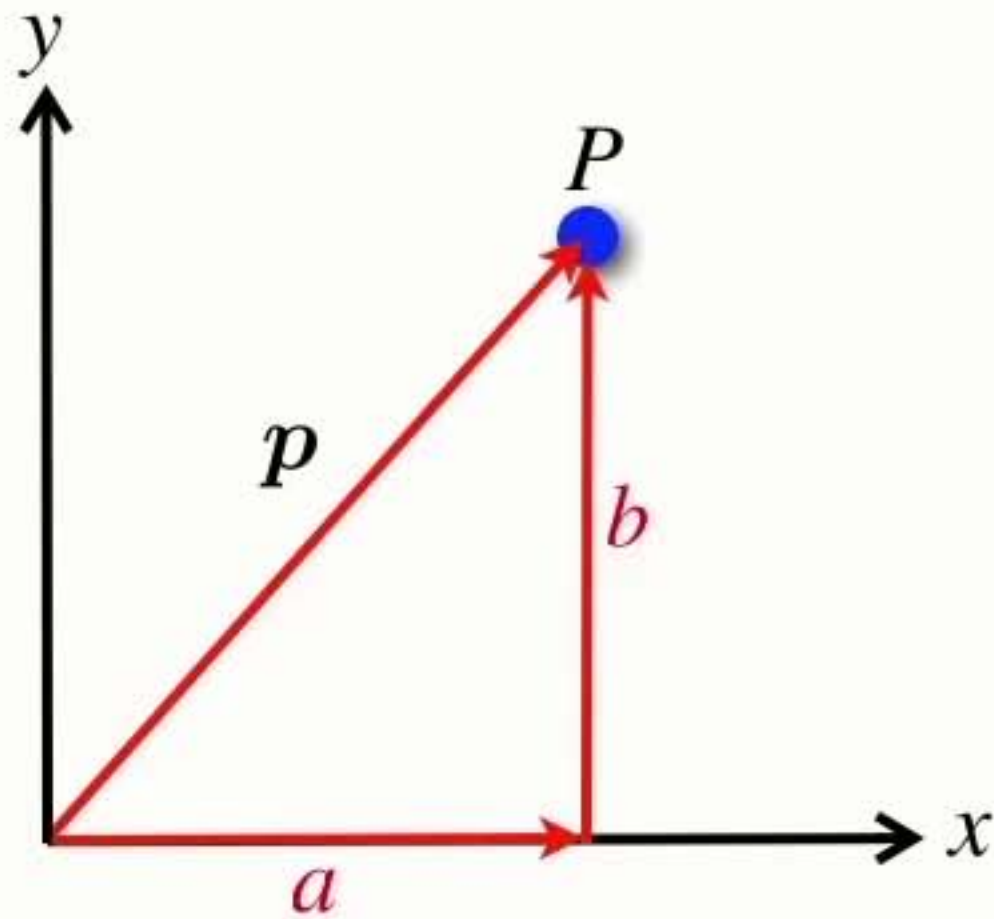
Vectors



Vectors

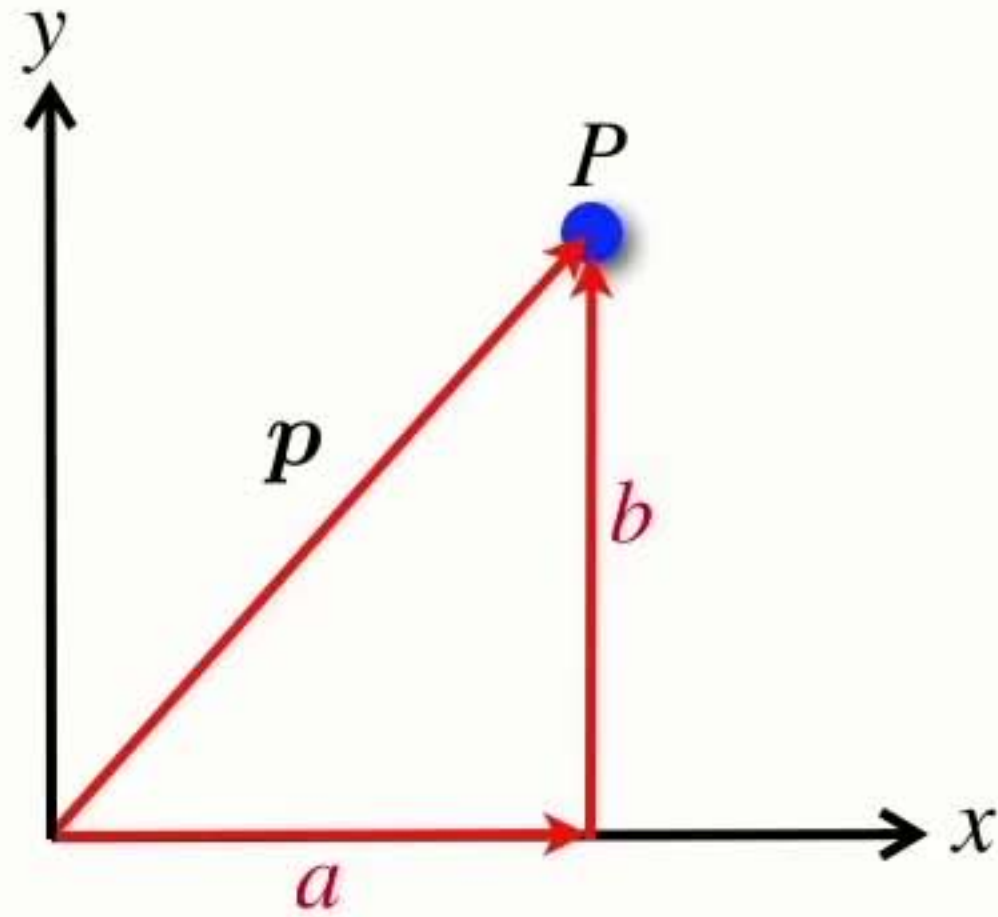


Vectors



$$p = \begin{bmatrix} a \\ b \end{bmatrix}$$

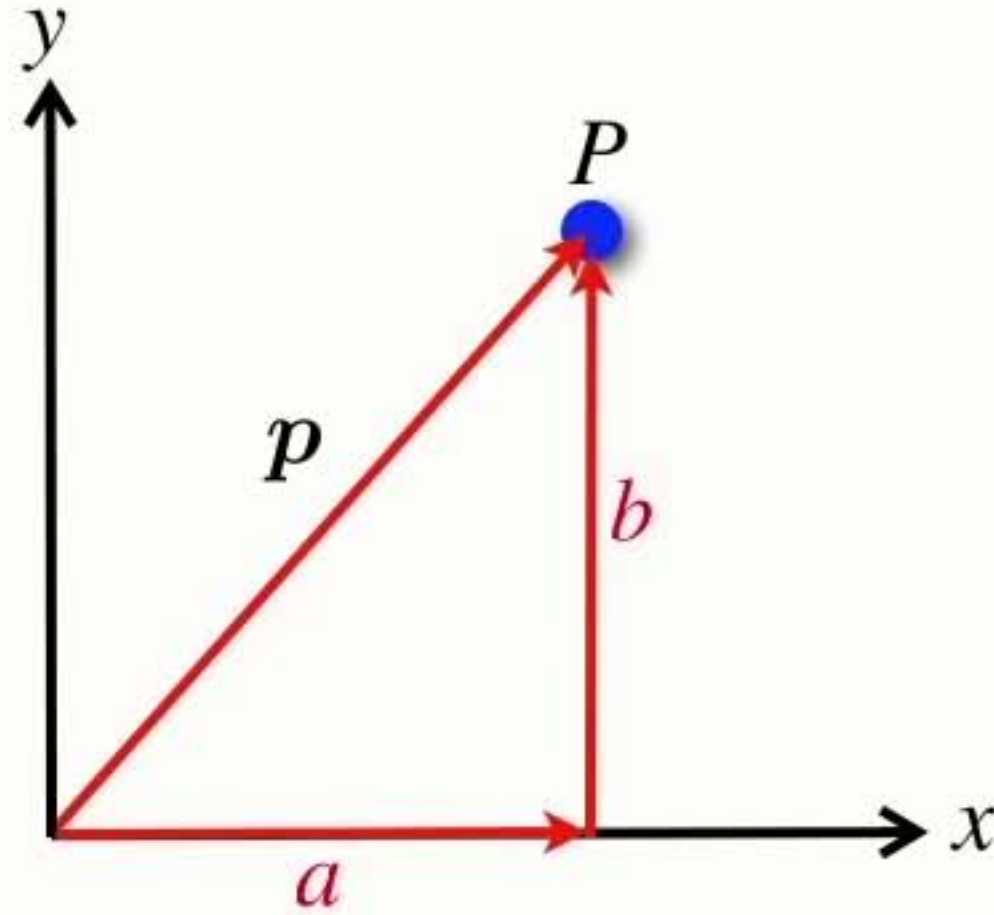
Vectors



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- A vector is an element of an n-dimensional vector space $\mathbf{p} \in \mathbb{R}^n$

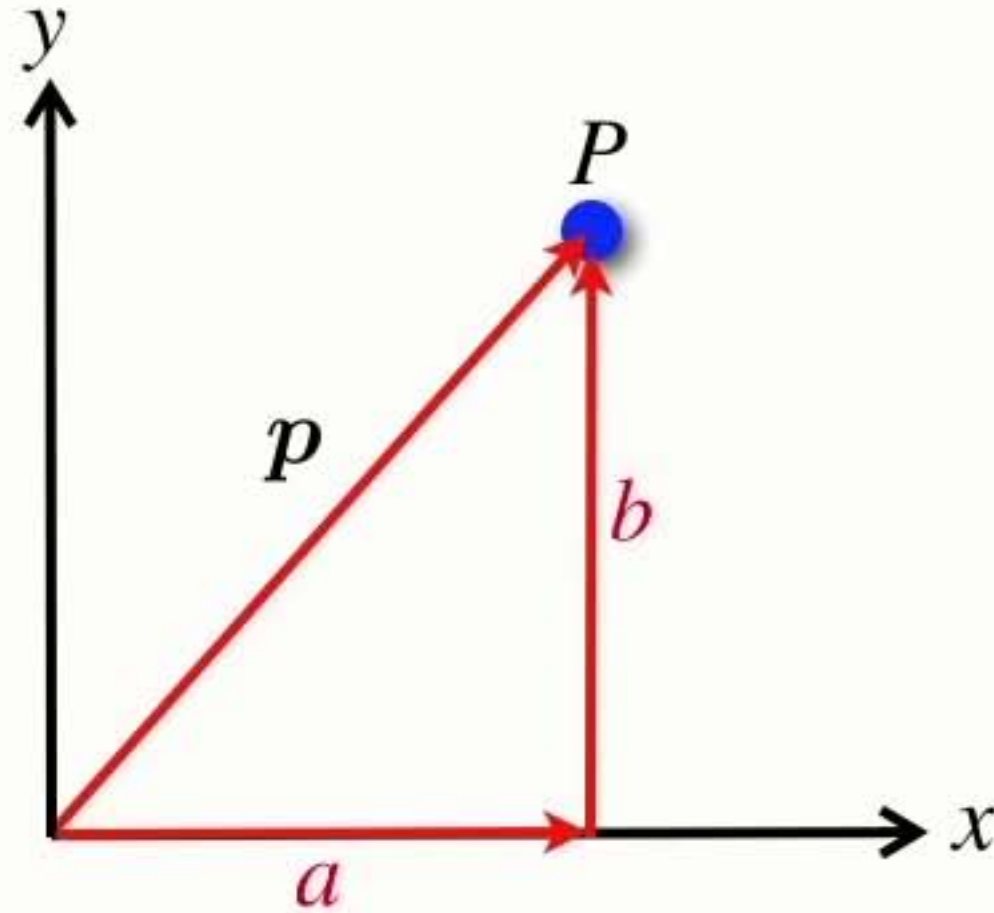
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- A vector is an element of an n-dimensional vector space $\mathbf{p} \in \mathbb{R}^n$
- Represented by an n-element column vector $[x_1, x_2 \cdots x_n]^T$

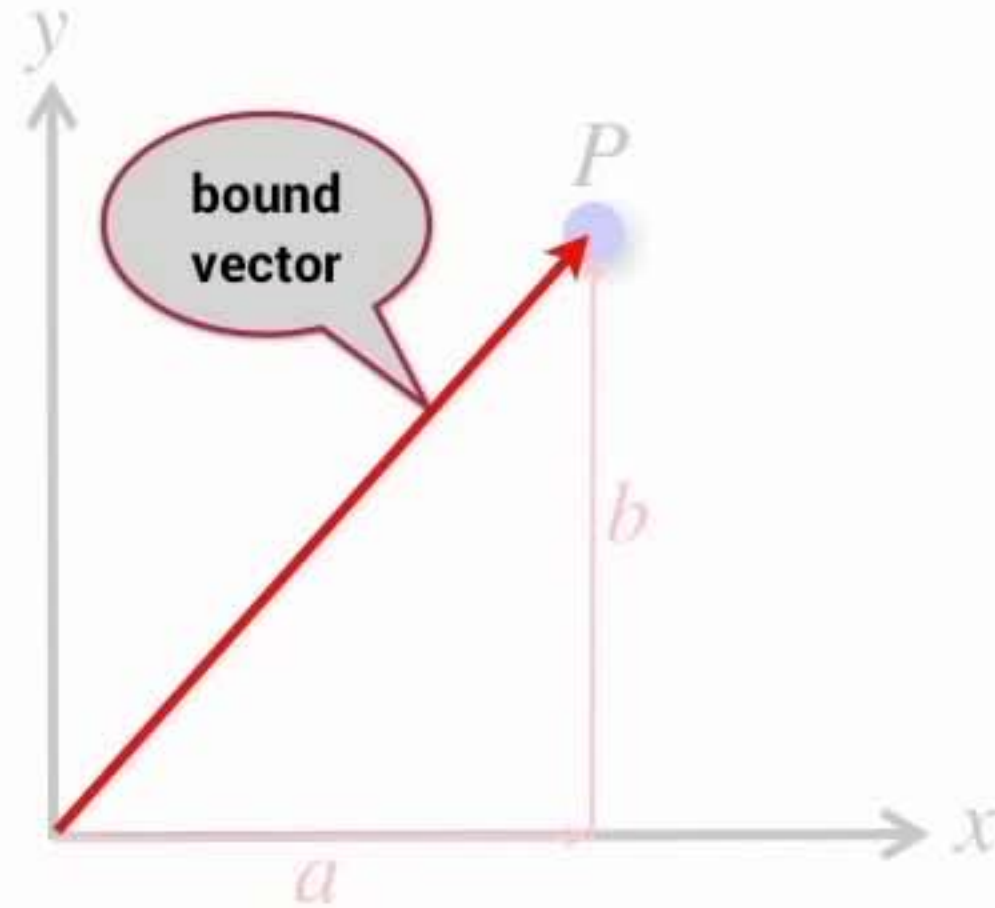
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- Represented by an n -element column vector $[x_1, x_2 \cdots x_n]^T$
- Can be considered a relative displacement

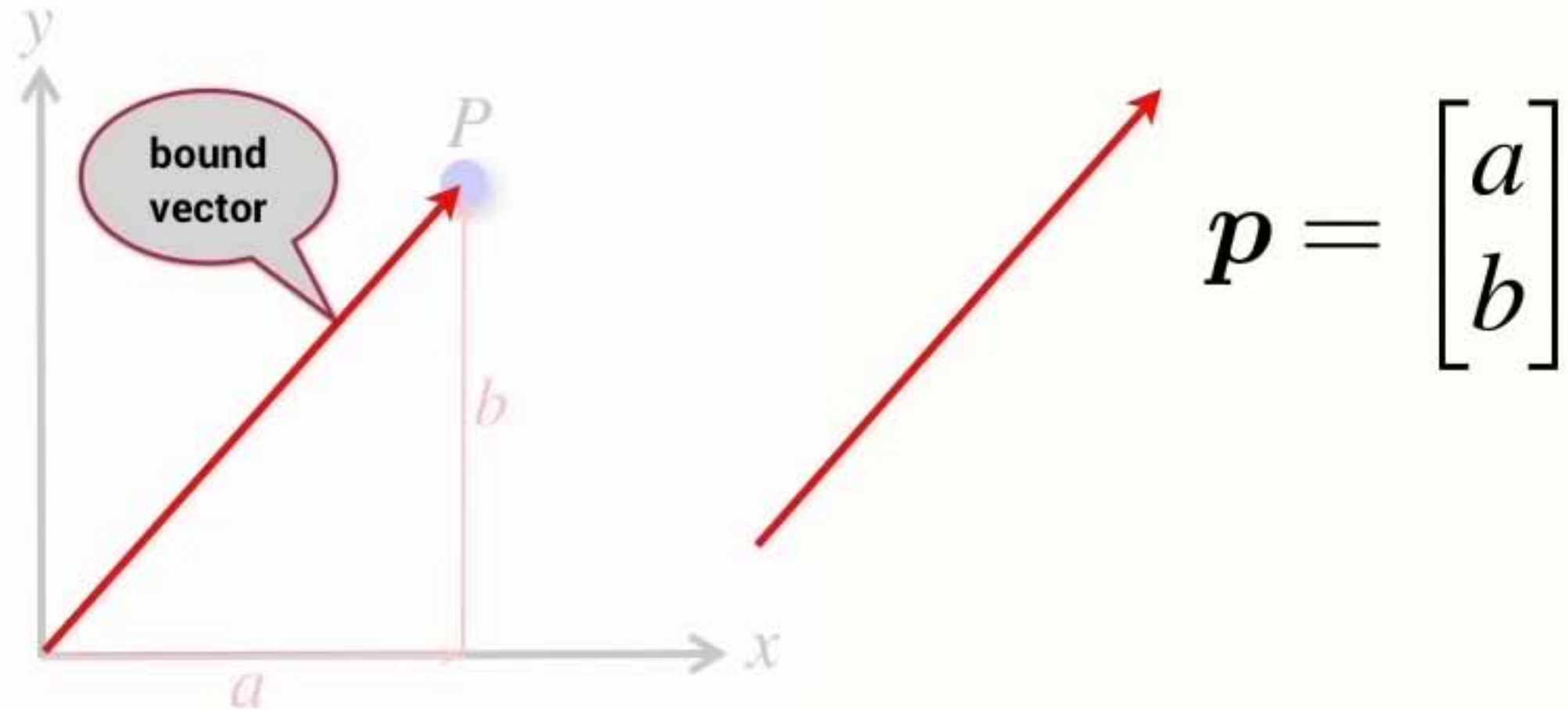
Vectors



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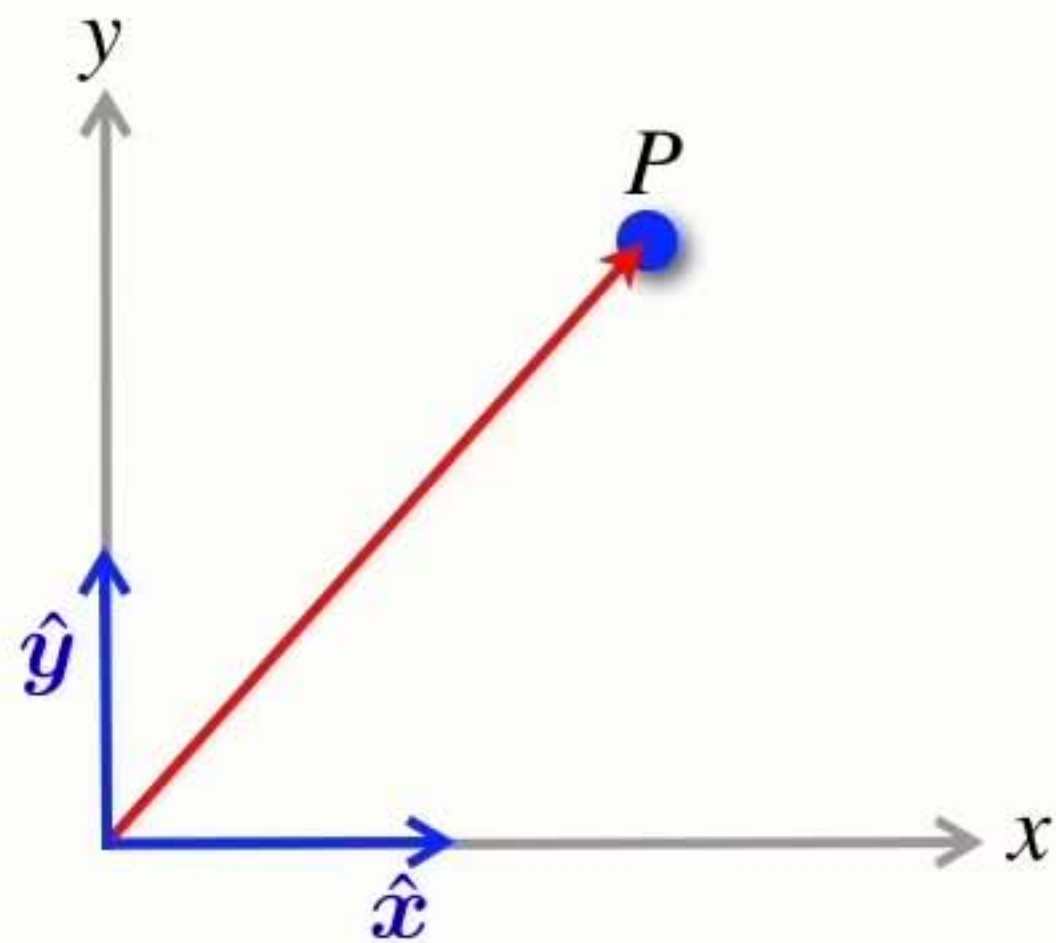
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- The vector has a particular starting point

Vectors



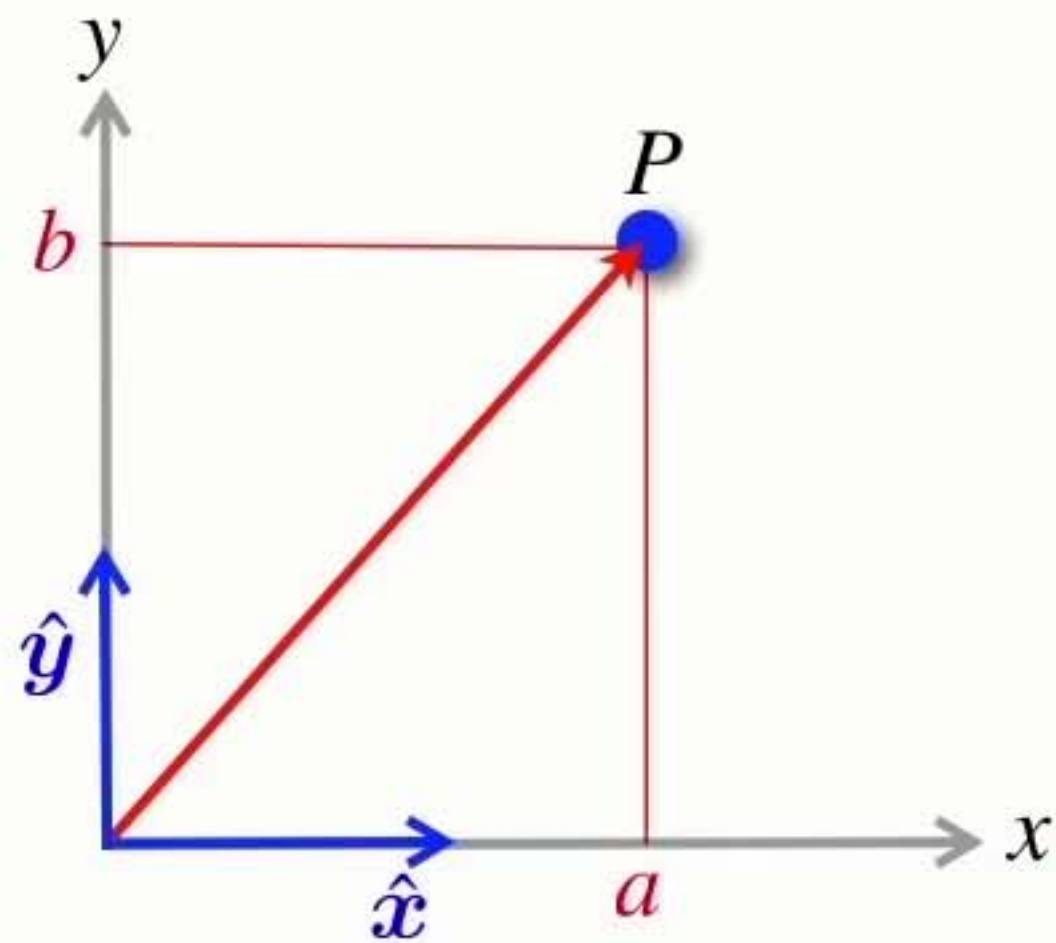
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Vectors



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Vectors



$$p = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$p = a\hat{x} + b\hat{y}$$

Points and vectors

- Points

- ➡ Define a location

- ➡ Cannot add or multiply points

Points and vectors

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- ➡ Define a location
- ➡ Cannot add or multiply points

- Vectors

- ➡ *Do not* define a location
 - specify how to get from one location to another
- ➡ Can add & subtract vectors
- ➡ Can multiply a vector by a scalar

Points and vectors

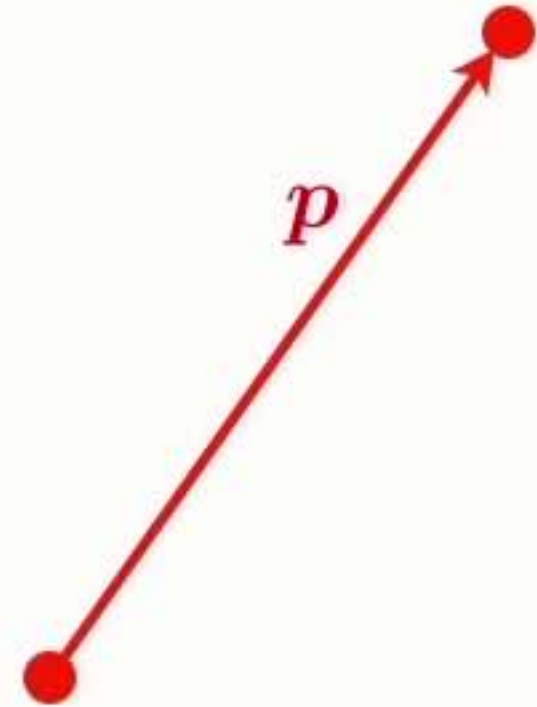
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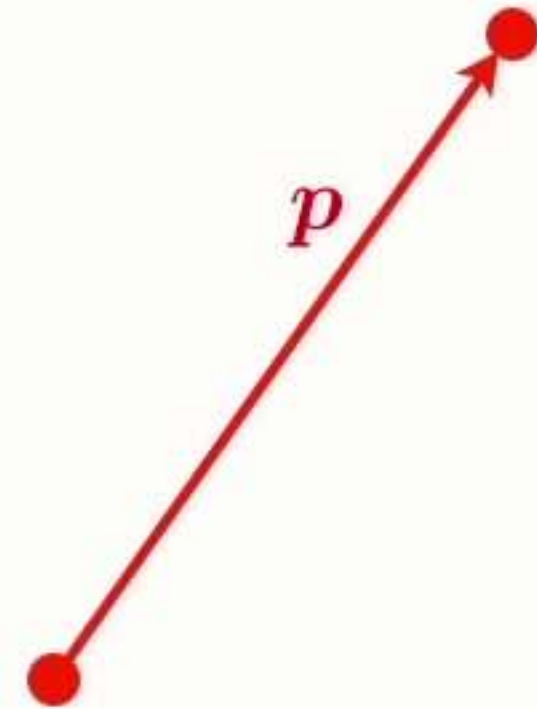
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- A point is defined by a vector displacement from another point (typically the origin of a coordinate frame)



Points and vectors

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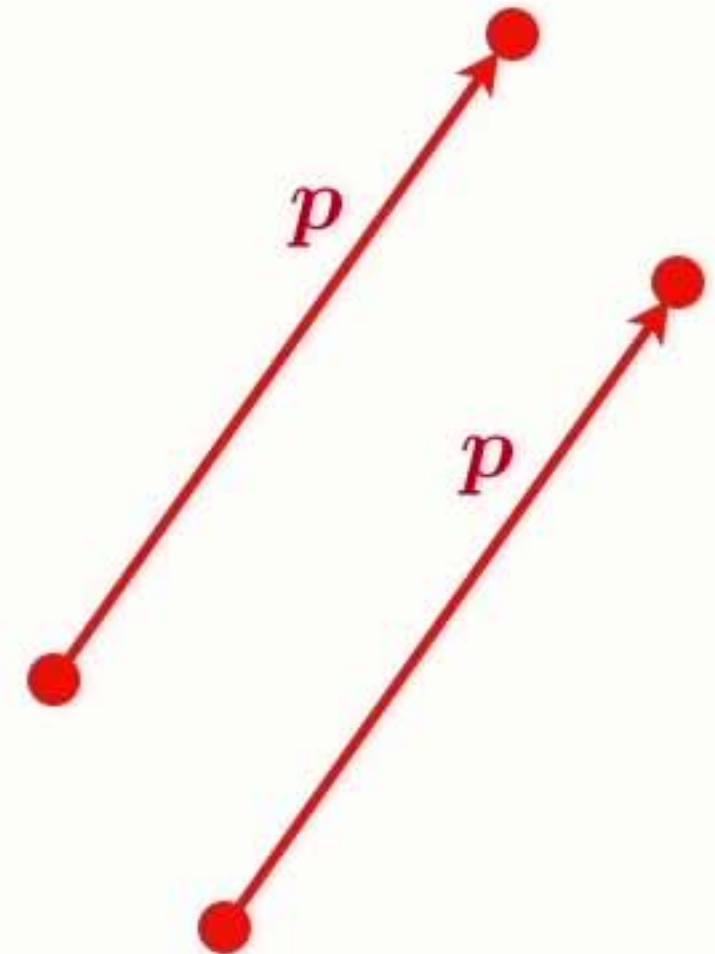
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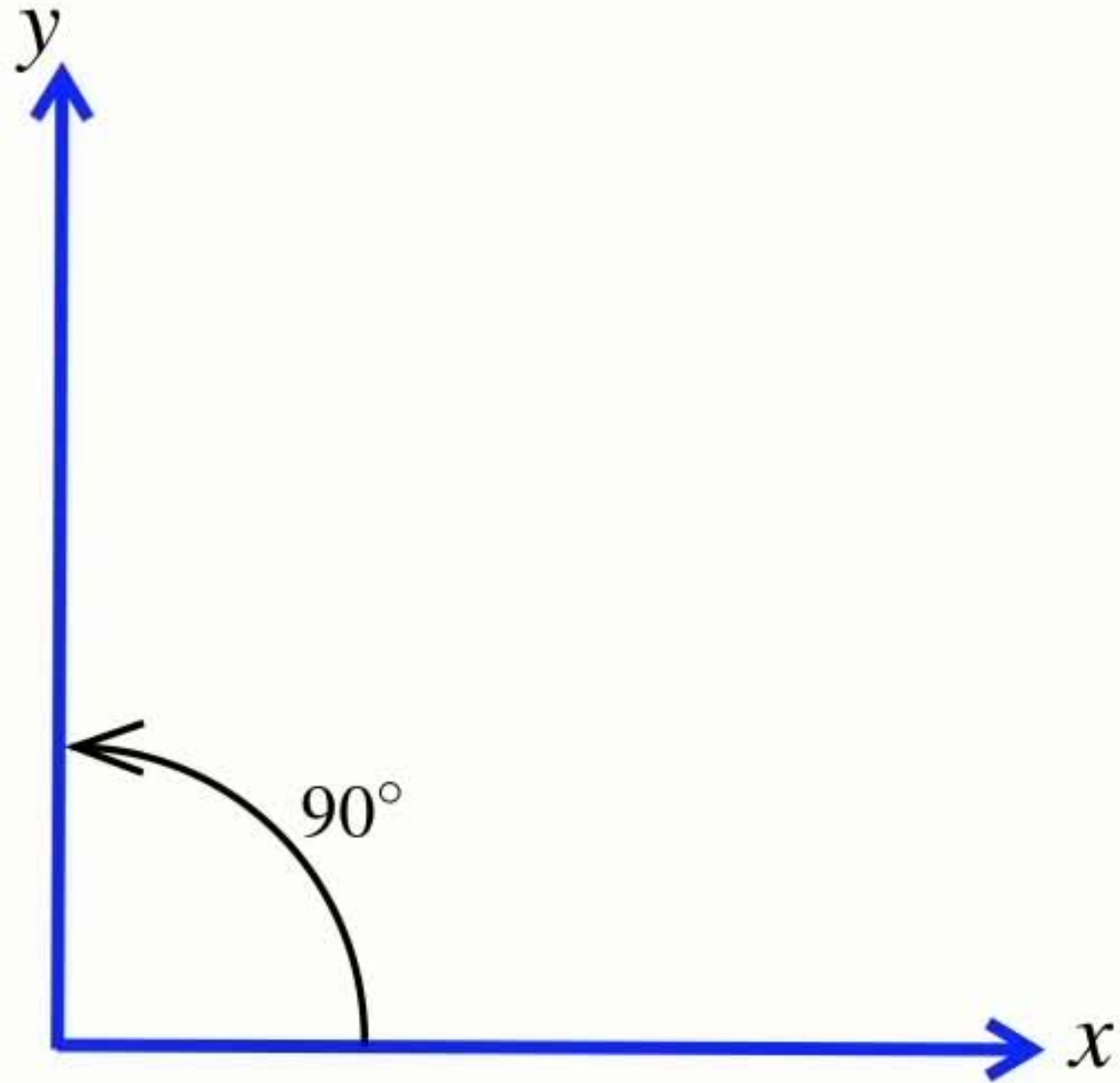
- Both can be represented by n real numbers



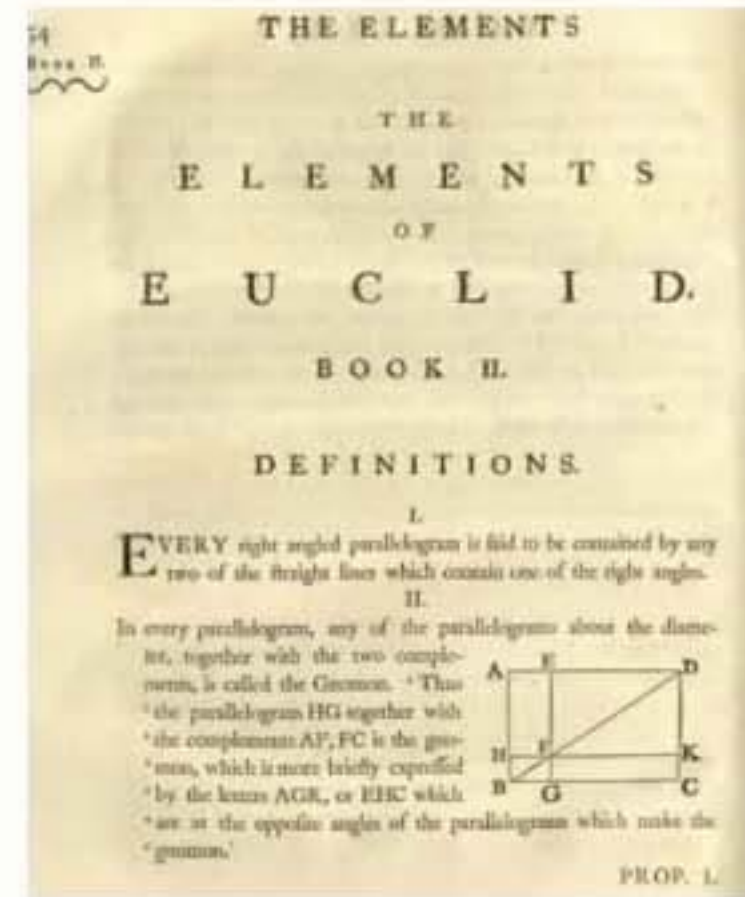
Right-handed **coordinate frame**



Right-handed **coordinate frame**

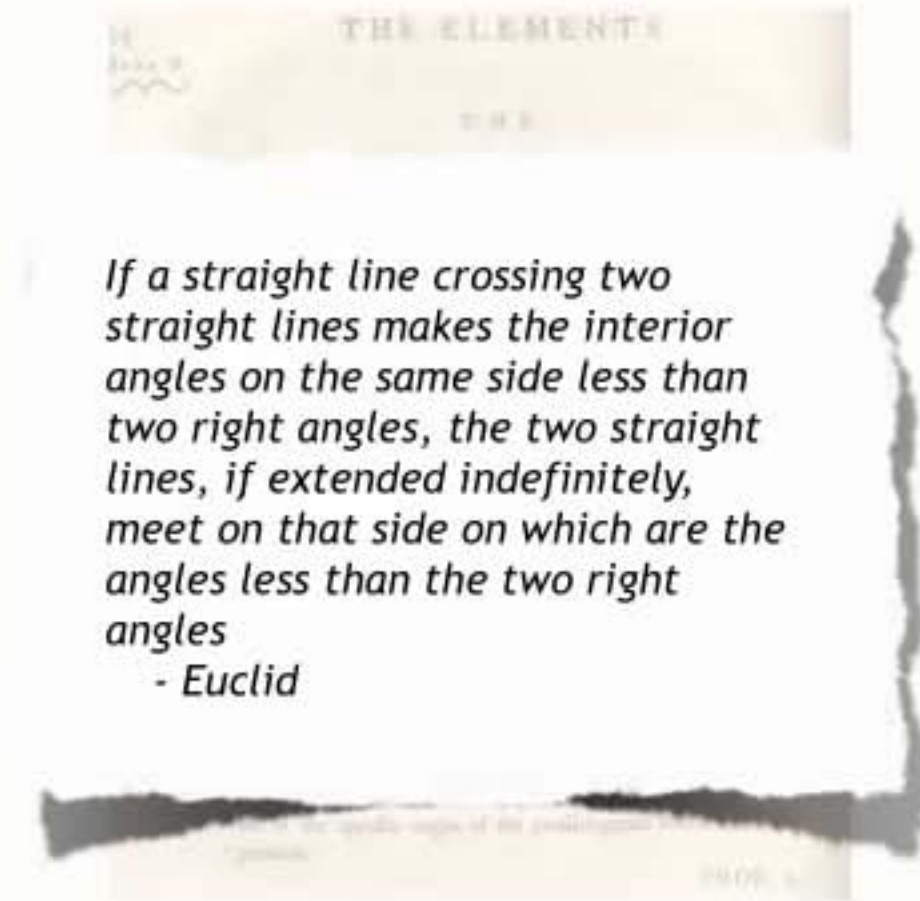


Euclidean geometry **applies to planes**



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- Euclid's geometry (postulate 5)
 - ➔ parallel lines never intersect



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 - ➔ it is a non-Euclidean geometry



Euclidean geometry **applies to planes**

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 - ➔ parallel lines never intersect
- On the surface of our planet, parallel lines at the equator will intersect at the poles
- Euclidean geometry does not apply
 - ➔ it is a non-Euclidean geometry
- Euclidean geometry is a good approximation over small areas



Section 2

Position and pose

Pose

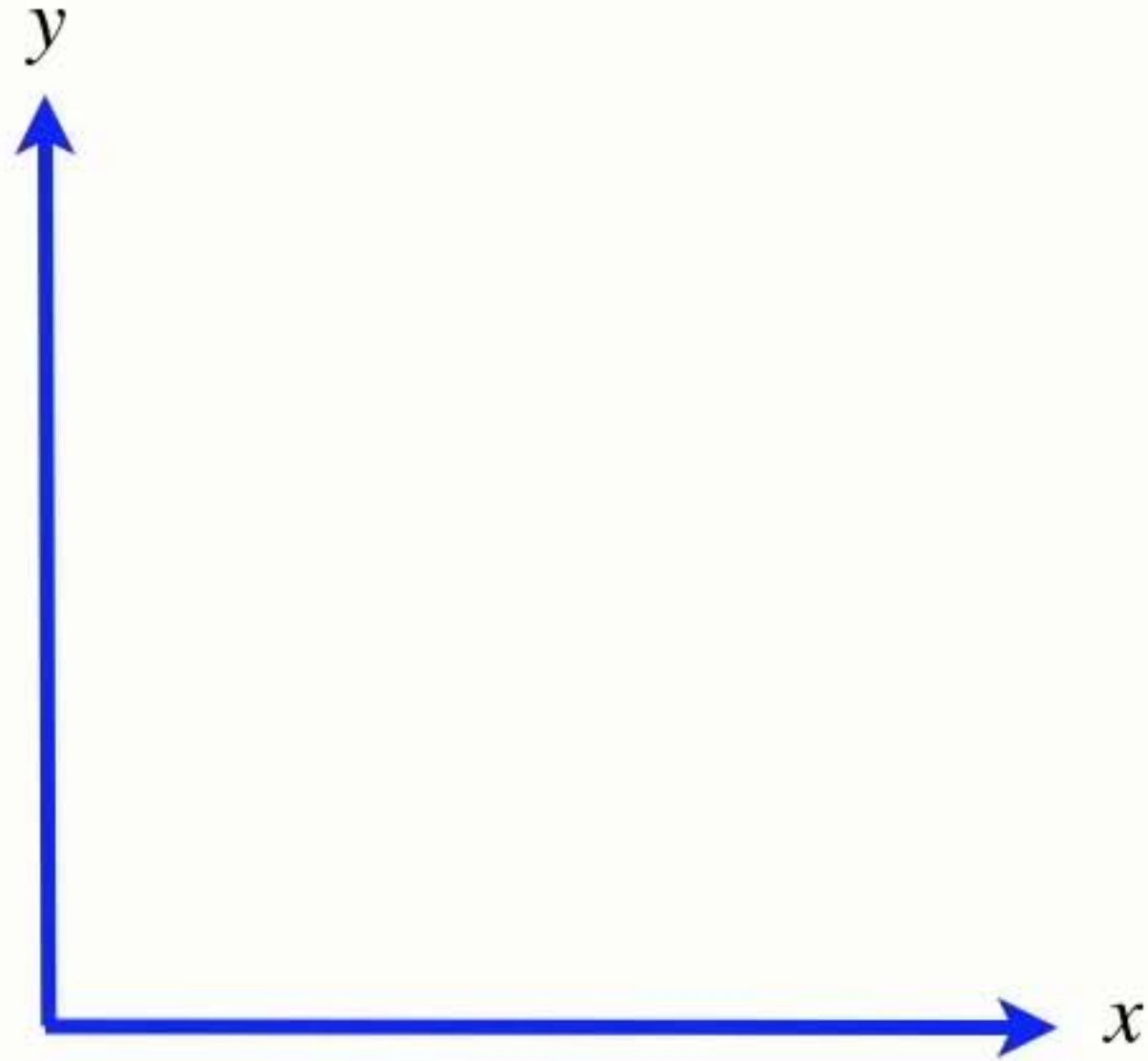
- It has 3 parameters:



pronounced ksi

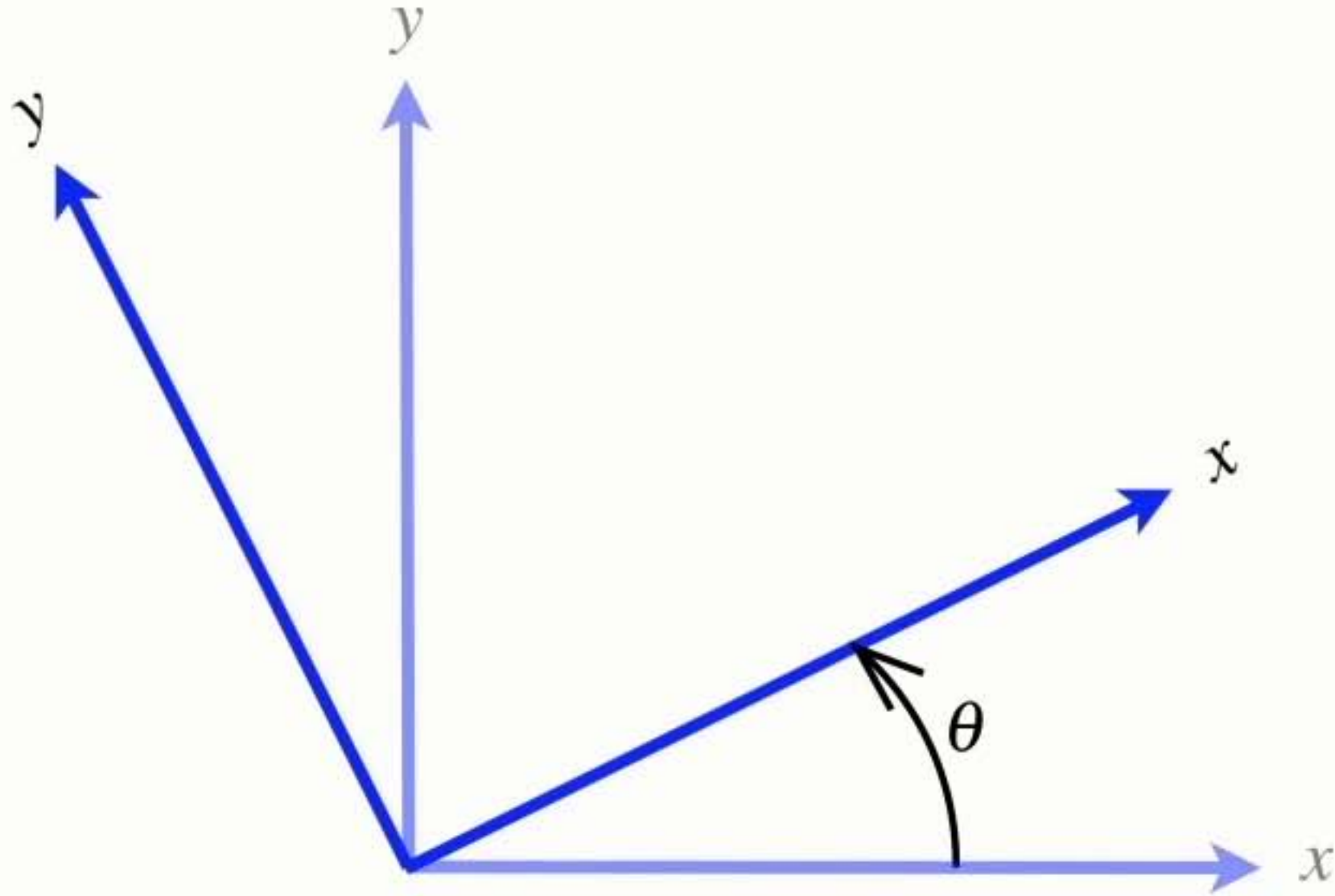
(x, y, θ)

Coordinate frame rotation convention



- Angles increase positively in the anti-clockwise direction

Coordinate frame rotation convention



- Angles increase positively in the anti-clockwise direction

Pose

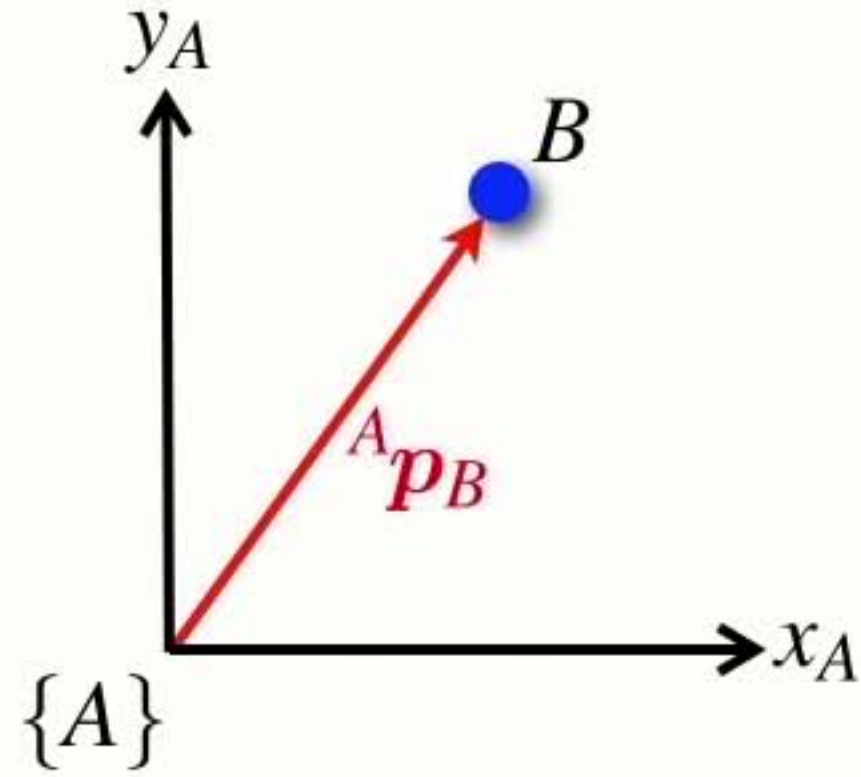
- It has 3 parameters:
- Think of it as just a move (x, y) and a twist (θ)



(x, y, θ)

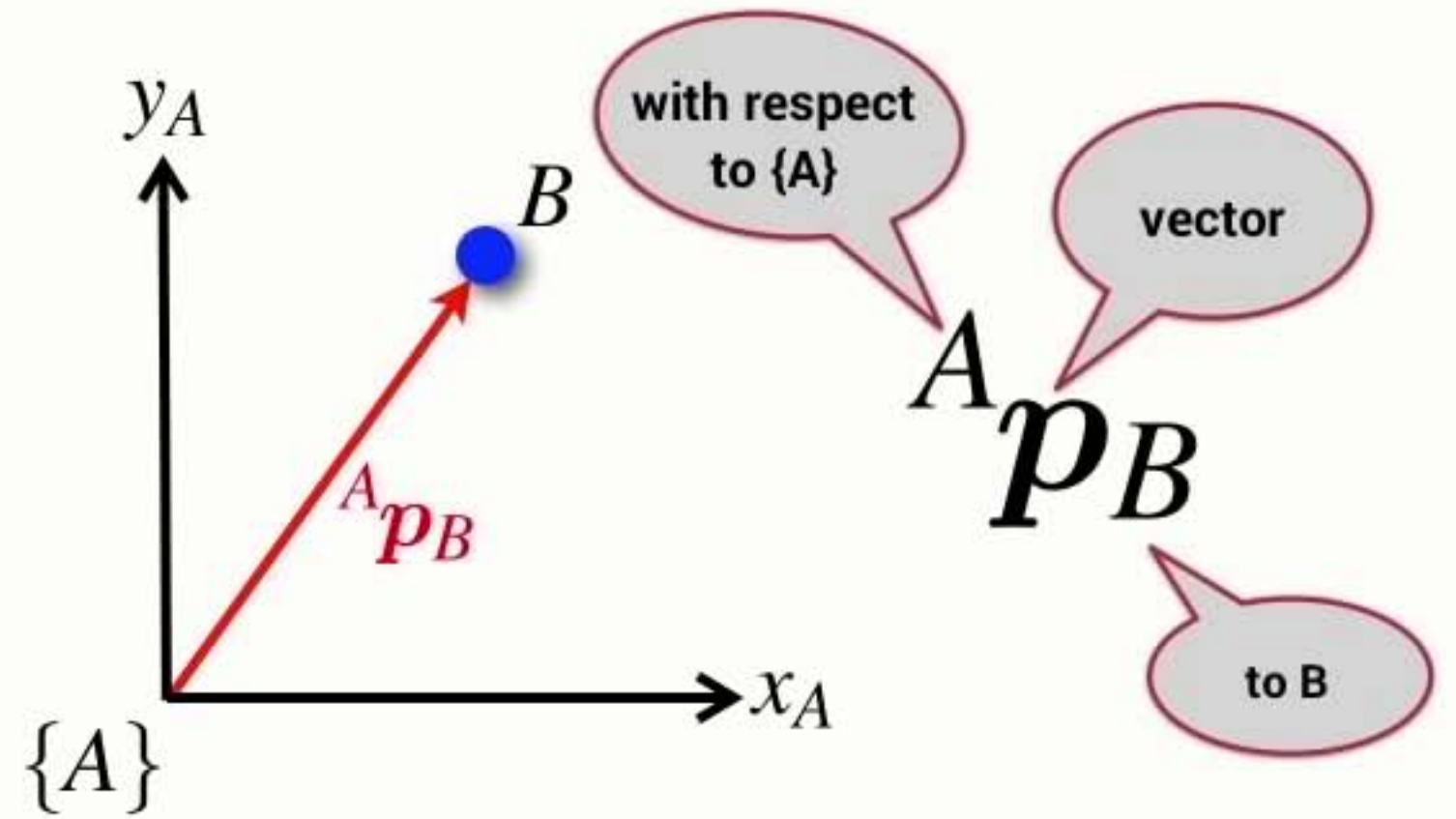
Summary

- A point is described by a vector with respect to a coordinate frame



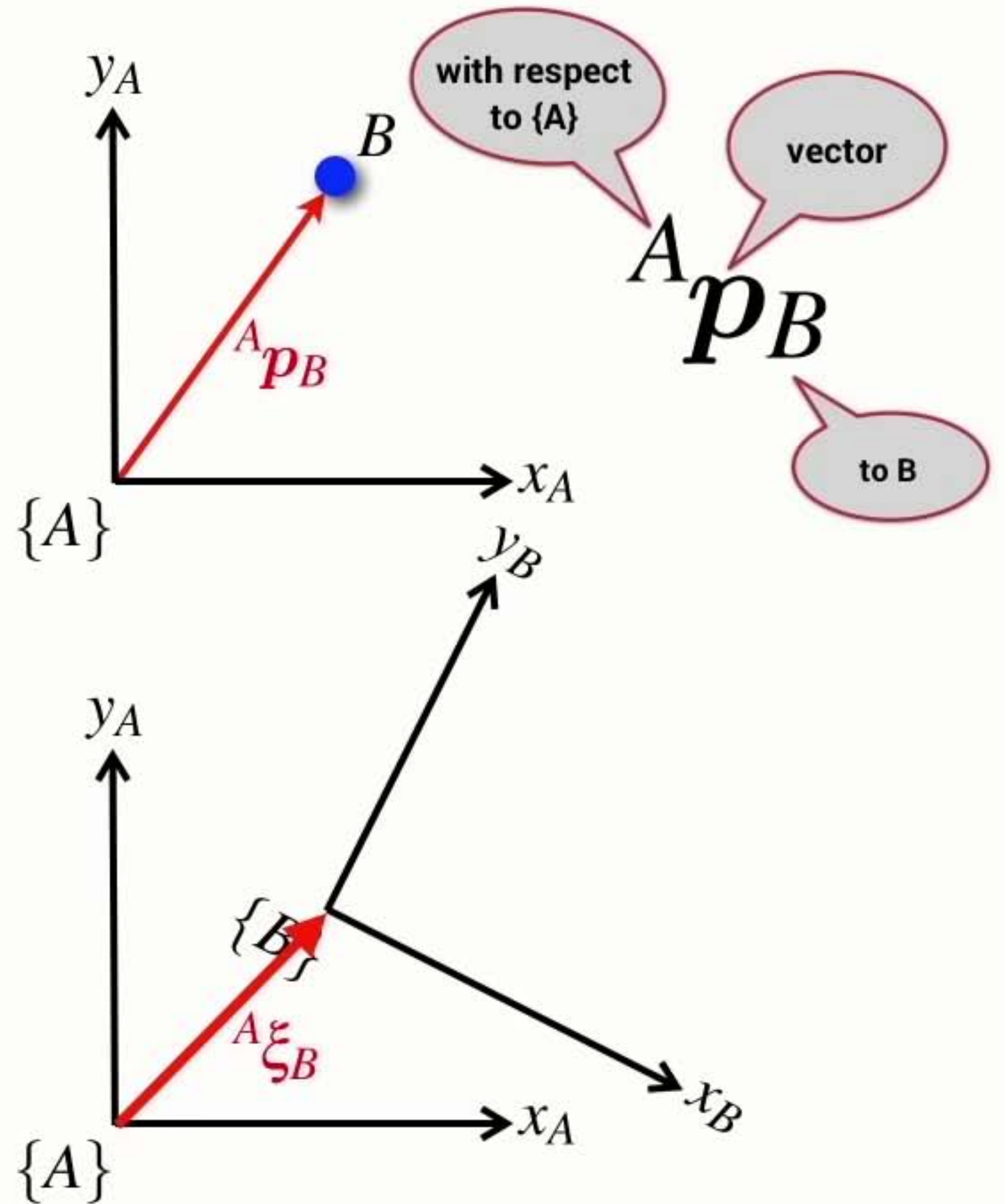
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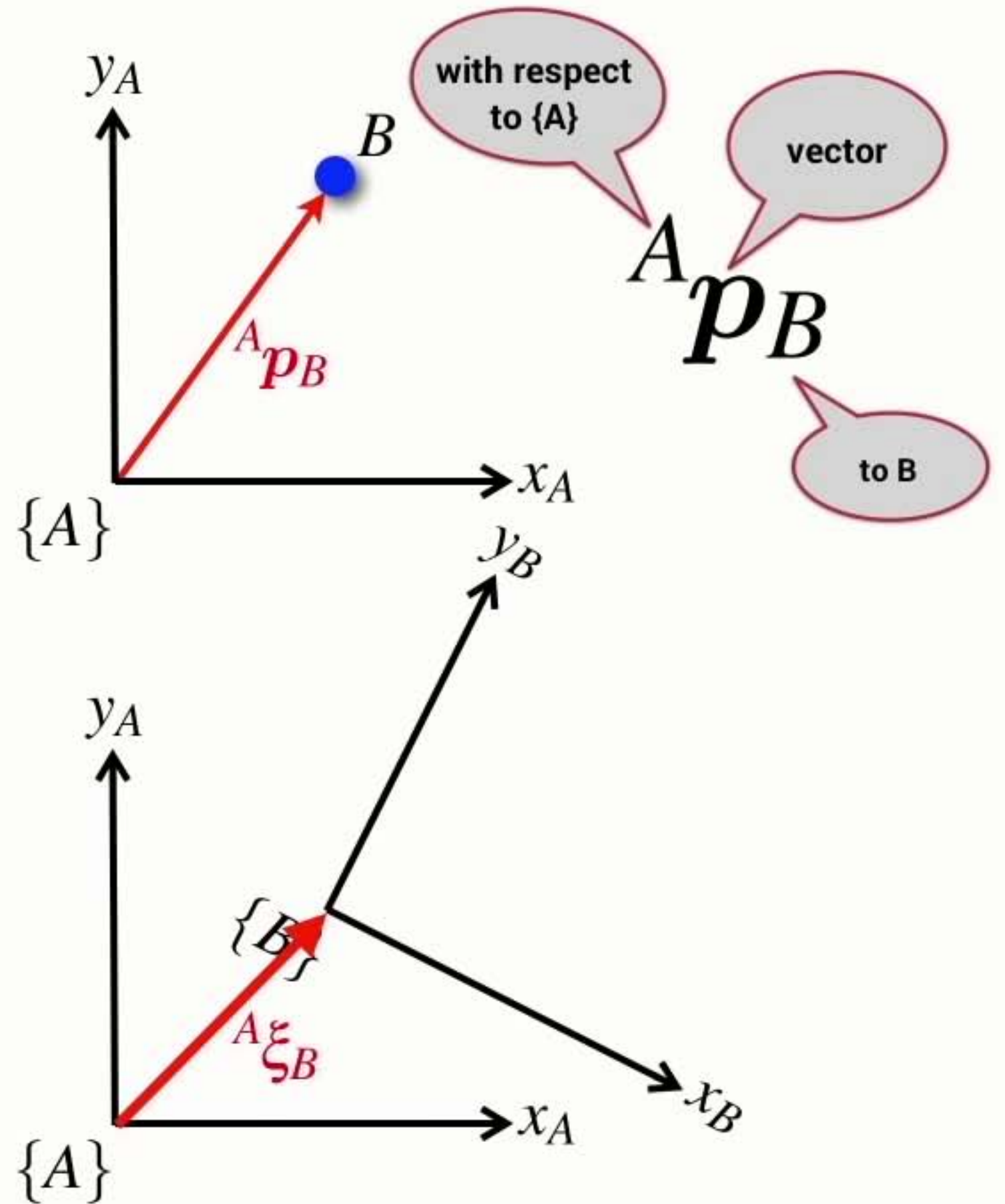
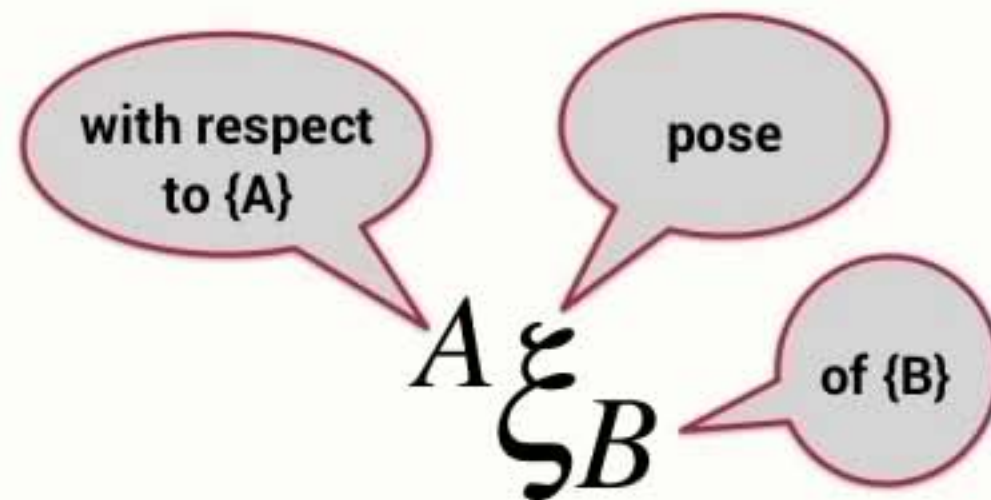
Summary

- A point is described by a vector with respect to a coordinate frame
- The pose of a coordinate frame can be described with respect to another coordinate frame



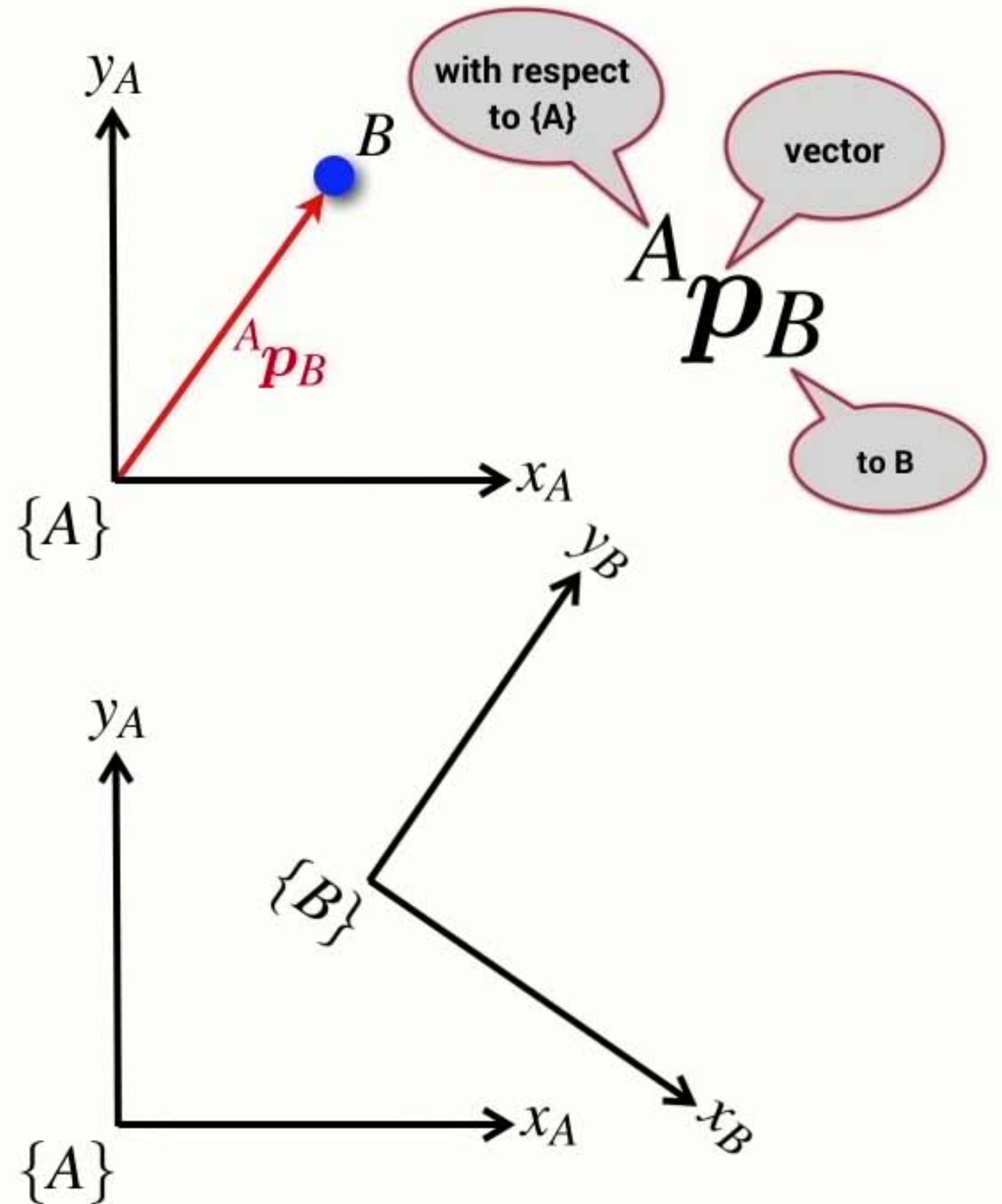
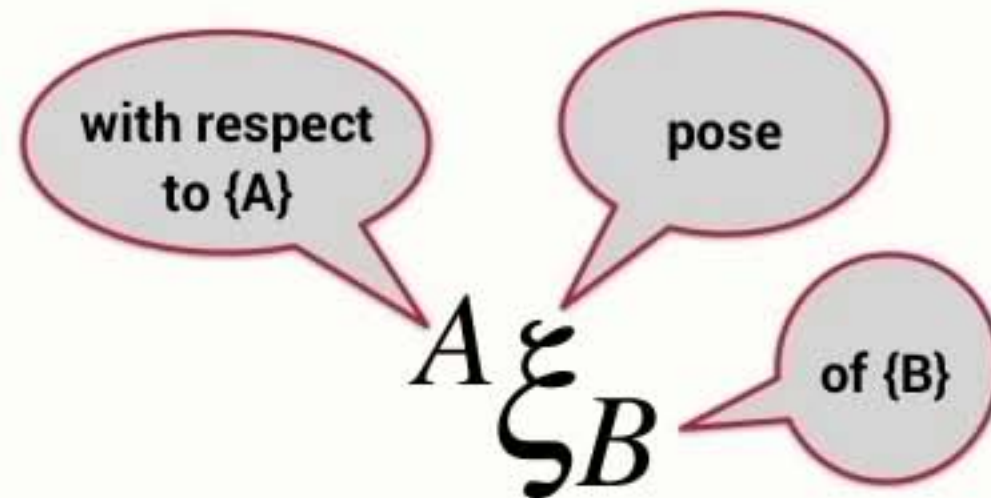
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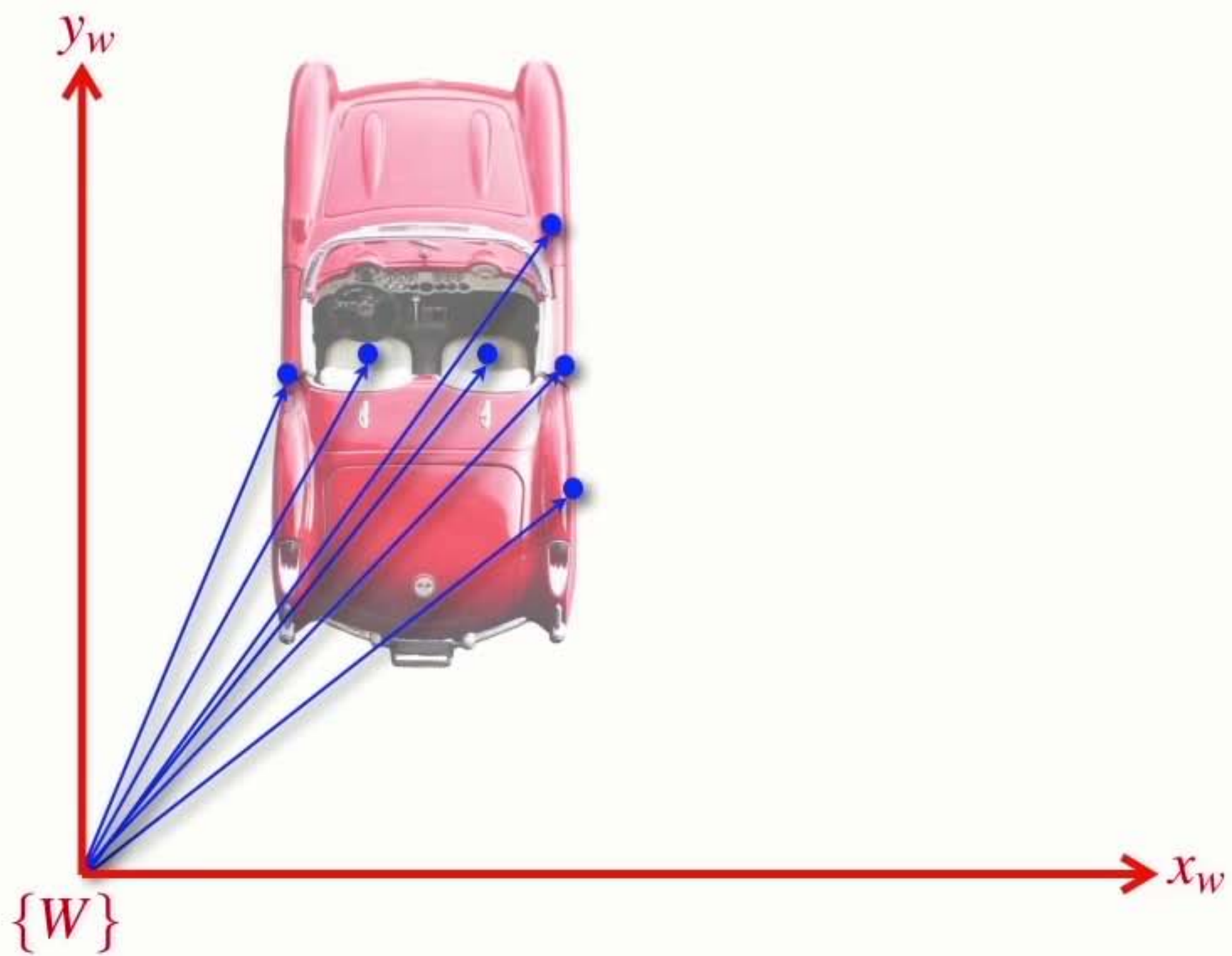
- A point is described by a vector with respect to a coordinate frame
- The pose of a coordinate frame can be described with respect to another coordinate frame
- Pose can be considered as the motion of a coordinate frame
 - ➔ a translation and a rotation

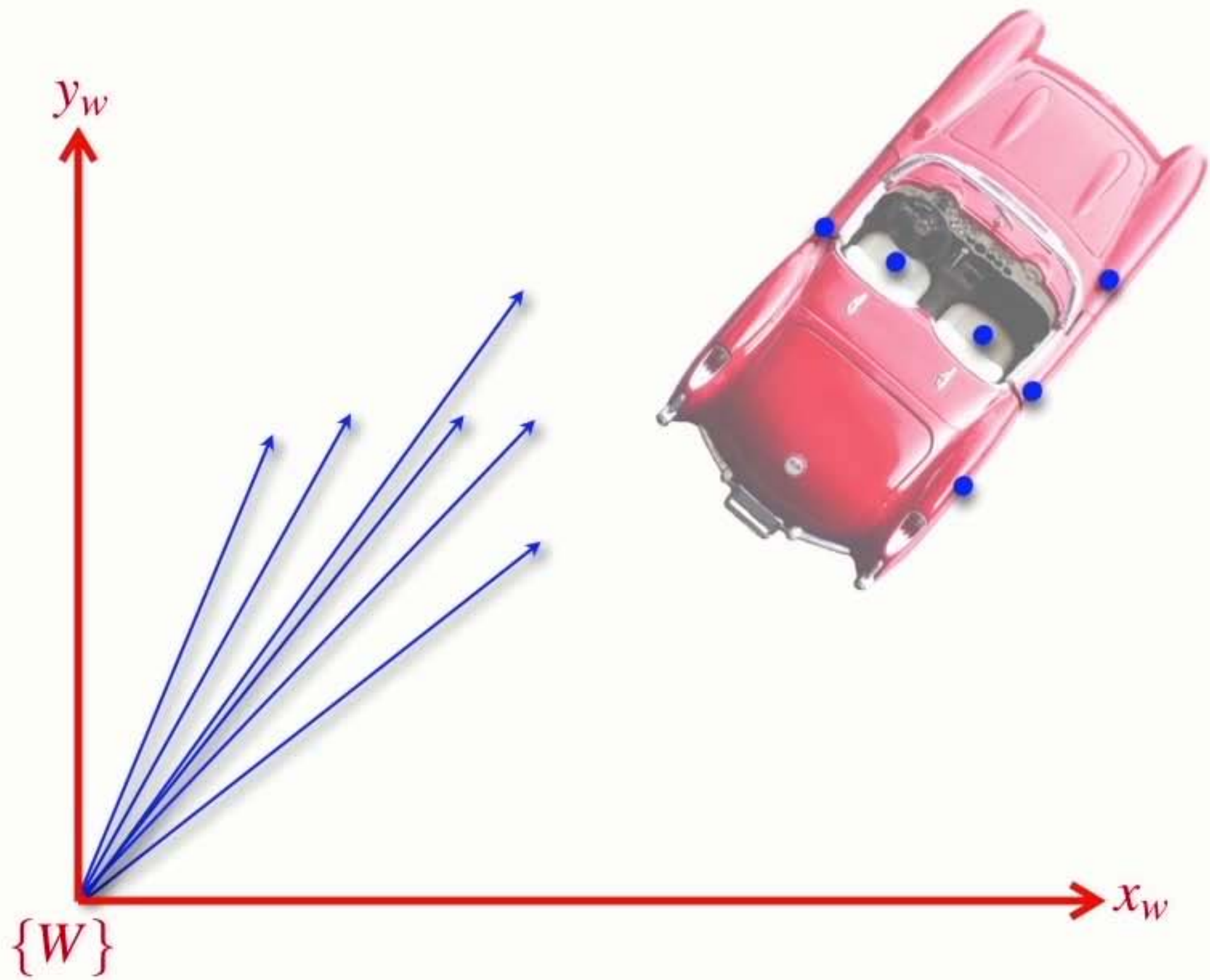


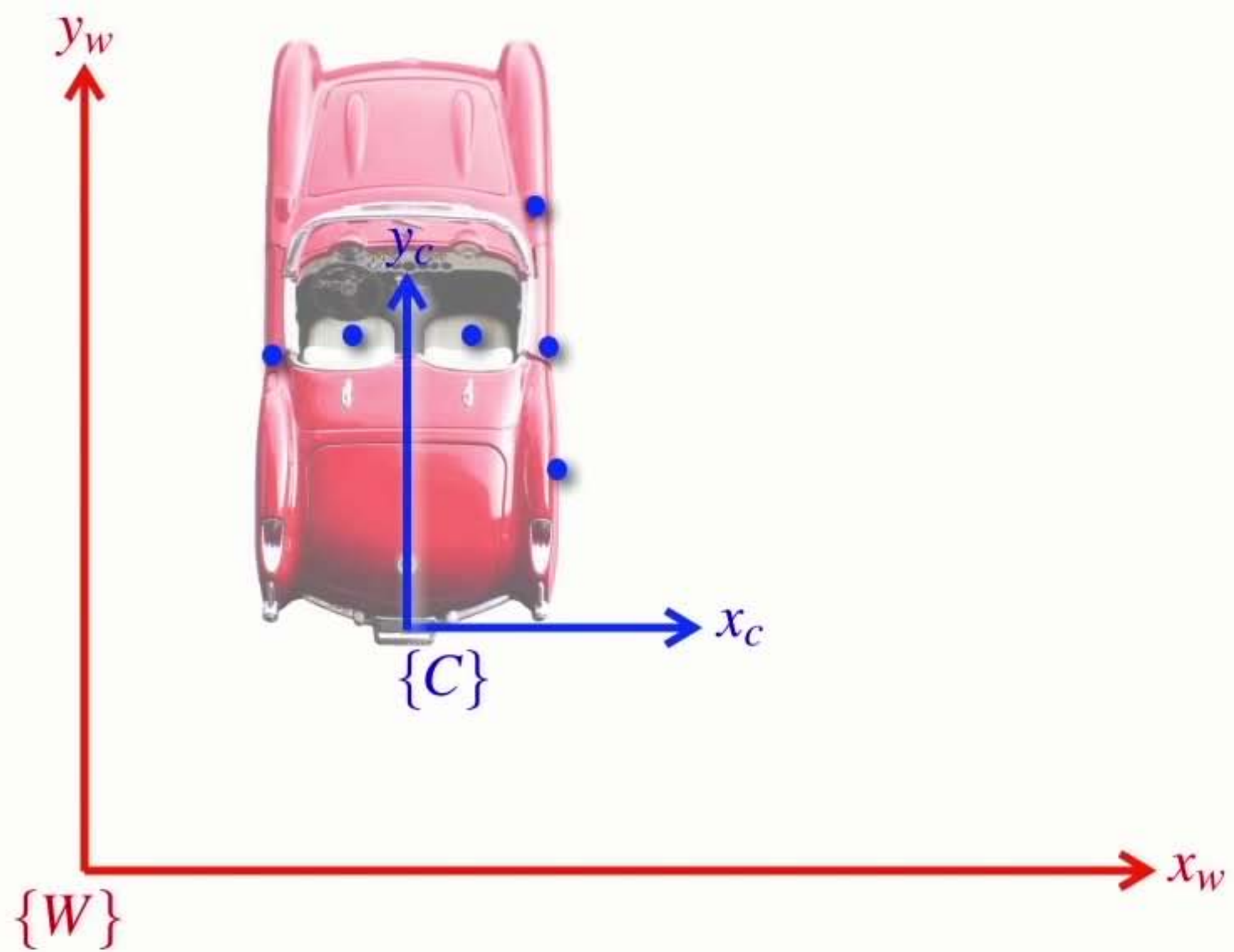
Section3 Relative positions

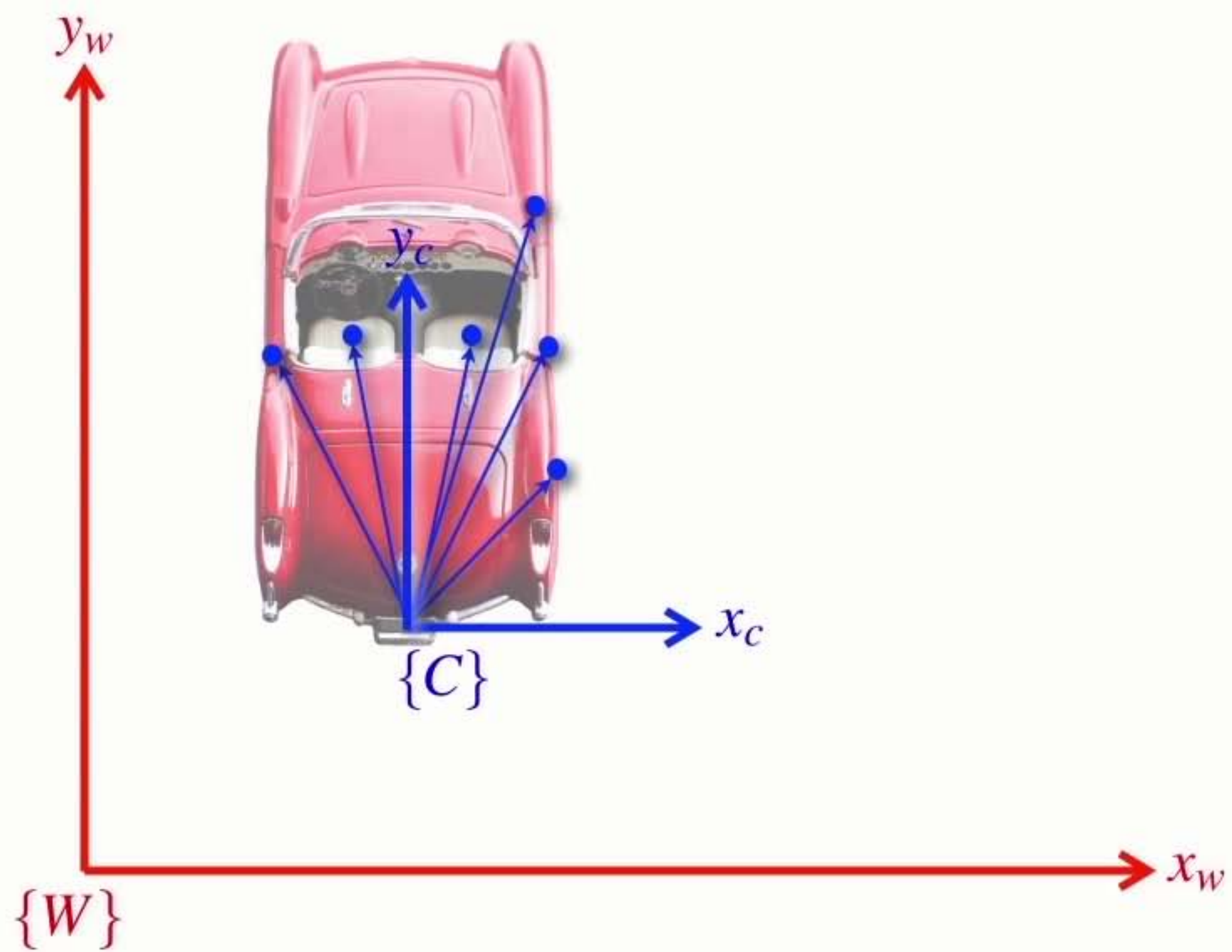


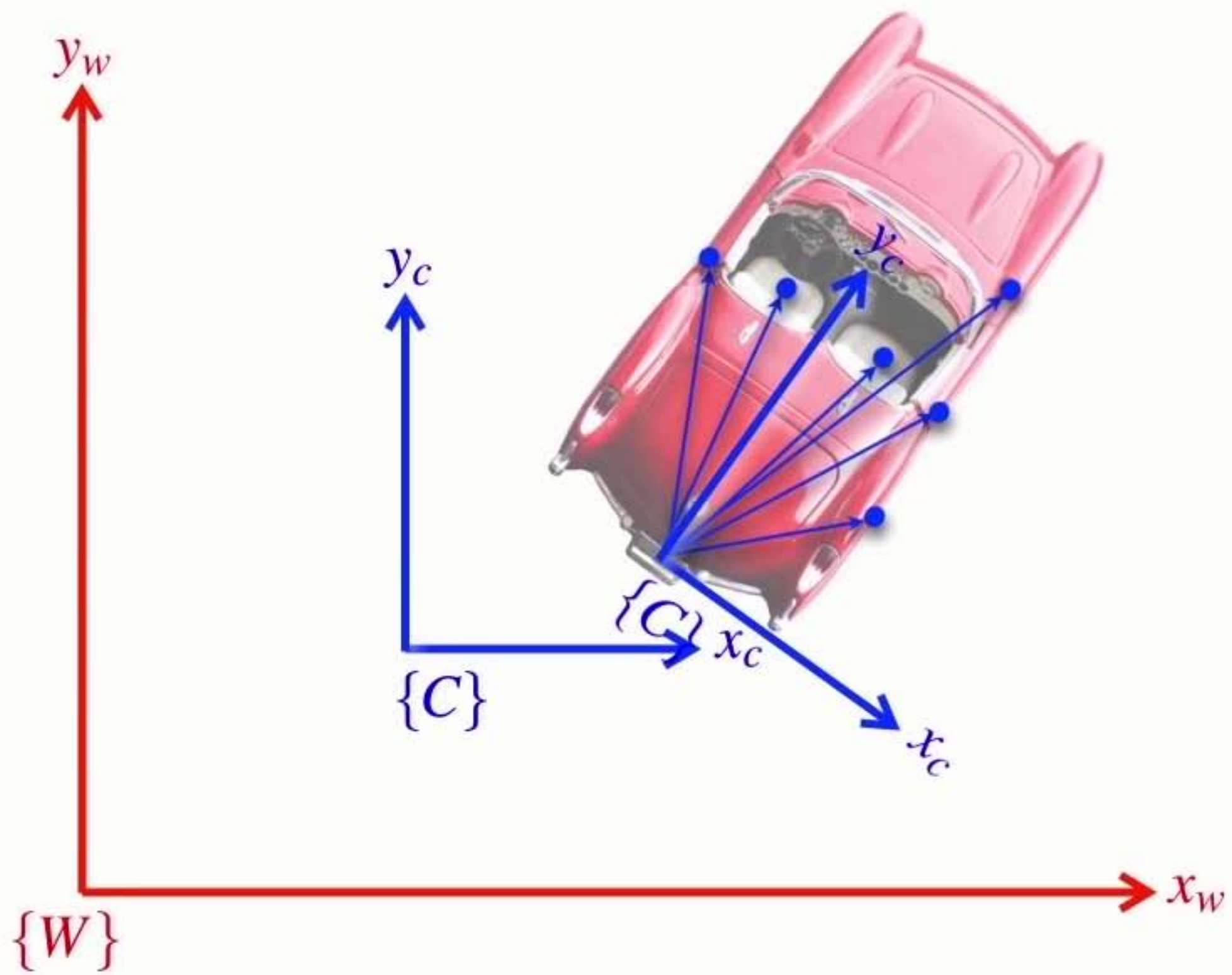


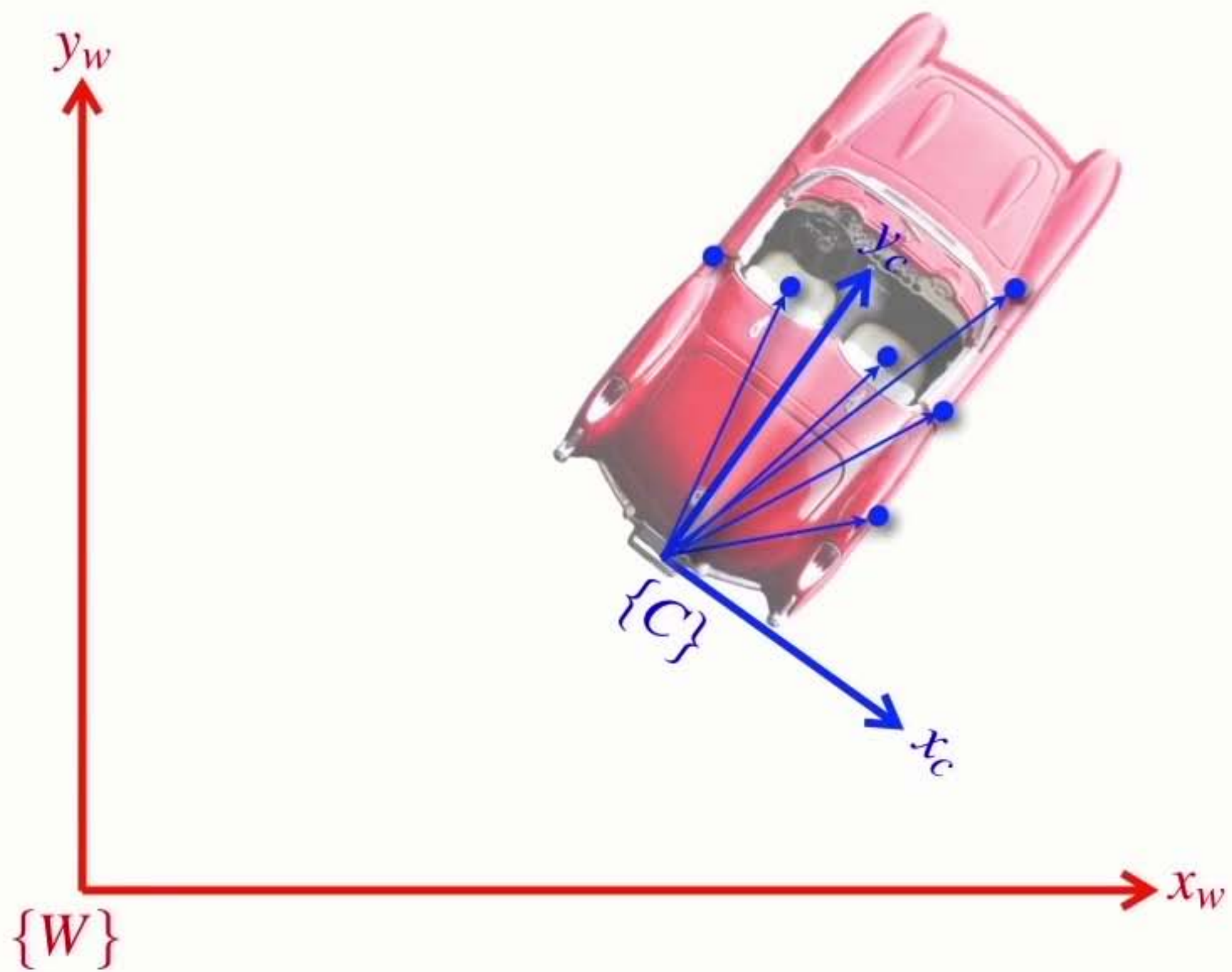


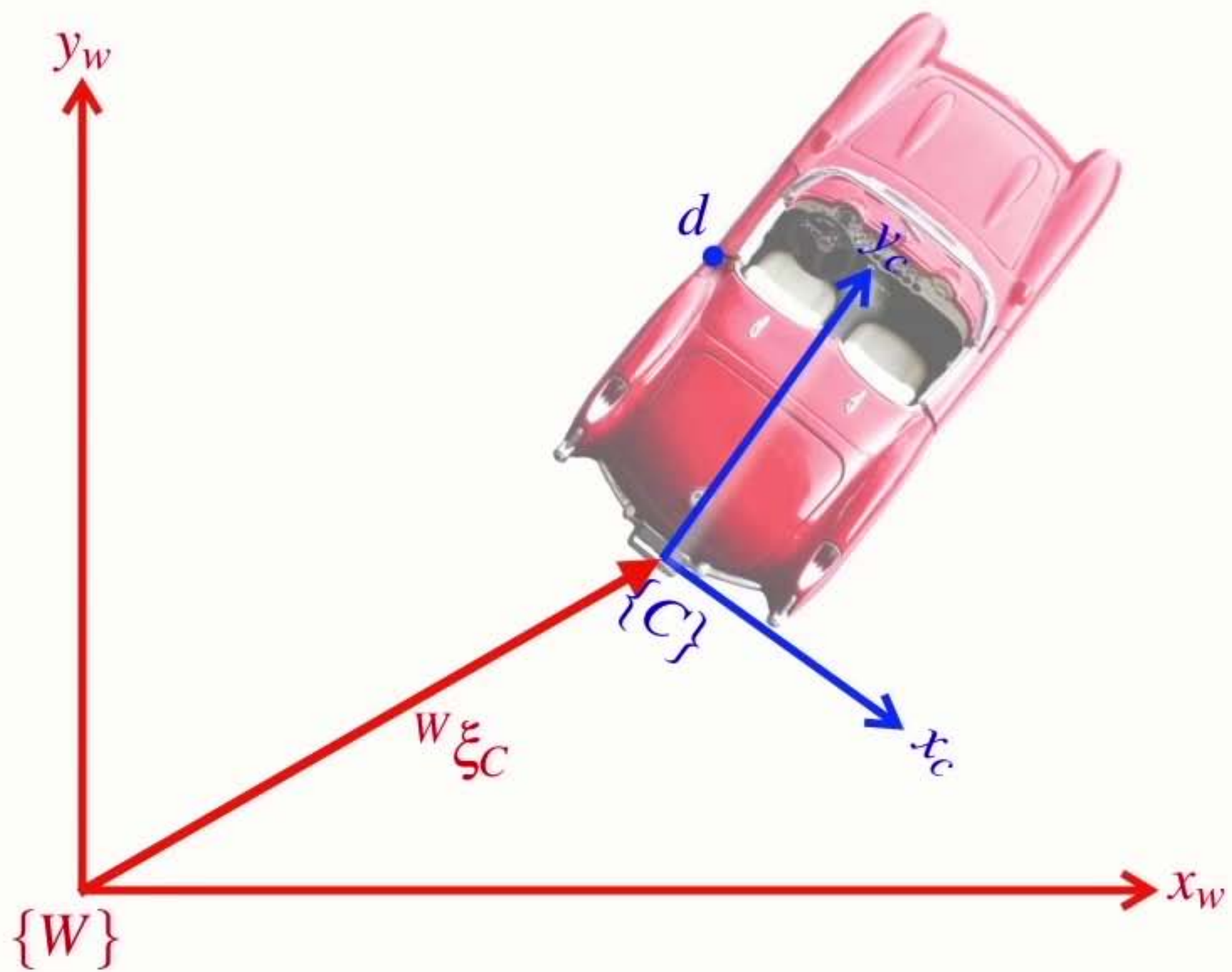


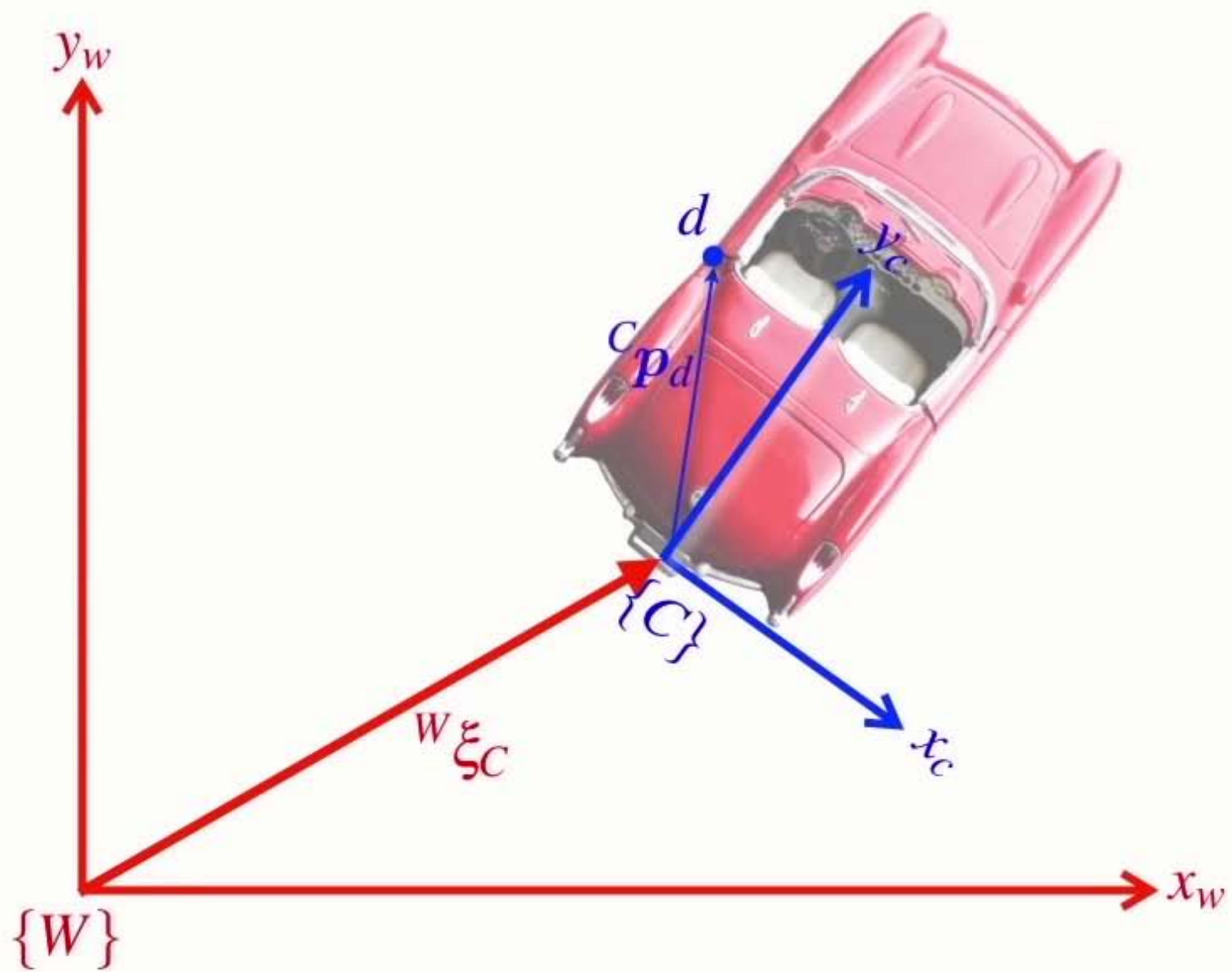


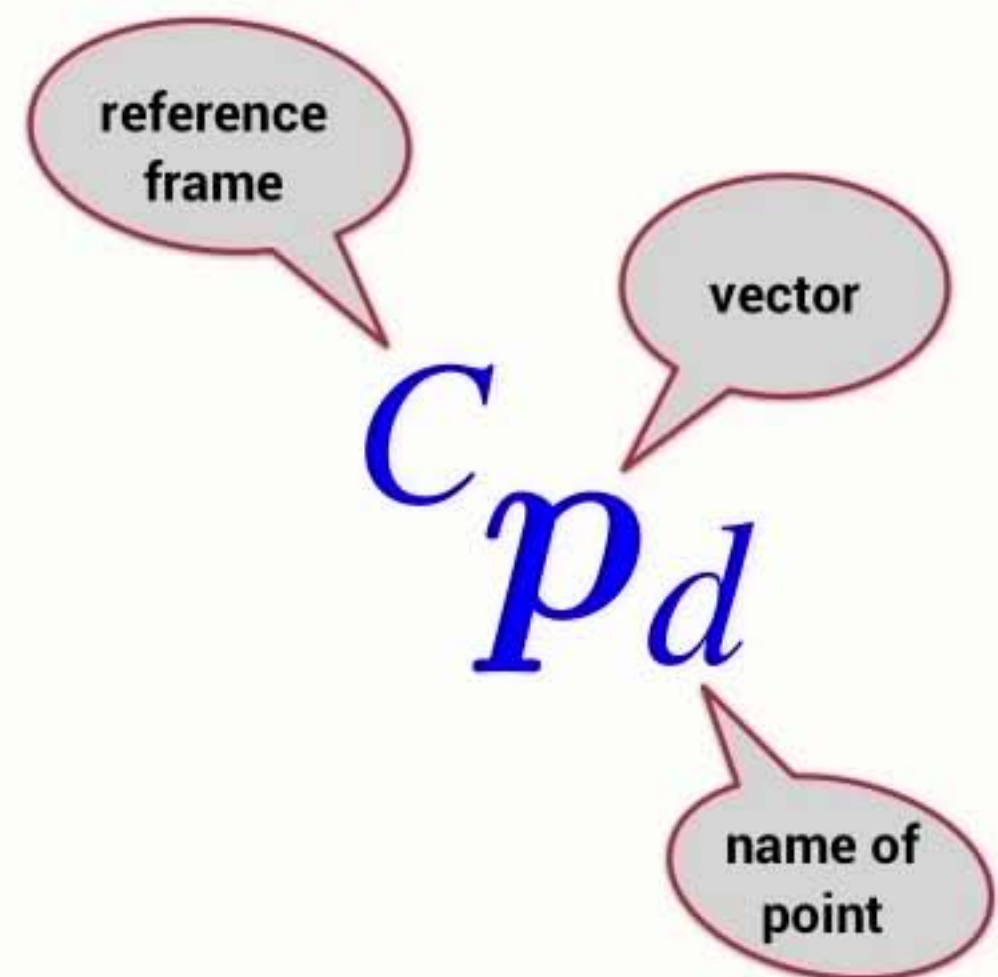
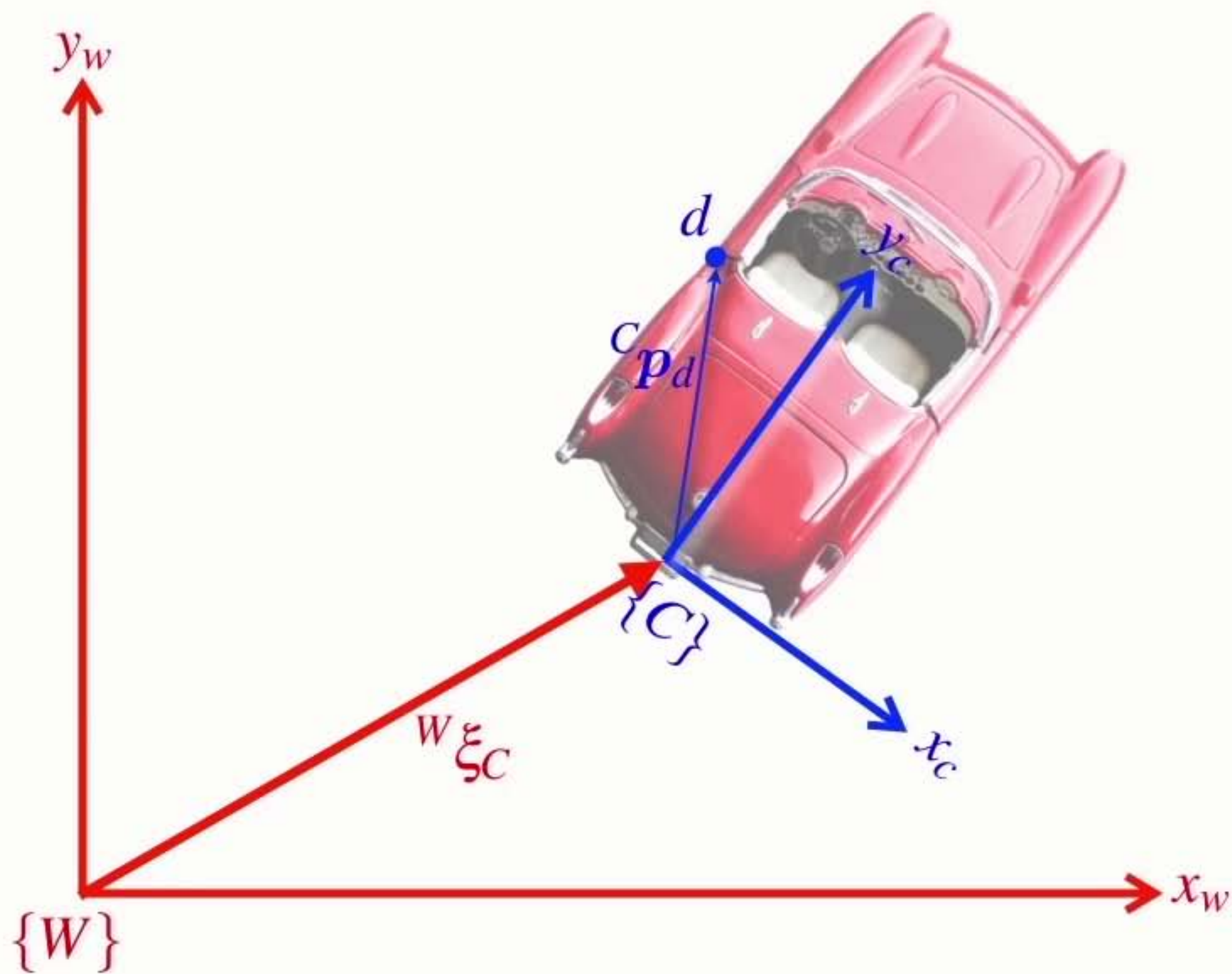


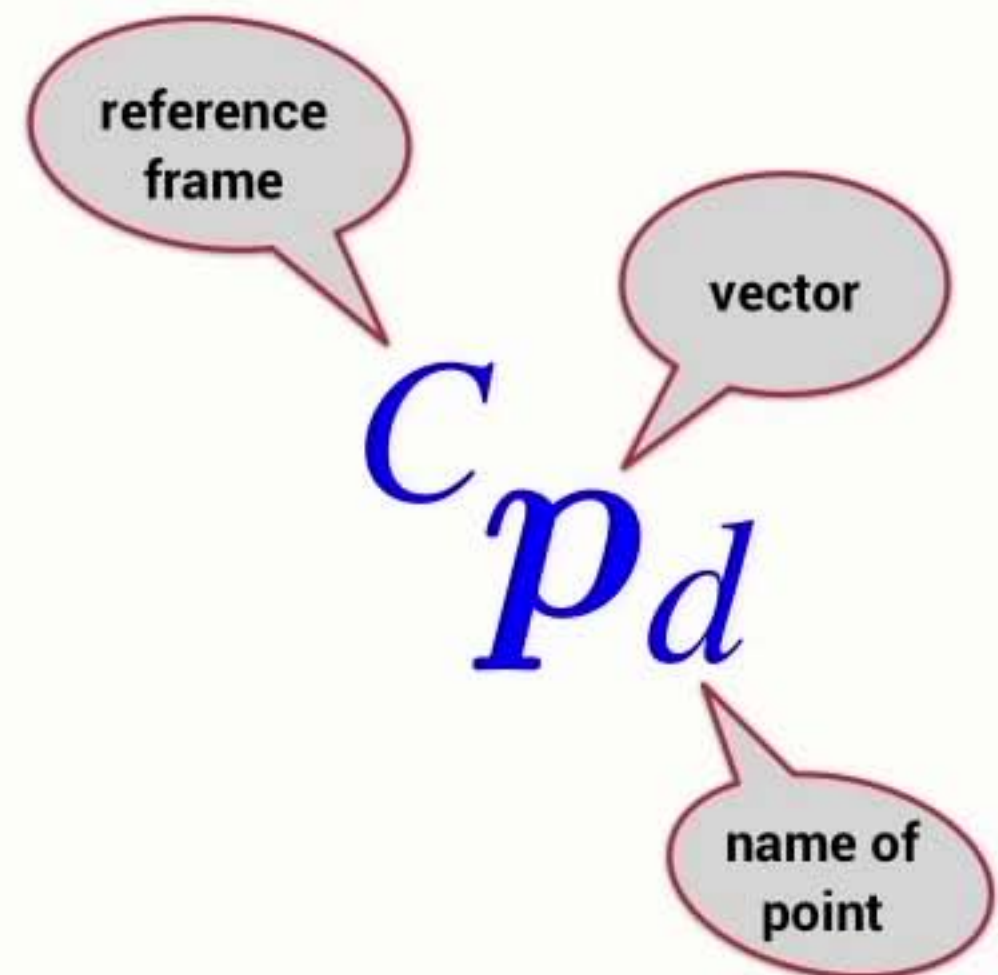
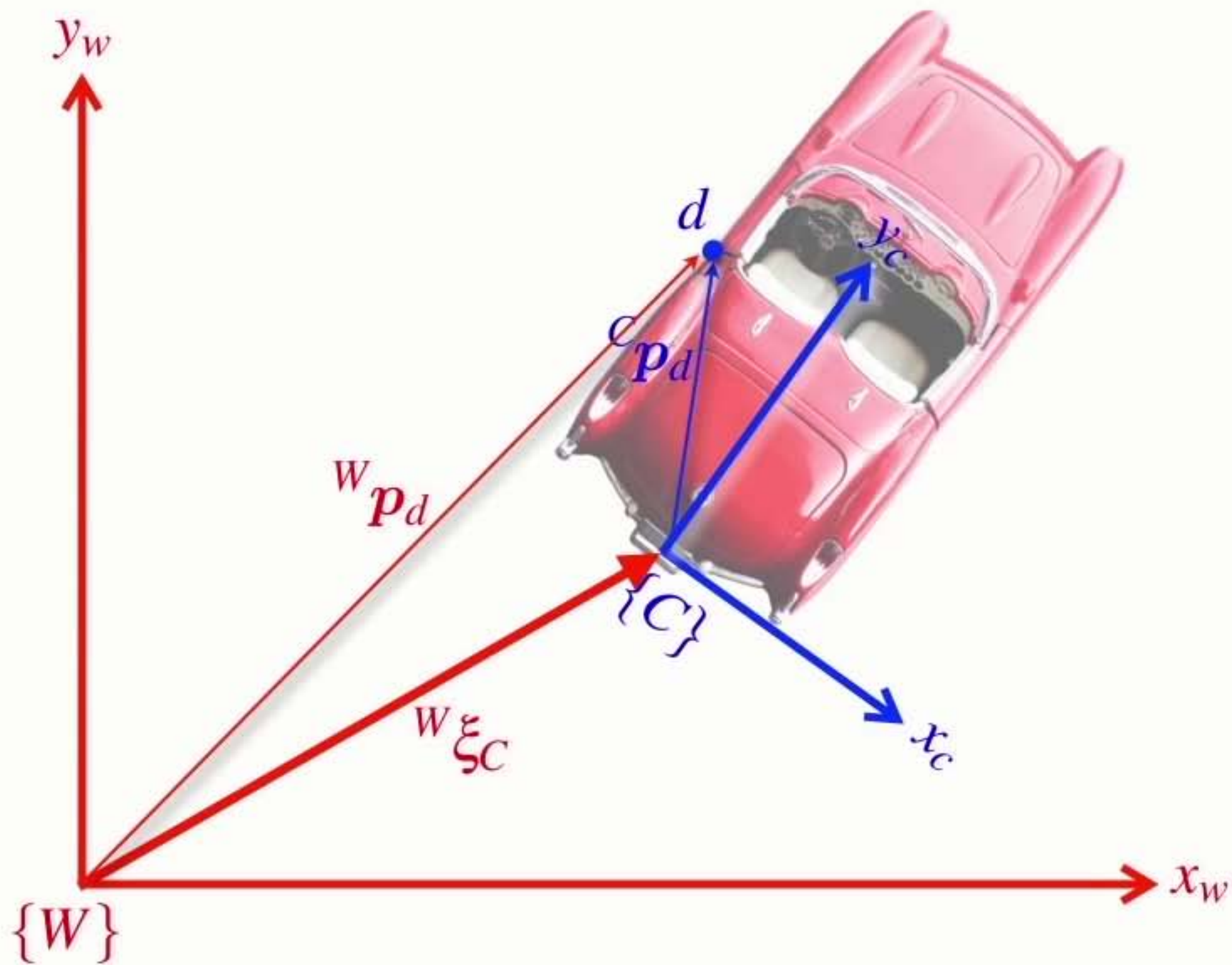


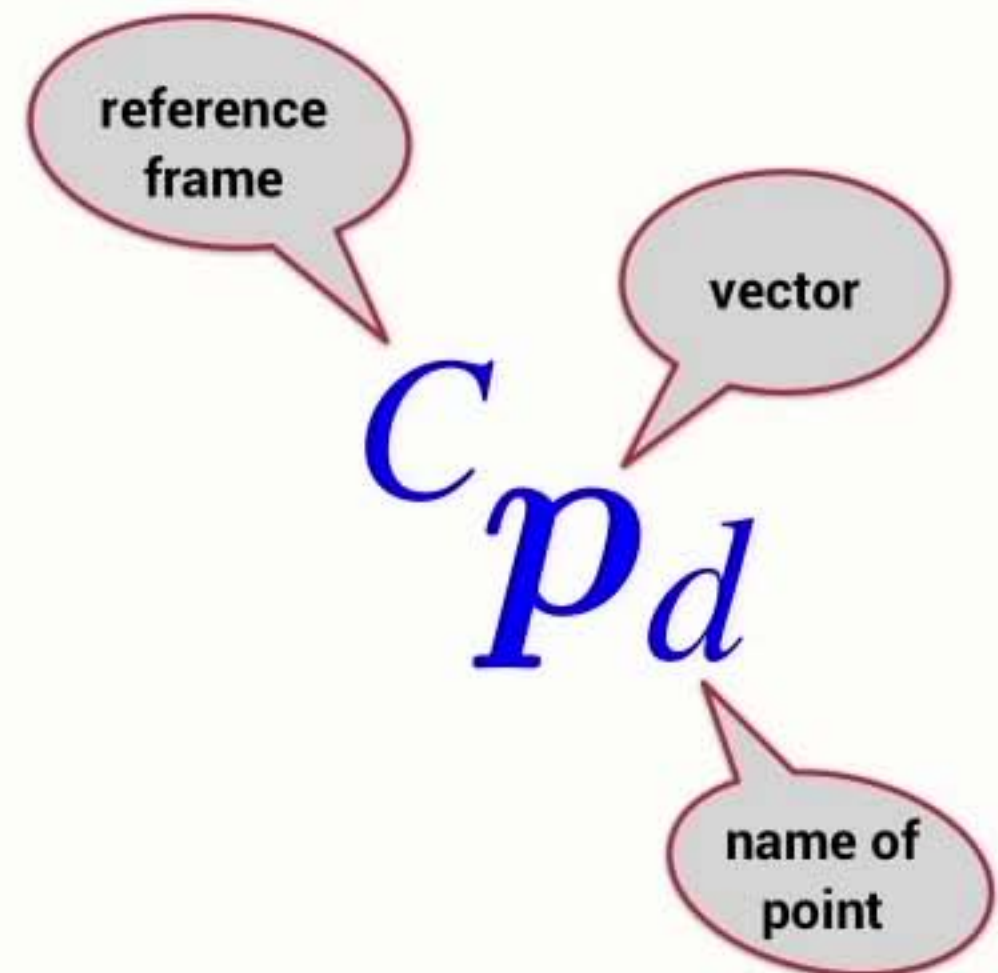
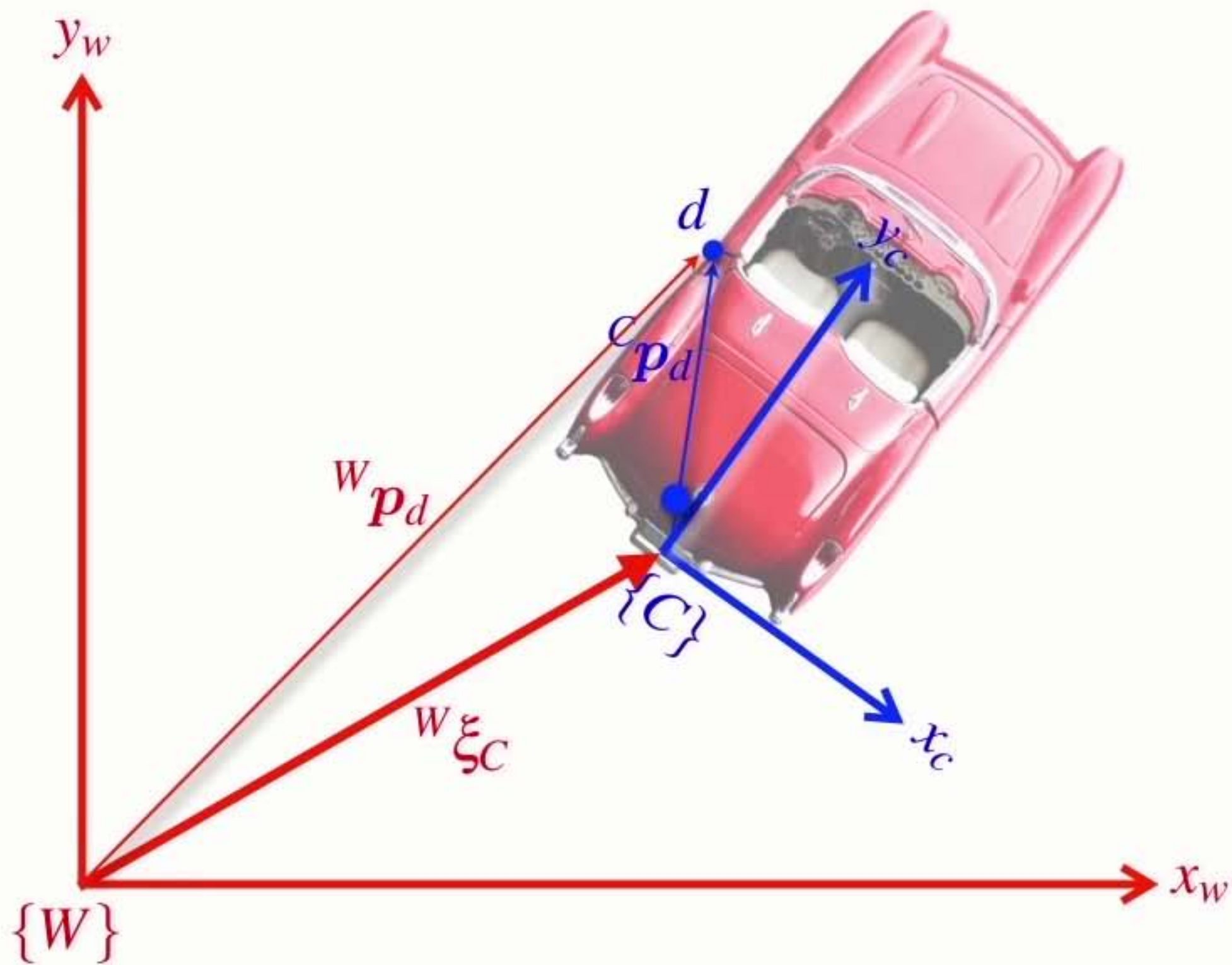


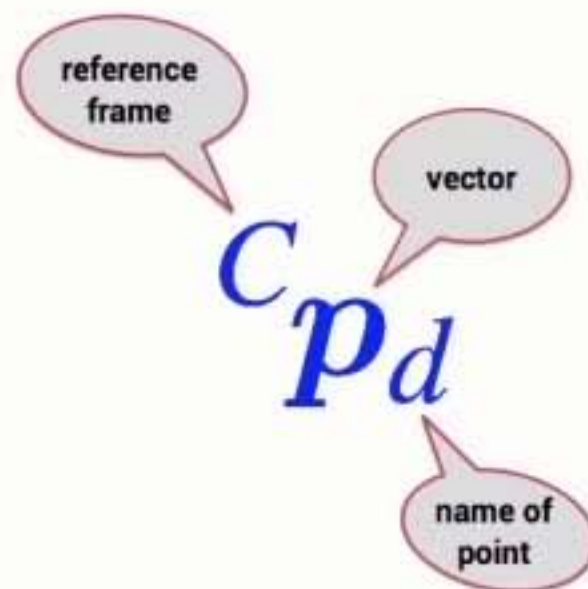
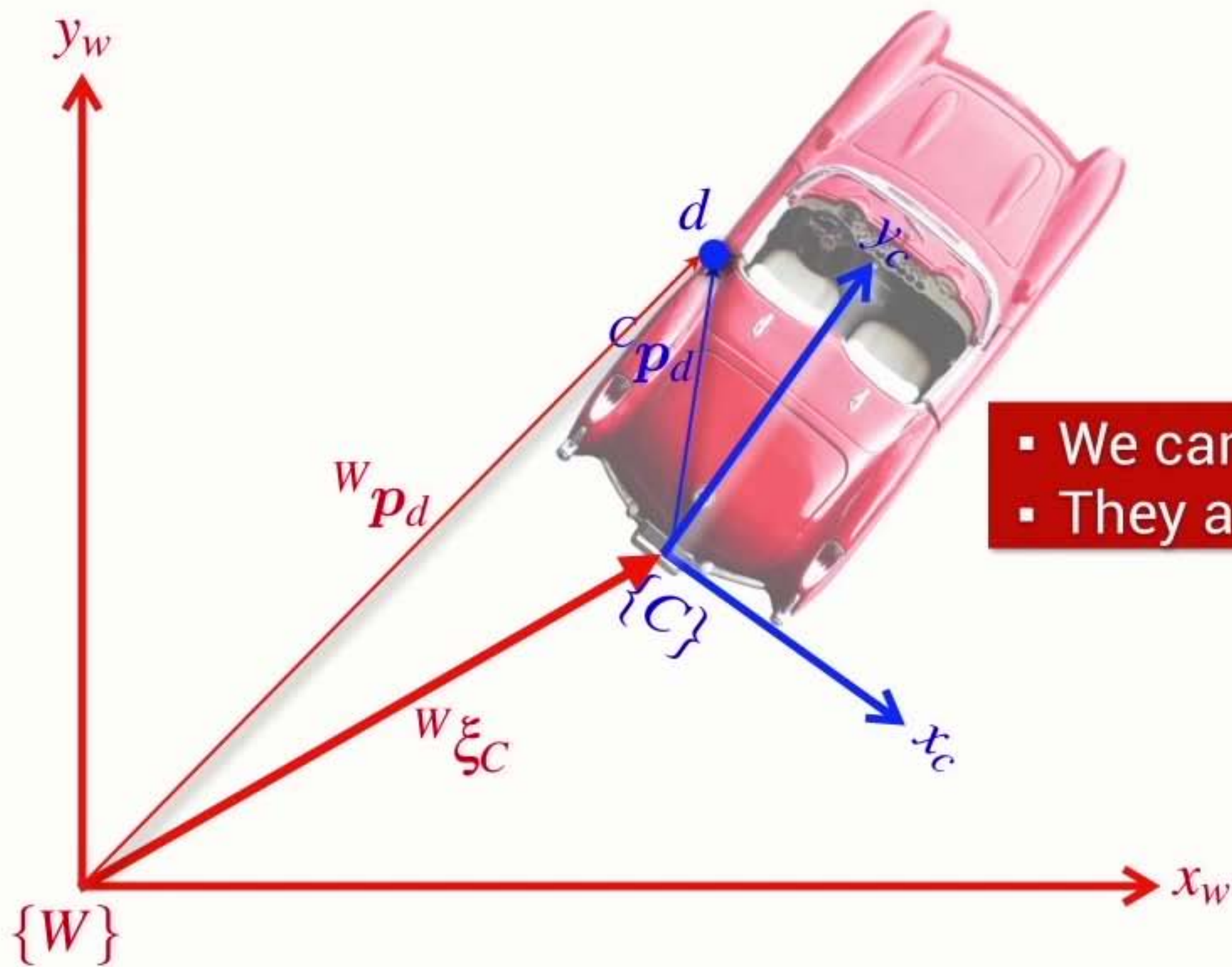




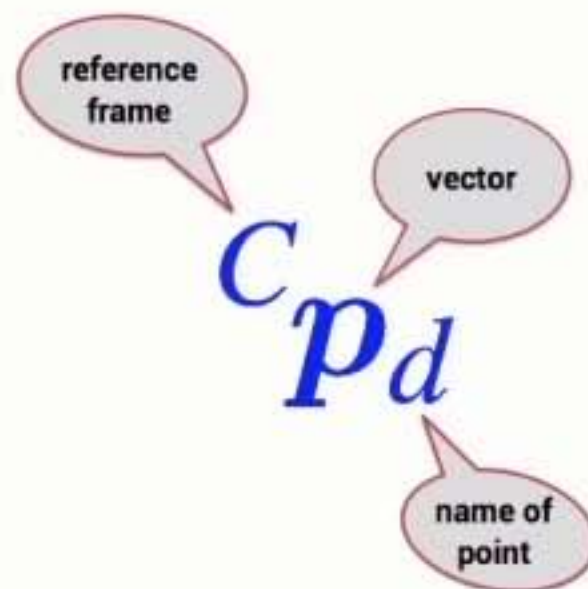
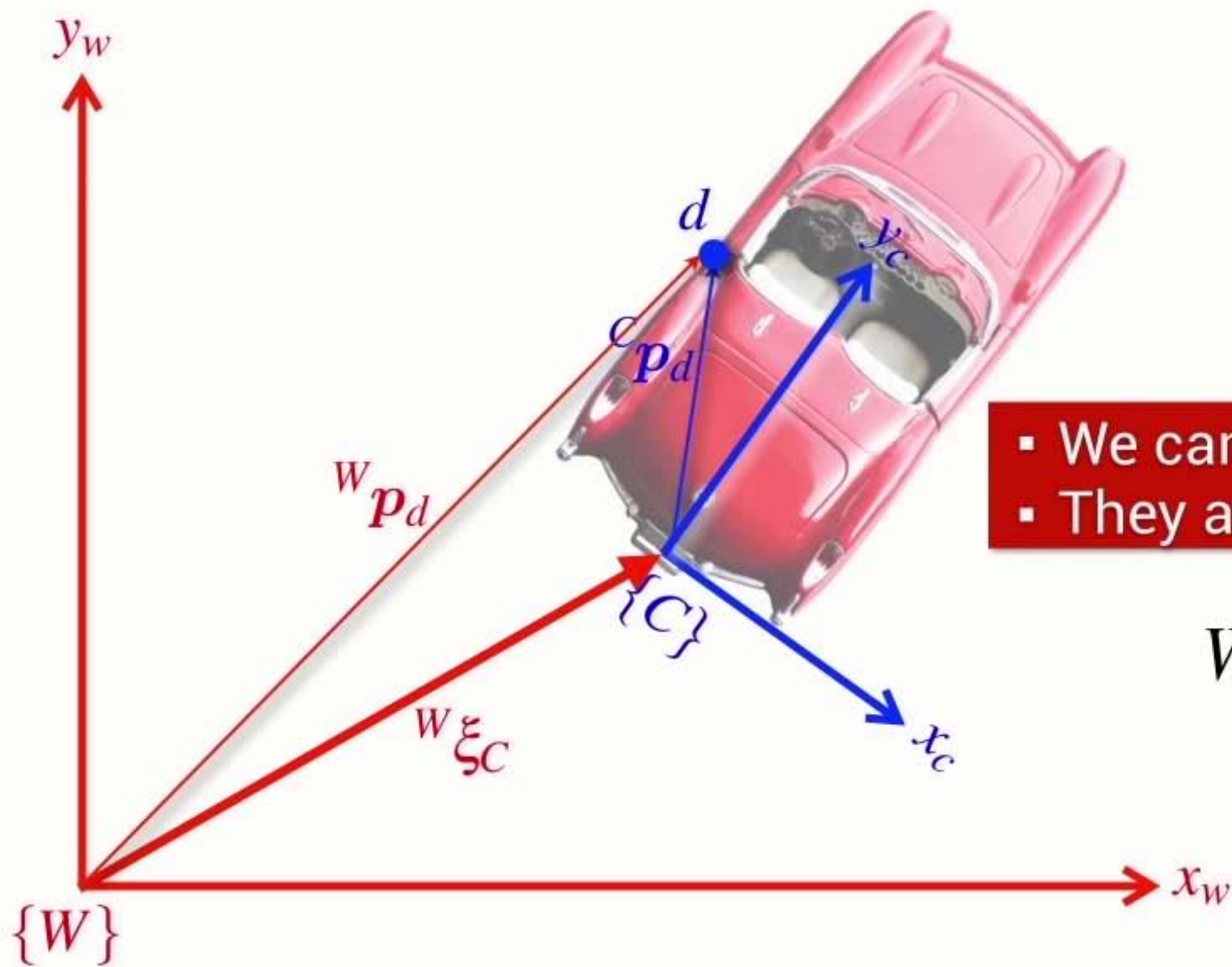






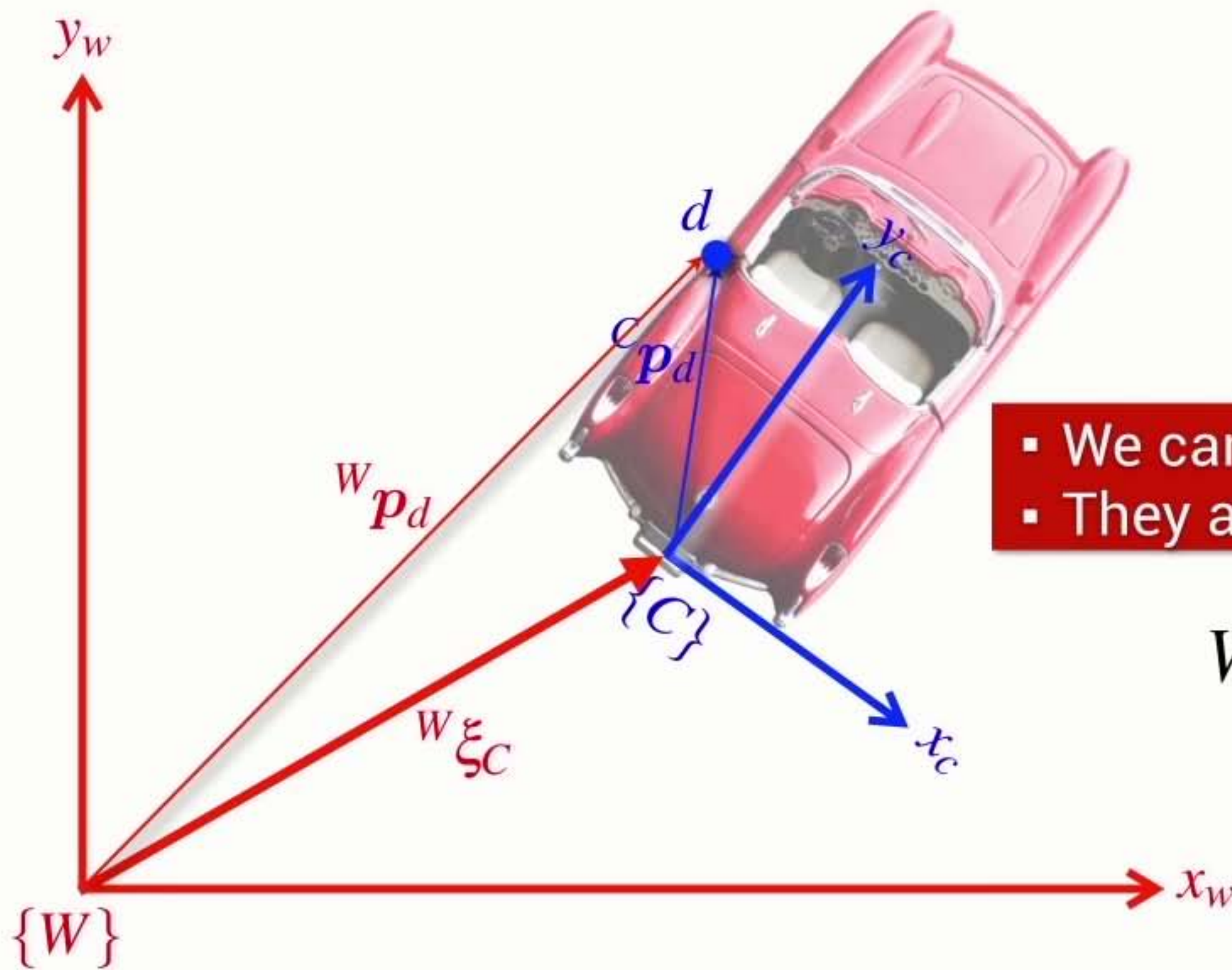


- We can't add **Poses** and **Vectors**
- They are different things



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$${}^W p_d = {}^W \xi_C \cdot {}^C p_d$$



- We can't add **Poses** and **Vectors**
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transform
vector from one frame
to another

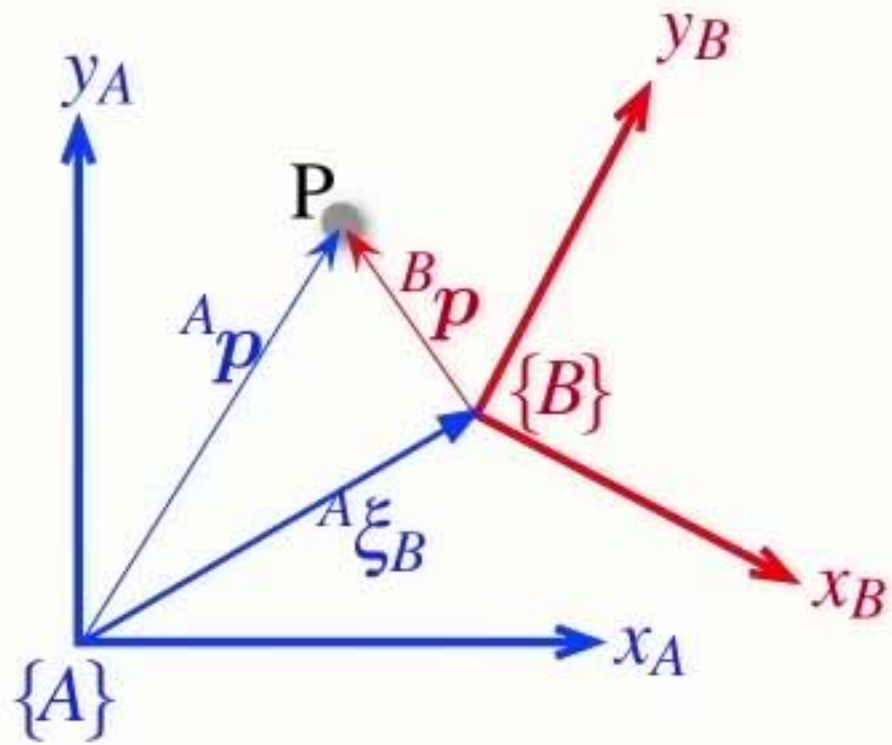
reference
frame

vector

name of
point

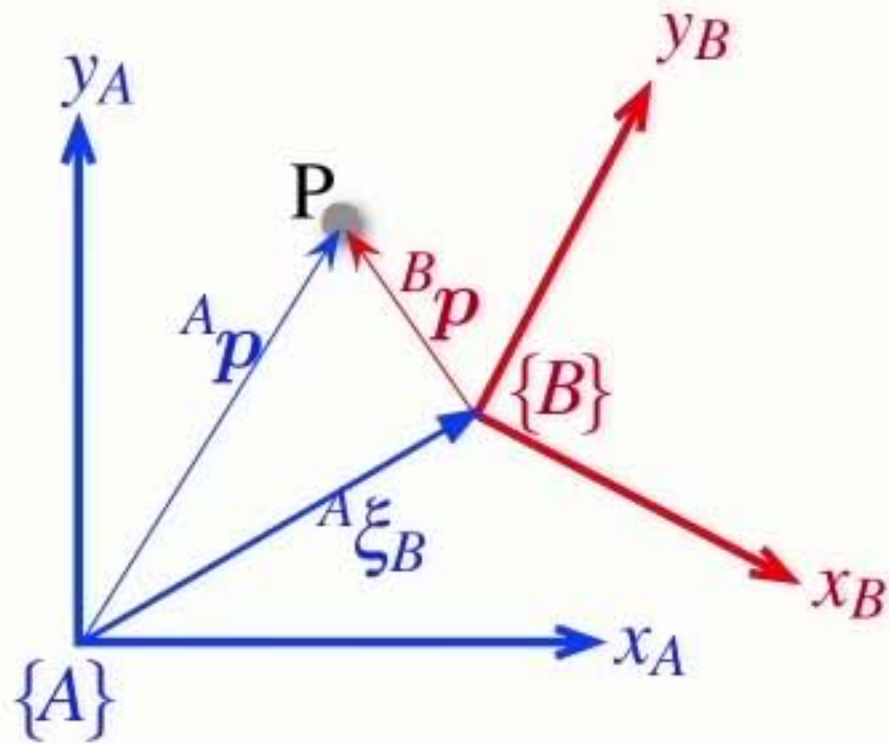
Section4 Relative poses

Relative points



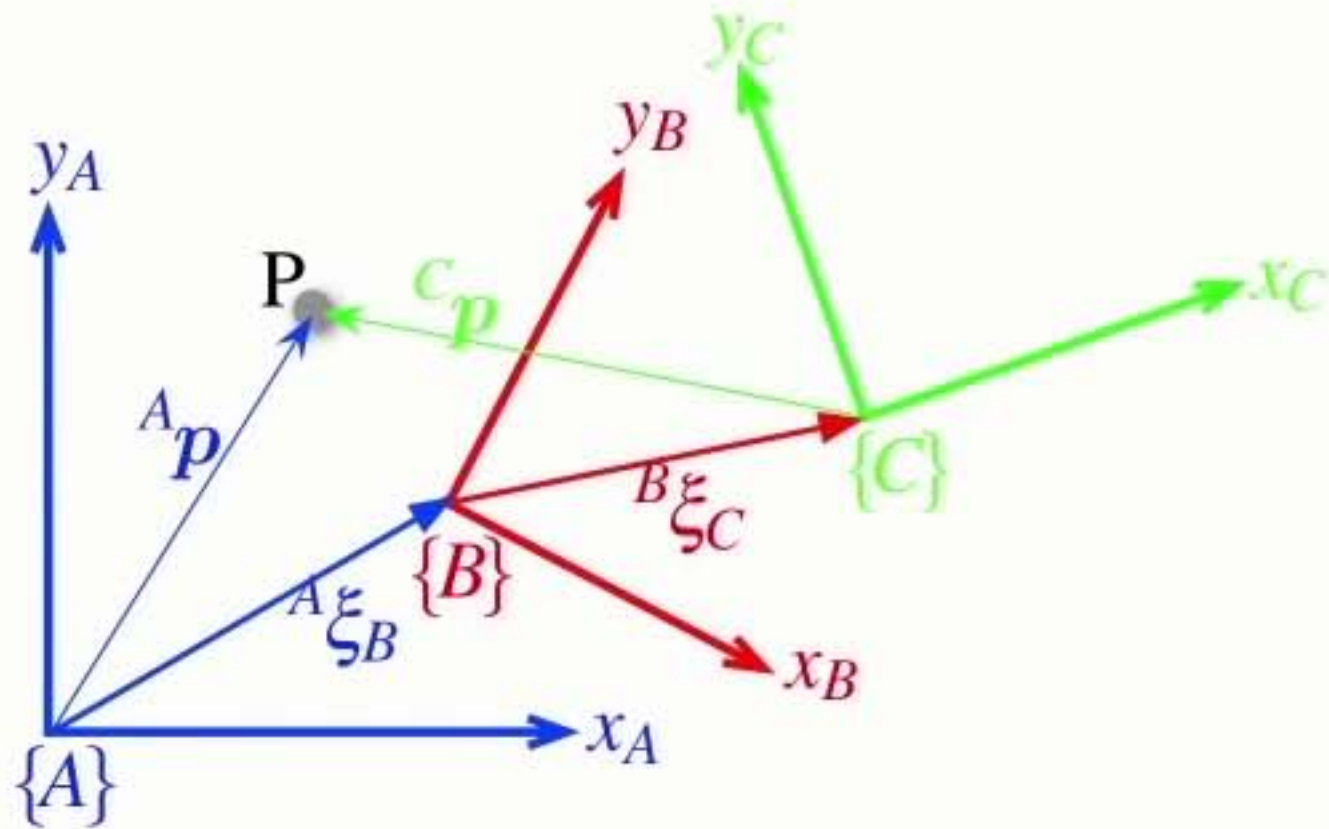
Relative **points**

$${}^A p = {}^A \xi_B \cdot {}^B p$$



Relative **points**

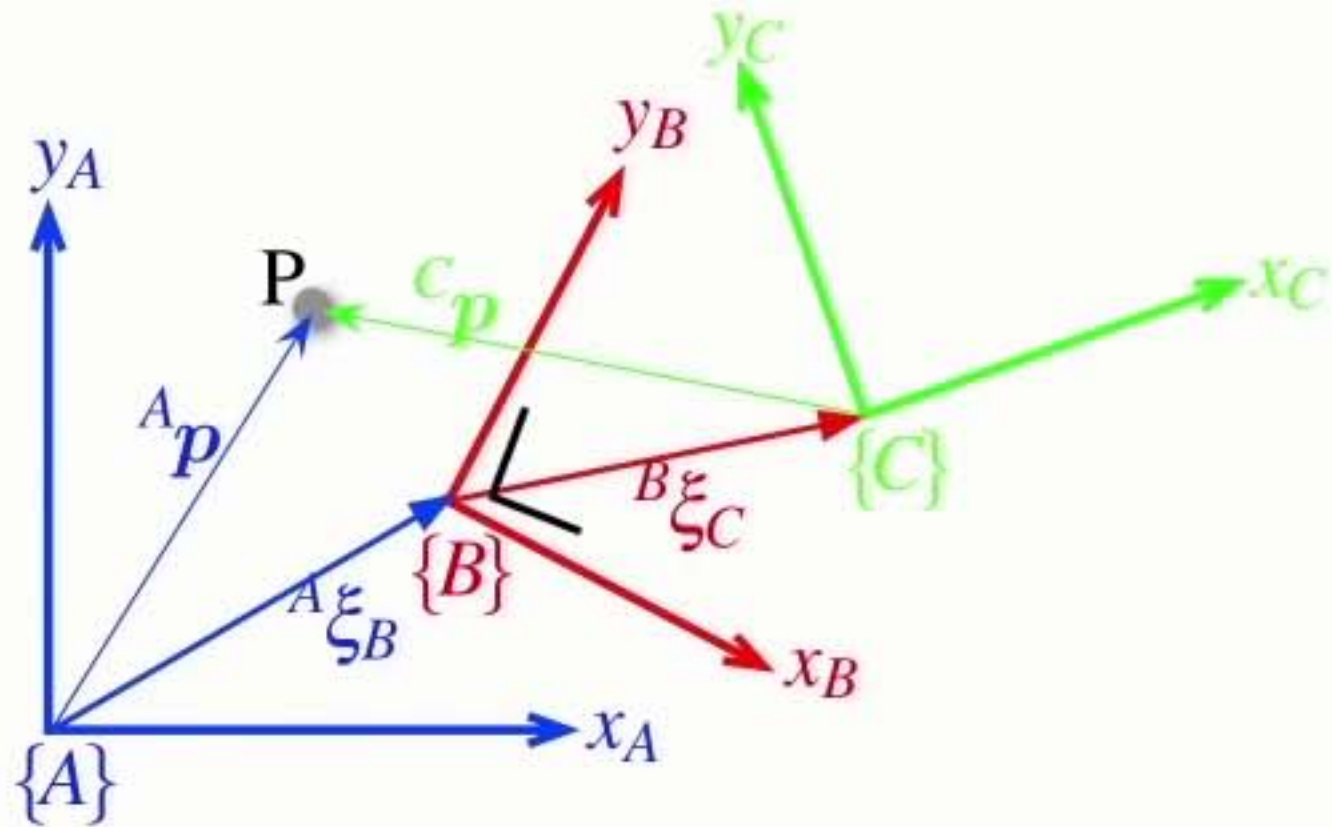
$${}^A p = {}^A \xi_B \cdot {}^B p$$



- Relative poses can be **compounded** or **composed**

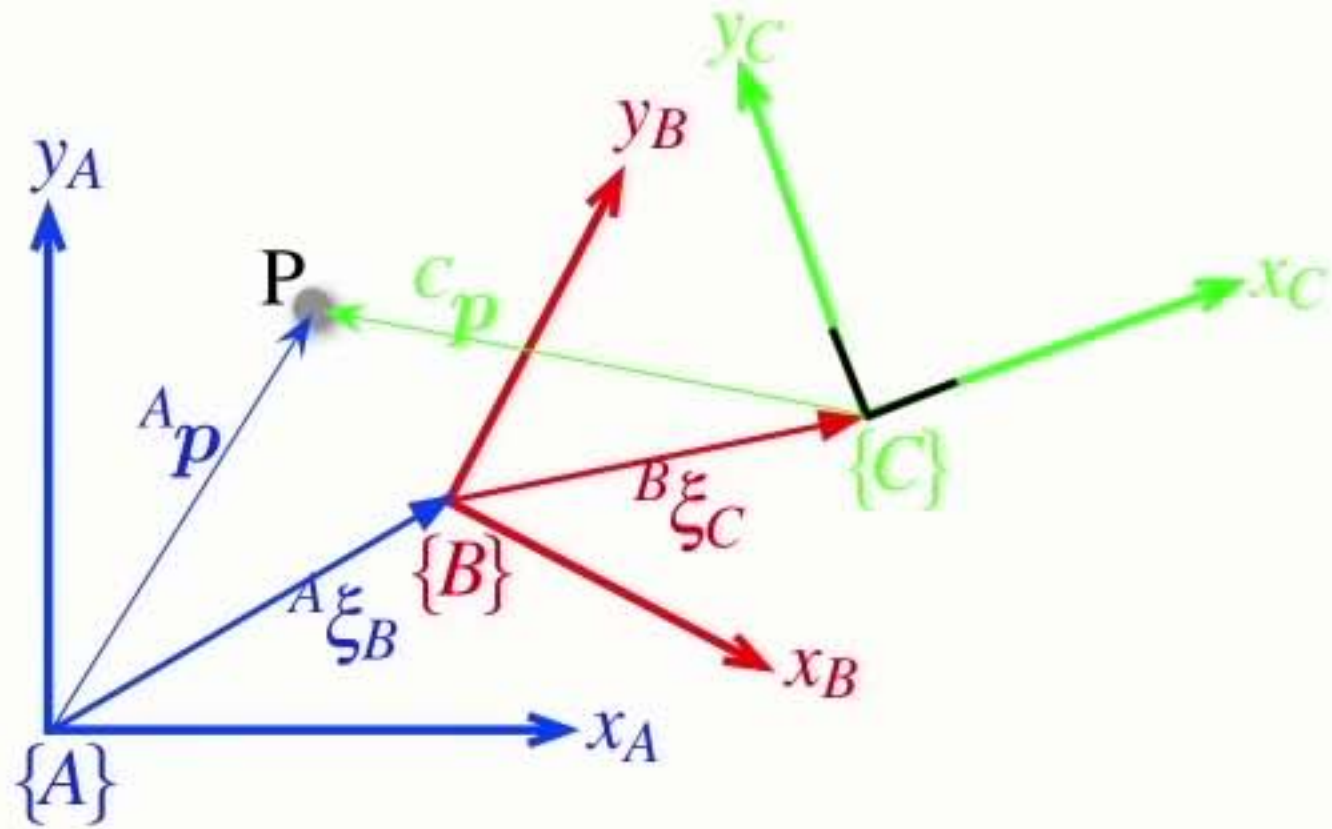
Relative **points**

$${}^A p = {}^A \xi_B \cdot {}^B p$$



- Relative poses can be **compounded** or **composed**

Relative **points**

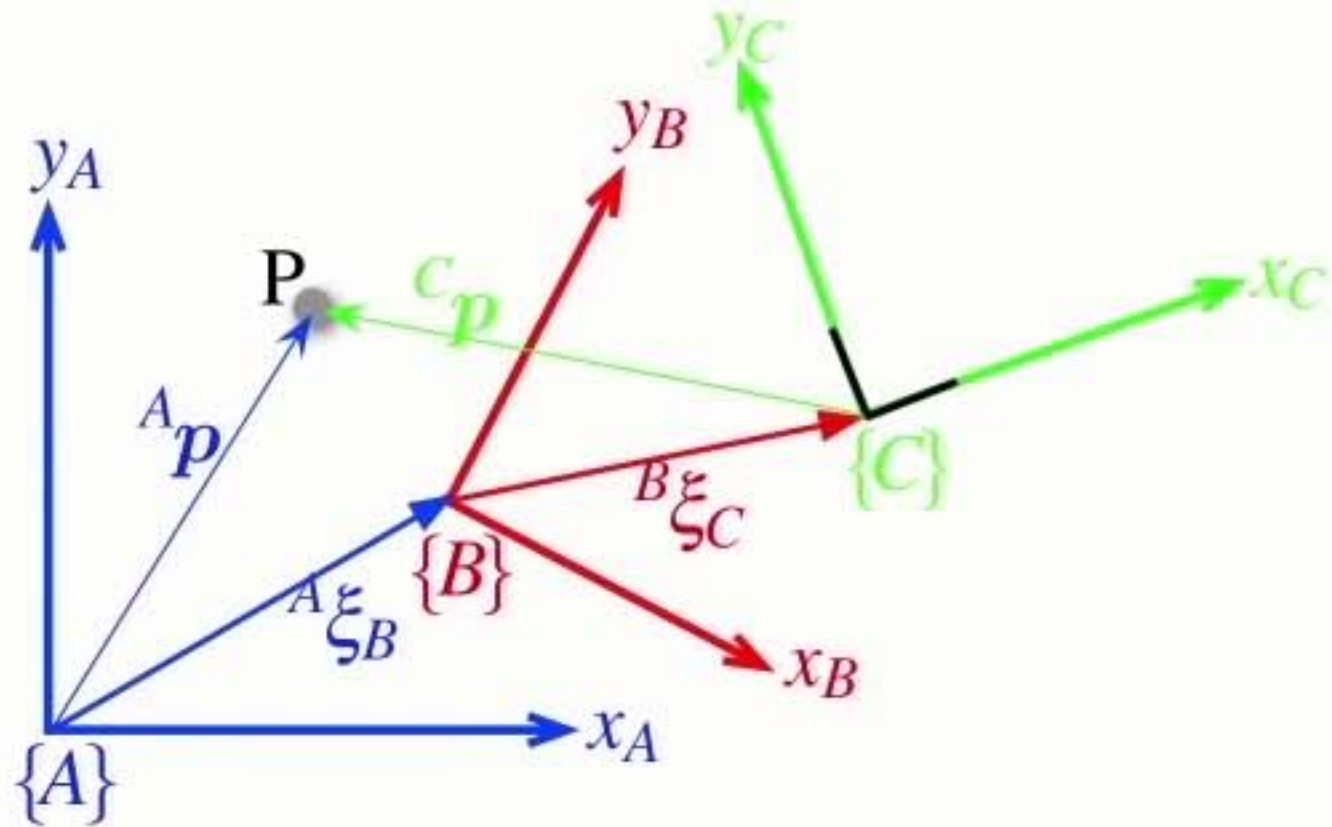


$${}^A p = {}^A \xi_B \cdot {}^B p$$

$${}^A \xi_C = {}^A \xi_B \oplus {}^B \xi_C$$

- Relative poses can be **compounded** or **composed**

Relative **points**

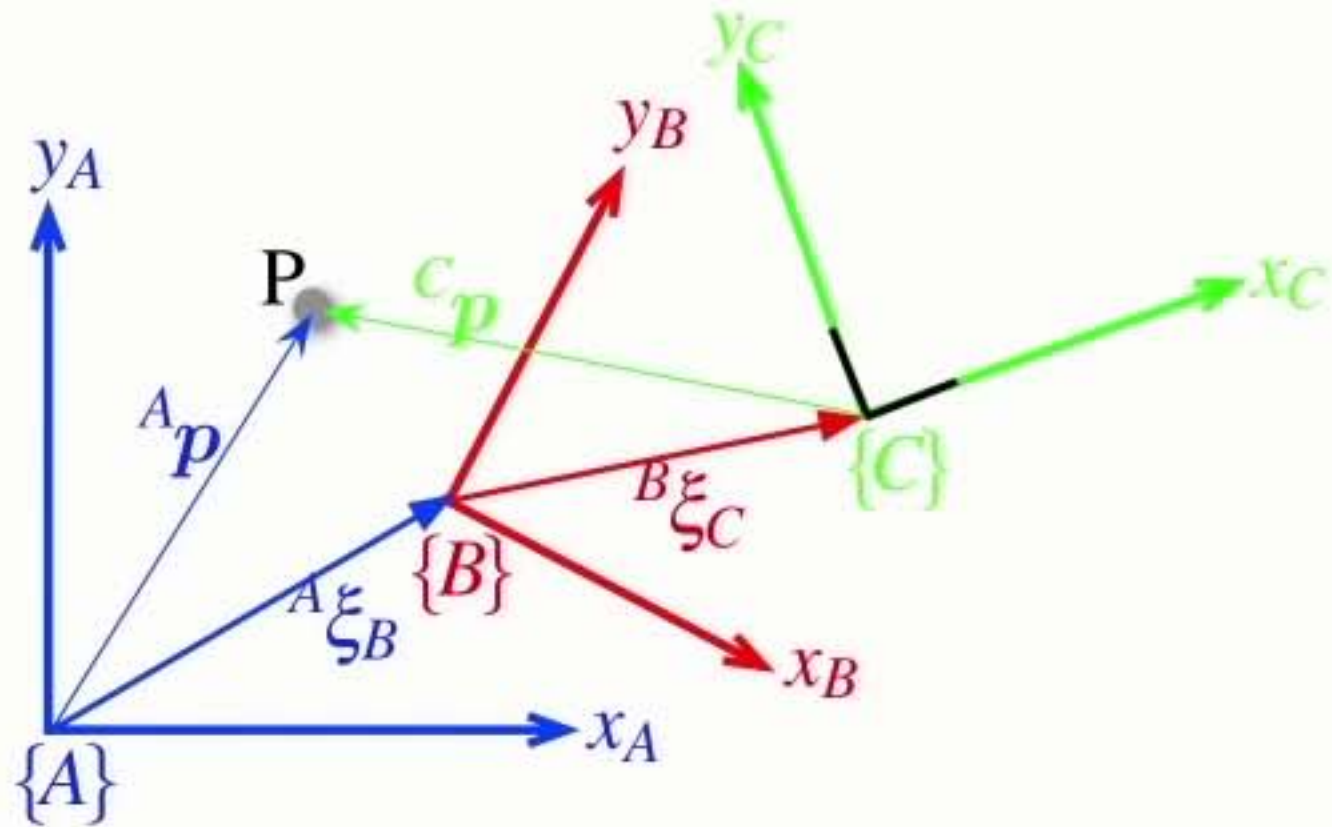


$${}^A p = {}^A \xi_B \cdot {}^B p$$

$${}^A \xi_C = {}^A \xi_B \oplus {}^B \xi_C$$

- Relative poses can be **compounded** or **composed**

Relative **points**



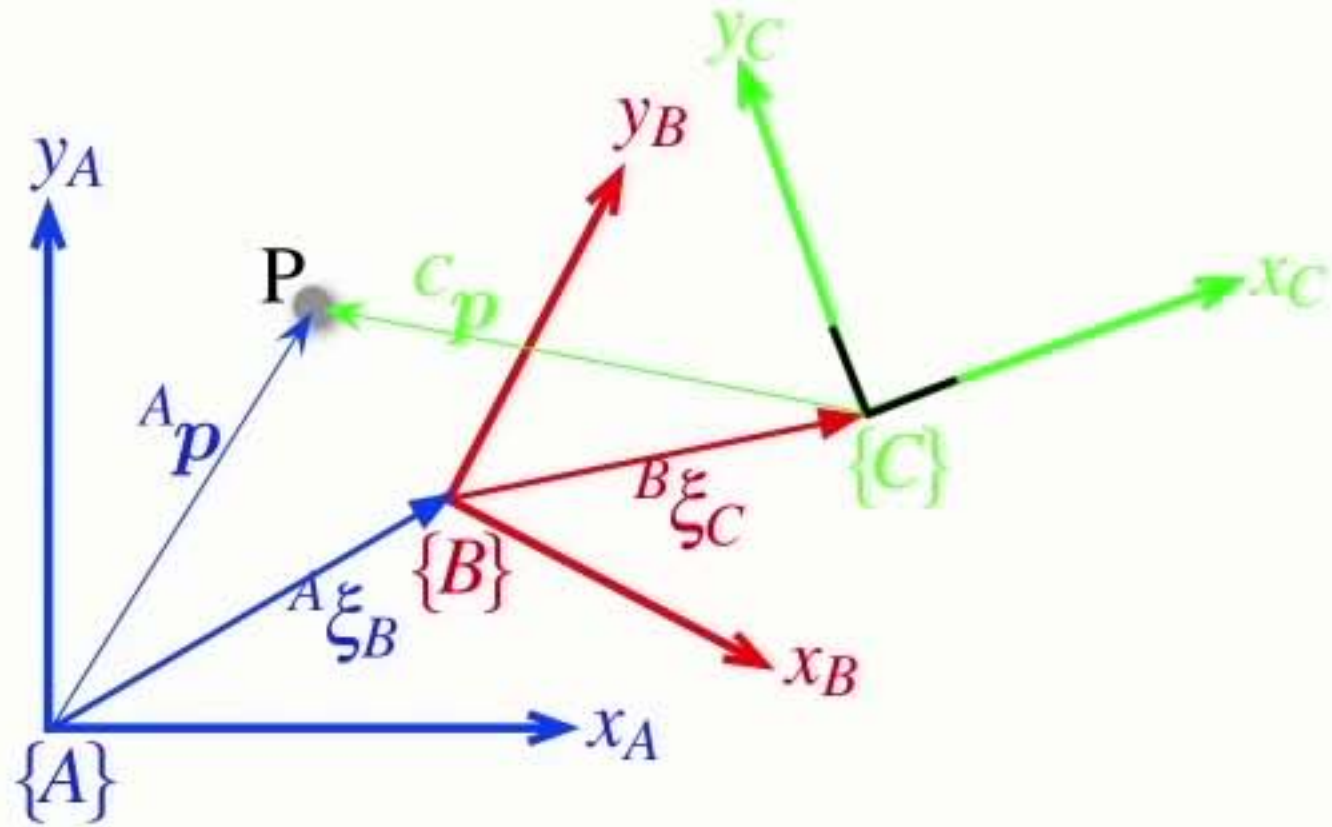
$${}^A p = {}^A \xi_B \cdot {}^B p$$

$${}^A \xi_C = {}^A \xi_B \oplus {}^B \xi_C$$

$${}^A p = ({}^A \xi_B \oplus {}^B \xi_C) \cdot {}^C p$$

- Relative poses can be **compounded** or **composed**

Relative **points**



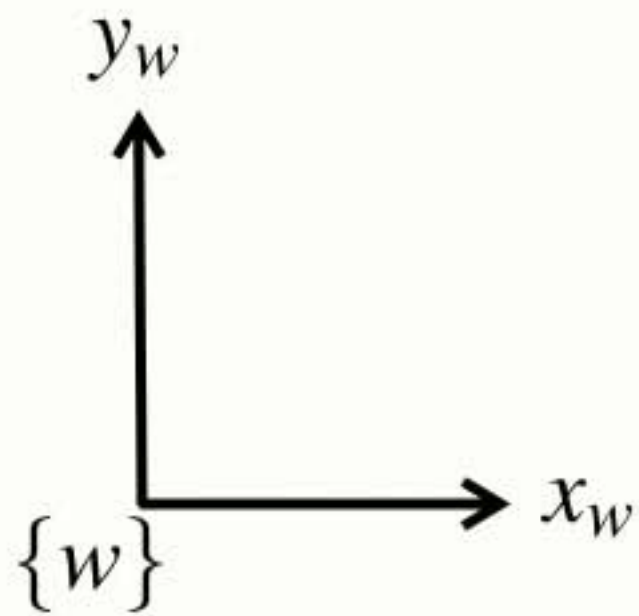
$${}^A p = {}^A \xi_B \cdot {}^B p$$

$${}^A \xi_C = {}^A \xi_B \oplus {}^B \xi_C$$

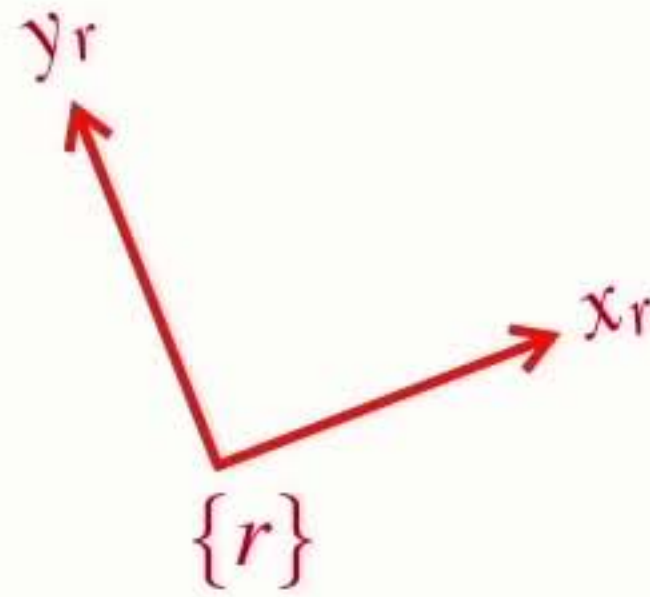
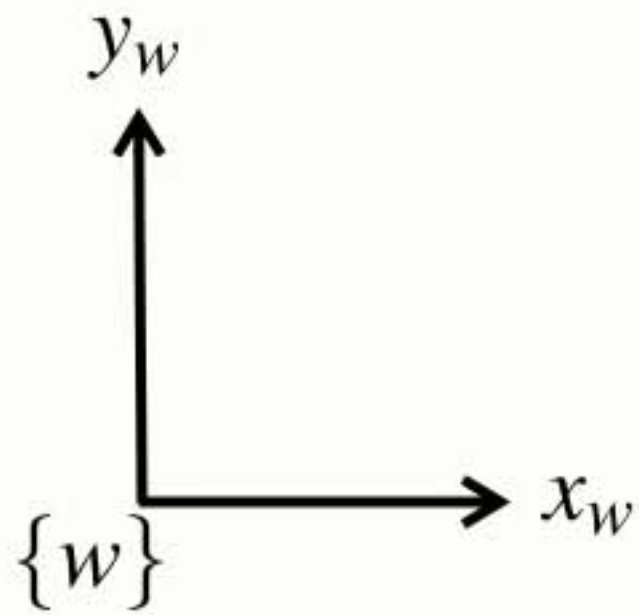
$${}^A p = ({}^A \xi_B \oplus {}^B \xi_C) \cdot {}^C p$$

- Relative poses can be **compounded** or **composed**
- We can extend this approach indefinitely...

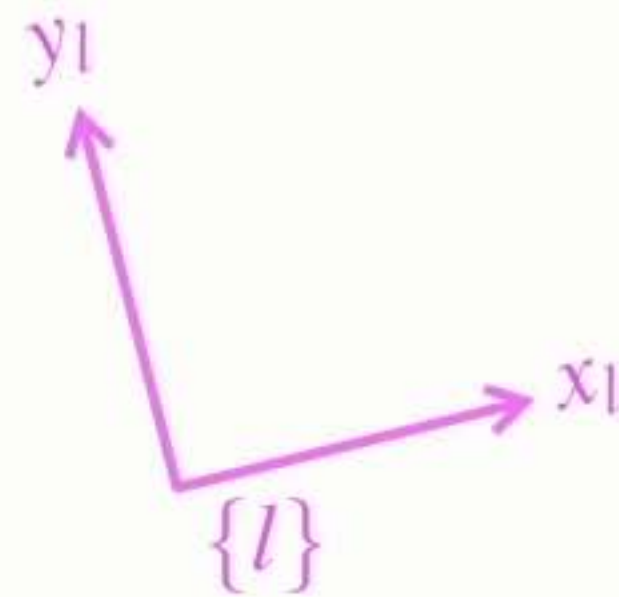
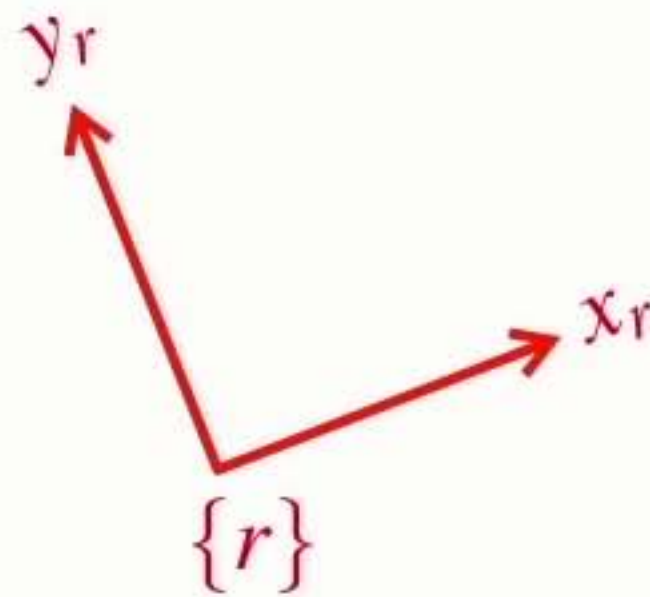
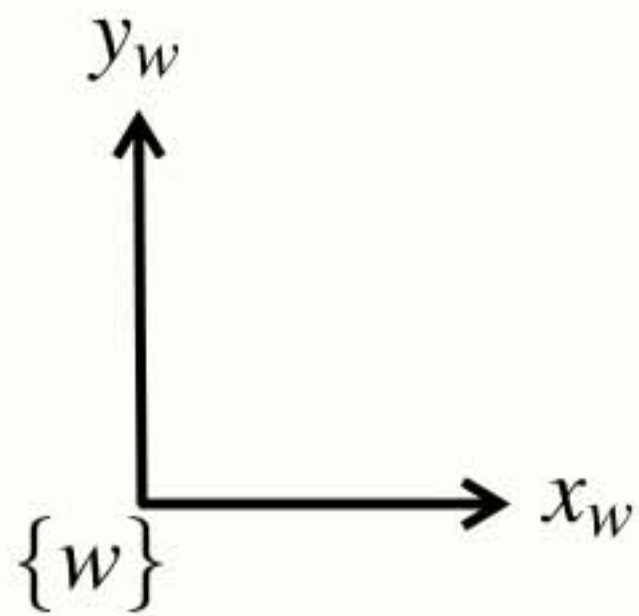
Complex **example**



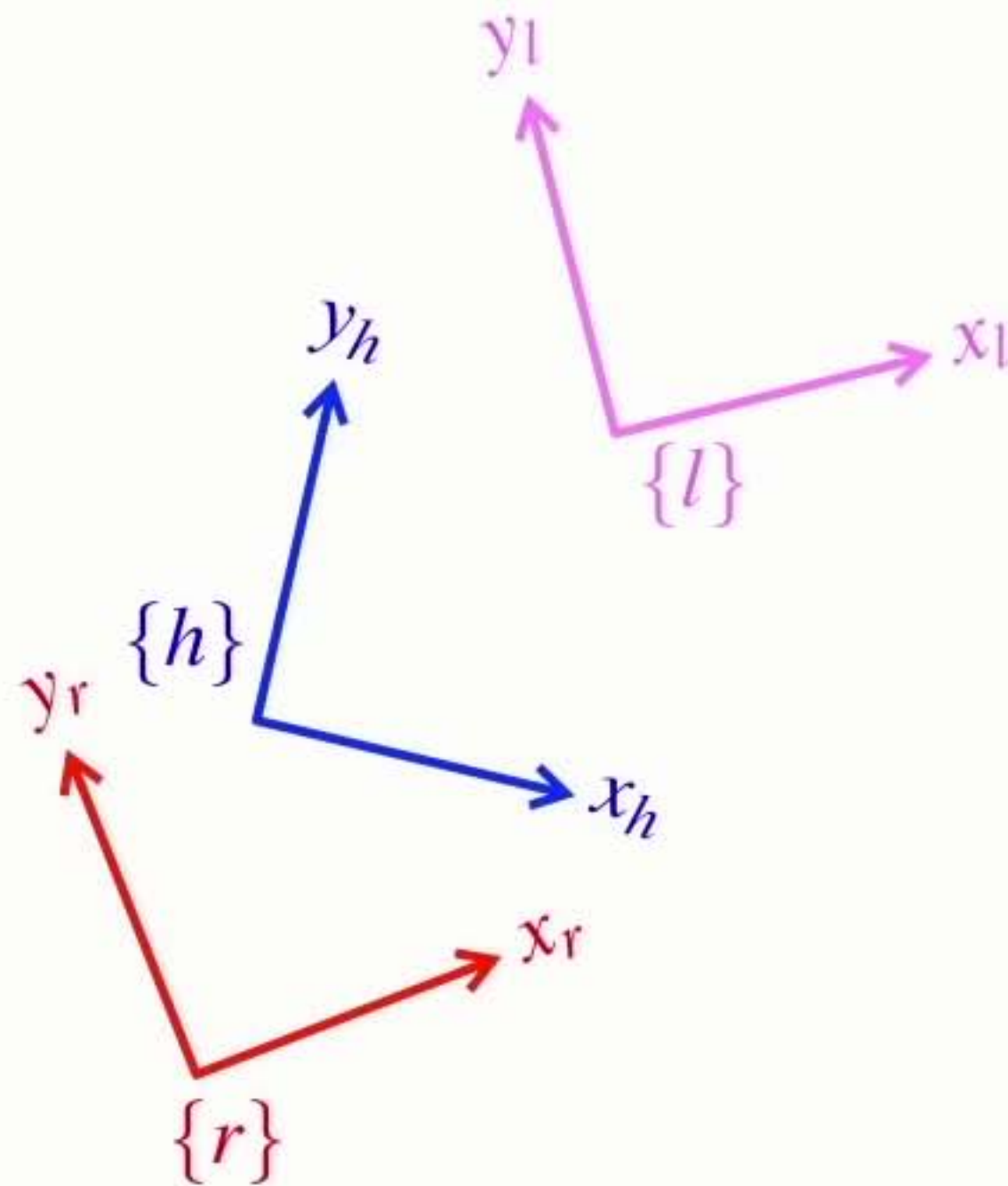
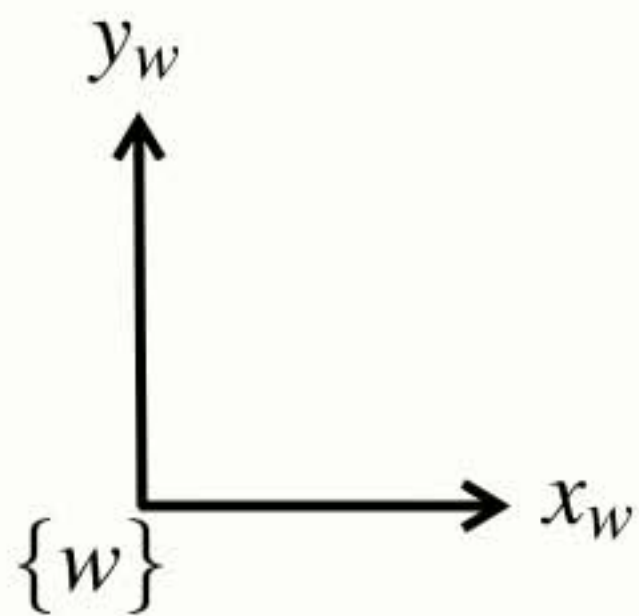
Complex **example**



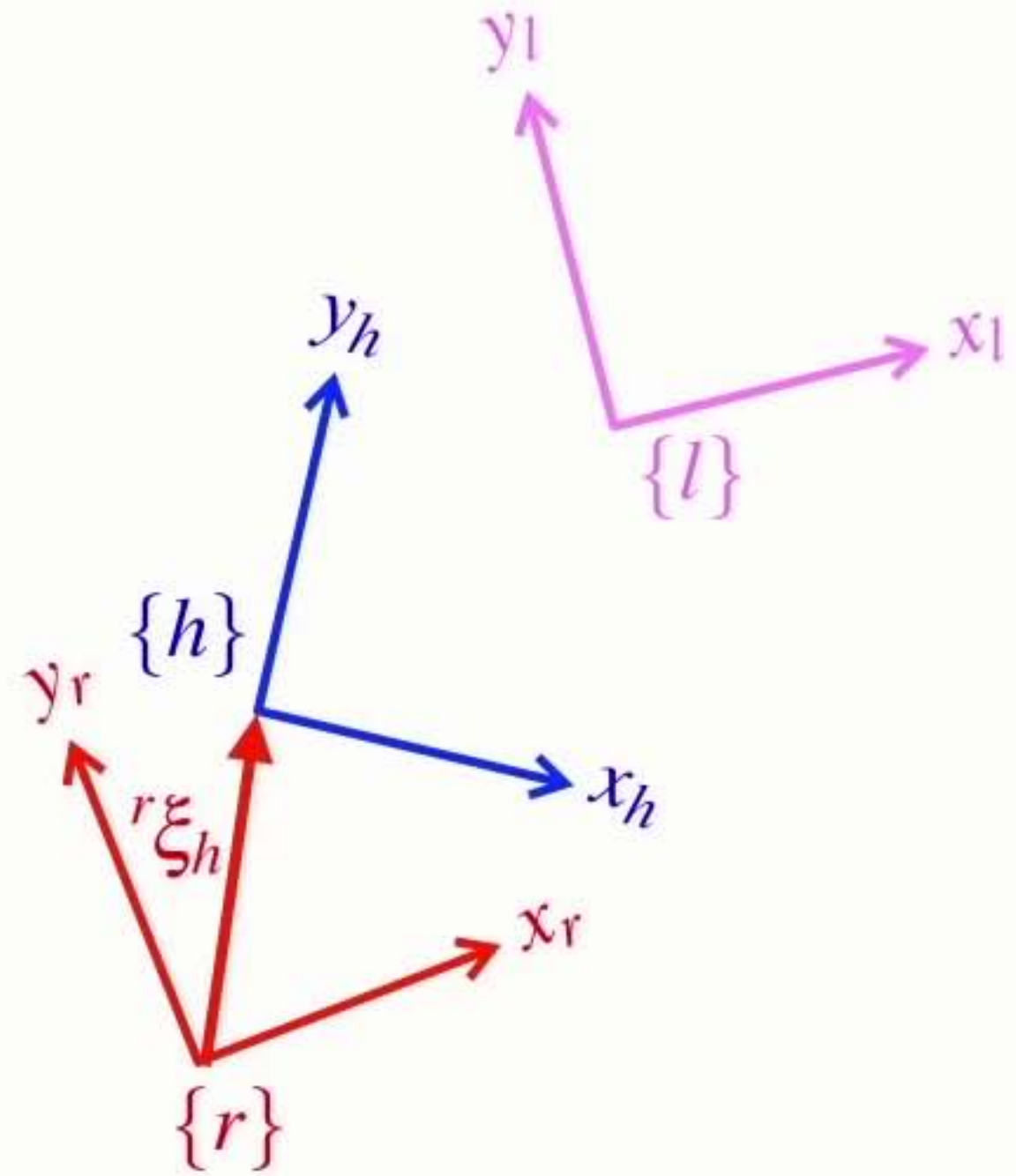
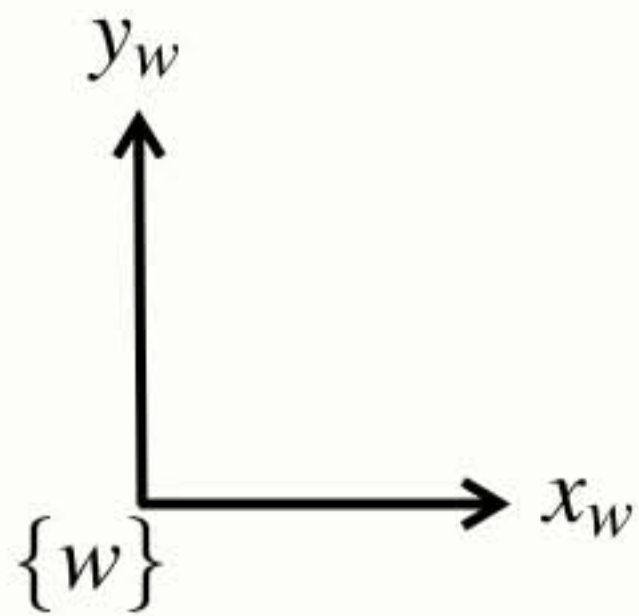
Complex **example**



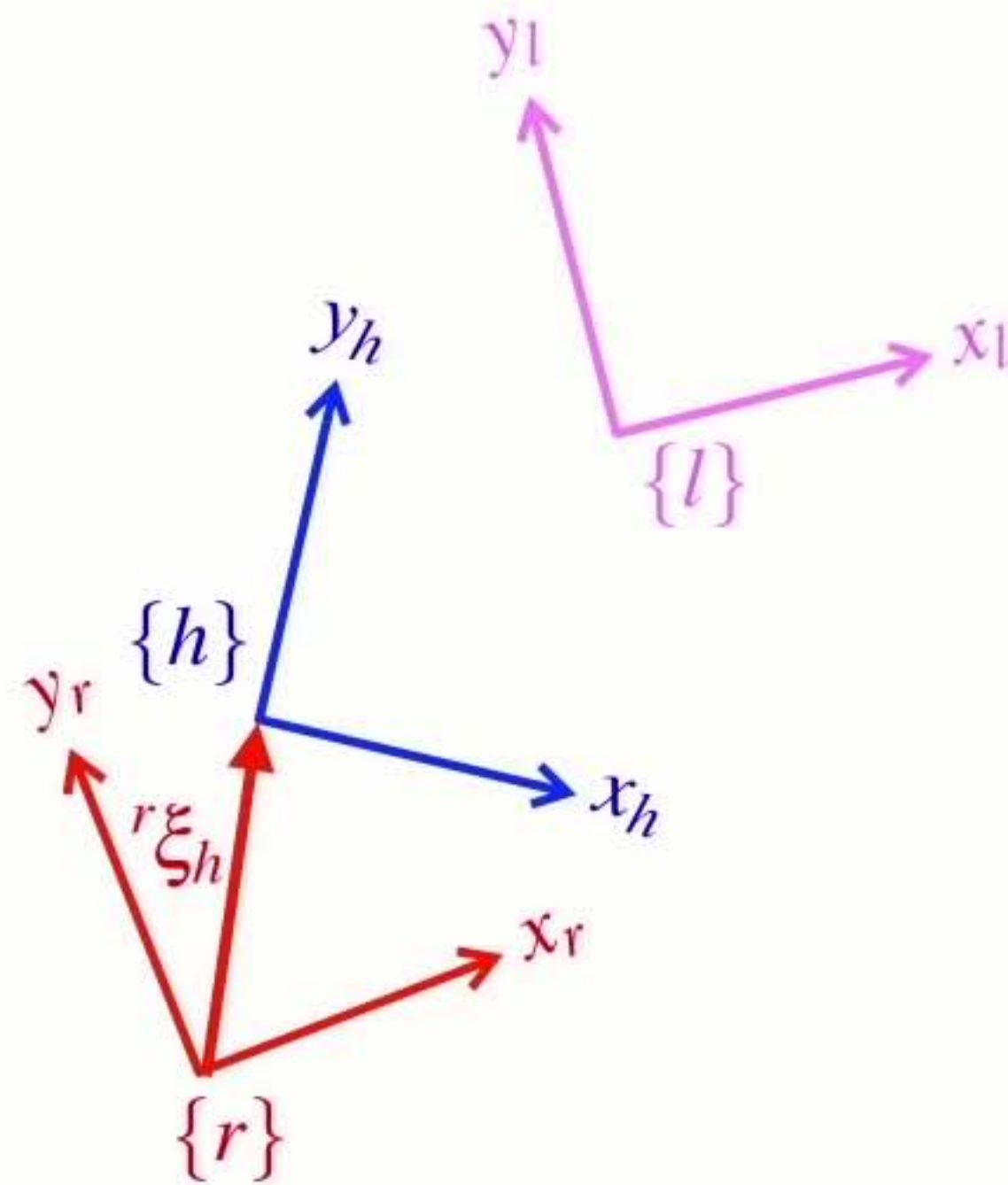
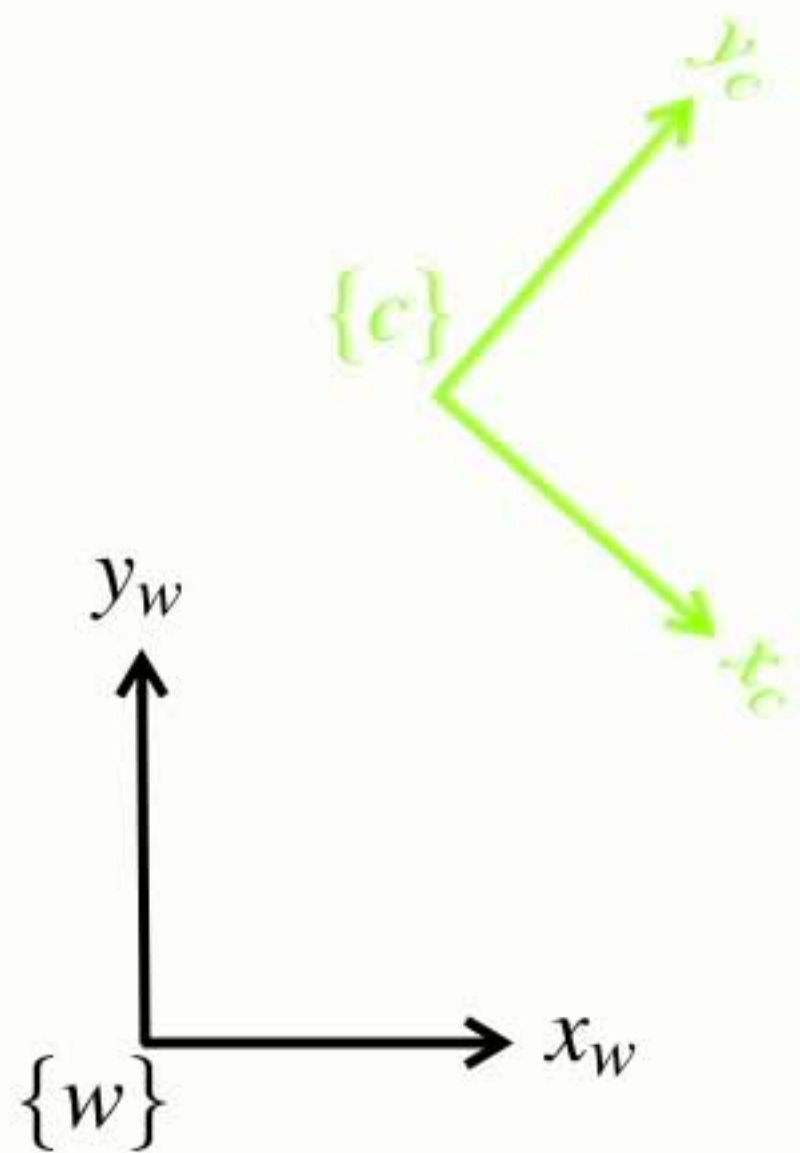
Complex **example**



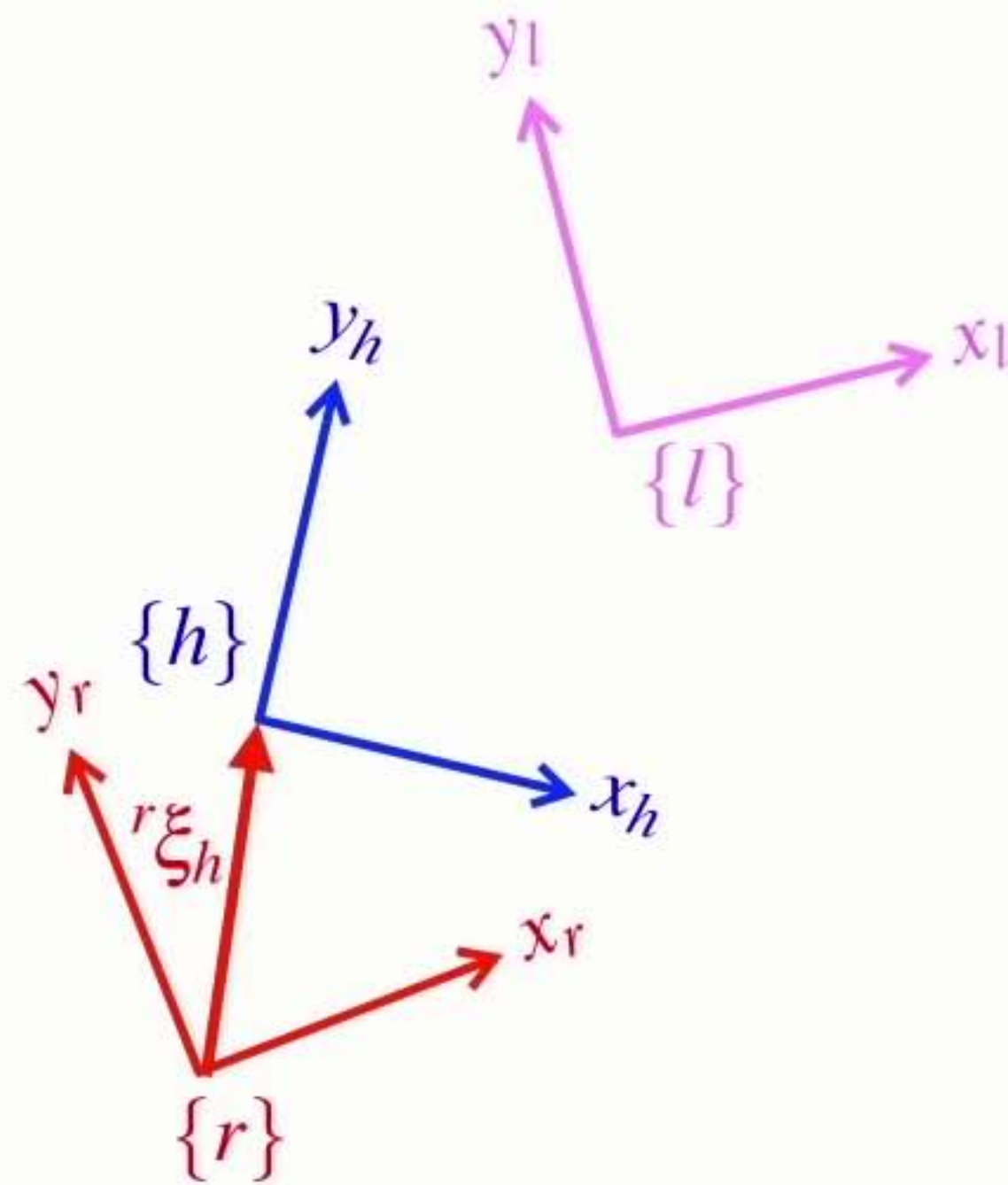
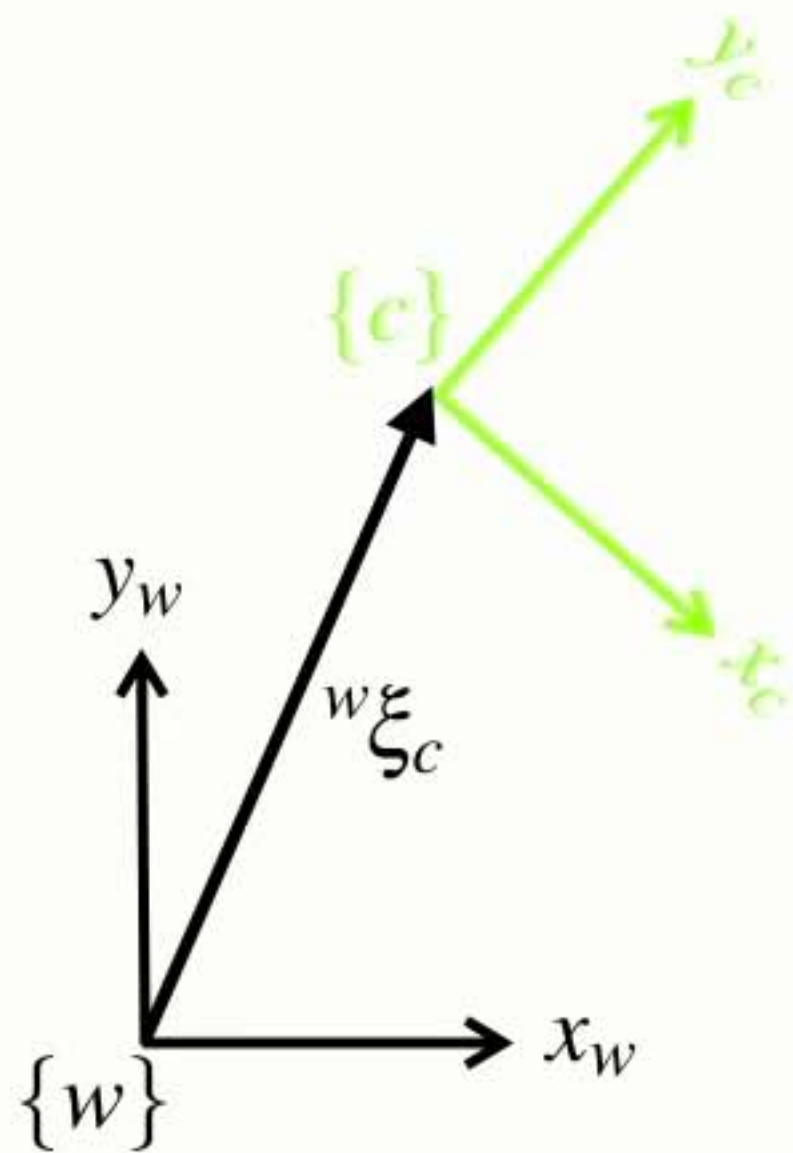
Complex **example**



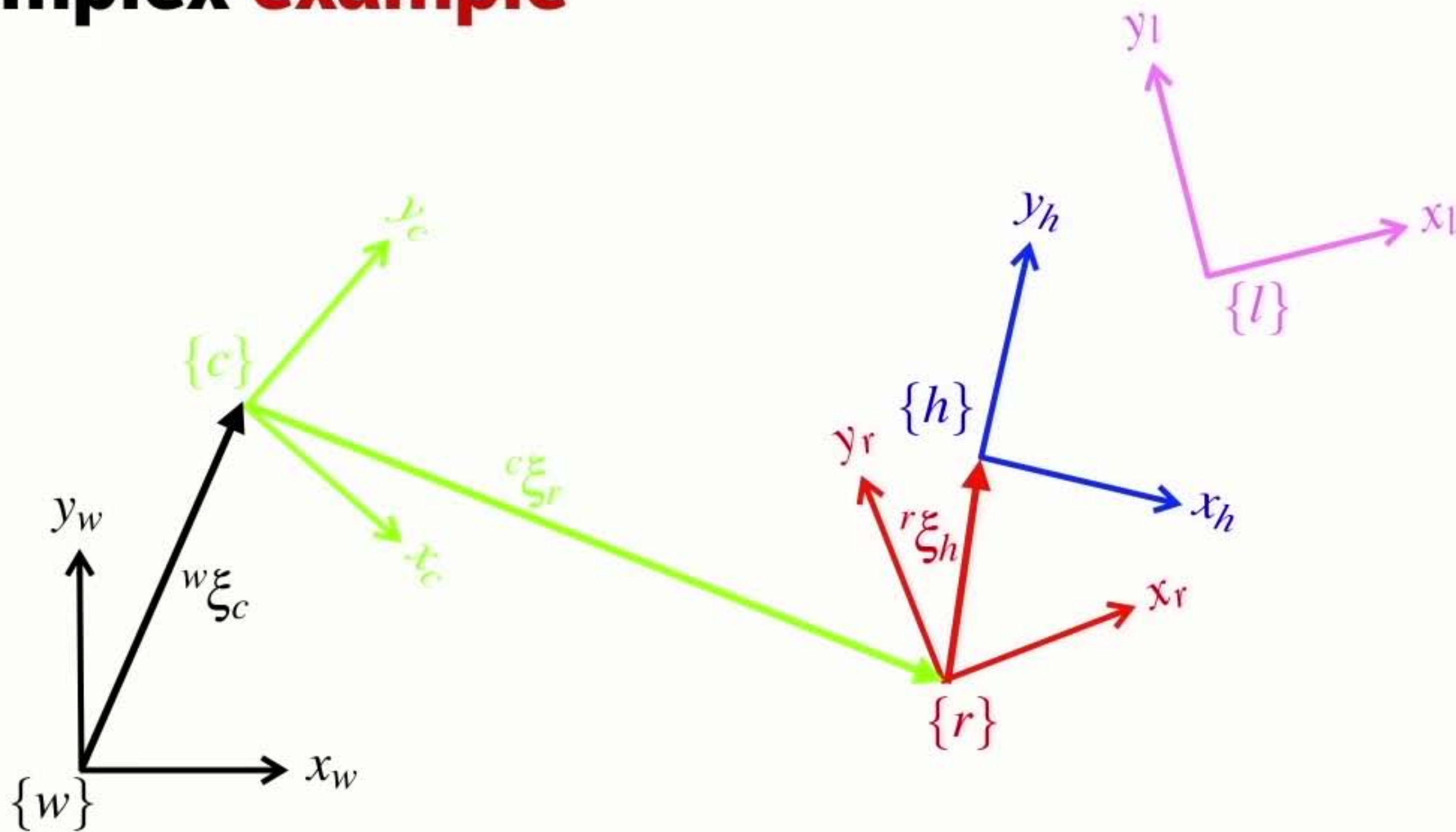
Complex example



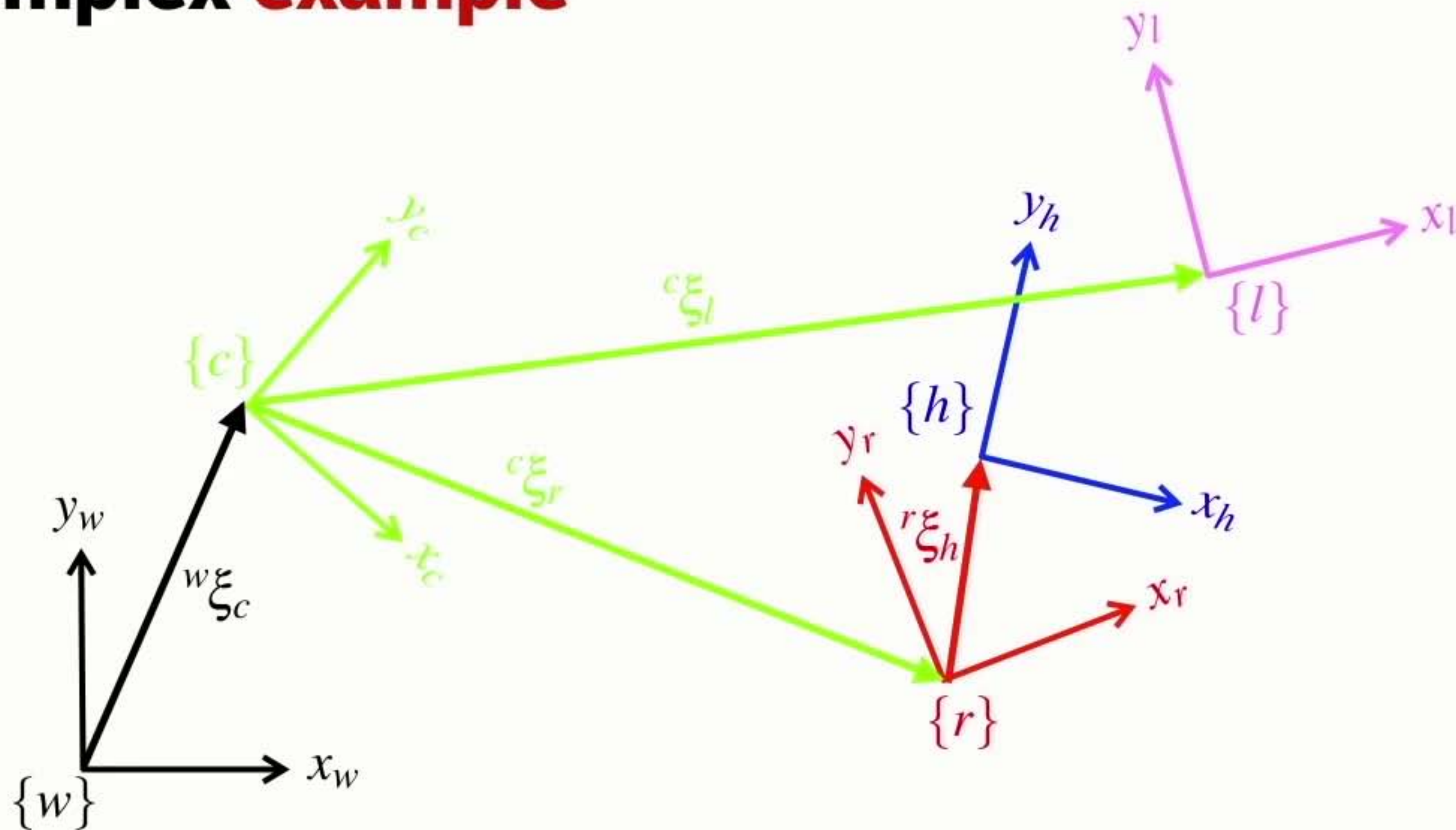
Complex **example**



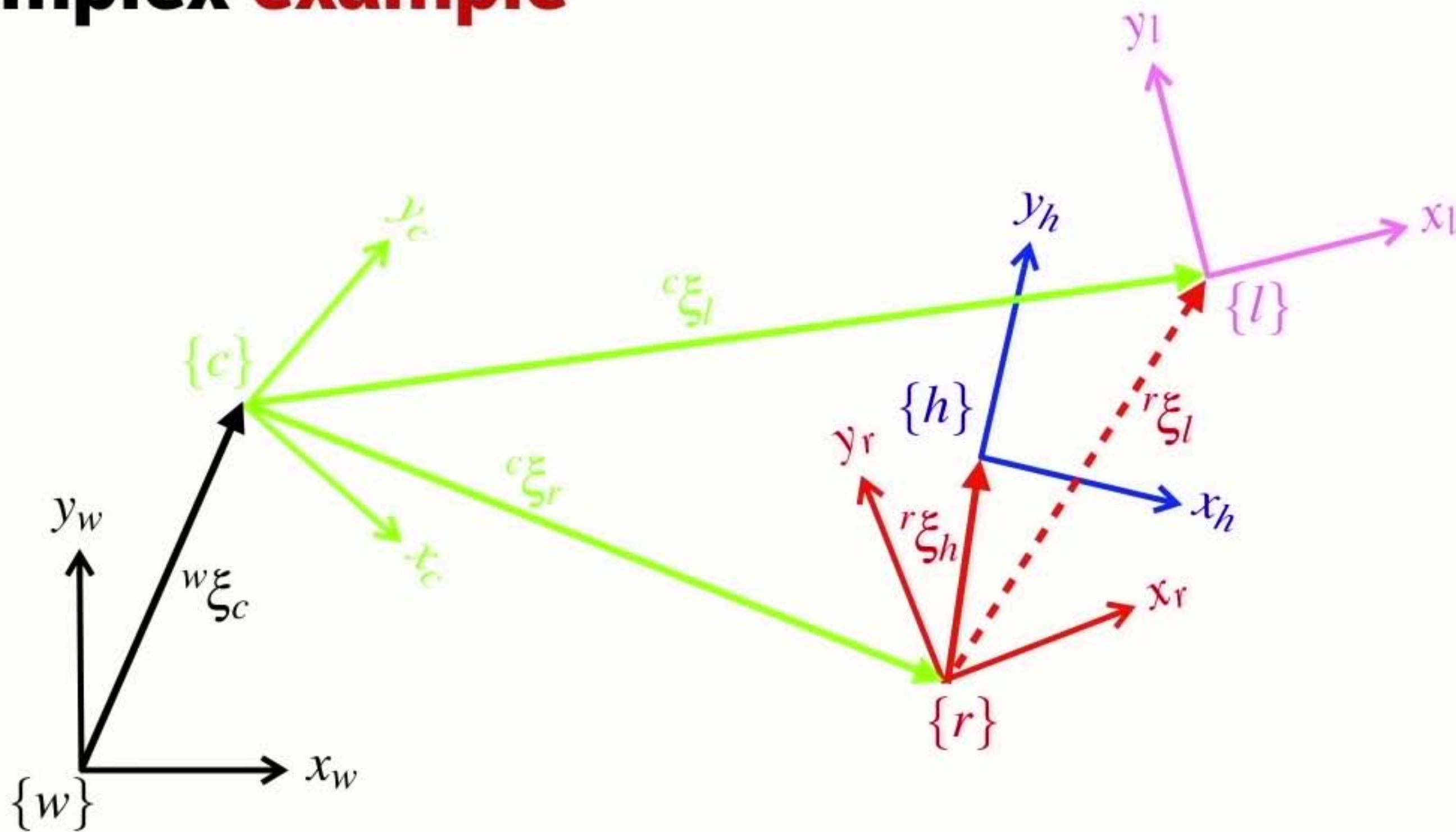
Complex example



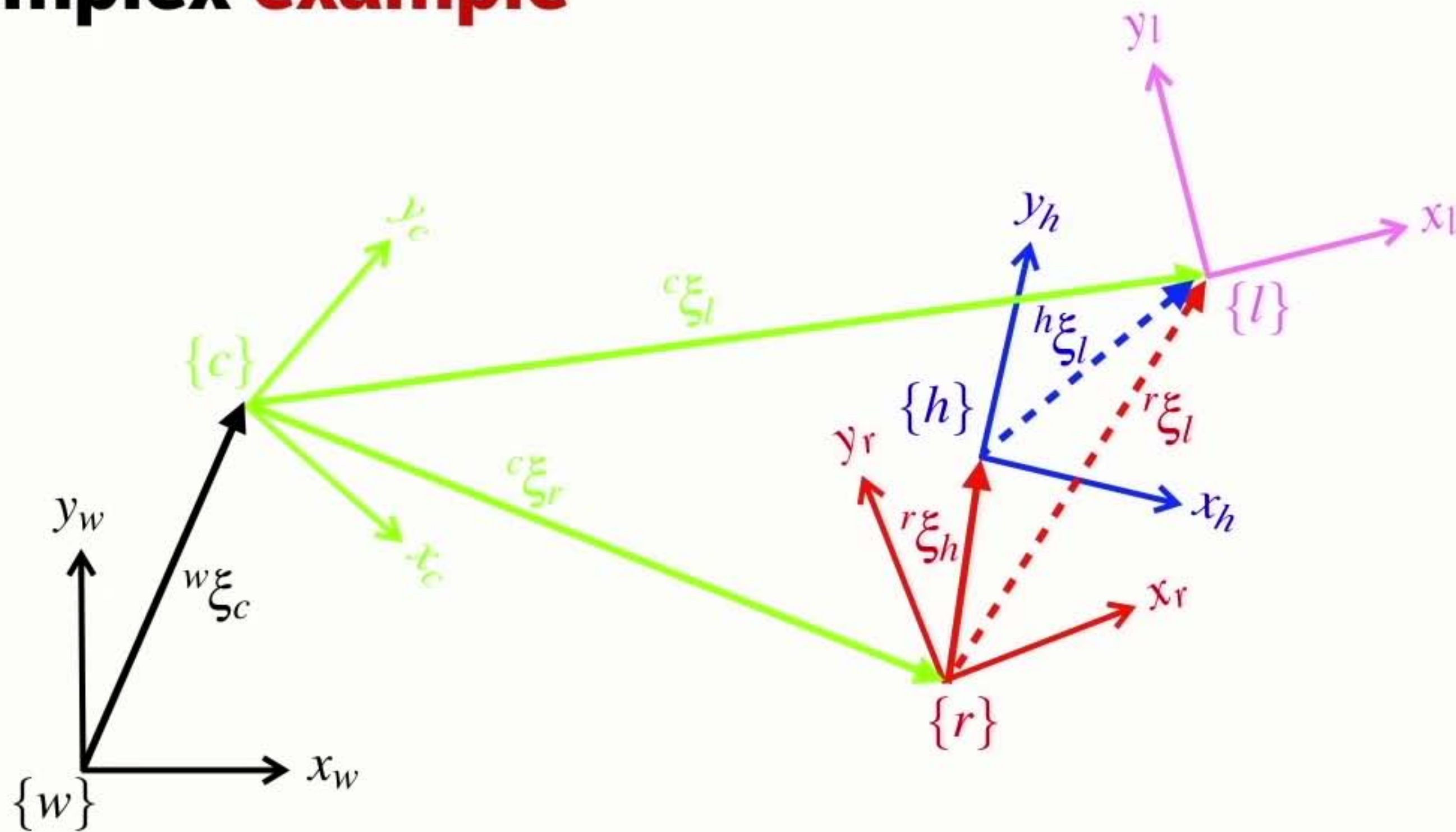
Complex example



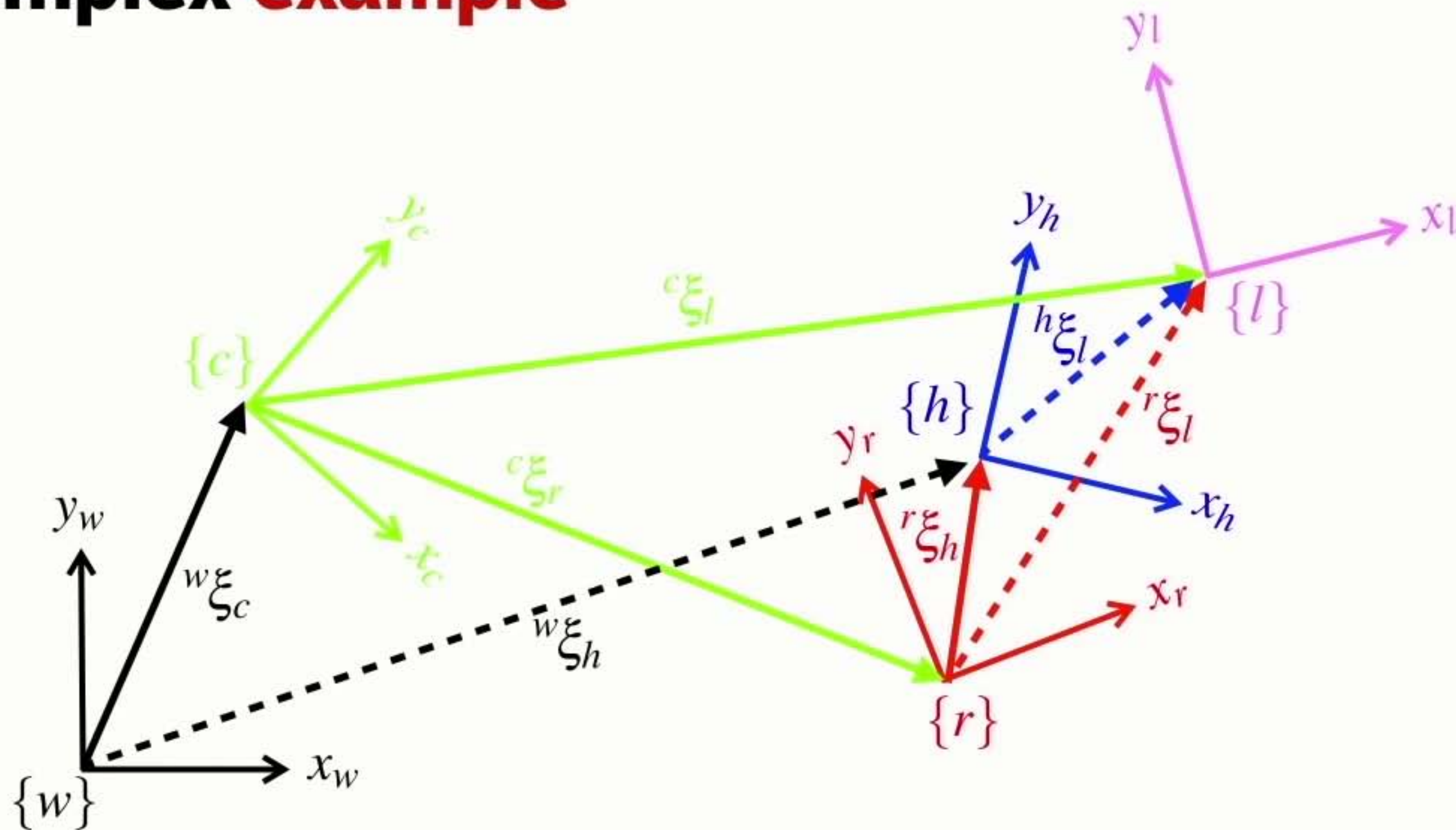
Complex example



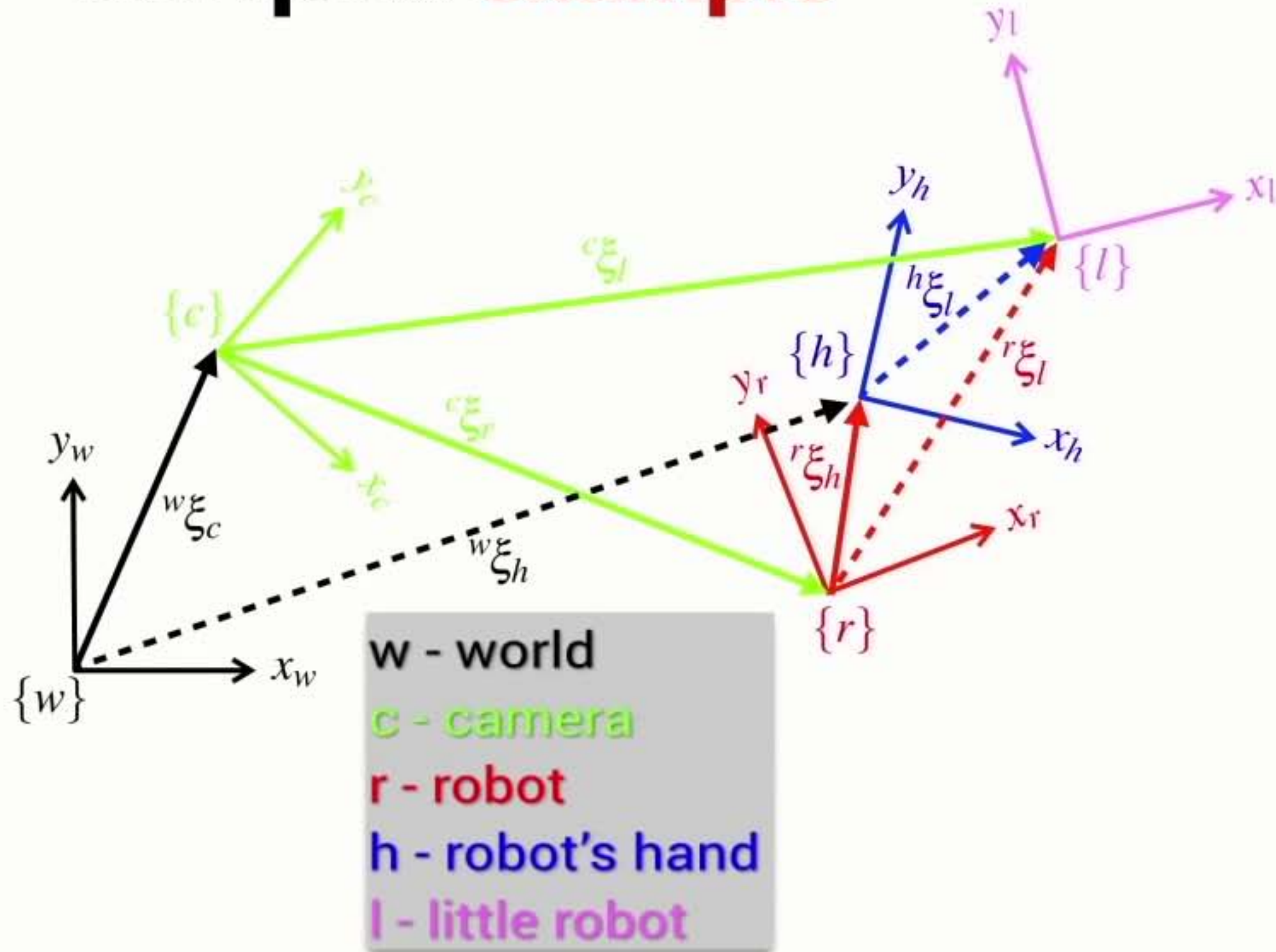
Complex example



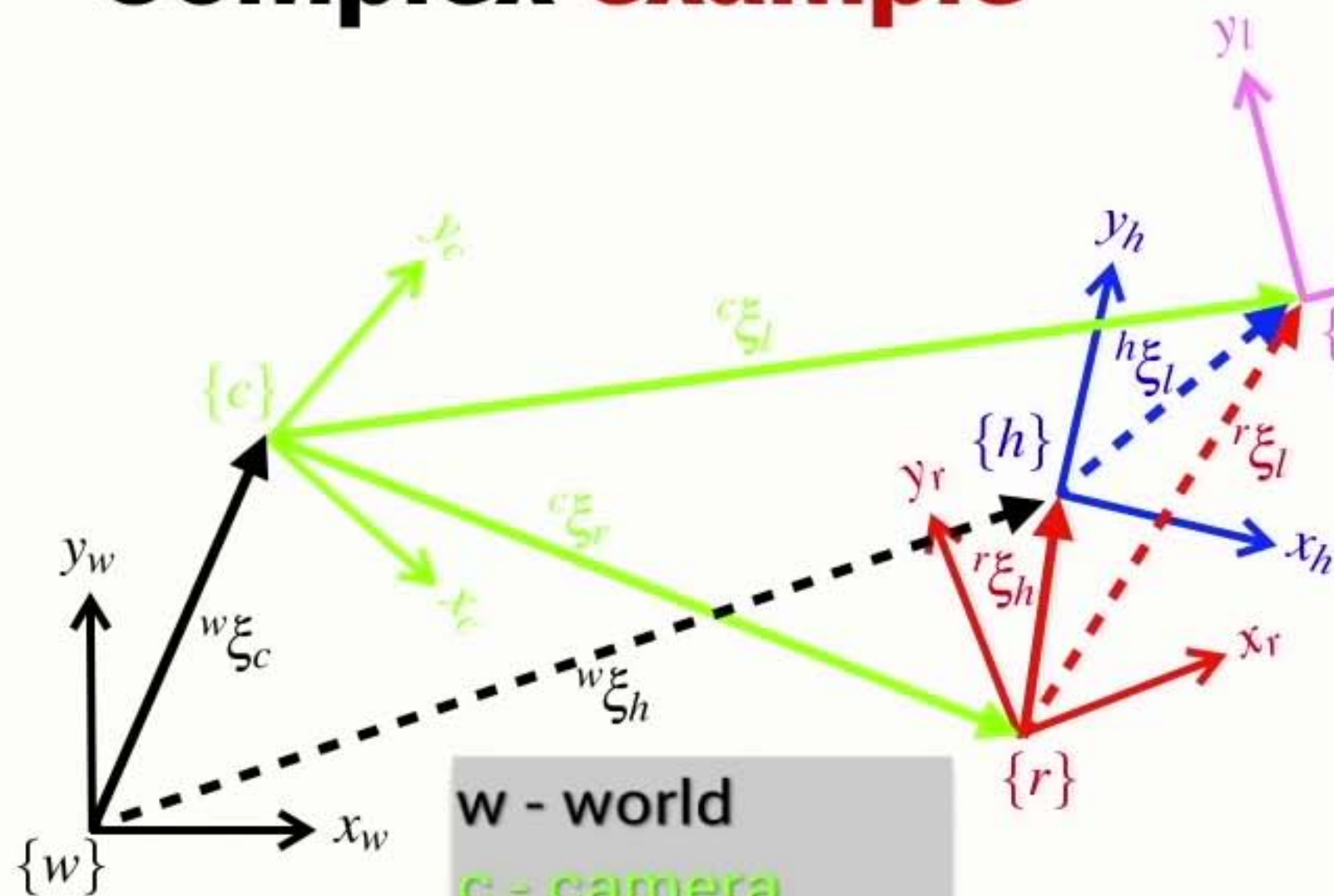
Complex example



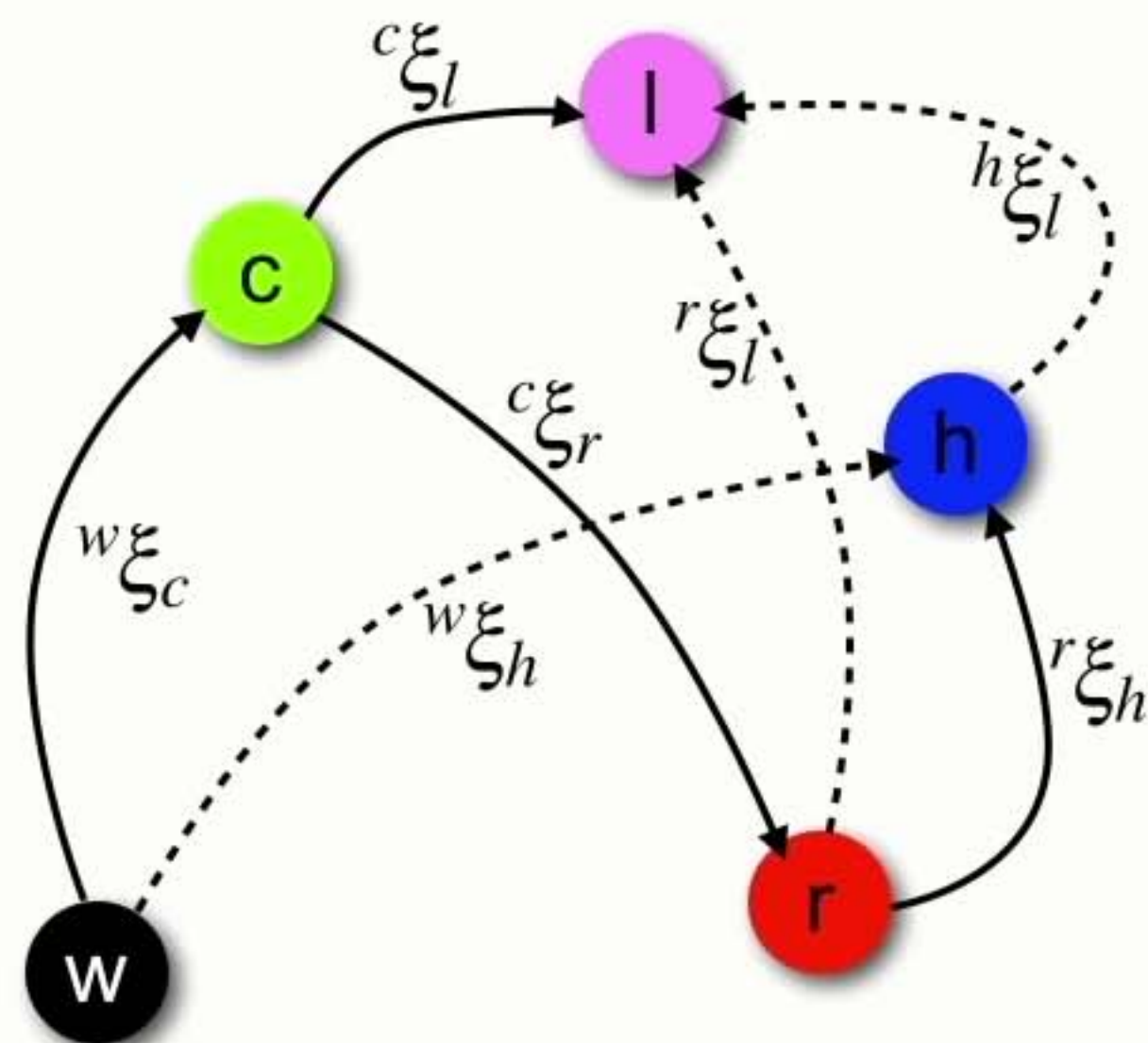
Complex example



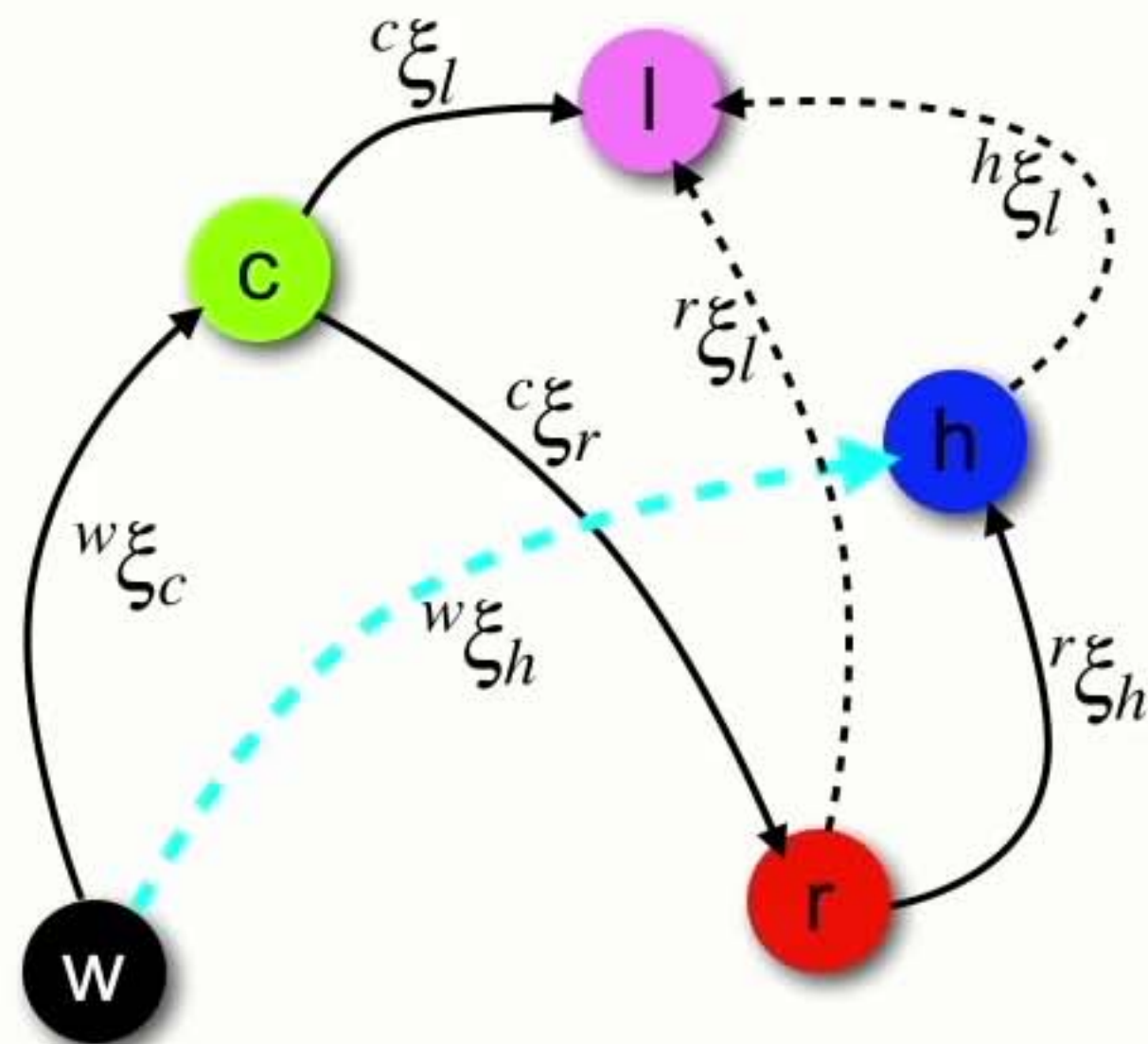
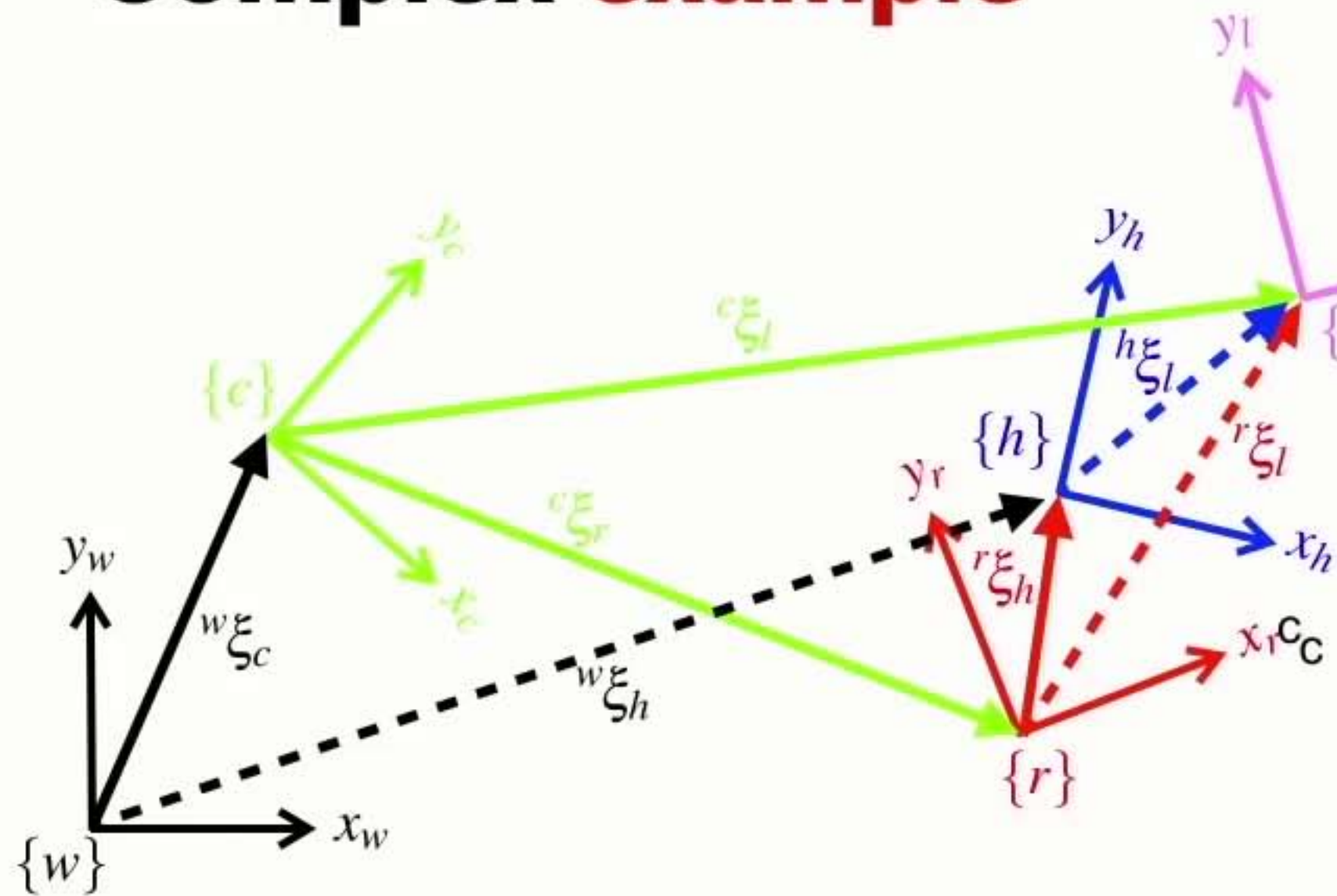
Complex example



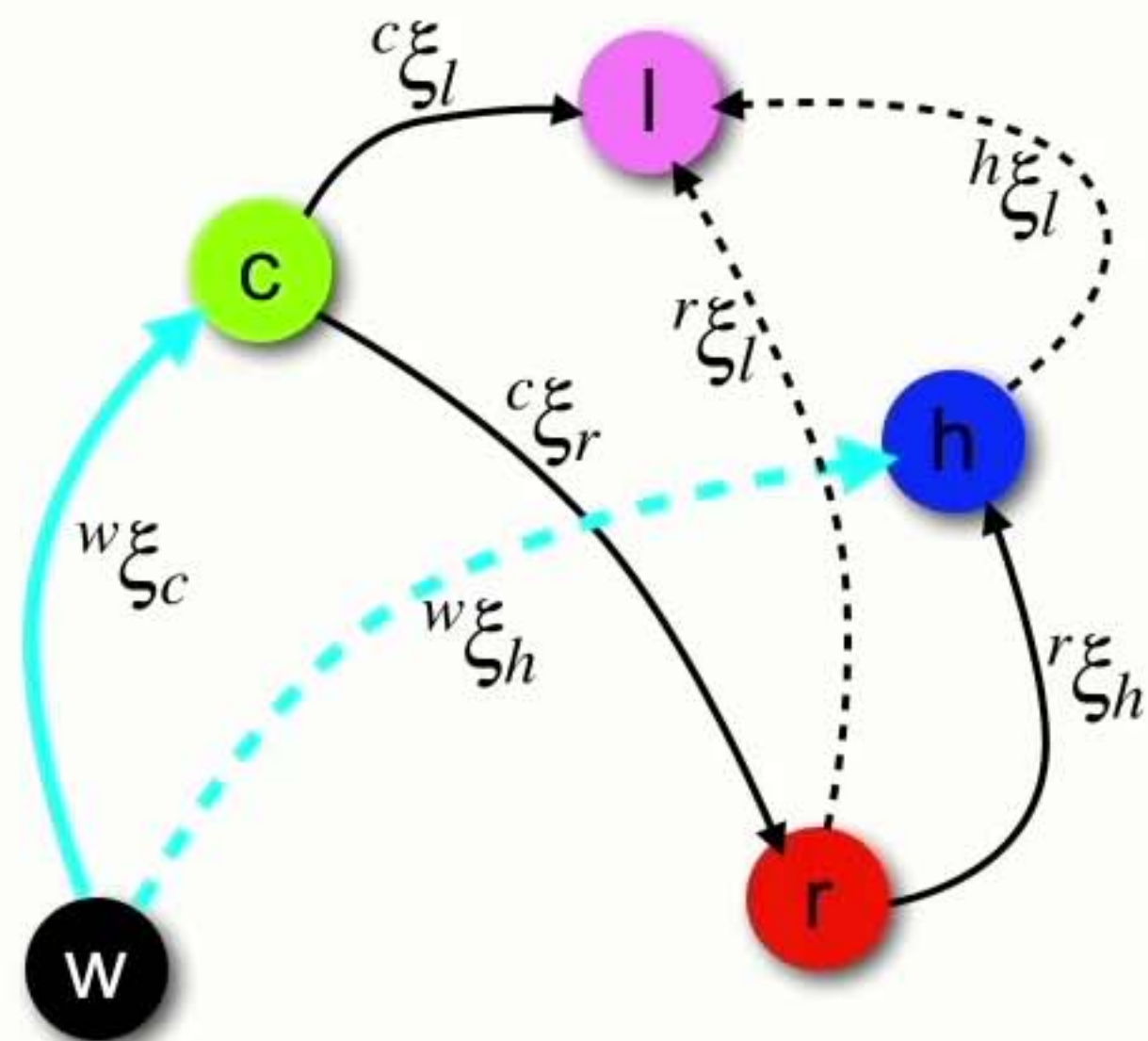
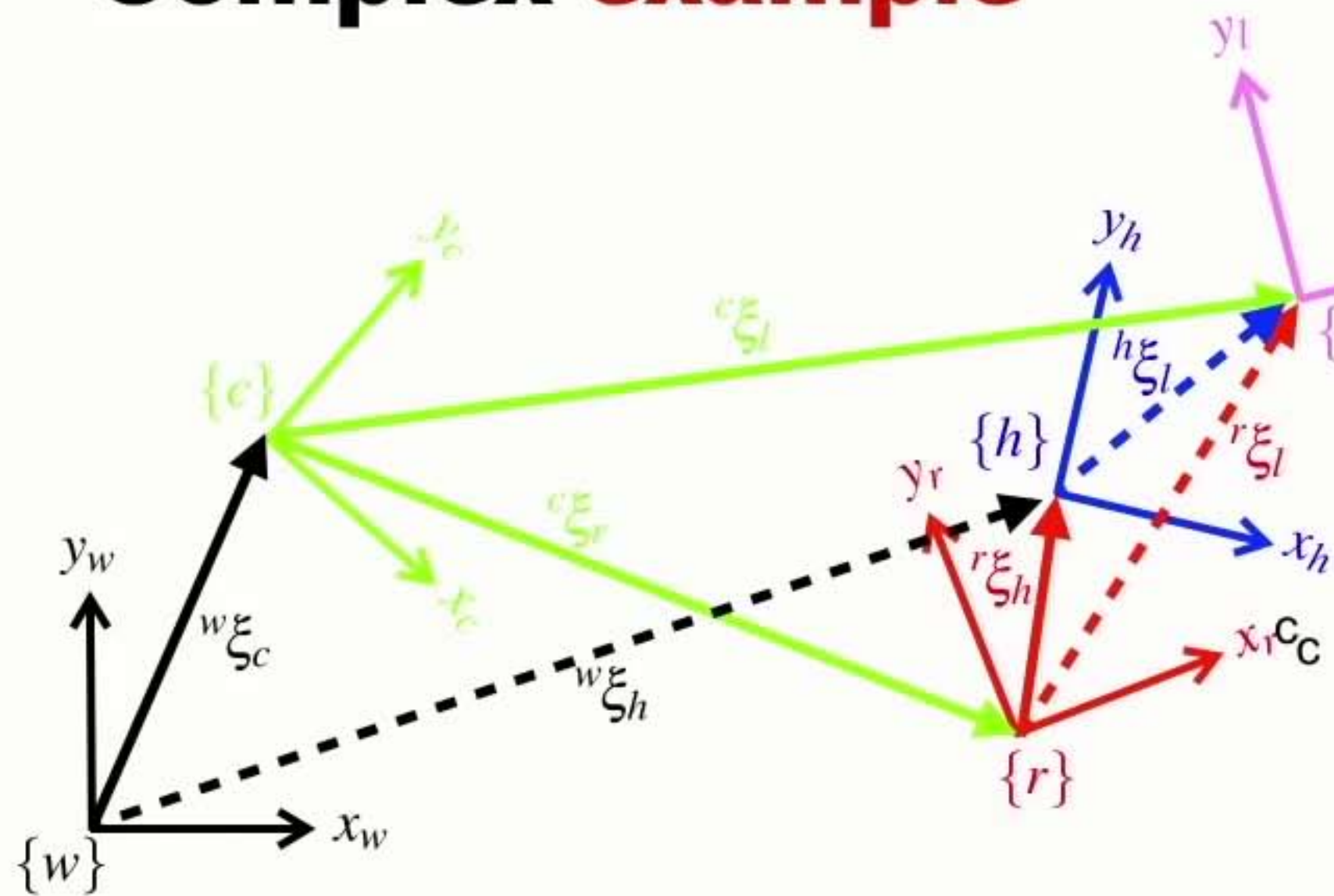
w - world
c - camera
r - robot
h - robot's hand
l - little robot



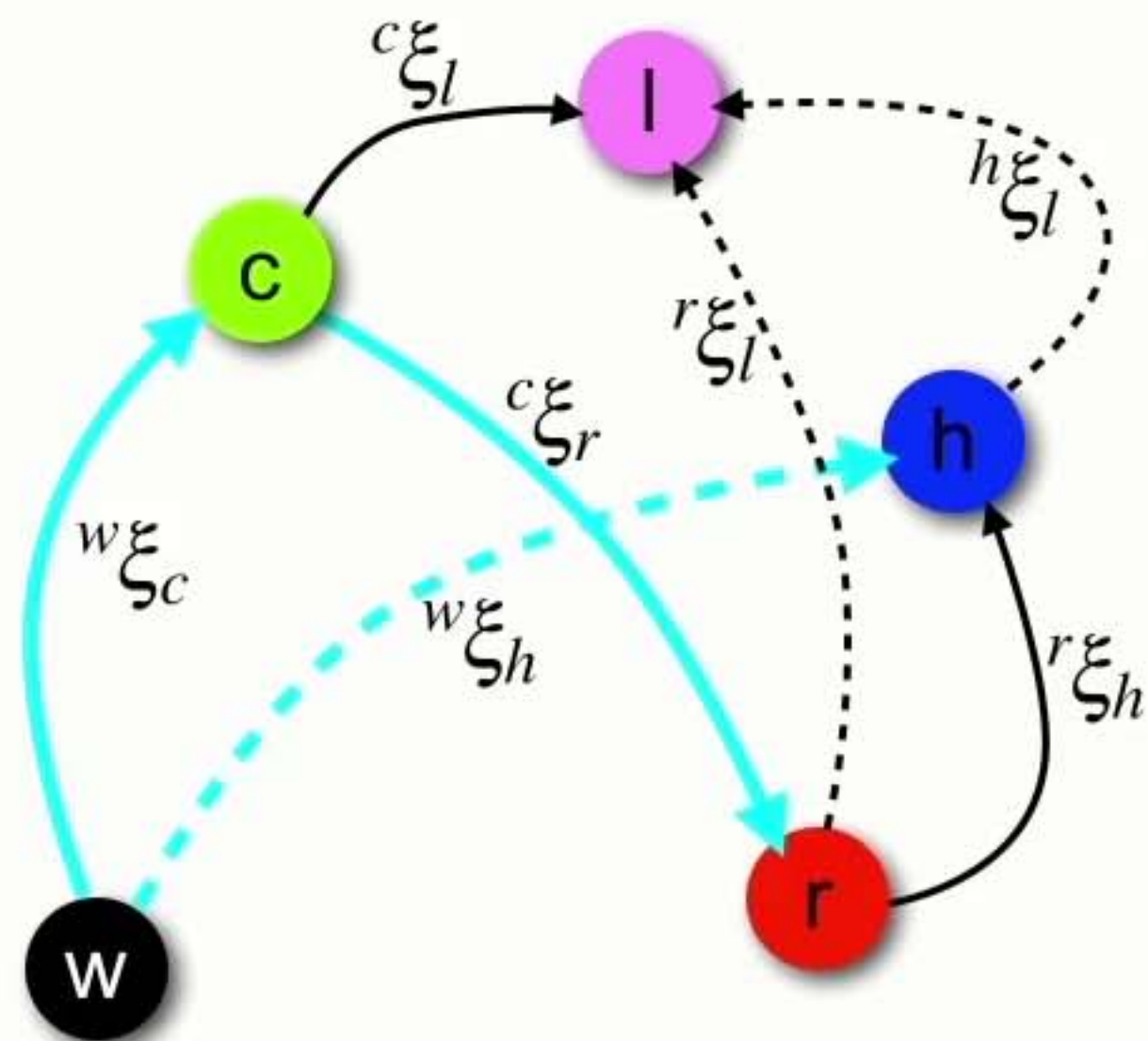
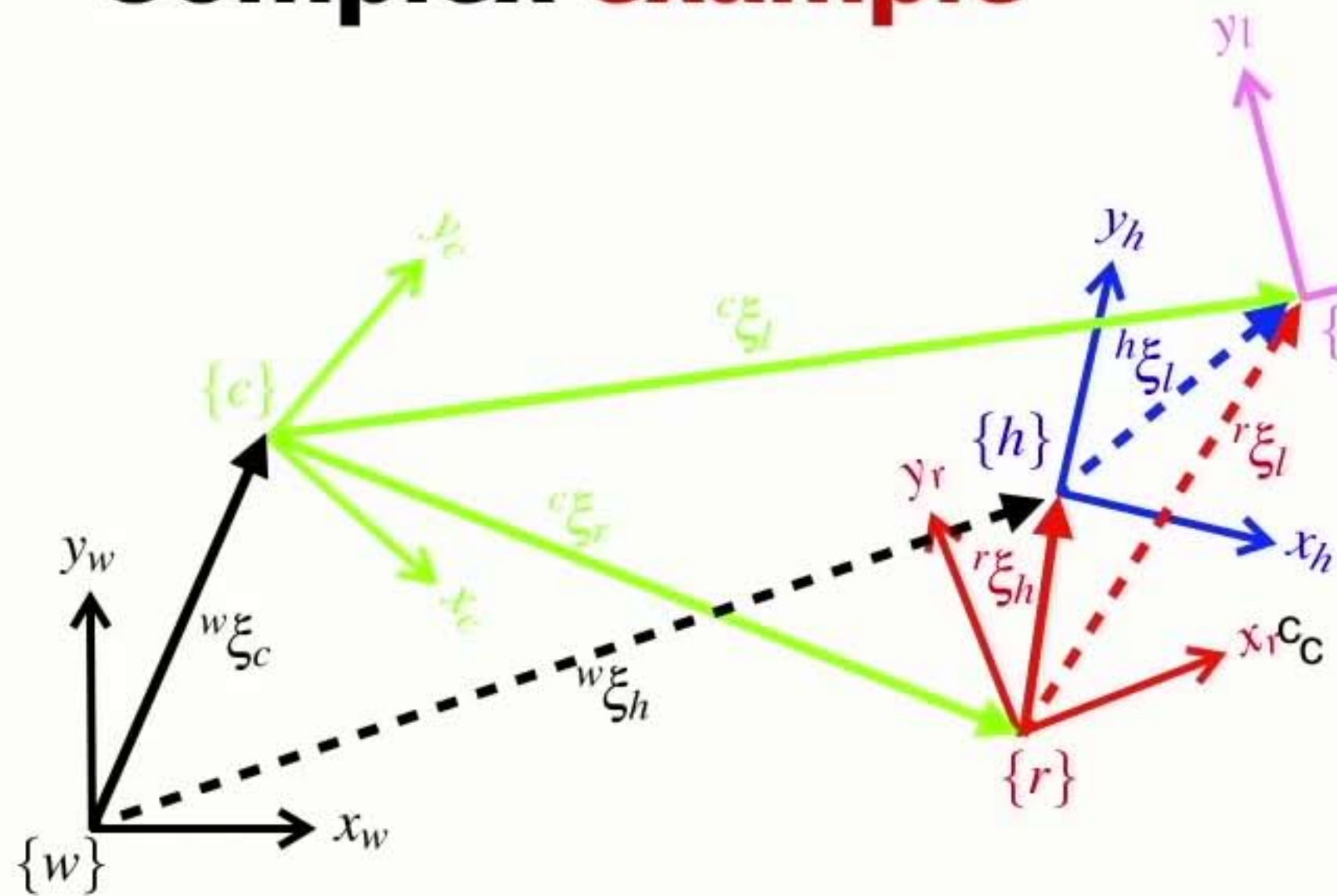
Complex example



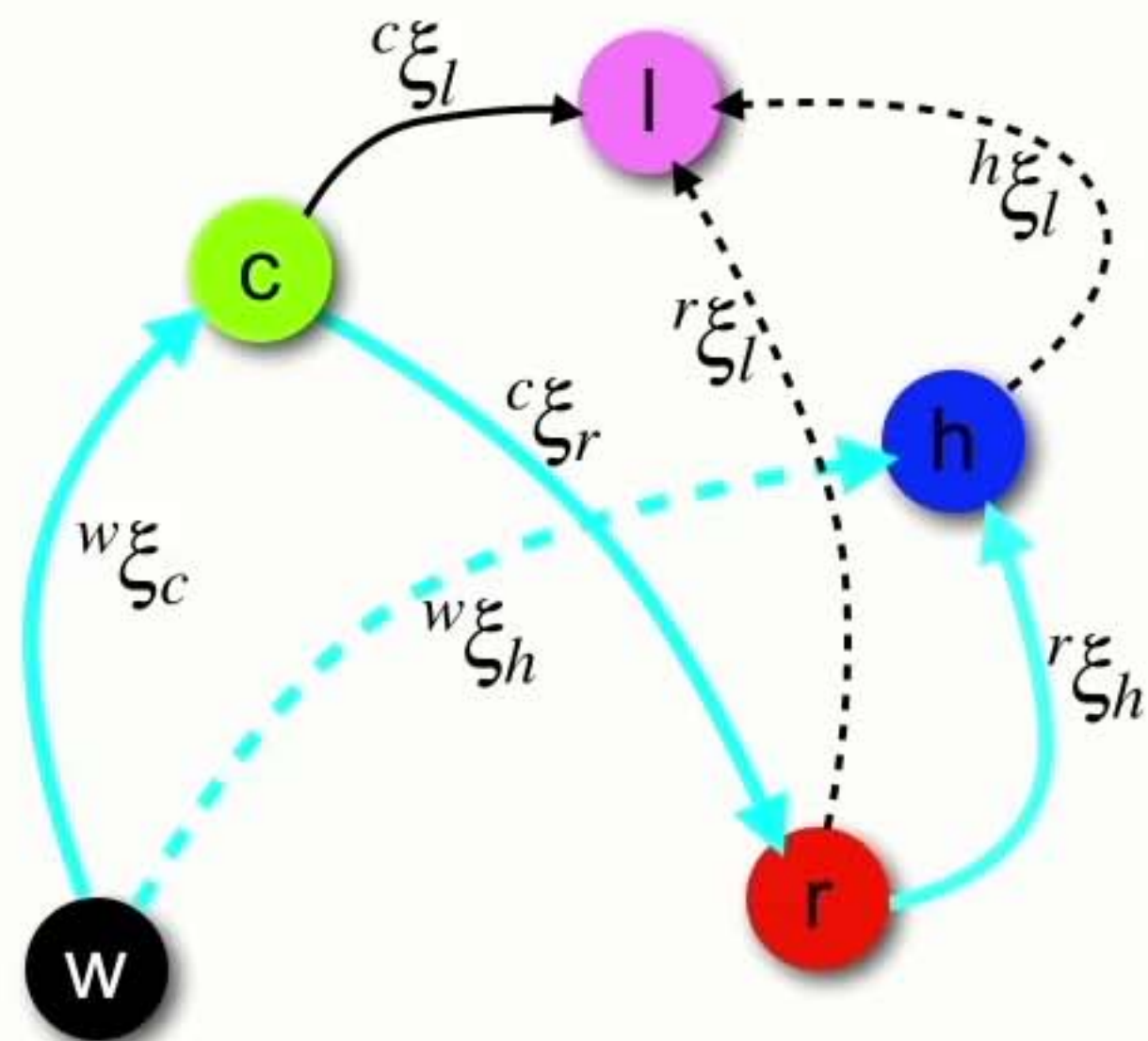
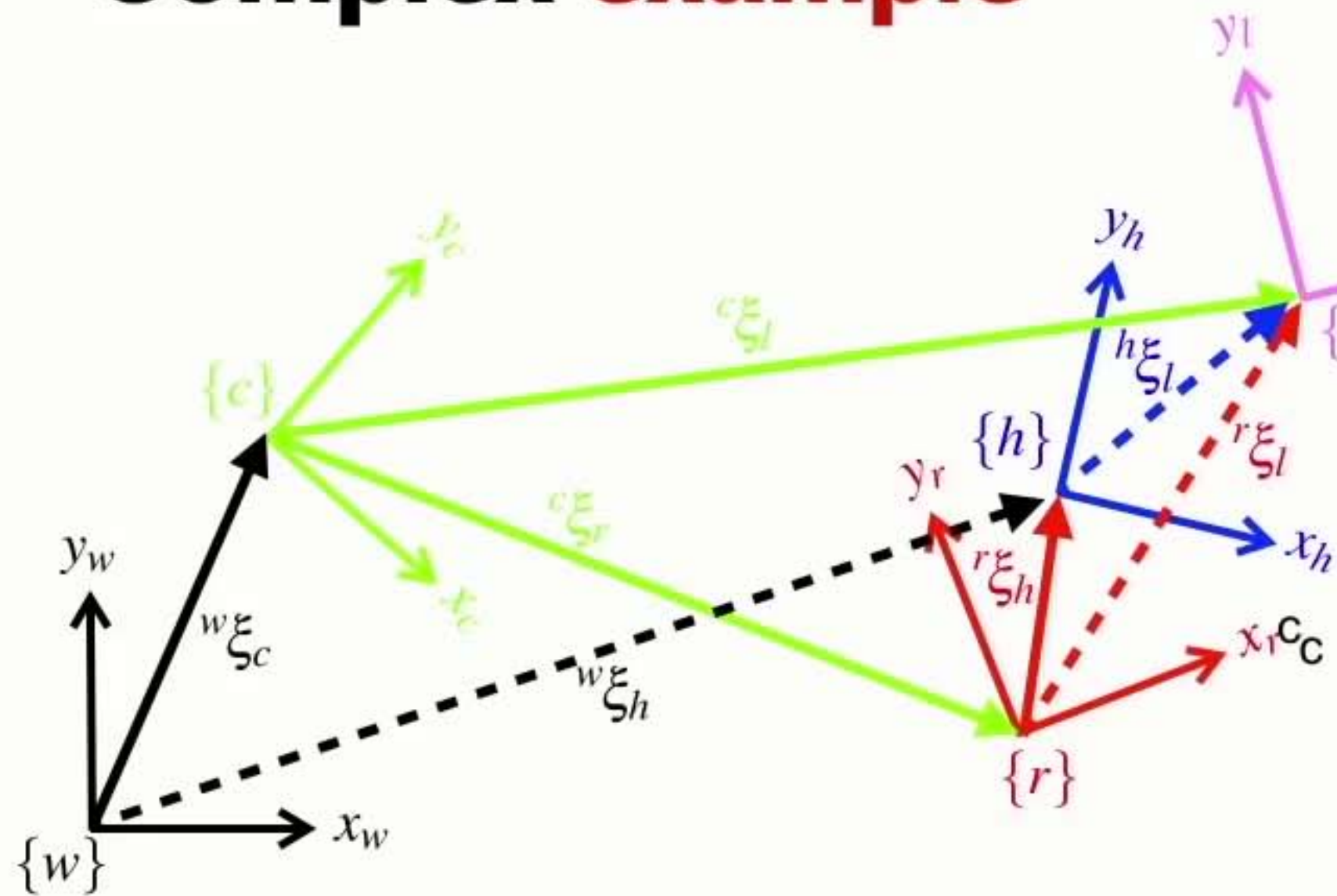
Complex example



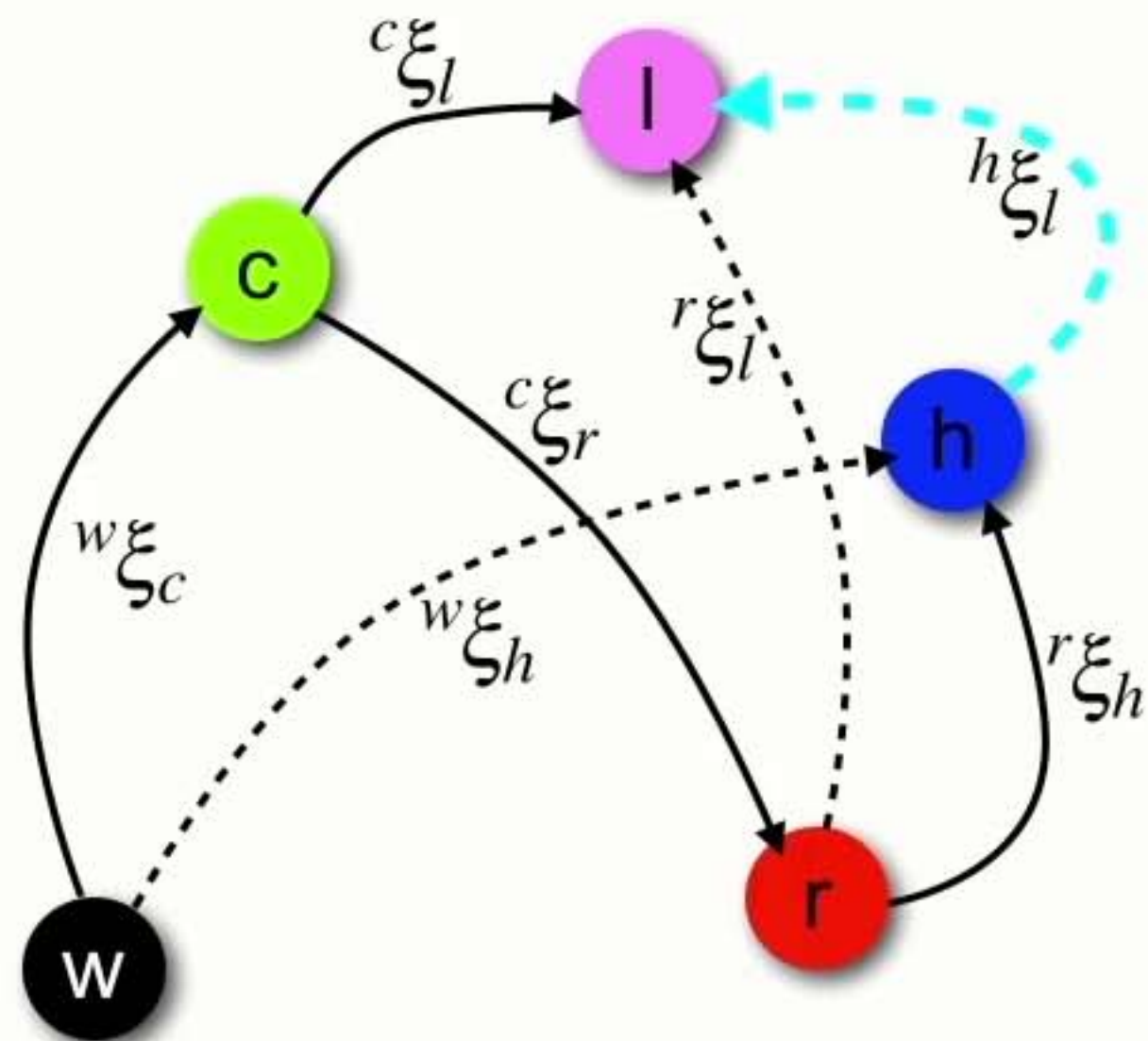
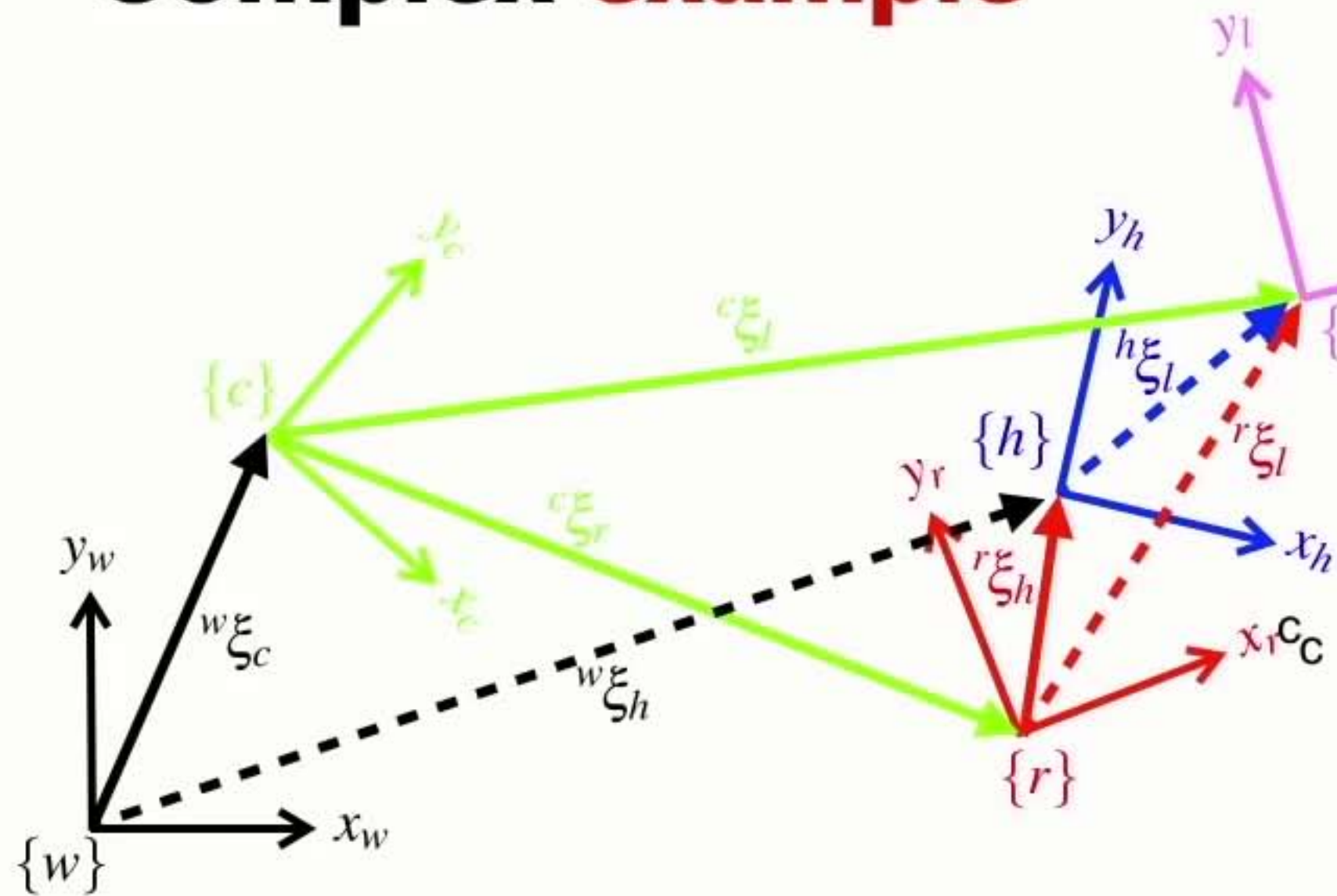
Complex example



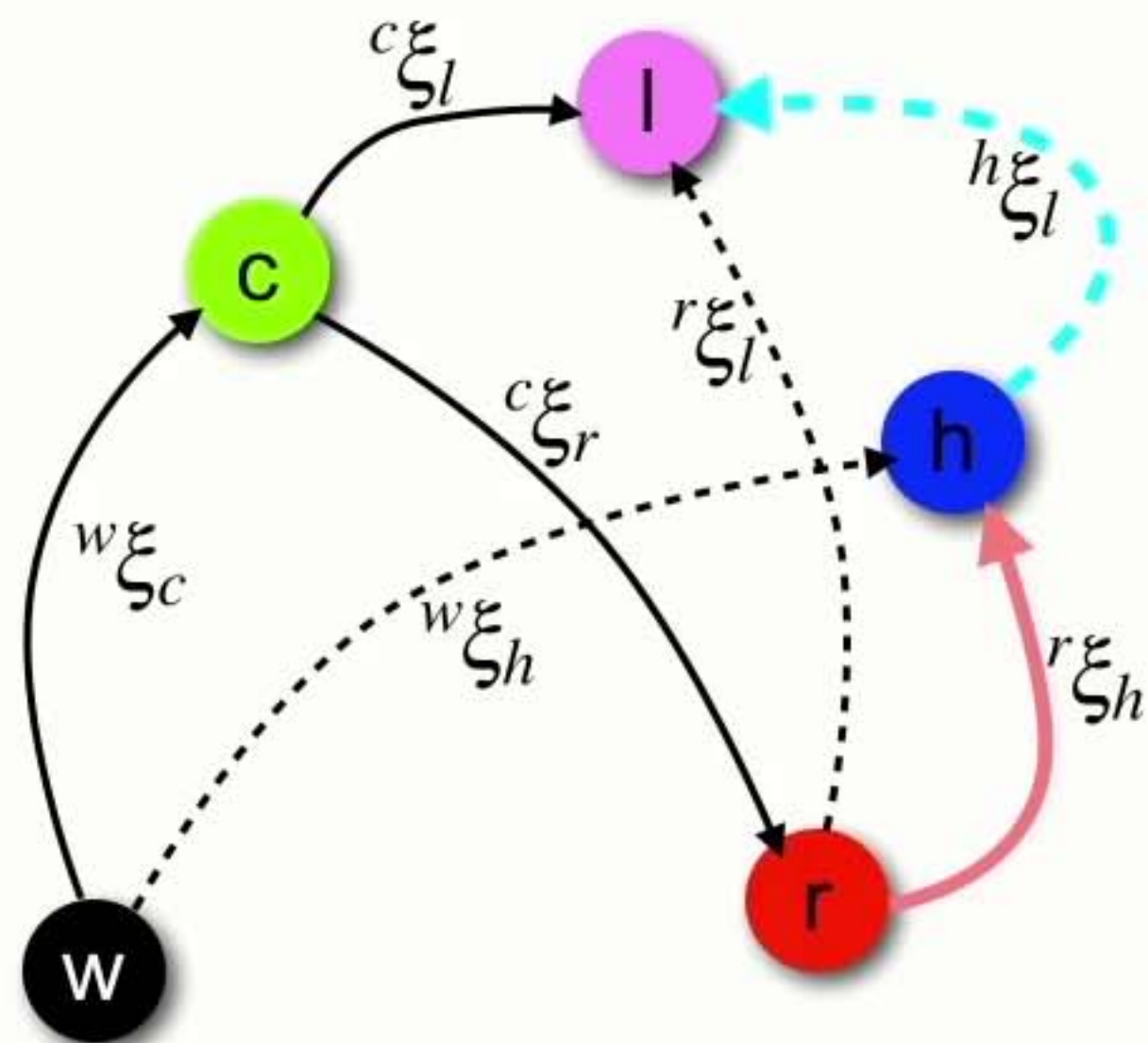
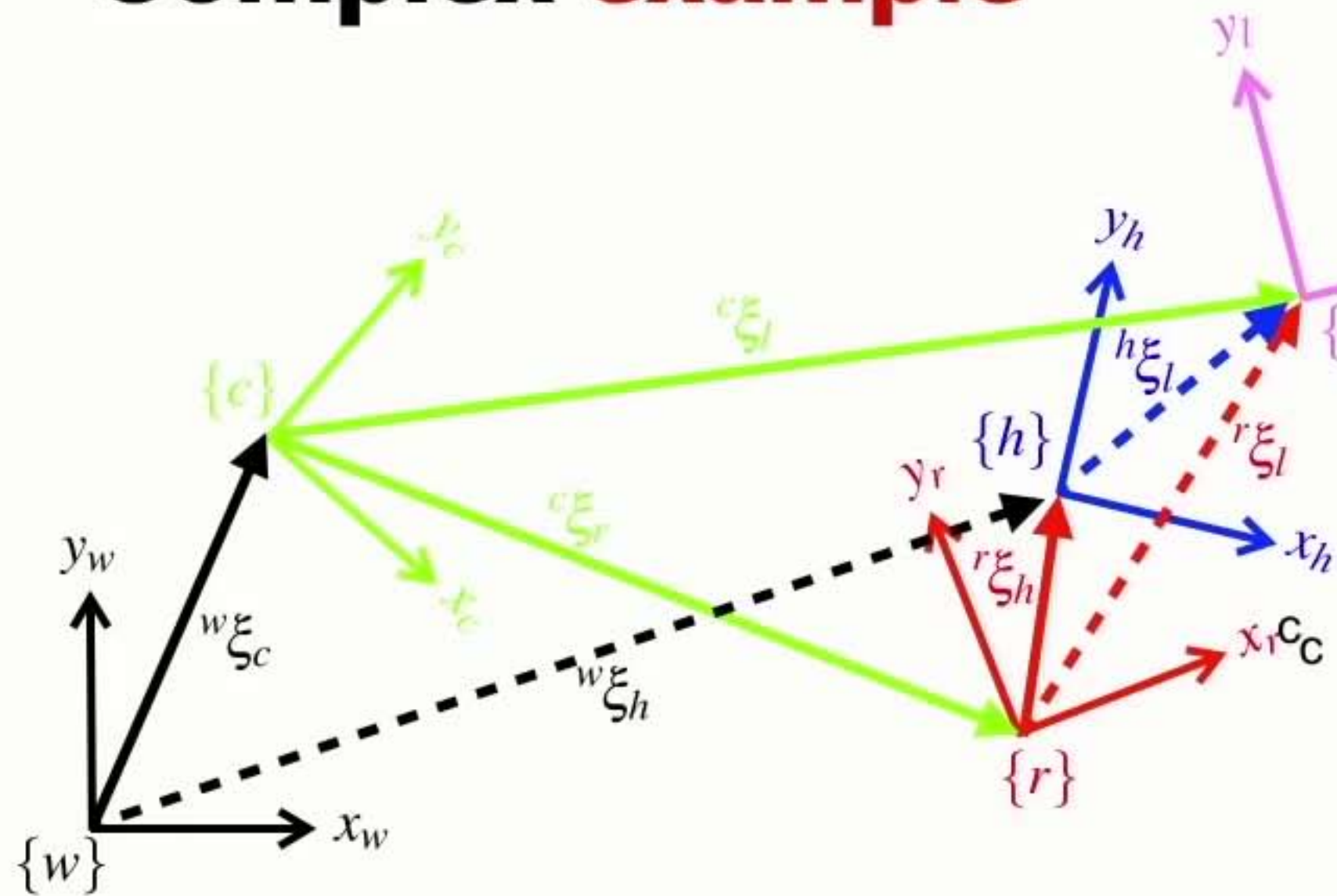
Complex example



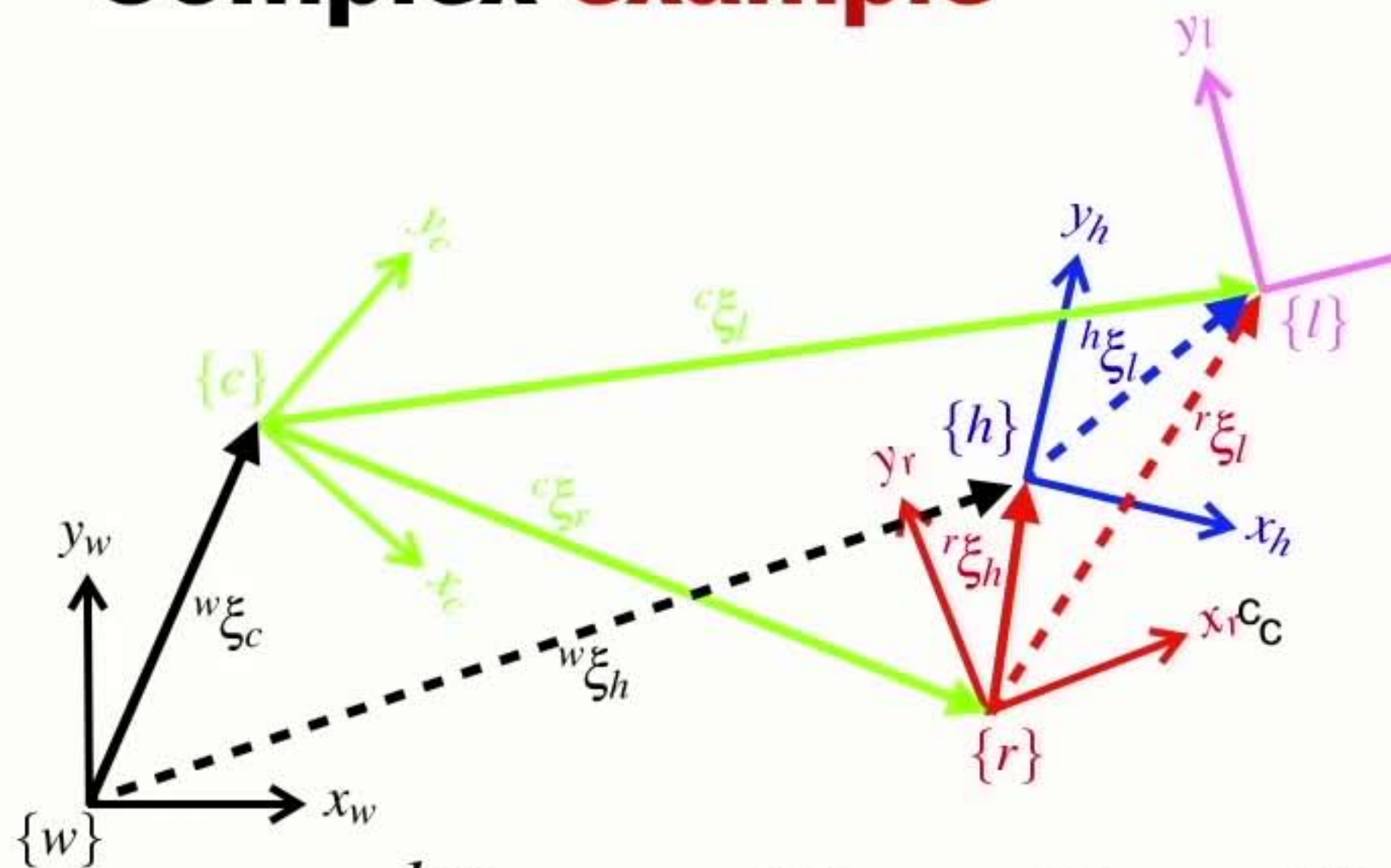
Complex example



Complex example

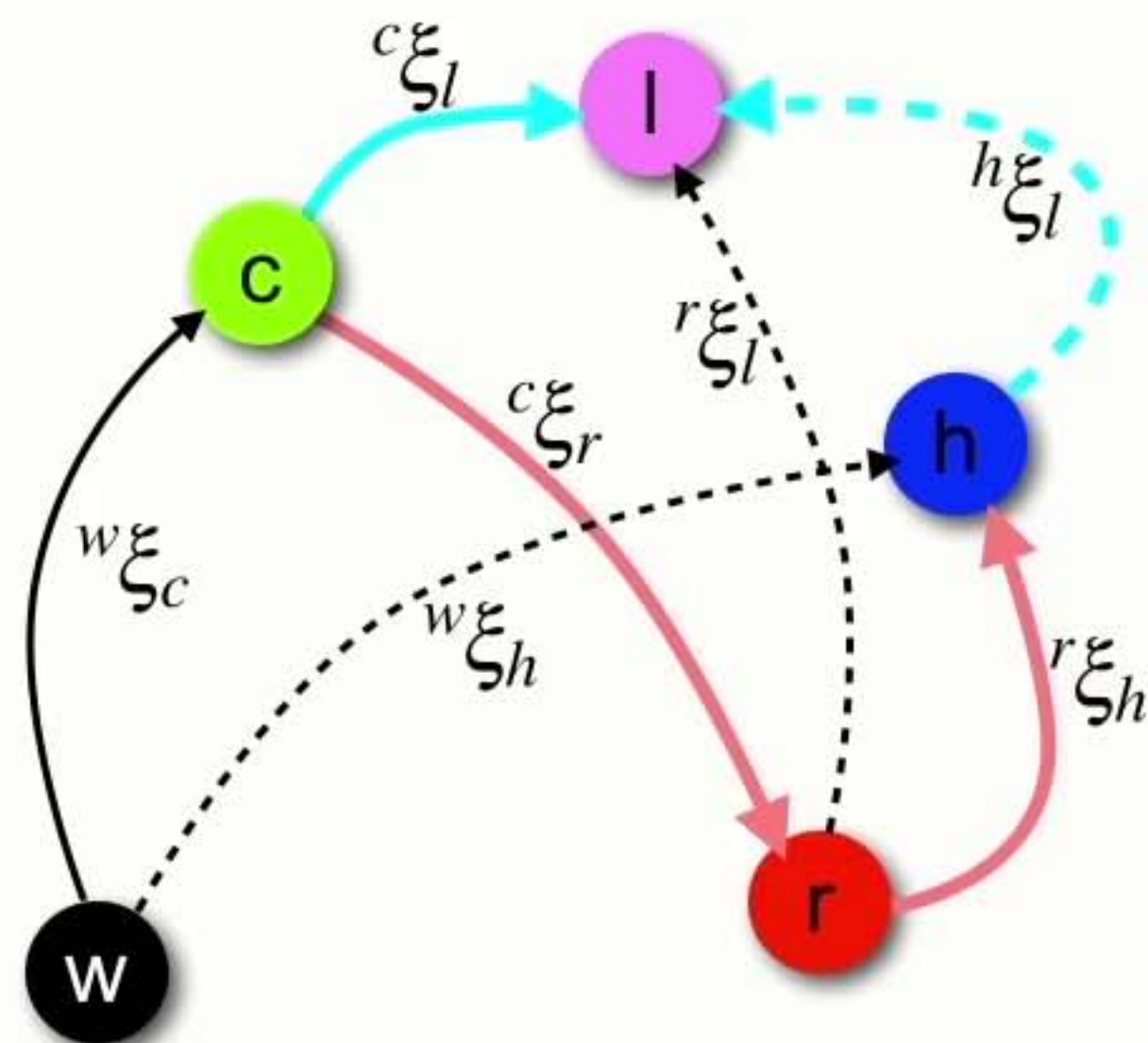


Complex example

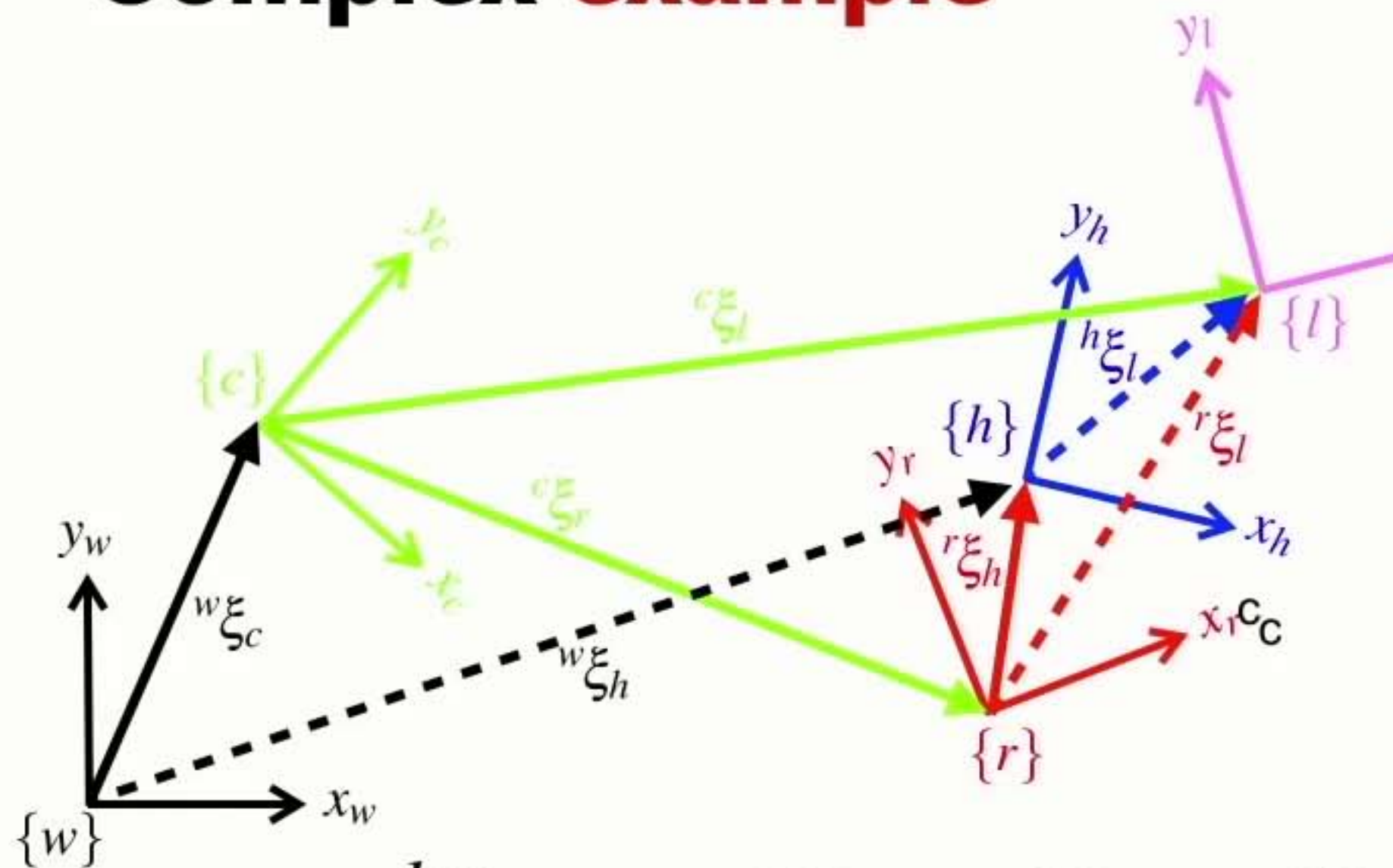


$$h_{\xi_l} = \ominus r_{\xi_h} \ominus c_{\xi_r} \oplus c_{\xi_l}$$

reverse direction

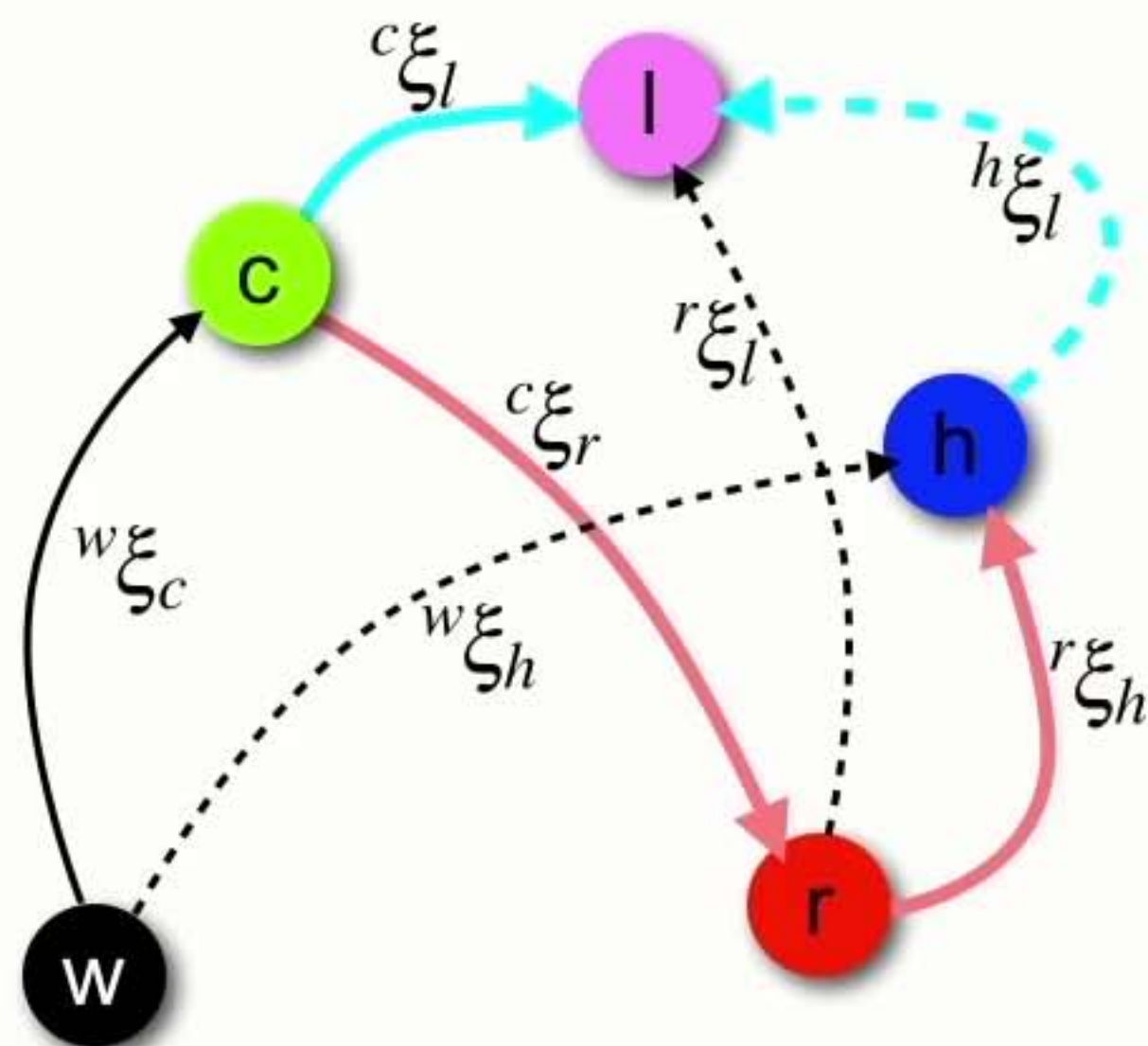


Complex example



$$h_{\xi_l}^{\xi} = \ominus r_{\xi_h}^{\xi} \ominus c_{\xi_r}^{\xi} \oplus c_{\xi_l}^{\xi}$$

reverse direction



Pose algebra

- There are just a few rules

$${}^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z$$

Pose algebra

- There are just a few rules

$$X_{\xi_Y}^{\xi} \oplus {}^Y_{\xi_Z}^{\xi} = X_{\xi_Z}^{\xi}$$

Pose algebra

- There are just a few rules

$$X^{\xi}_{\zeta_Y} \oplus Y^{\xi}_{\zeta_Z} = X^{\xi}_{\zeta_Z}$$

Pose algebra

- There are just a few rules

$$X \overset{\xi}{\underset{Y}{\zeta}} \oplus Y \overset{\xi}{\underset{Z}{\zeta}} = X \overset{\xi}{\underset{Z}{\zeta}}$$

Pose algebra

- There are just a few rules

$$X^{\xi}_{\zeta_Y} \oplus Y^{\xi}_{\zeta_Z} = X^{\xi}_{\zeta_Z}$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

Pose algebra

- There are just a few rules

$${}^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus^X\xi_Y = {}^Y\xi_X$$

Pose algebra

- There are just a few rules

$${}^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus^X\xi_Y = {}^Y\xi_X$$

$$\xi \ominus \xi = 0 \quad \ominus\xi \oplus \xi = 0$$

Pose algebra

- There are just a few rules

$$X \overset{\xi}{\underset{Y}{\zeta}} \oplus Y \overset{\xi}{\underset{Z}{\zeta}} = X \overset{\xi}{\underset{Z}{\zeta}}$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus^X \xi_Y = {}^Y \xi_X$$

$$\xi \ominus \xi = 0 \quad \ominus \xi \oplus \xi = 0$$

$$\xi \ominus 0 = \xi \quad \xi \oplus 0 = \xi$$

Pose algebra

- There are just a few rules

$${}^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z$$

$${}^Xp = {}^X\xi_Y \bullet {}^Yp$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus {}^X\xi_Y = {}^Y\xi_X$$

$$\xi \ominus \xi = 0 \quad \ominus \xi \oplus \xi = 0$$

$$\xi \ominus 0 = \xi \quad \xi \oplus 0 = \xi$$

Pose algebra

- There are just a few rules

$${}^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z$$

$${}^Xp = {}^X\xi_Y \bullet {}^Yp$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus {}^X\xi_Y = {}^Y\xi_X$$

$$\xi \ominus \xi = 0 \quad \ominus \xi \oplus \xi = 0$$

$$\xi \ominus 0 = \xi \quad \xi \oplus 0 = \xi$$

Pose algebra

- There are just a few rules

$$X \xi_Y \oplus Y \xi_Z = X \xi_Z$$

$$X p = X \xi_Y \bullet Y p$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\ominus^X \xi_Y = {}^Y \xi_X$$

$$\xi \ominus \xi = 0 \quad \ominus \xi \oplus \xi = 0$$

$$\xi \ominus 0 = \xi \quad \xi \oplus 0 = \xi$$

Section 5

Developing a real representation of pose

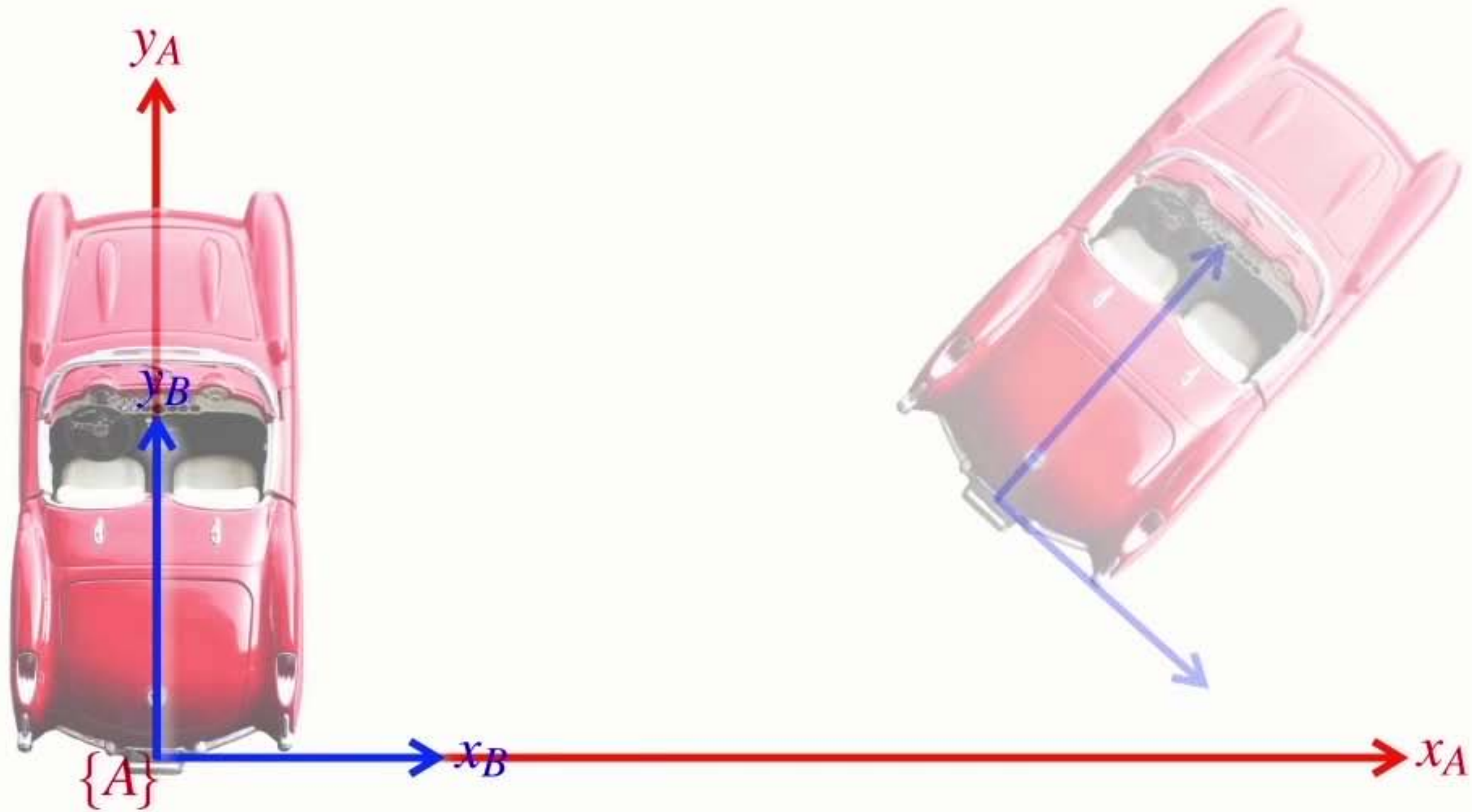
Pose



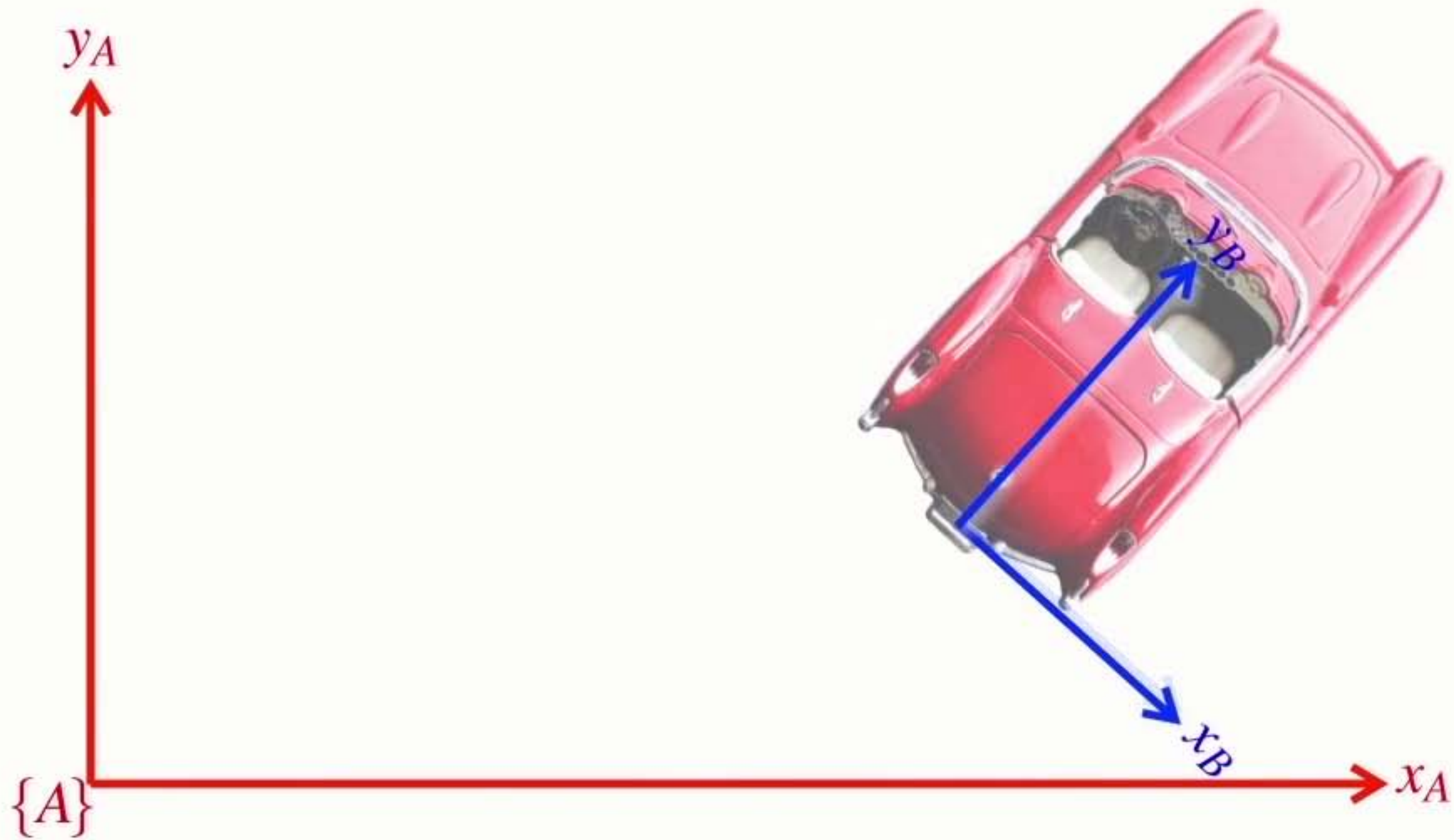
pronounced **ksi**

- It has 3 parameters: (x, y, θ)

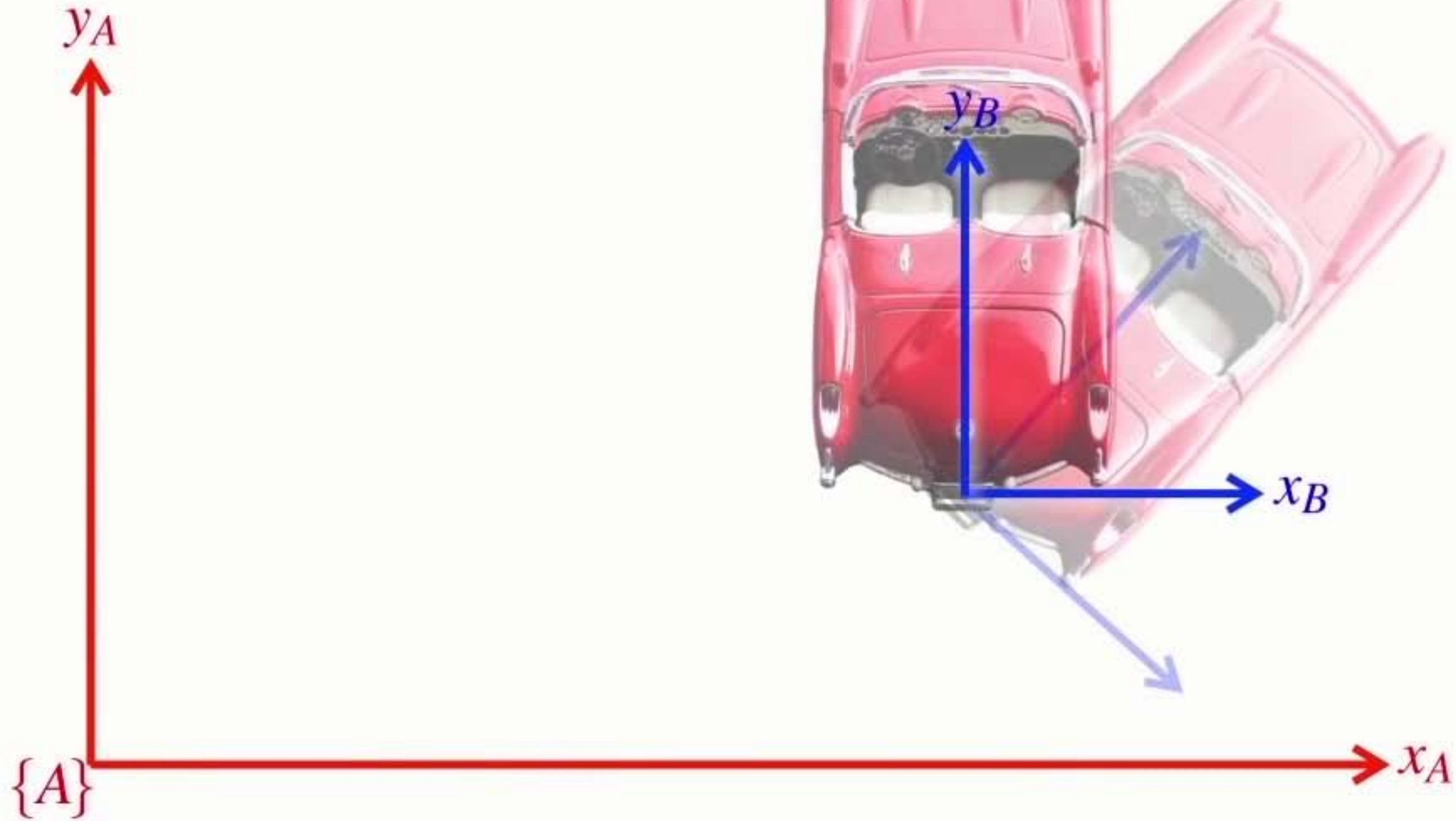
Separating **rotation** and **translation**



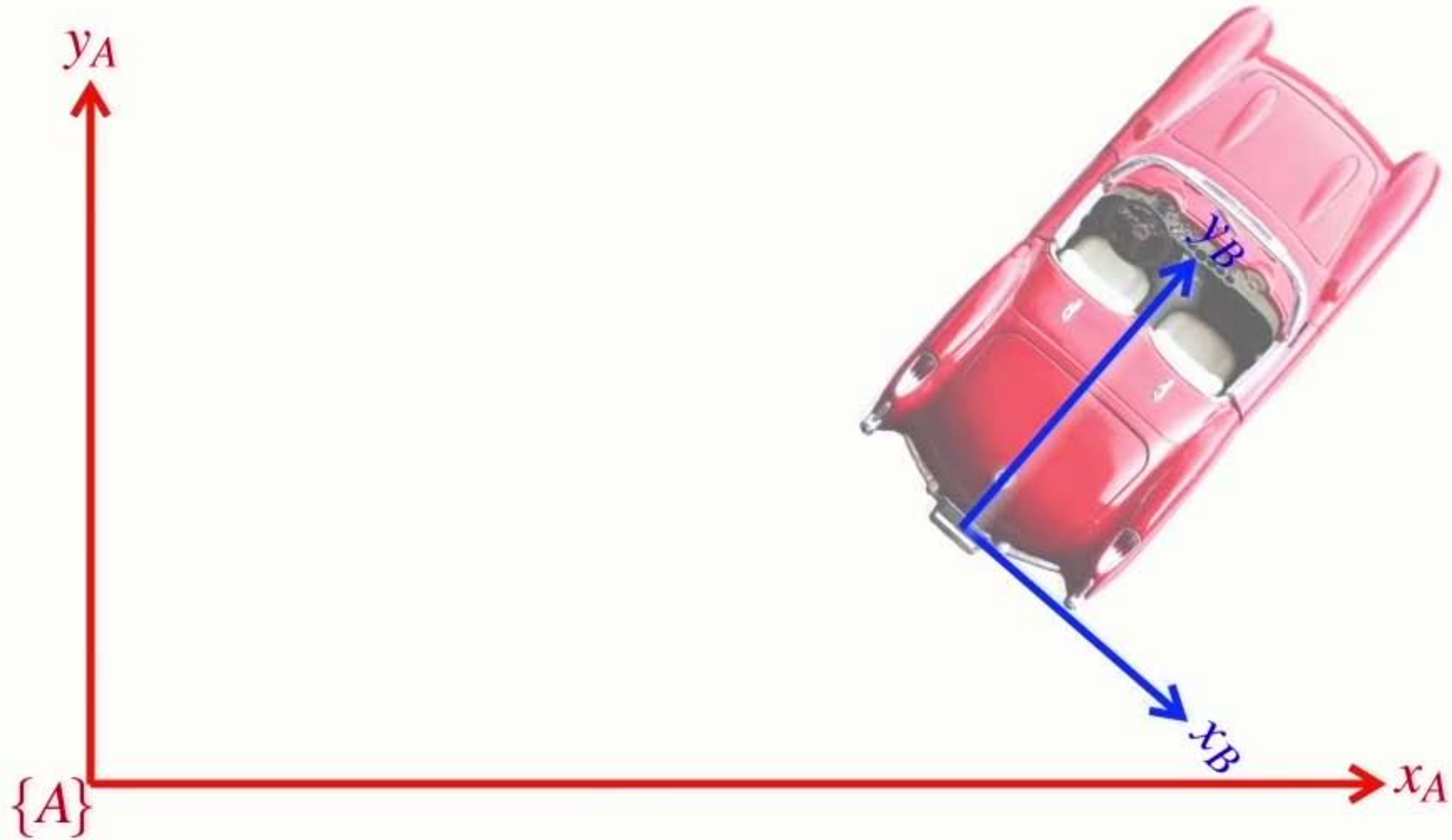
Separating **rotation** and **translation**



First a **translation**

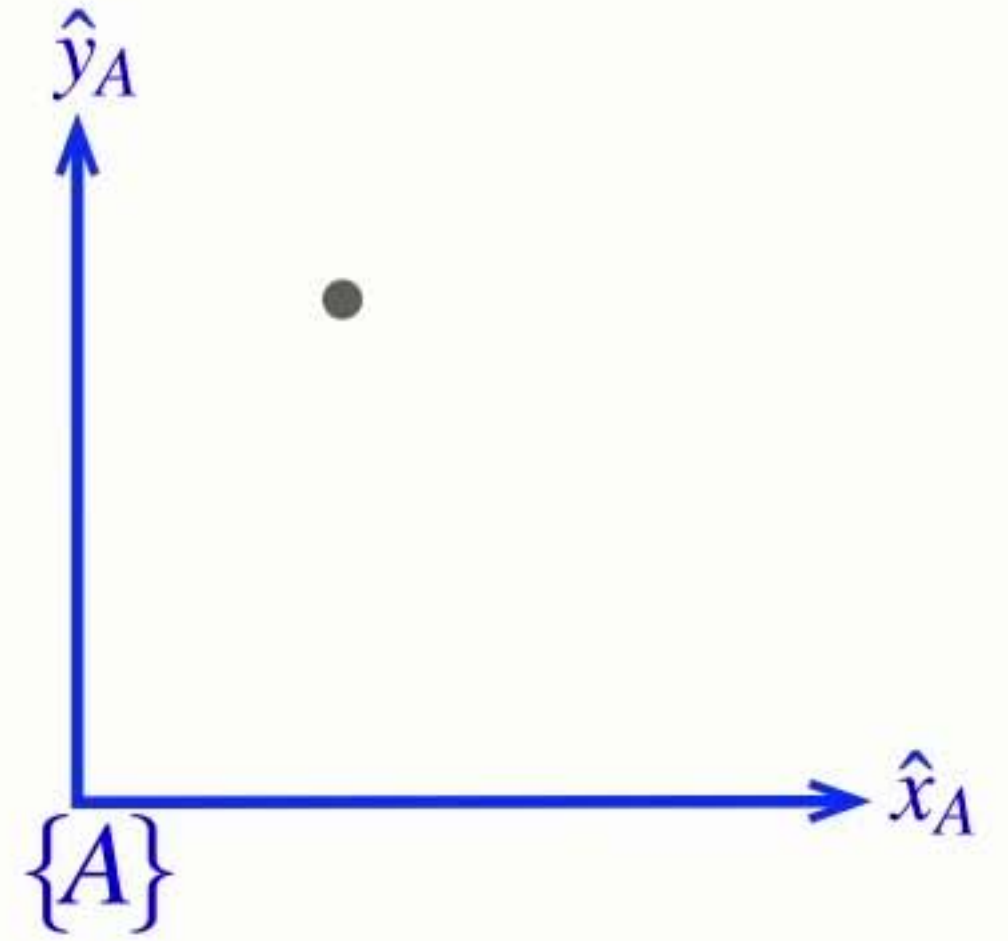


Then **rotation**

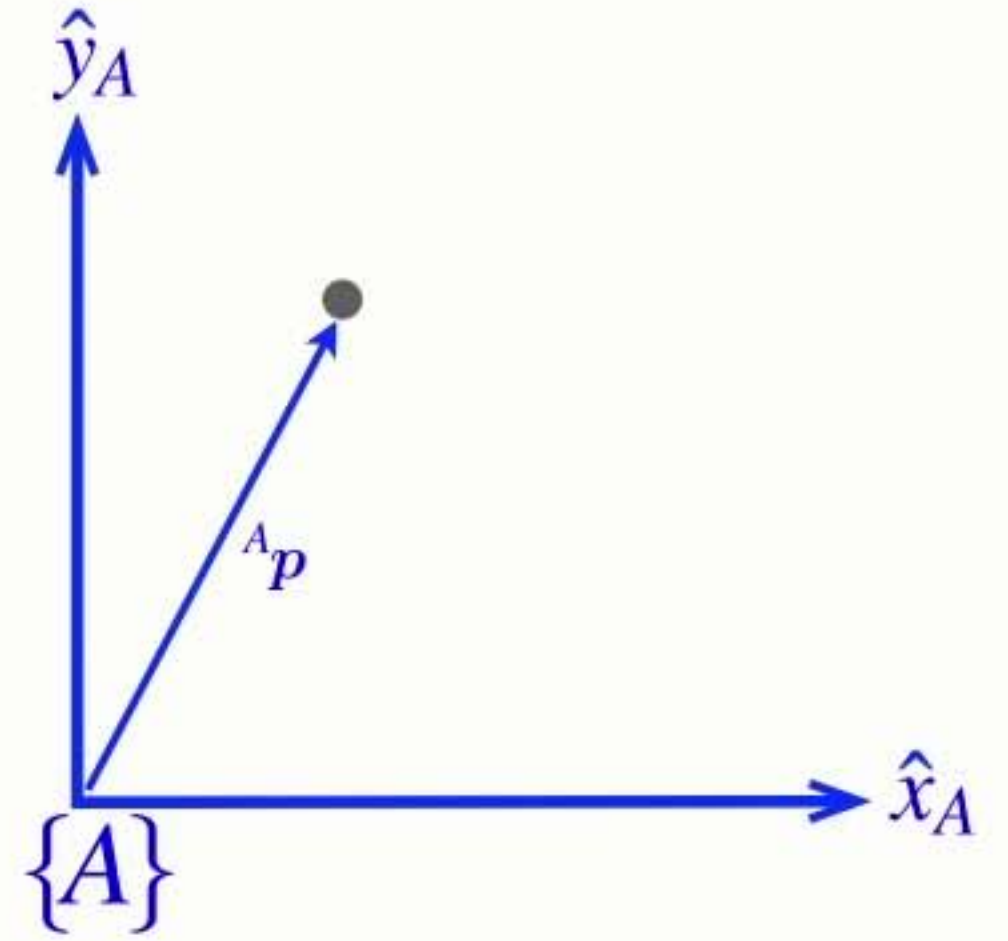


Section 6 Describing rotation

Understanding **rotation**

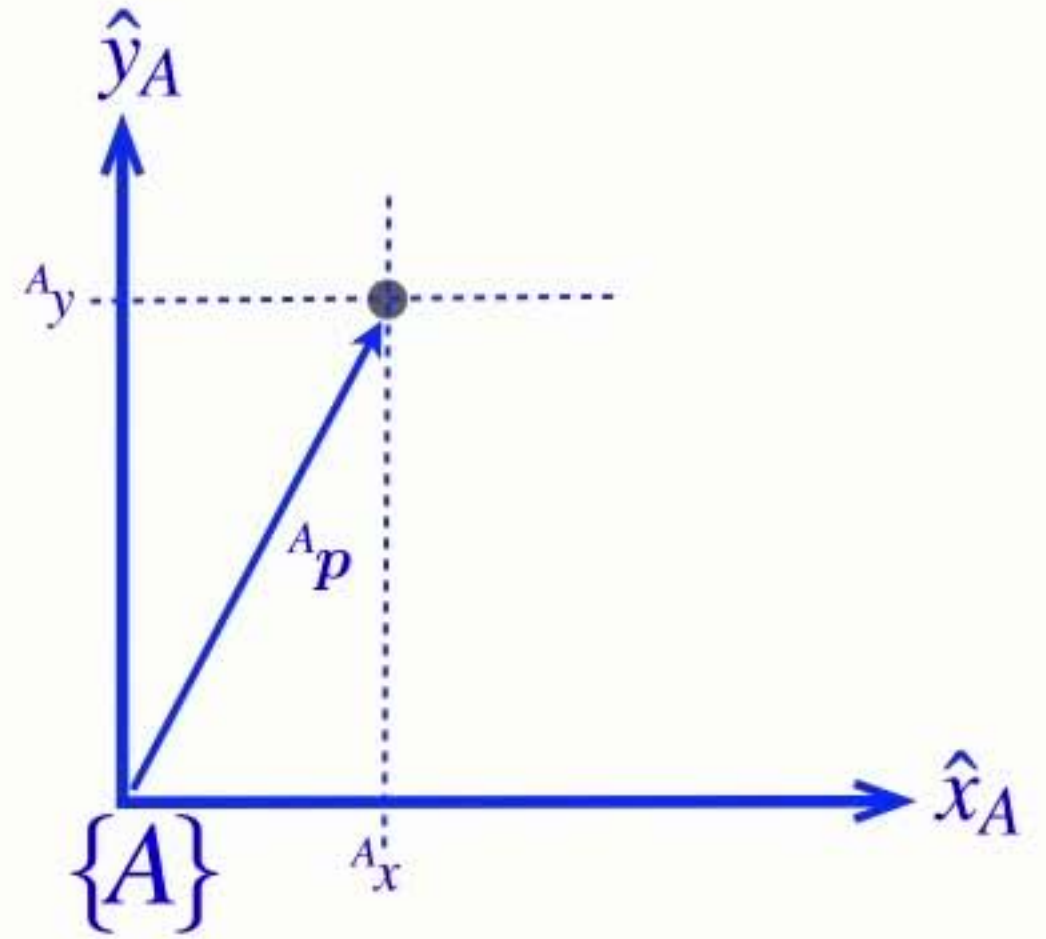


Understanding **rotation**



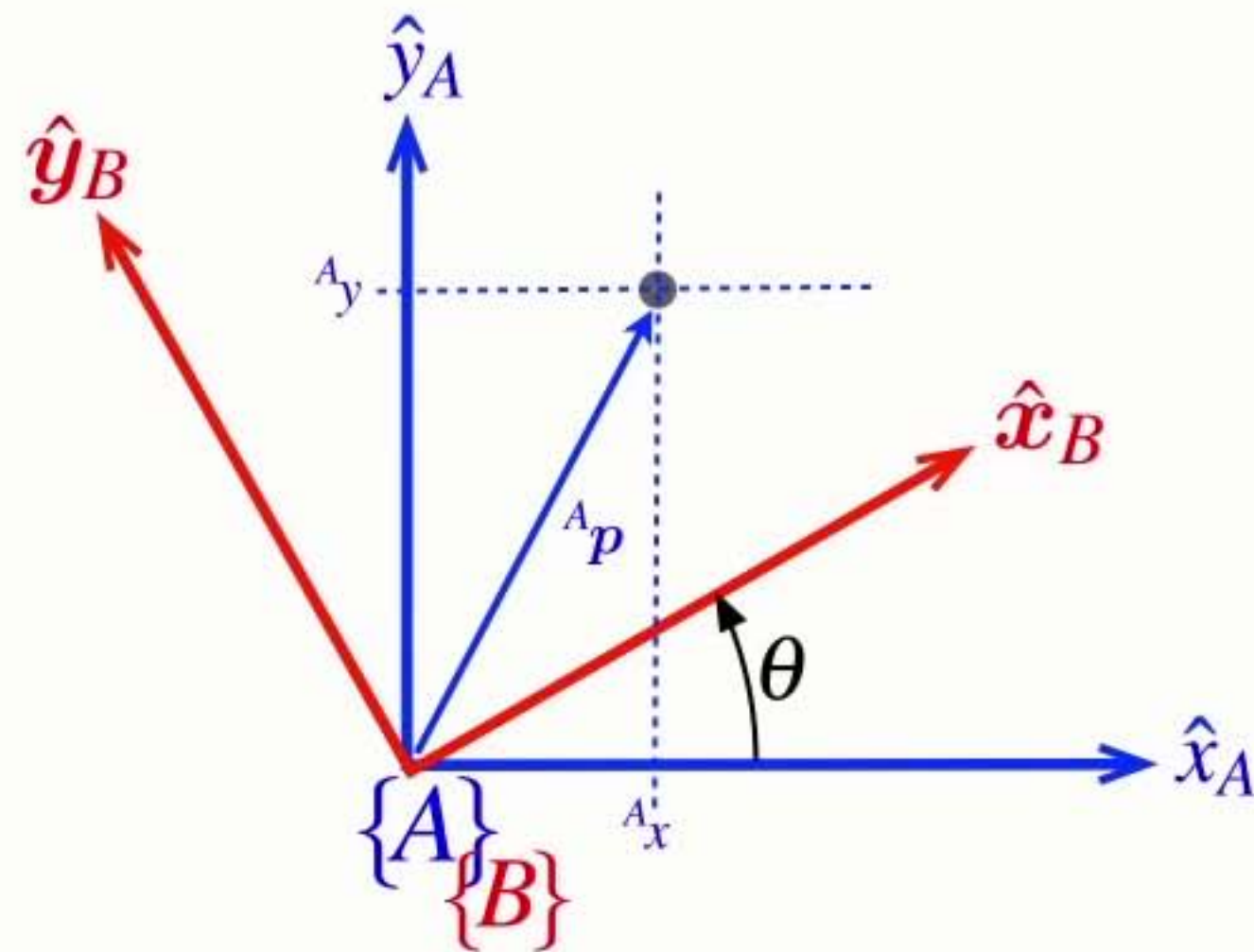
Understanding **rotation**

$${}^A\mathbf{p} = {}^A_x\hat{\mathbf{x}}_A + {}^A_y\hat{\mathbf{y}}_A$$



Understanding **rotation**

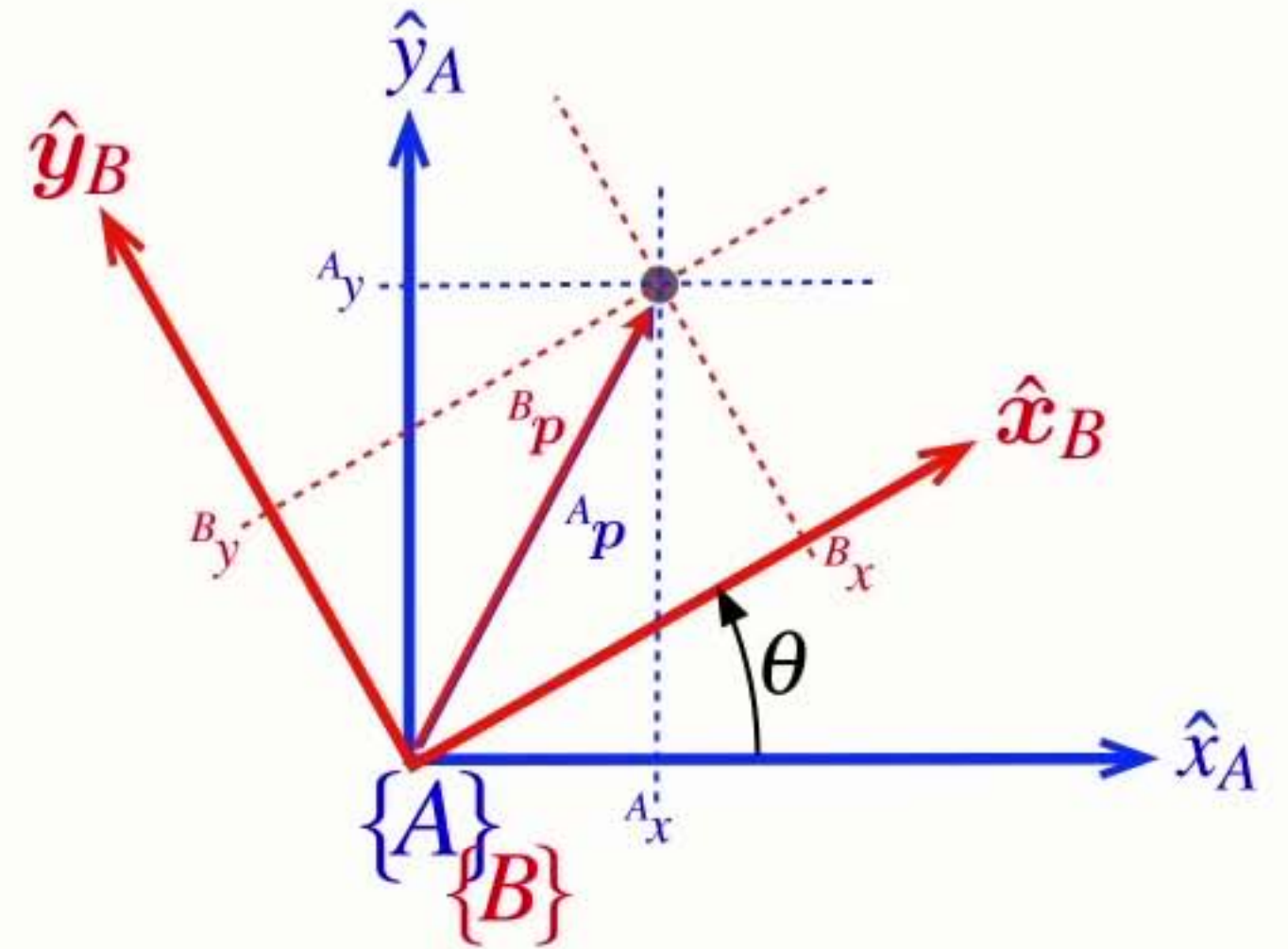
$${}^A\mathbf{p} = {}^A_x\hat{\mathbf{x}}_A + {}^A_y\hat{\mathbf{y}}_A$$



Understanding **rotation**

$${}^A\mathbf{p} = {}^A_x \hat{\mathbf{x}}_A + {}^A_y \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B_x \hat{\mathbf{x}}_B + {}^B_y \hat{\mathbf{y}}_B$$

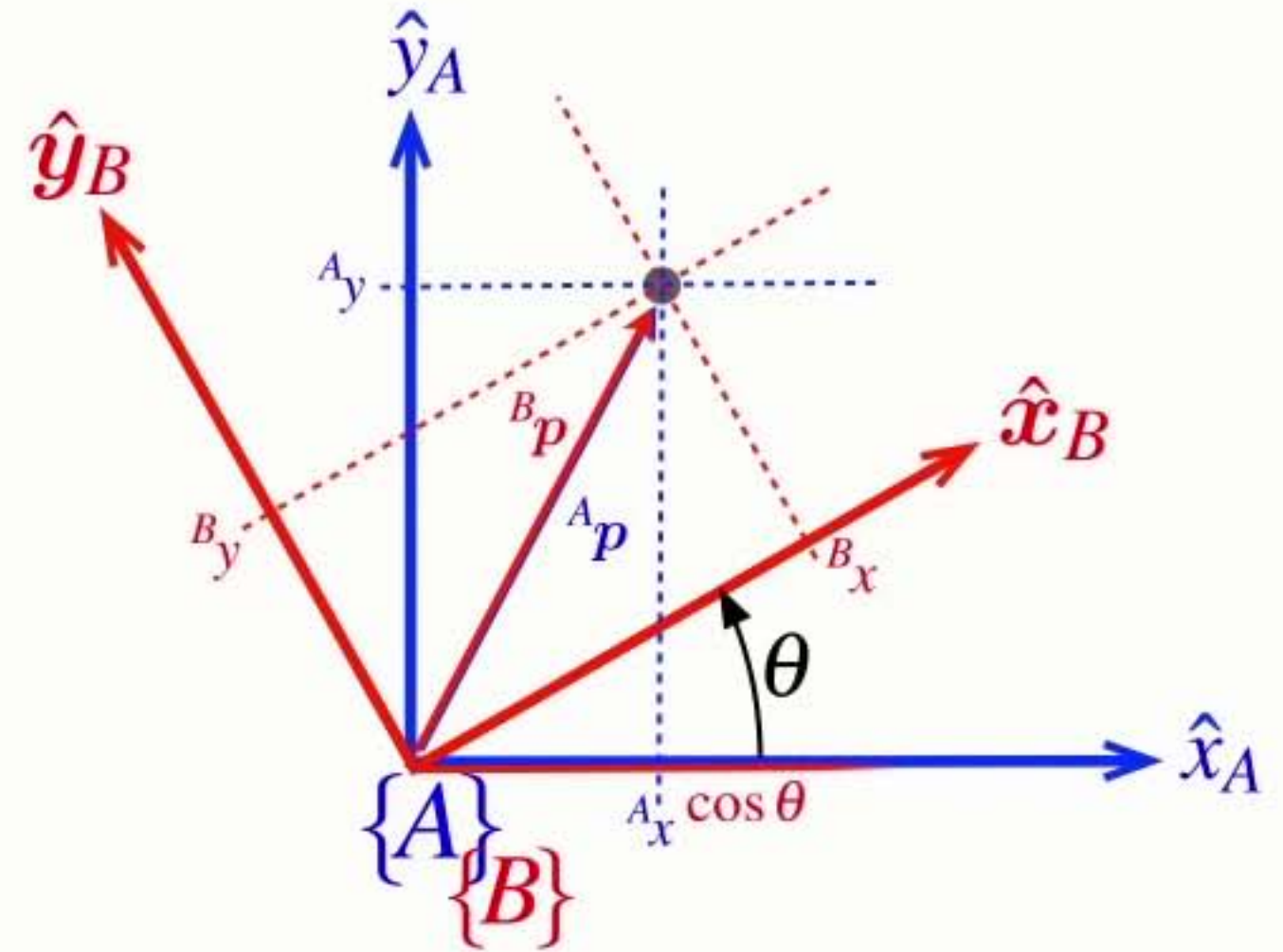


Understanding **rotation**

$${}^A\mathbf{p} = {}^A x \hat{\mathbf{x}}_A + {}^A y \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B x \hat{\mathbf{x}}_B + {}^B y \hat{\mathbf{y}}_B$$

$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$



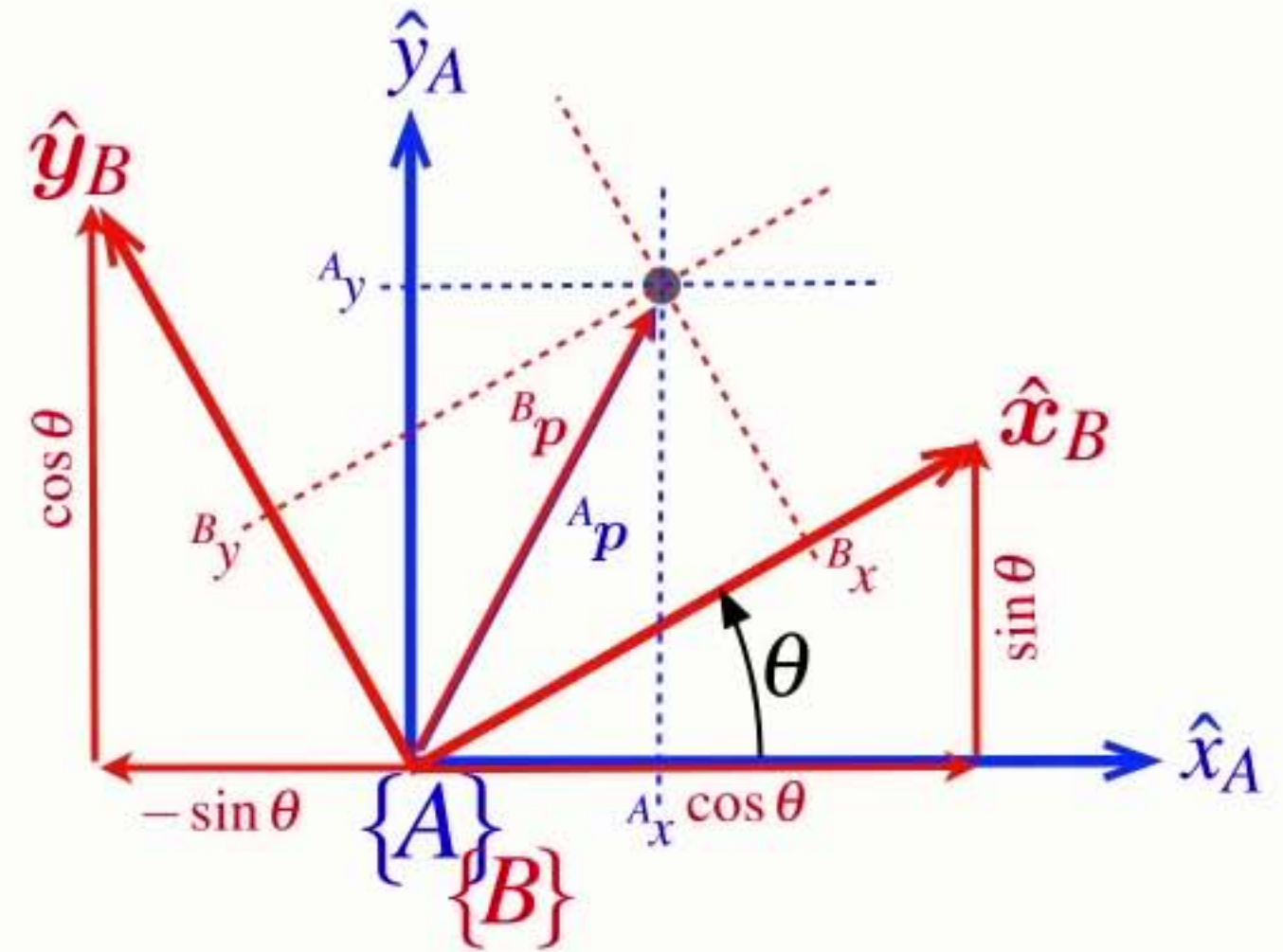
Understanding **rotation**

$${}^A\mathbf{p} = {}^A_x \hat{\mathbf{x}}_A + {}^A_y \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B_x \hat{\mathbf{x}}_B + {}^B_y \hat{\mathbf{y}}_B$$

$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$

$$\hat{\mathbf{y}}_B = -\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A$$



Understanding **rotation**

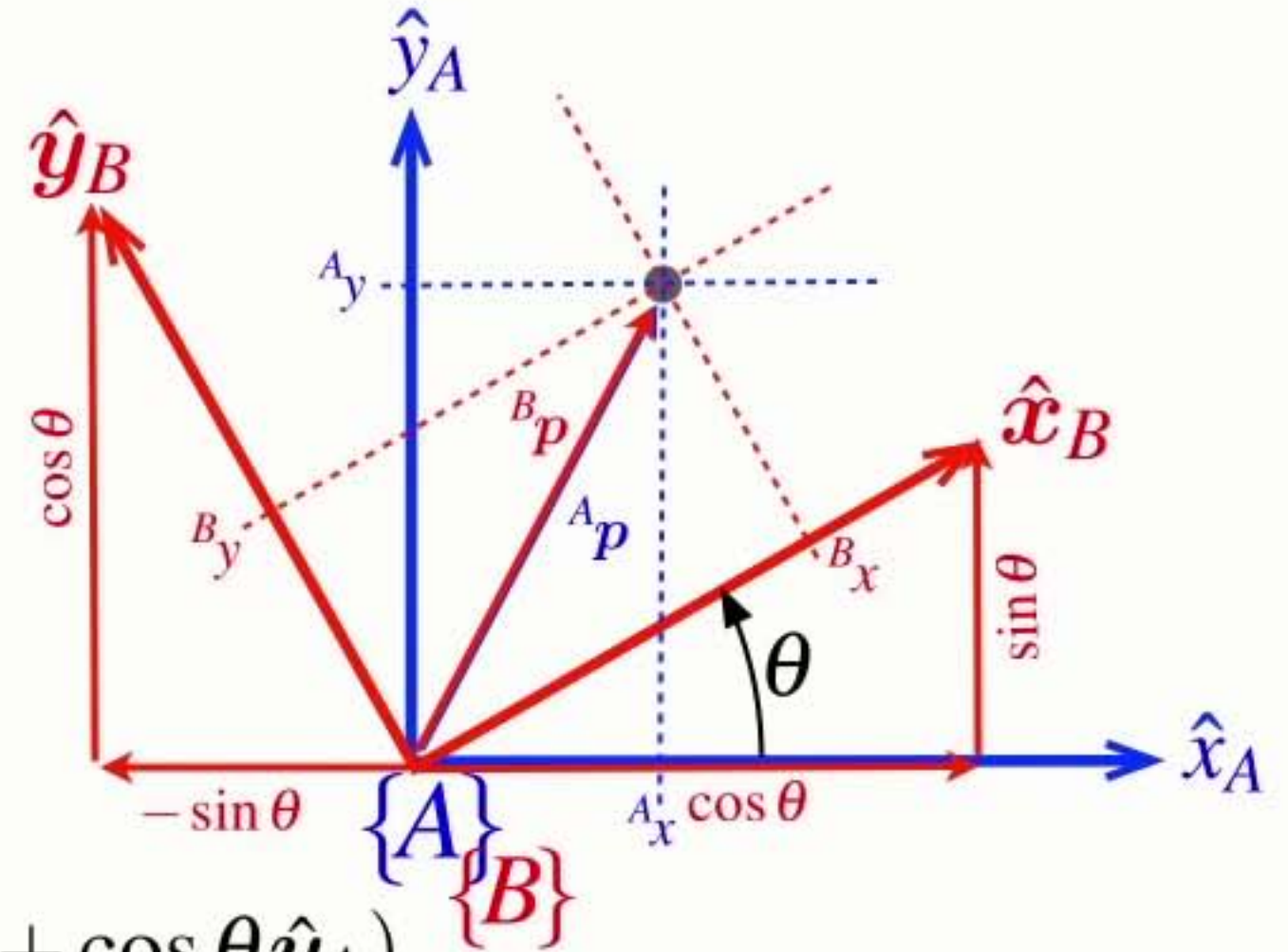
$${}^A\mathbf{p} = {}^A x \hat{\mathbf{x}}_A + {}^A y \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B x \hat{\mathbf{x}}_B + {}^B y \hat{\mathbf{y}}_B$$

$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$

$$\hat{\mathbf{y}}_B = -\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B x (\cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A) + {}^B y (-\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A)$$



Understanding **rotation**

$${}^A\mathbf{p} = {}^A_x \hat{\mathbf{x}}_A + {}^A_y \hat{\mathbf{y}}_A$$

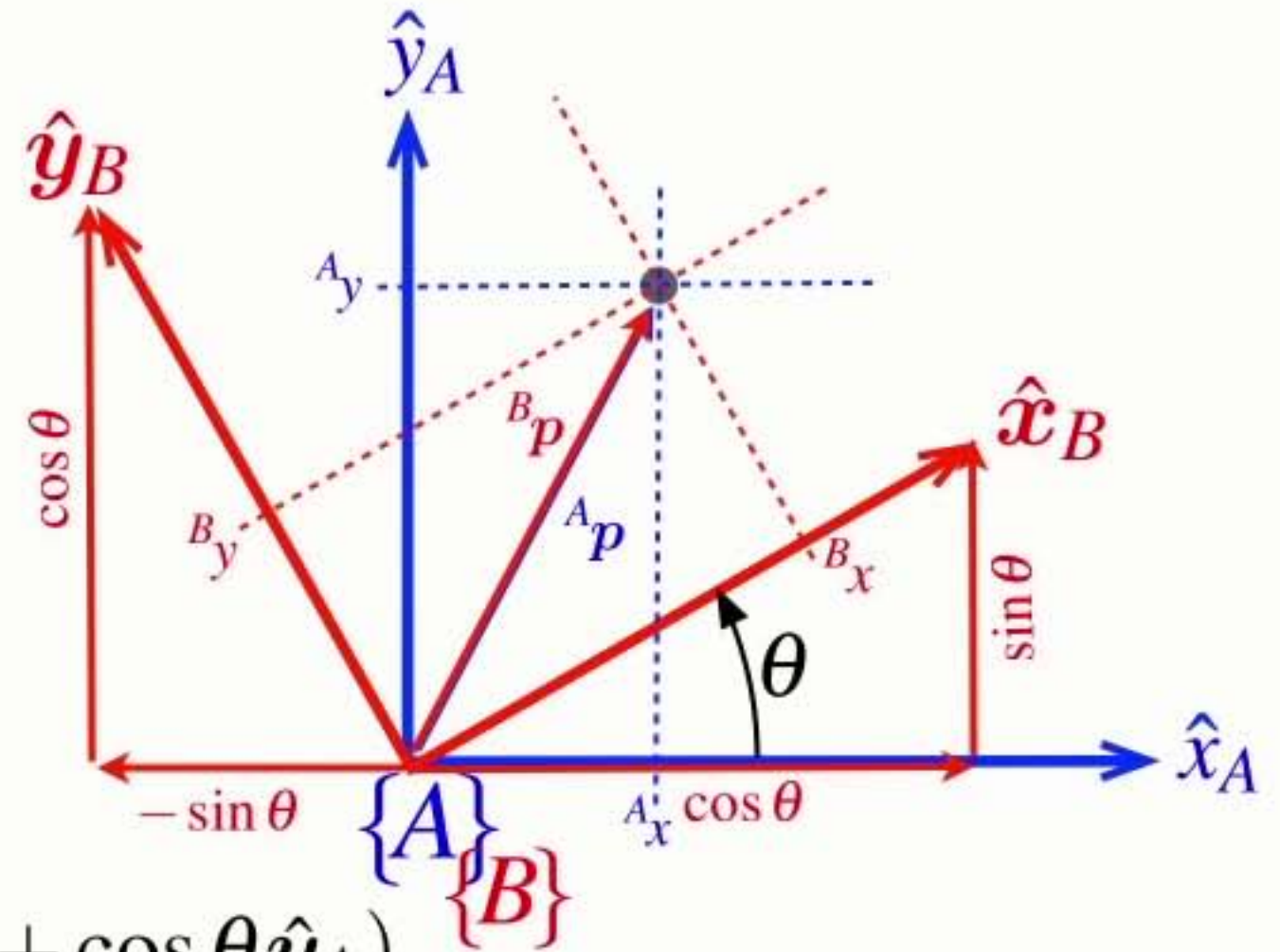
$${}^B\mathbf{p} = {}^B_x \hat{\mathbf{x}}_B + {}^B_y \hat{\mathbf{y}}_B$$

$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$

$$\hat{\mathbf{y}}_B = -\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B_x (\cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A) + {}^B_y (-\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A)$$

$${}^B\mathbf{p} = ({}^B_x \cos \theta - {}^B_y \sin \theta) \hat{\mathbf{x}}_A + ({}^B_x \sin \theta + {}^B_y \cos \theta) \hat{\mathbf{y}}_A$$



Understanding **rotation**

$$\textcircled{{}^A\mathbf{p}} = {}^A x \hat{\mathbf{x}}_A + {}^A y \hat{\mathbf{y}}_A$$

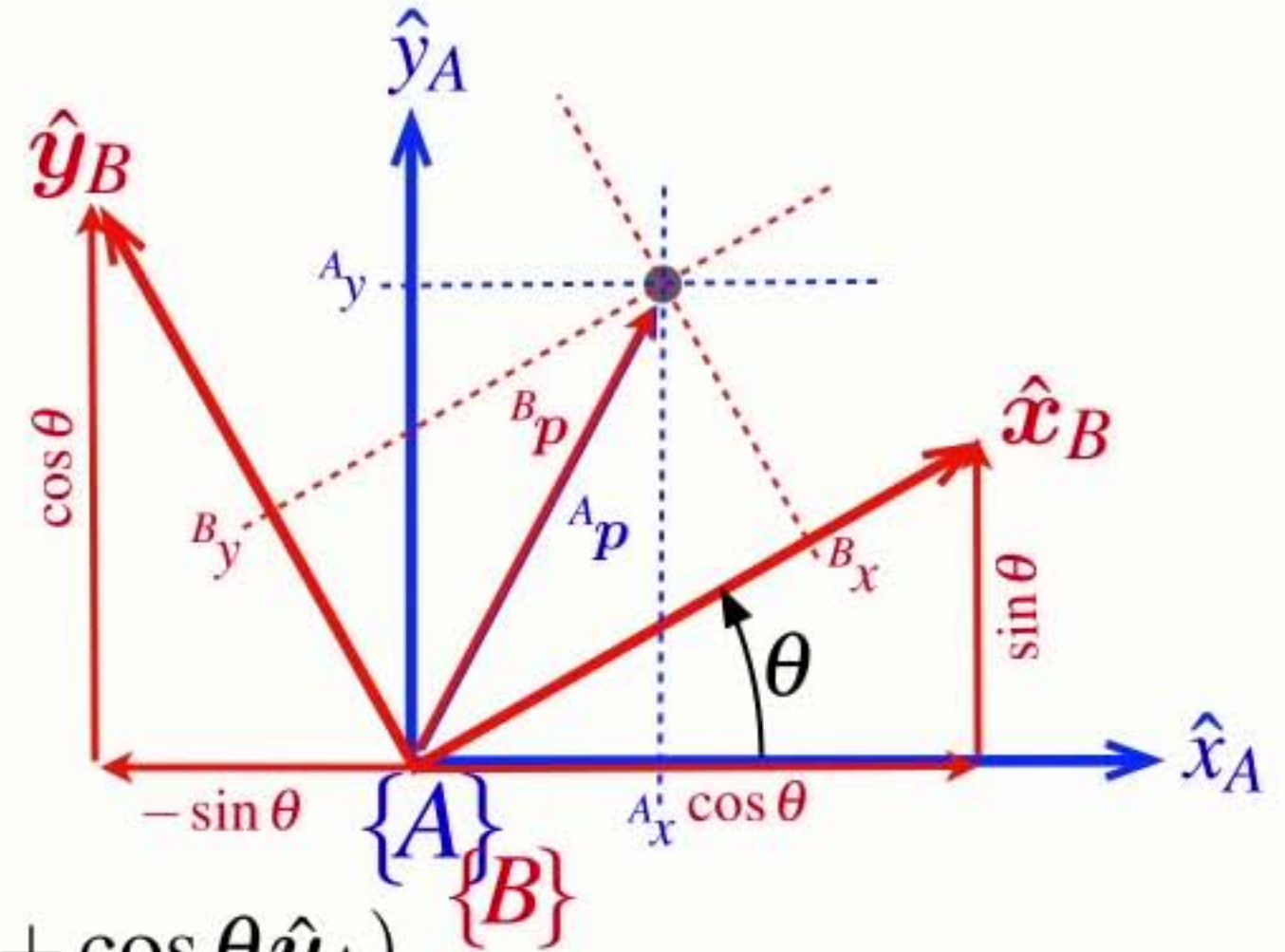
$${}^B\mathbf{p} = {}^B x \hat{\mathbf{x}}_B + {}^B y \hat{\mathbf{y}}_B$$

$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$

$$\hat{\mathbf{y}}_B = -\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B x (\cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A) + {}^B y (-\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A)$$

$$\textcircled{{}^B\mathbf{p}} = ({}^B x \cos \theta - {}^B y \sin \theta) \hat{\mathbf{x}}_A + ({}^B x \sin \theta + {}^B y \cos \theta) \hat{\mathbf{y}}_A$$



Understanding rotation

$${}^A\mathbf{p} = {}^A_x \hat{\mathbf{x}}_A + {}^A_y \hat{\mathbf{y}}_A$$

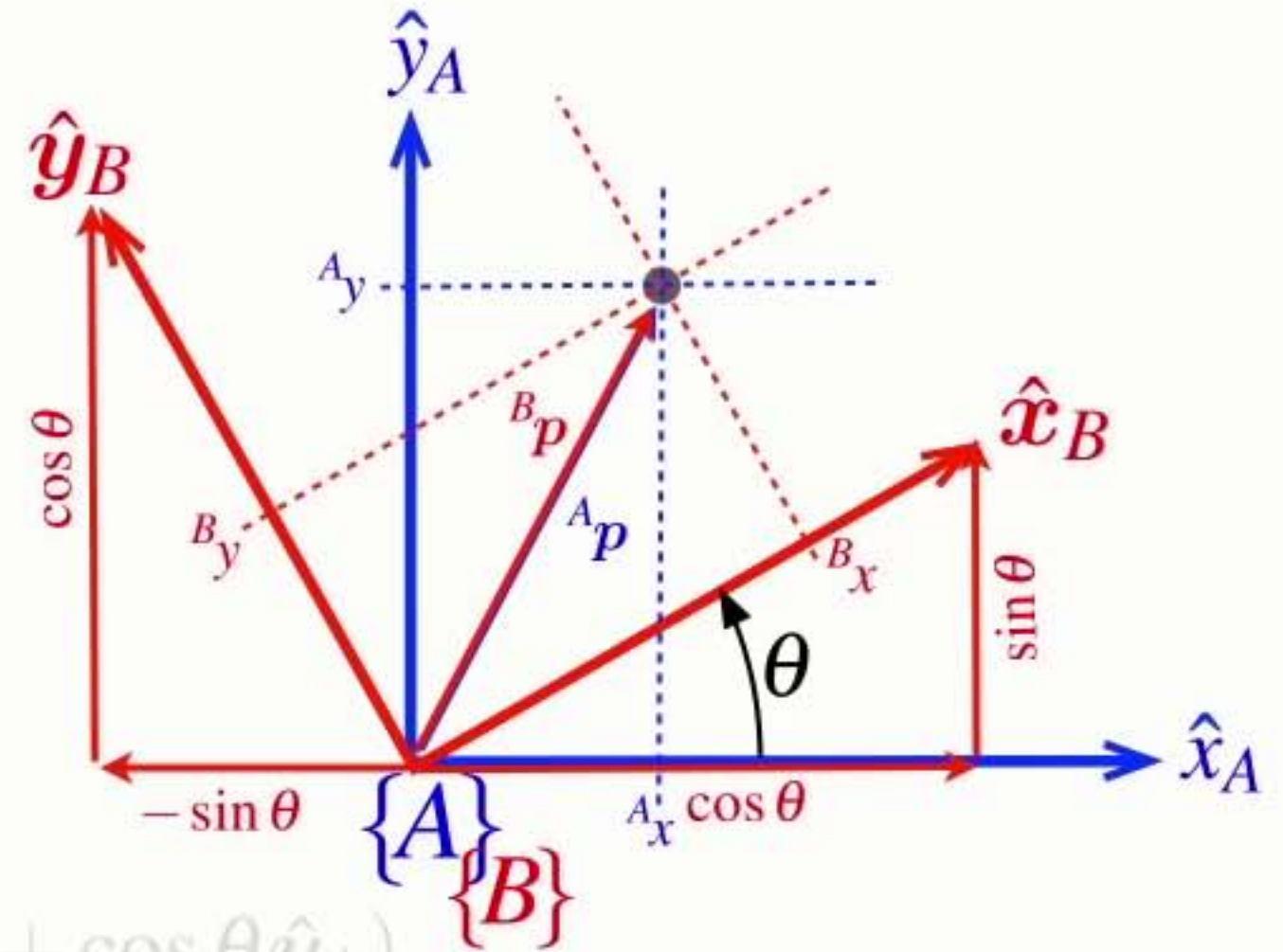
$${}^B\mathbf{p} = {}^B_x \hat{\mathbf{x}}_B + {}^B_y \hat{\mathbf{y}}_B$$

$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$

$$\hat{\mathbf{y}}_B = -\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B_x (\cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A) + {}^B_y (-\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A)$$

$${}^B\mathbf{p} = ({}^B_x \cos \theta - {}^B_y \sin \theta) \hat{\mathbf{x}}_A + ({}^B_x \sin \theta + {}^B_y \cos \theta) \hat{\mathbf{y}}_A$$



Understanding rotation

$${}^A\mathbf{p} = {}^A_x \hat{\mathbf{x}}_A + {}^A_y \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B_x \hat{\mathbf{x}}_B + {}^B_y \hat{\mathbf{y}}_B$$

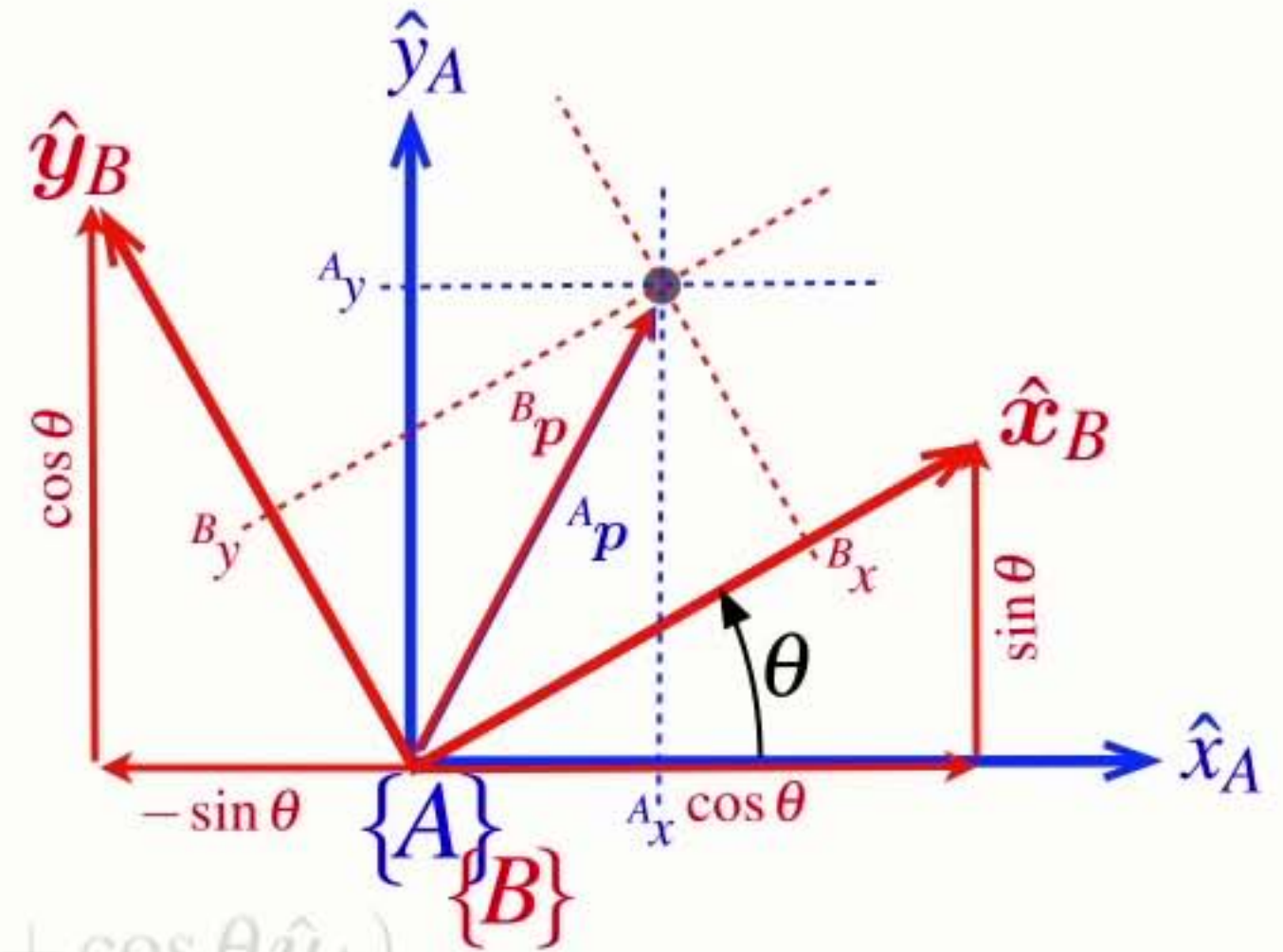
$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$

$$\hat{\mathbf{y}}_B = -\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B_x (\cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A) + {}^B_y (-\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A)$$

$${}^B\mathbf{p} = ({}^B_x \cos \theta - {}^B_y \sin \theta) \hat{\mathbf{x}}_A + ({}^B_x \sin \theta + {}^B_y \cos \theta) \hat{\mathbf{y}}_A$$

$${}^A_x = {}^B_x \cos \theta - {}^B_y \sin \theta$$



Understanding rotation

$$\textcircled{A} \mathbf{p} = A_x \hat{\mathbf{x}}_A + A_y \hat{\mathbf{y}}_A$$

$$B \mathbf{p} = B_x \hat{\mathbf{x}}_B + B_y \hat{\mathbf{y}}_B$$

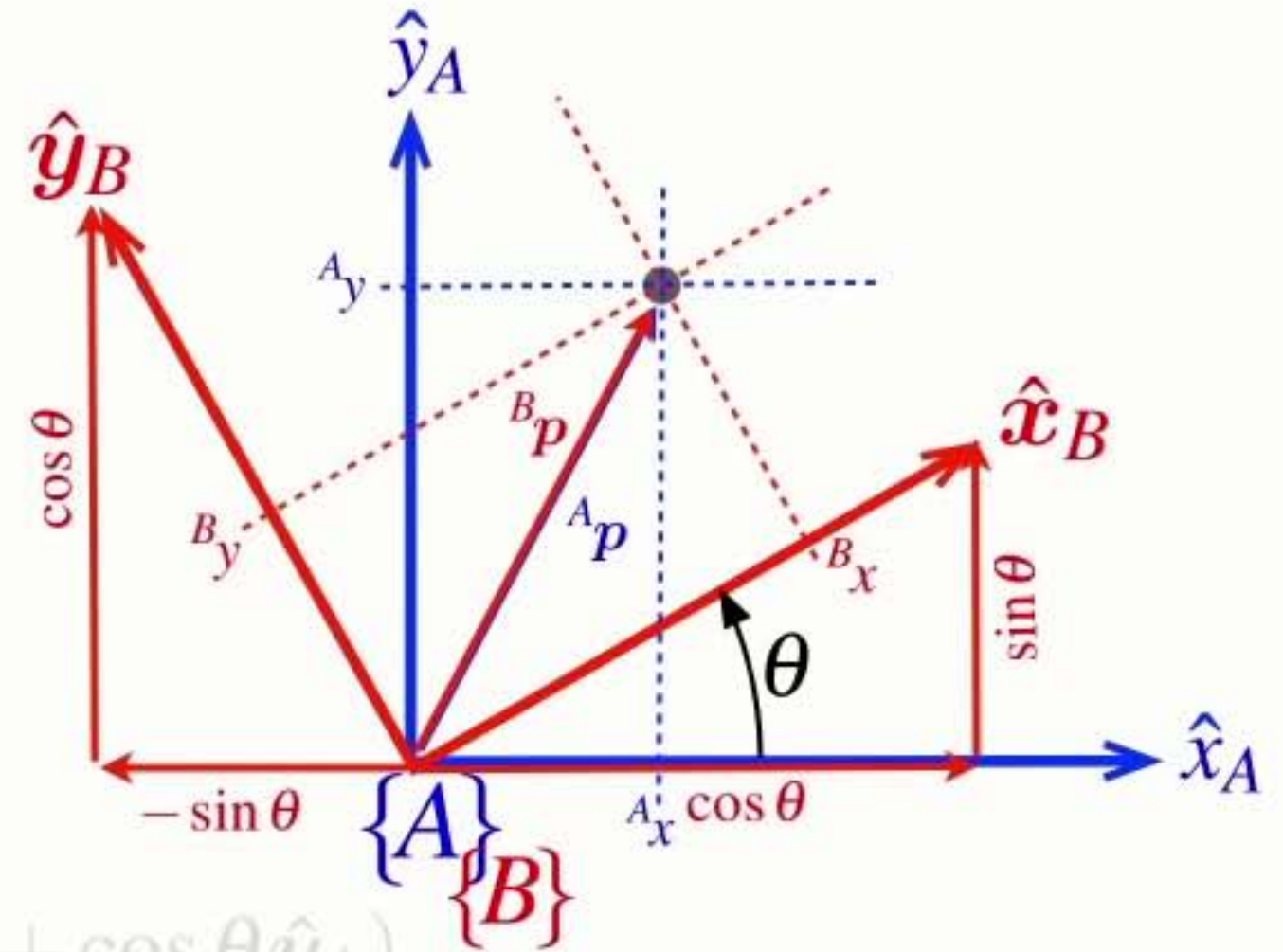
$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$

$$\hat{\mathbf{y}}_B = -\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A$$

$$B \mathbf{p} = B_x (\cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A) + B_y (-\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A)$$

$$\textcircled{B} \mathbf{p} = (B_x \cos \theta - B_y \sin \theta) \hat{\mathbf{x}}_A + (B_x \sin \theta + B_y \cos \theta) \hat{\mathbf{y}}_A$$

$$A_x = B_x \cos \theta - B_y \sin \theta$$



Understanding rotation

$${}^A\mathbf{p} = {}^A_x \hat{\mathbf{x}}_A + {}^A_y \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B_x \hat{\mathbf{x}}_B + {}^B_y \hat{\mathbf{y}}_B$$

$$\hat{\mathbf{x}}_B = \cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A$$

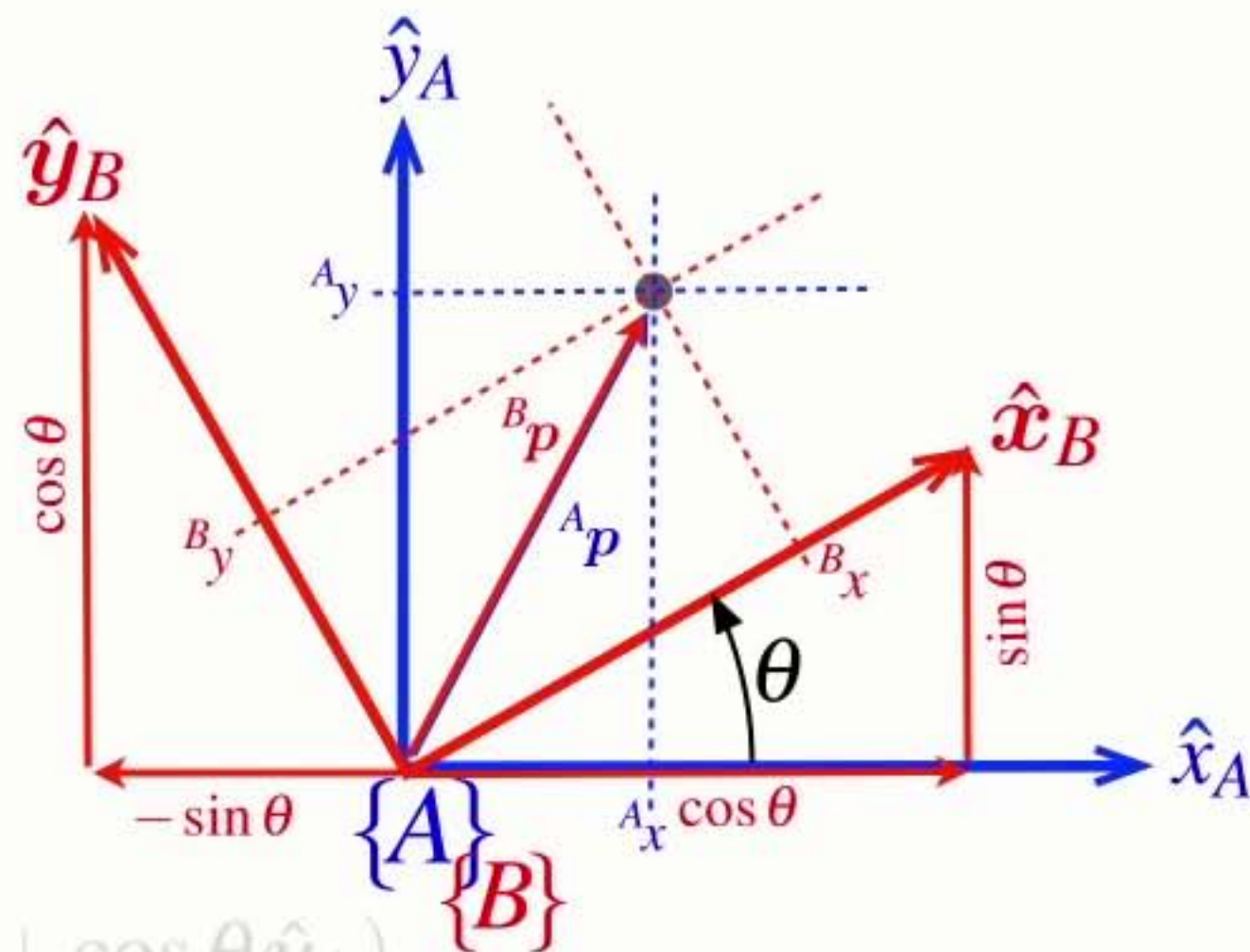
$$\hat{\mathbf{y}}_B = -\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A$$

$${}^B\mathbf{p} = {}^B_x (\cos \theta \hat{\mathbf{x}}_A + \sin \theta \hat{\mathbf{y}}_A) + {}^B_y (-\sin \theta \hat{\mathbf{x}}_A + \cos \theta \hat{\mathbf{y}}_A)$$

$${}^B\mathbf{p} = ({}^B_x \cos \theta - {}^B_y \sin \theta) \hat{\mathbf{x}}_A + ({}^B_x \sin \theta + {}^B_y \cos \theta) \hat{\mathbf{y}}_A$$

$${}^A_x = {}^B_x \cos \theta - {}^B_y \sin \theta$$

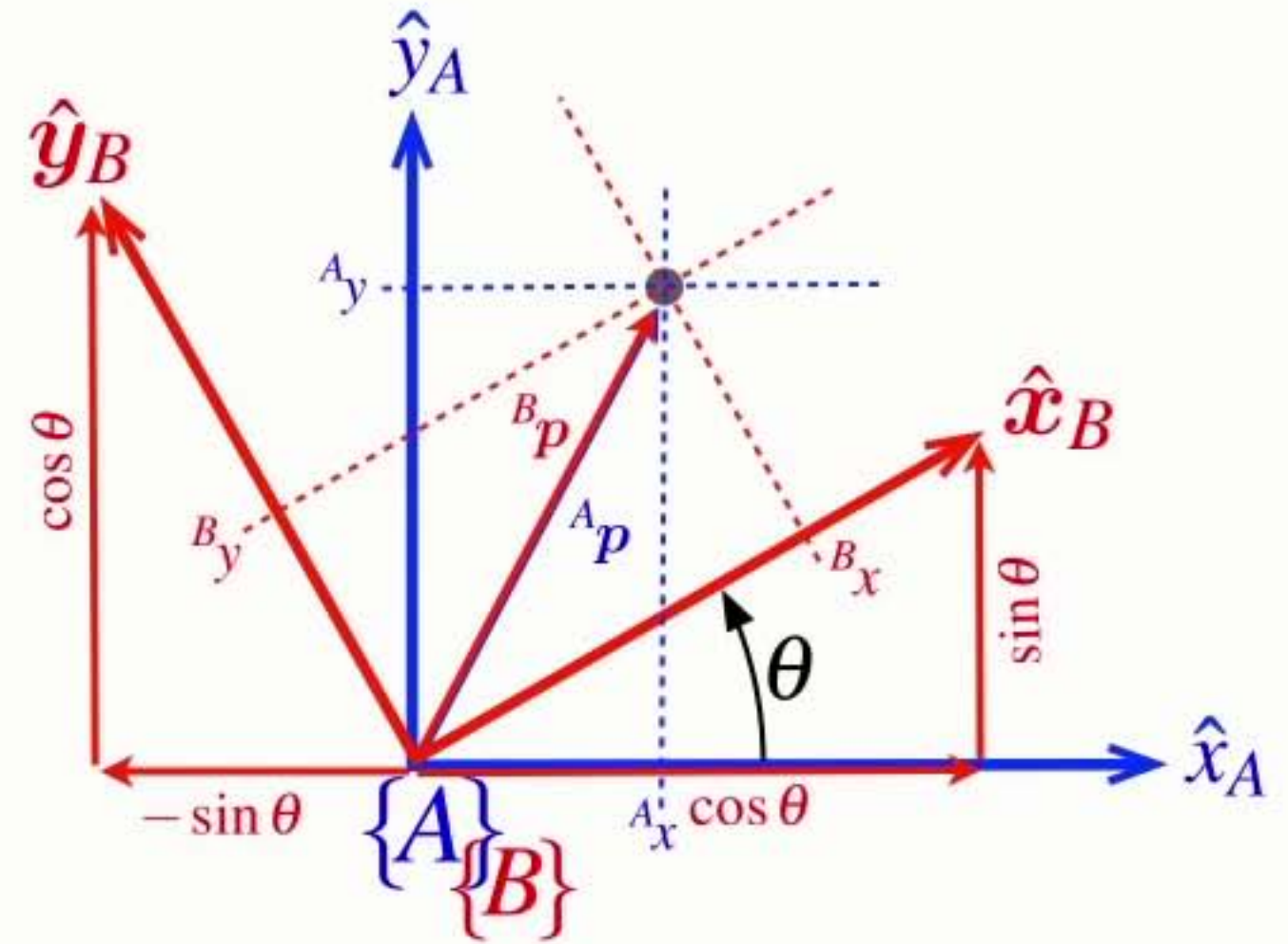
$${}^A_y = {}^B_x \sin \theta + {}^B_y \cos \theta$$



Consider **rotation** first

$$A_x = B_x \cos \theta - B_y \sin \theta$$

$$A_y = B_x \sin \theta + B_y \cos \theta$$

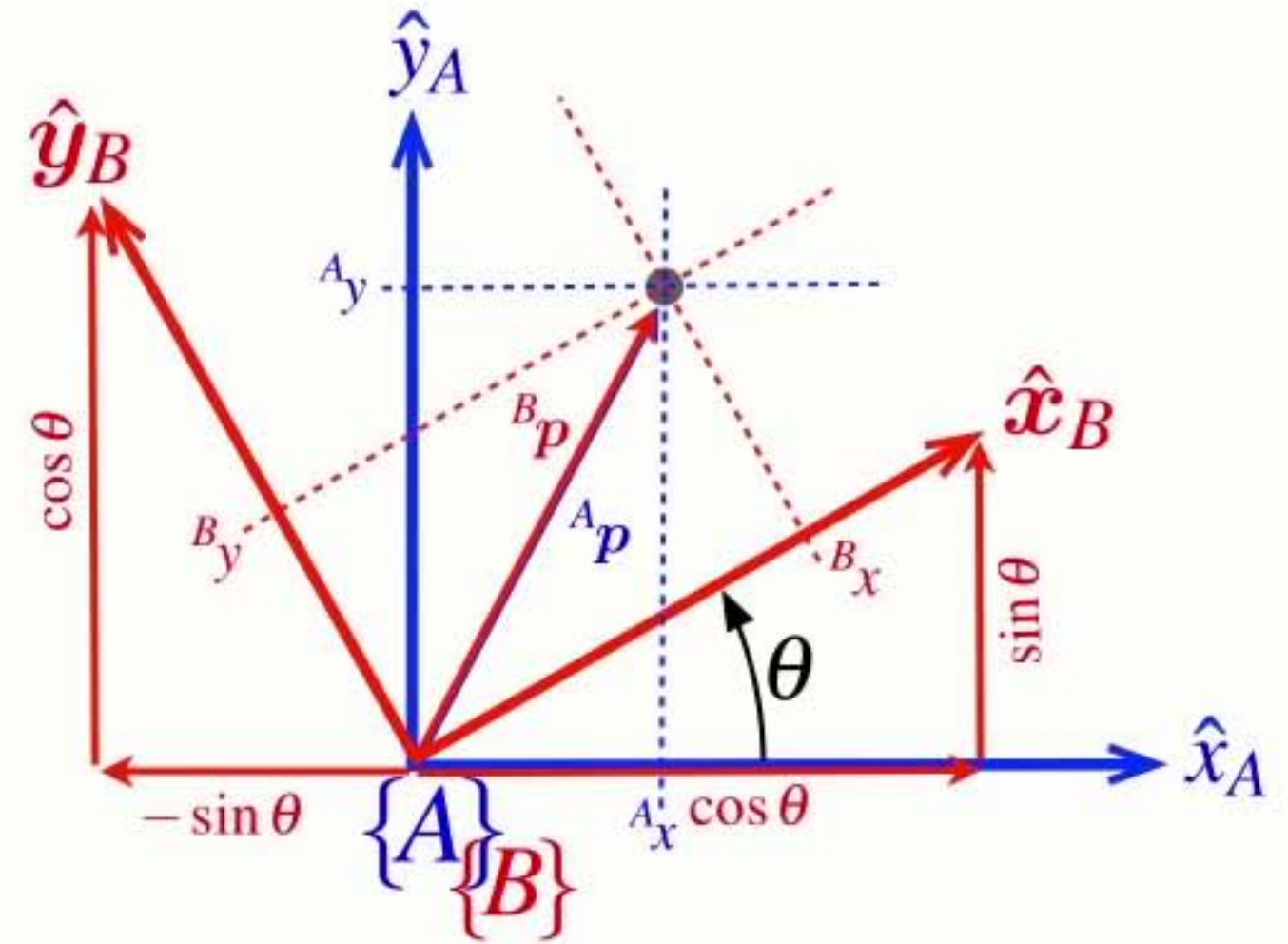


Consider **rotation** first

$$A_x = B_x \cos \theta - B_y \sin \theta$$

$$A_y = B_x \sin \theta + B_y \cos \theta$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} B_x \\ B_y \end{pmatrix}$$



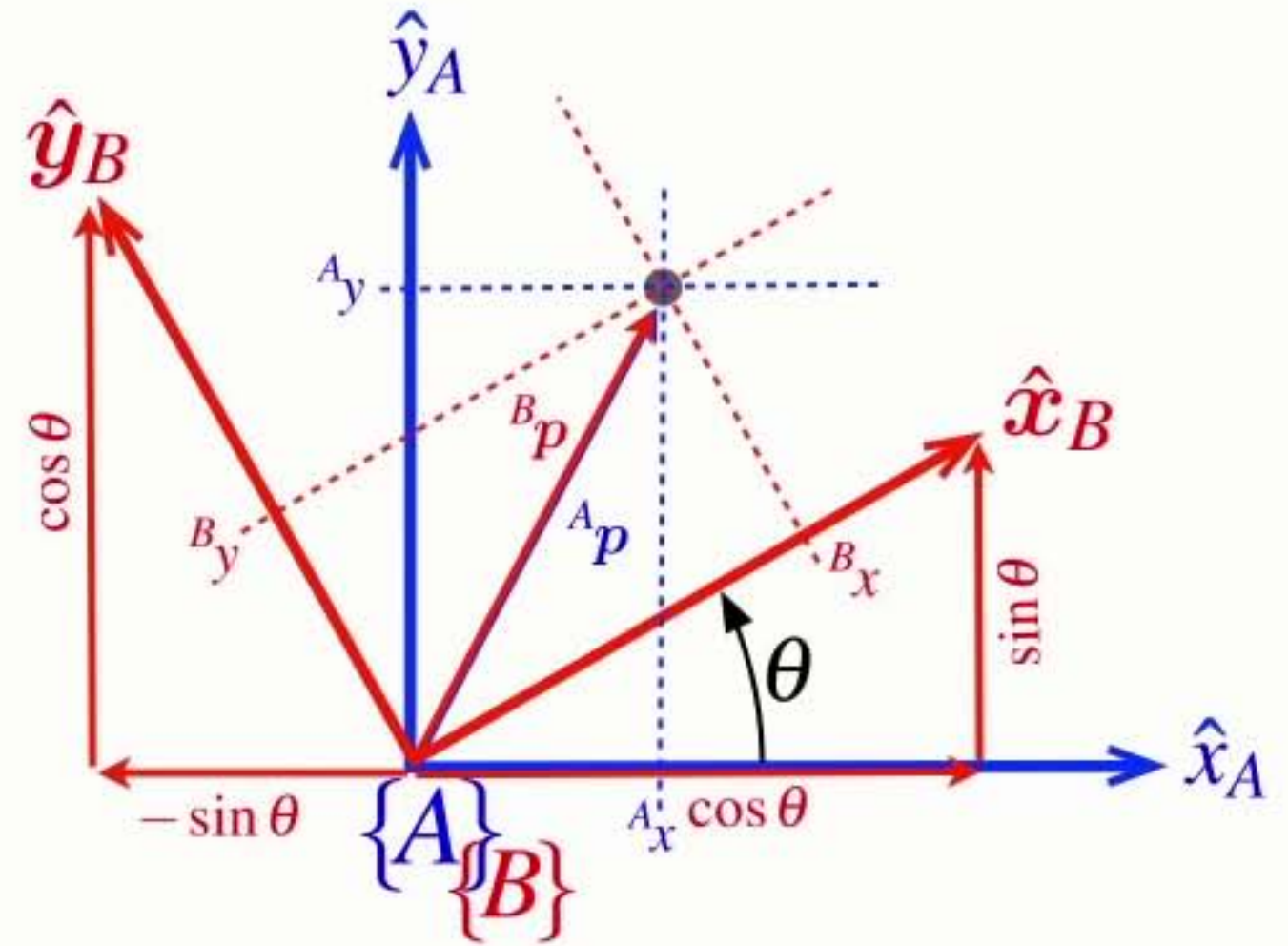
Consider **rotation** first

$$A_x = B_x \cos \theta - B_y \sin \theta$$

$$A_y = B_x \sin \theta + B_y \cos \theta$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} B_x \\ B_y \end{pmatrix}$$

$${}^A \mathbf{p} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} {}^B \mathbf{p}$$



Consider **rotation** first

$$A_x = B_x \cos \theta - B_y \sin \theta$$

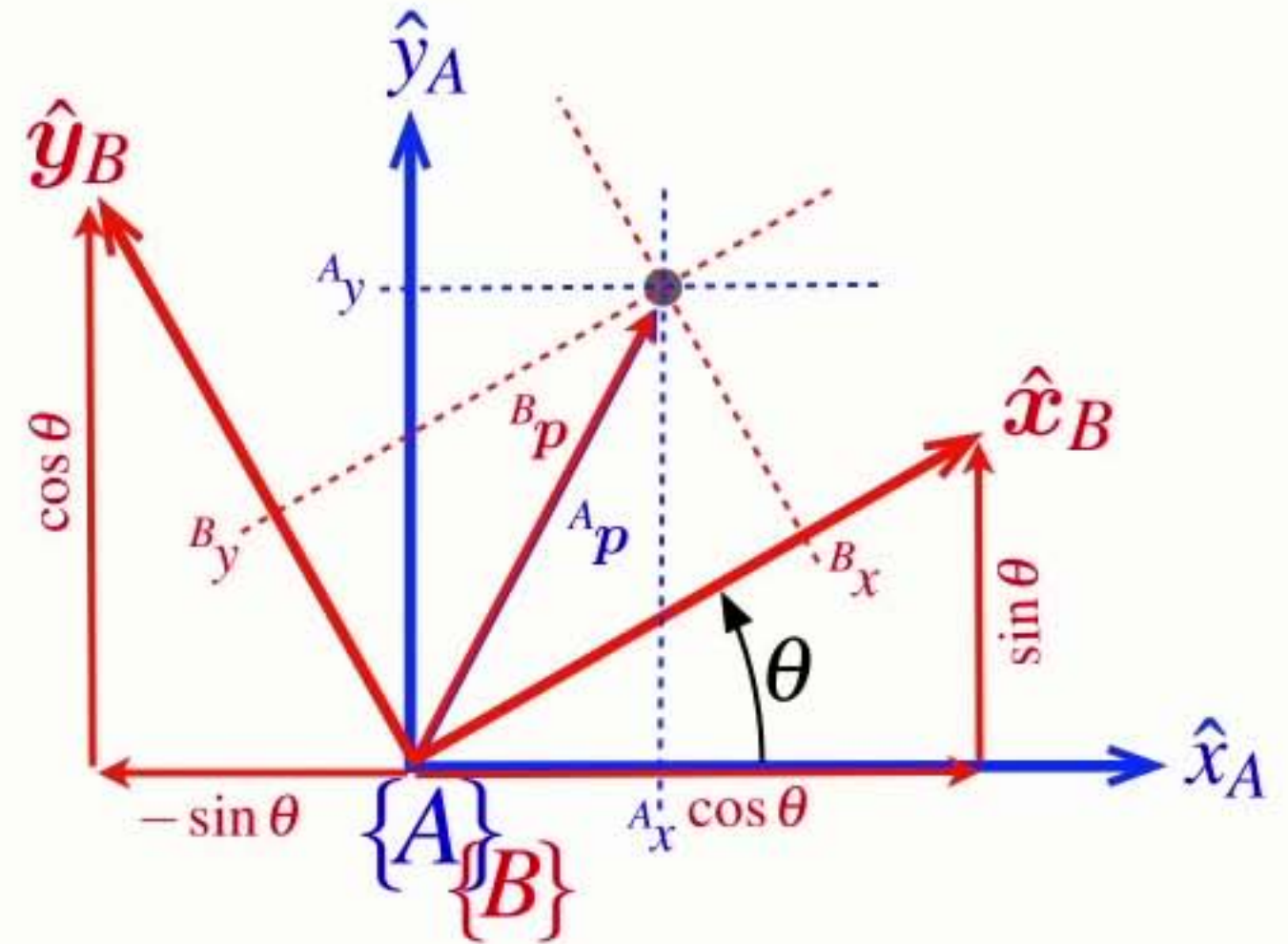
$$A_y = B_x \sin \theta + B_y \cos \theta$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} B_x \\ B_y \end{pmatrix}$$

$${}^A \mathbf{p} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} {}^B \mathbf{p}$$

$${}^A \mathbf{p} = \mathbf{R} {}^B \mathbf{p}$$

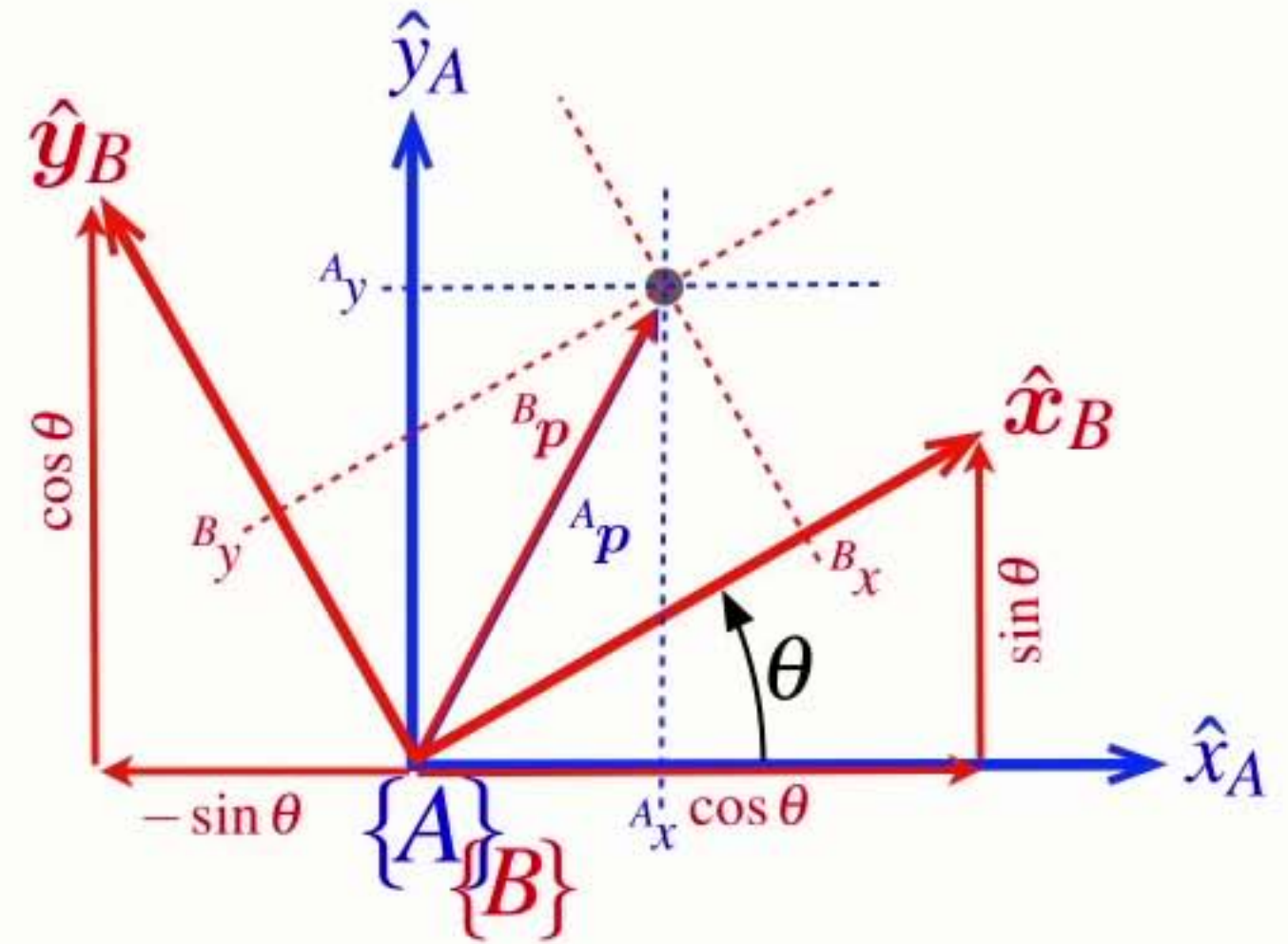
$$\mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



Consider **rotation first**

$${}^A p = \mathbf{R} {}^B p \quad \mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

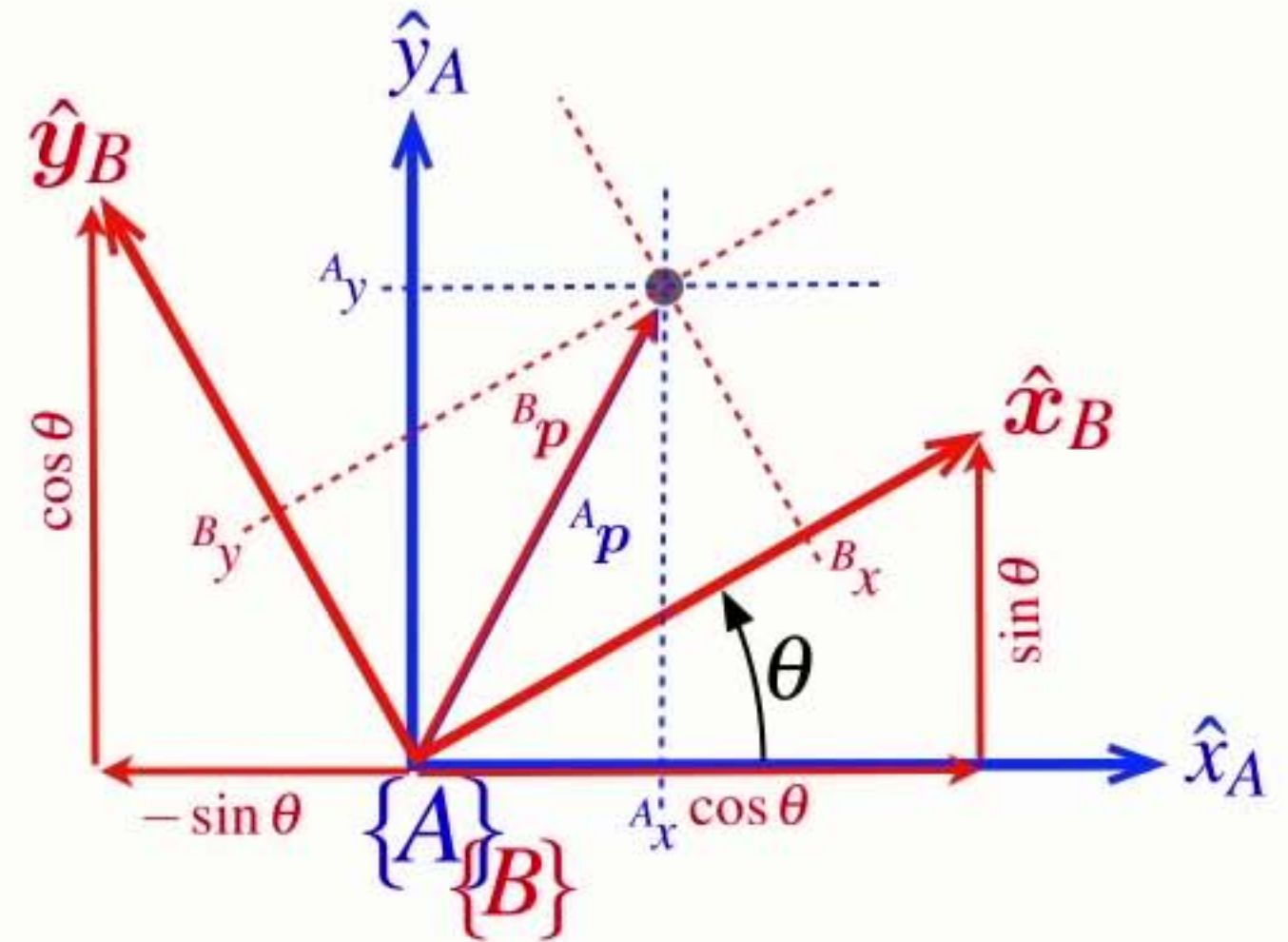
$${}^A p = {}^A \mathbf{R}_B {}^B p \quad {}^A \mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



Consider **rotation** first

$${}^A p = \mathbf{R} {}^B p \quad \mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$${}^A p = {}^A \mathbf{R}_B {}^B p \quad {}^A \mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

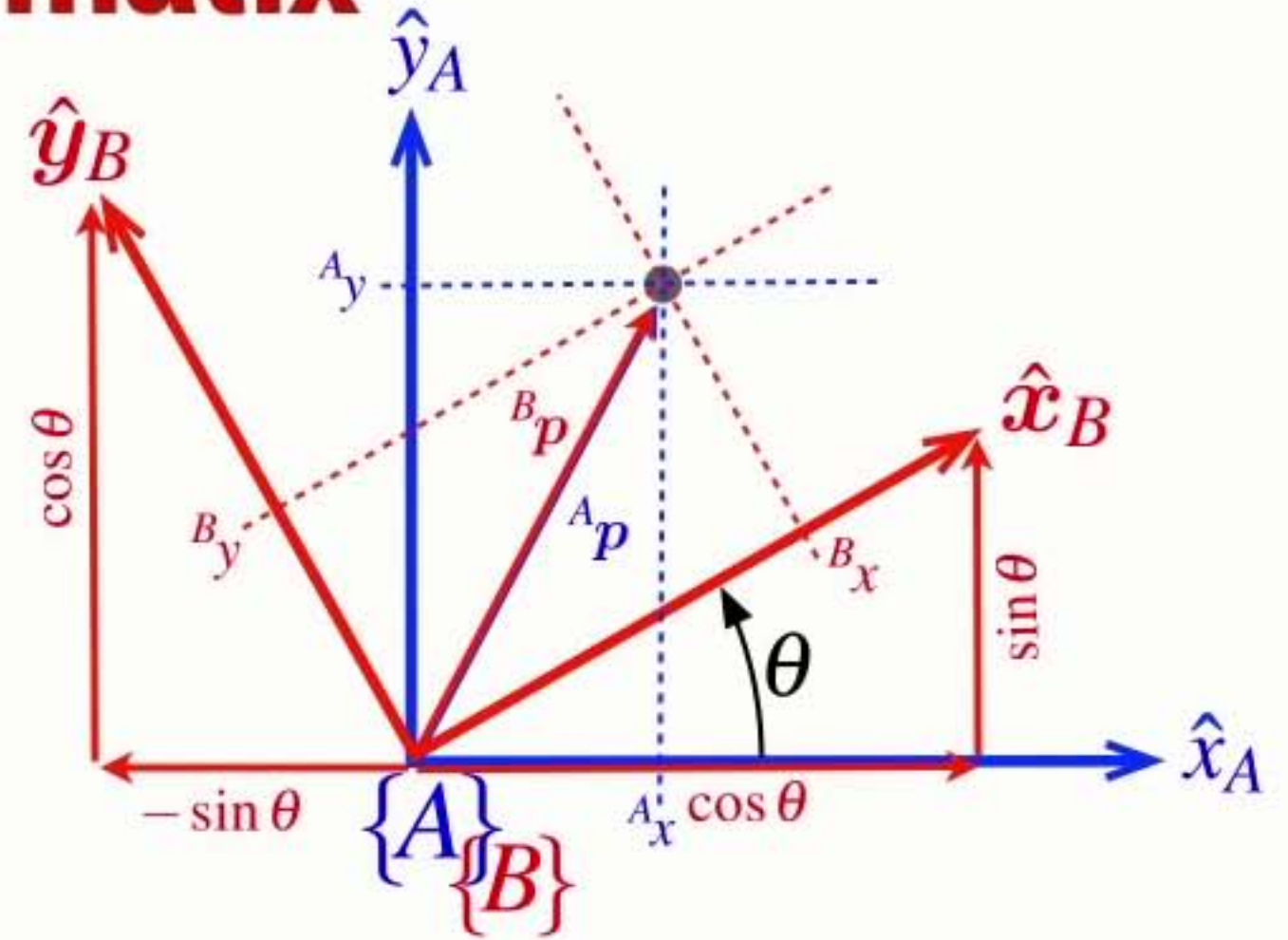


- $\mathbf{R}$ is a rotation matrix that:
 - ➔ rotates vectors from frame $\{B\}$ to frame $\{A\}$

$A \mathbf{R}_B$

Properties of the **rotation matrix**

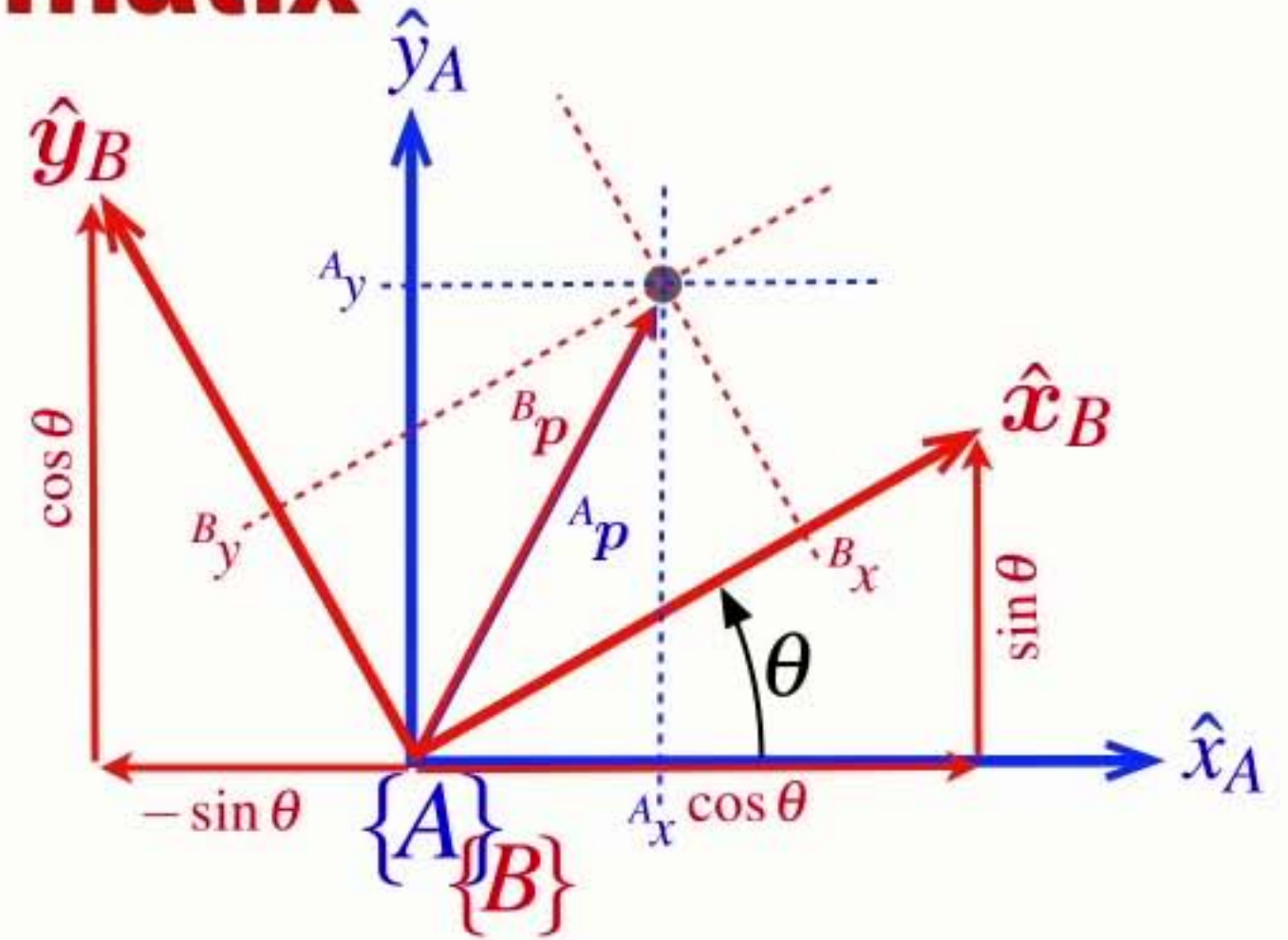
$${}^A\mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



Properties of the **rotation matrix**

$${}^A\mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

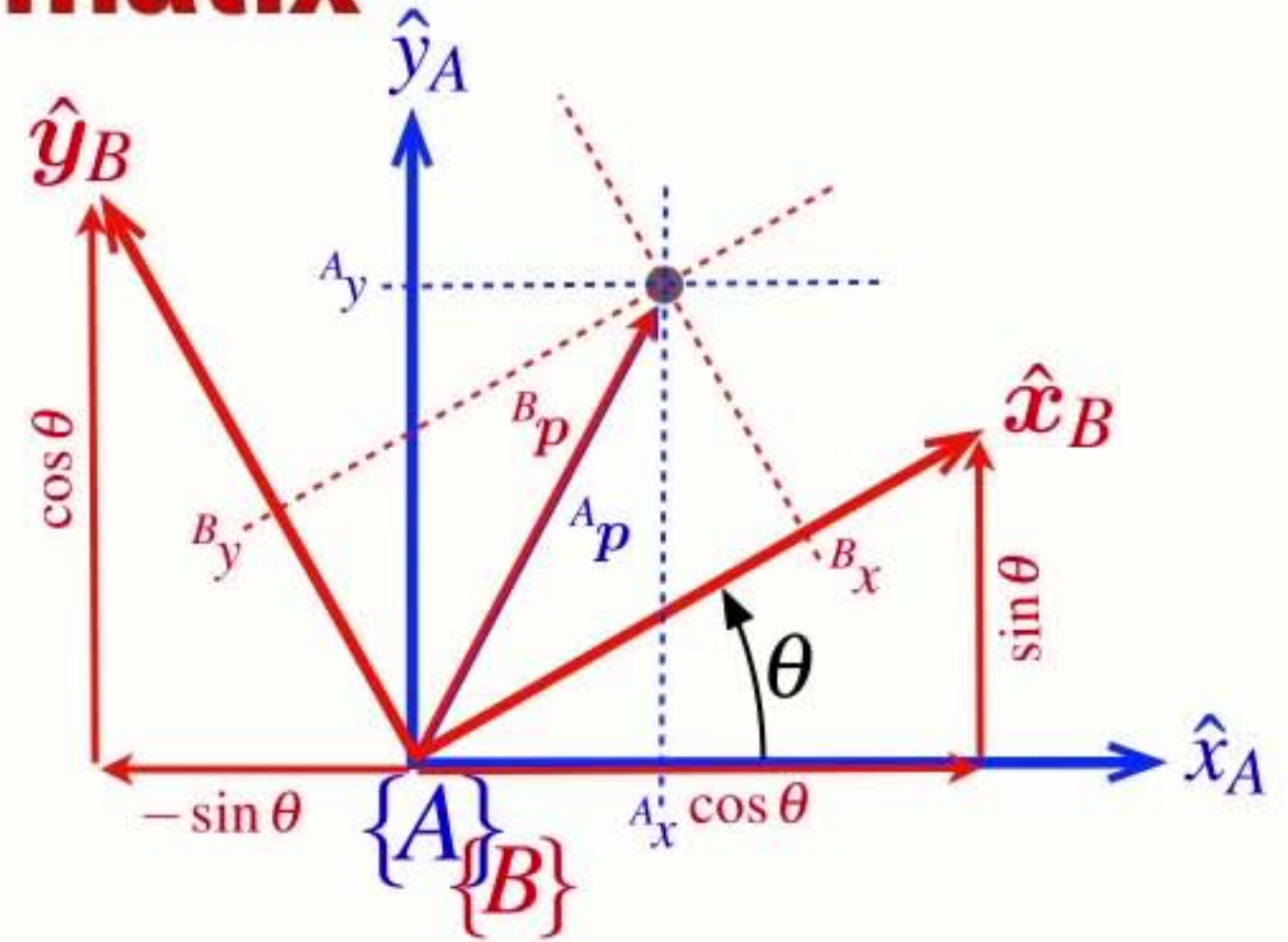
- An orthogonal (orthonormal) matrix



Properties of the **rotation matrix**

$${}^A\mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

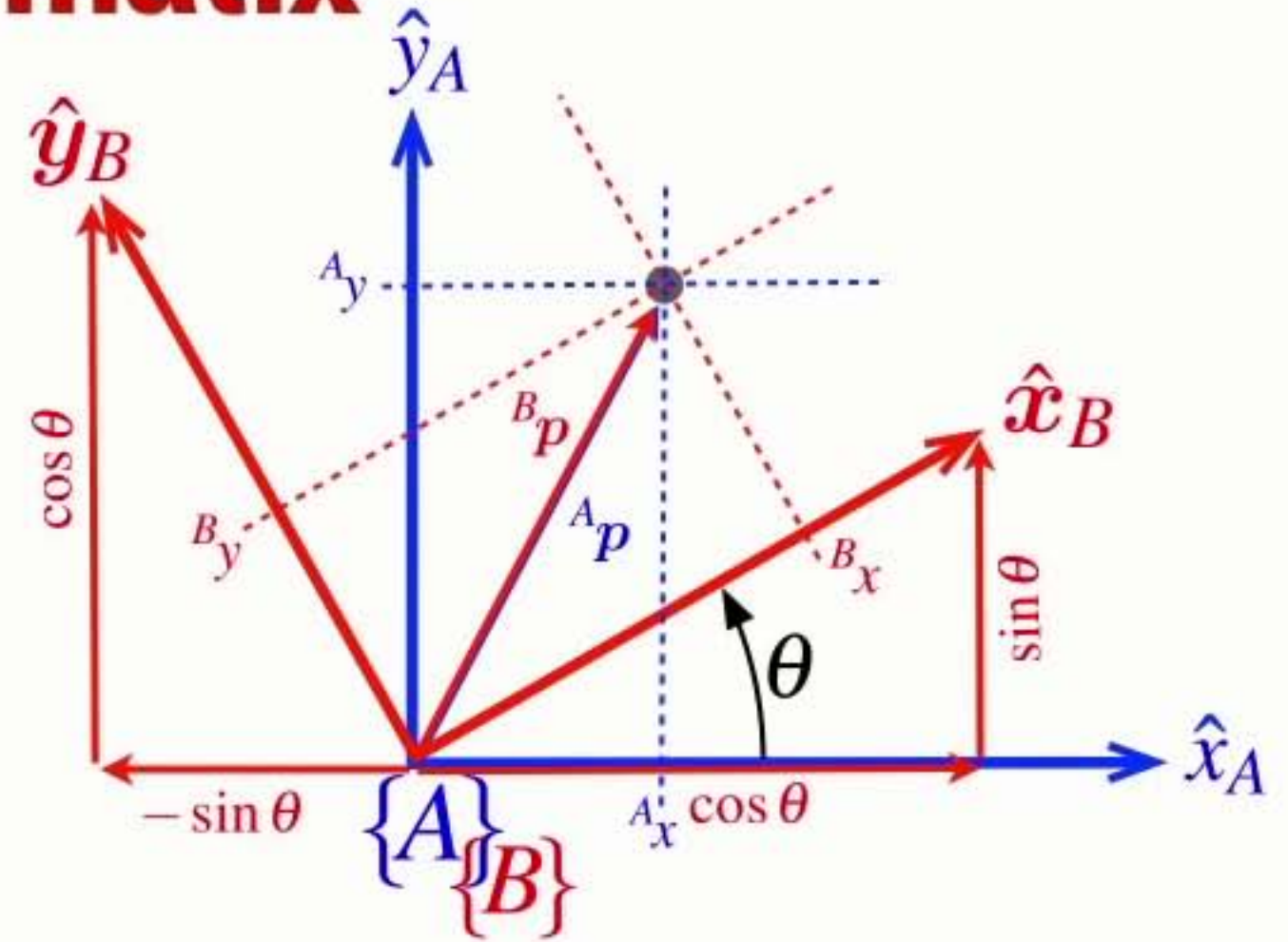
- An orthogonal (orthonormal) matrix
 - ➡ Each column is a unit length vector



Properties of the **rotation matrix**

$${}^A\mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

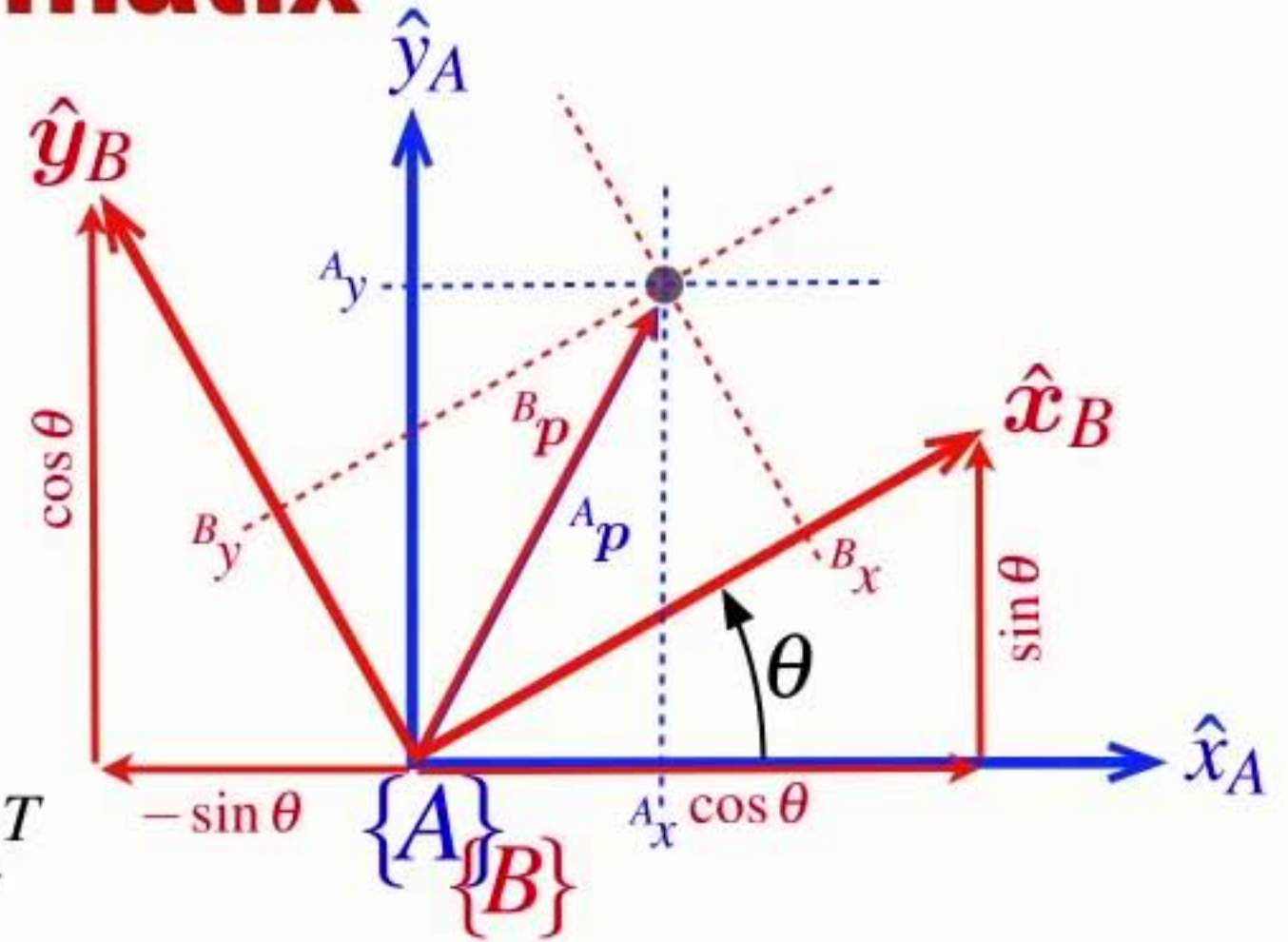
- An orthogonal (orthonormal) matrix
 - ➔ Each column is a unit length vector
 - ➔ Each column is orthogonal to all other columns



Properties of the **rotation matrix**

$${}^A\mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

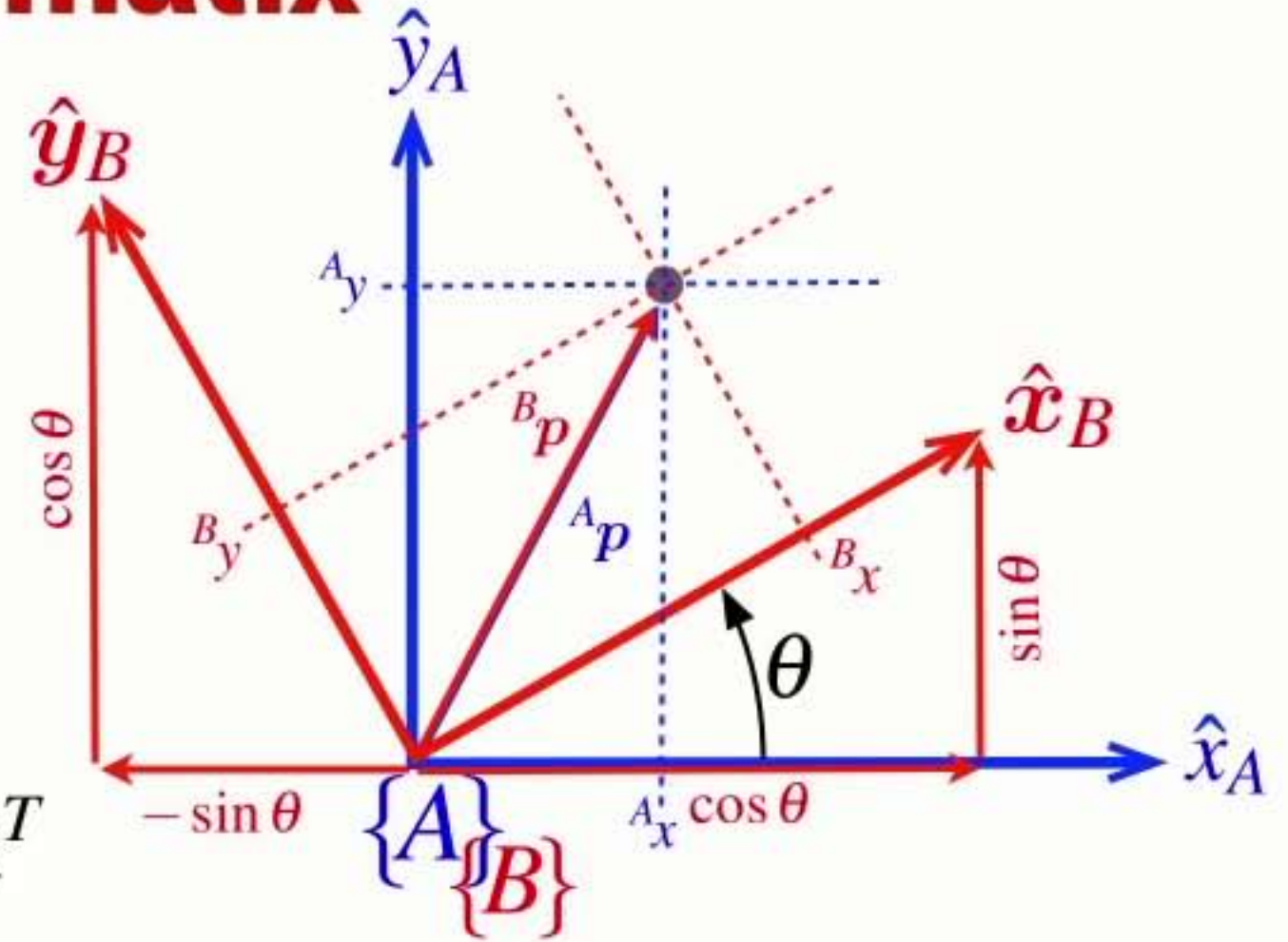
- An orthogonal (orthonormal) matrix
 - ➔ Each column is a unit length vector
 - ➔ Each column is orthogonal to all other columns
- The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^T$



Properties of the **rotation matrix**

$${}^A\mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

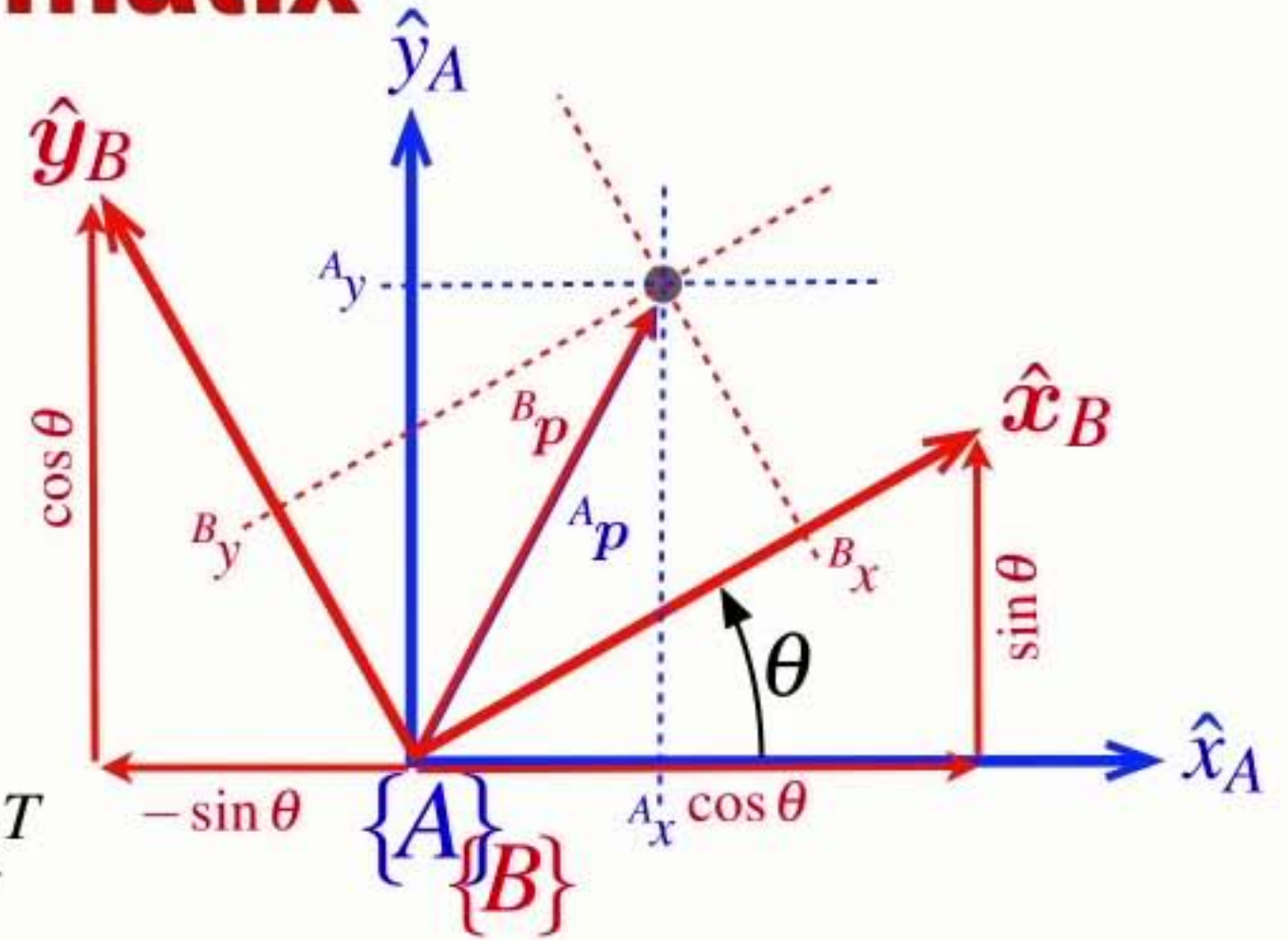
- An orthogonal (orthonormal) matrix
 - ➔ Each column is a unit length vector
 - ➔ Each column is orthogonal to all other columns
- The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^T$
- The determinant is +1 $\det(\mathbf{R}) = 1$
 - ➔ the length of a vector is unchanged by rotation



Properties of the **rotation matrix**

$${}^A\mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

- An orthogonal (orthonormal) matrix
 - ➔ Each column is a unit length vector
 - ➔ Each column is orthogonal to all other columns
- The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^T$
- The determinant is +1 $\det(\mathbf{R}) = 1$
 - ➔ the length of a vector is unchanged by rotation
- Rotation matrices belong to the Special Orthogonal group of dimension 2 $\mathbf{R} \in SO(2)$



Inverse rotation

$${}^A\boldsymbol{p} = {}^A\mathbf{R}_B {}^B\boldsymbol{p}$$

Inverse rotation

$${}^A\boldsymbol{p} = {}^A\mathbf{R}_B {}^B\boldsymbol{p}$$

$${}^B\boldsymbol{p} = {}^A\mathbf{R}_B^{-1} {}^A\boldsymbol{p}$$

Inverse rotation

- To rotate a vector from frame {A} to frame {B} we use the inverse rotation matrix
 - ➡ The inverse is simply the transpose

$${}^A\mathbf{p} = {}^A\mathbf{R}_B {}^B\mathbf{p}$$

$${}^B\mathbf{p} = {}^A\mathbf{R}_B^{-1} {}^A\mathbf{p}$$

Inverse rotation

- To rotate a vector from frame {A} to frame {B} we use the inverse rotation matrix
 - ➡ The inverse is simply the transpose

$${}^A\mathbf{p} = {}^A\mathbf{R}_B {}^B\mathbf{p}$$

$${}^B\mathbf{p} = {}^A\mathbf{R}_B^{-1} {}^A\mathbf{p}$$

$${}^B\mathbf{p} = {}^B\mathbf{R}_A {}^A\mathbf{p}$$

Inverse rotation

- To rotate a vector from frame {A} to frame {B} we use the inverse rotation matrix
 - ➡ The inverse is simply the transpose

$${}^A p = {}^A \mathbf{R}_B {}^B p$$

$${}^B p = {}^A \mathbf{R}_B^{-1} {}^A p$$

$${}^B p = {}^B \mathbf{R}_A {}^A p$$

$${}^B \mathbf{R}_A = {}^A \mathbf{R}_B^{-1}$$

Table 2.1.
Summary of the various
concrete representations of
pose ξ introduced in this chapter

Representation	$\oplus$	$\ominus$	transl.	rotn.	dim	MATLAB
$(x, y, \theta) \in \mathbb{R}^2 \times \mathbb{S}$			✓	✓	2D	
$T \in SE(2)$	$T_1 T_2$	T^{-1}	✓	×	2D	<code>se2(x, y)</code>
$R \in SO(2)$	$R_1 R_2$	R^T	×	✓	2D	<code>se2(0, 0, th)</code>
$T \in SE(2)$	$T_1 T_2$	T^{-1}	✓	✓	2D	<code>se2(x, y, th)</code>
$(x, y, z, \Gamma) \in \mathbb{R}^3 \times \mathbb{S}^3$			✓	✓	3D	
$R \in SO(3)$	$R_1 R_2$	R^T	×	✓	3D	<code>rotx, roty, ...</code>
$\Gamma \in \mathbb{S}^3$	×			✓	3D	<code>tr2eul, eul2tr</code>
$\Gamma \in \mathbb{S}^3$	×			✓	3D	<code>tr2rpy1, rpy2tr</code>
$T \in SE(3)$	$T_1 T_2$	T^{-1}	✓	✓	3D	<code>transl(x, y, z)</code>
$\dot{q} \in \mathbb{Q}$	$\dot{q}_1 \dot{q}_2$	$\dot{q}^{-1}$	×	✓	3D	<code>quaternion</code>

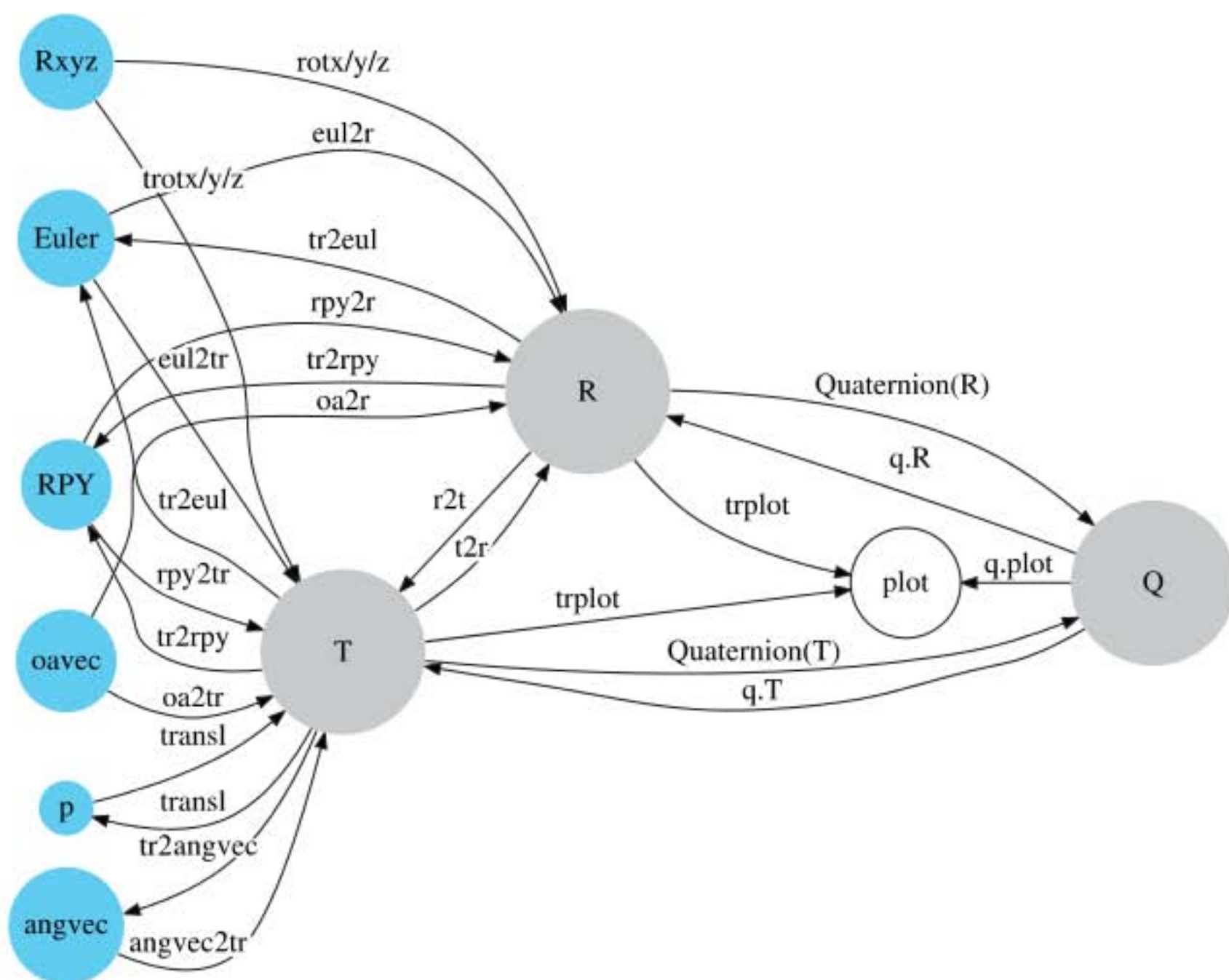


Fig. 2.15.
Conversion between rotational
representations

RVC 1, Ch 2

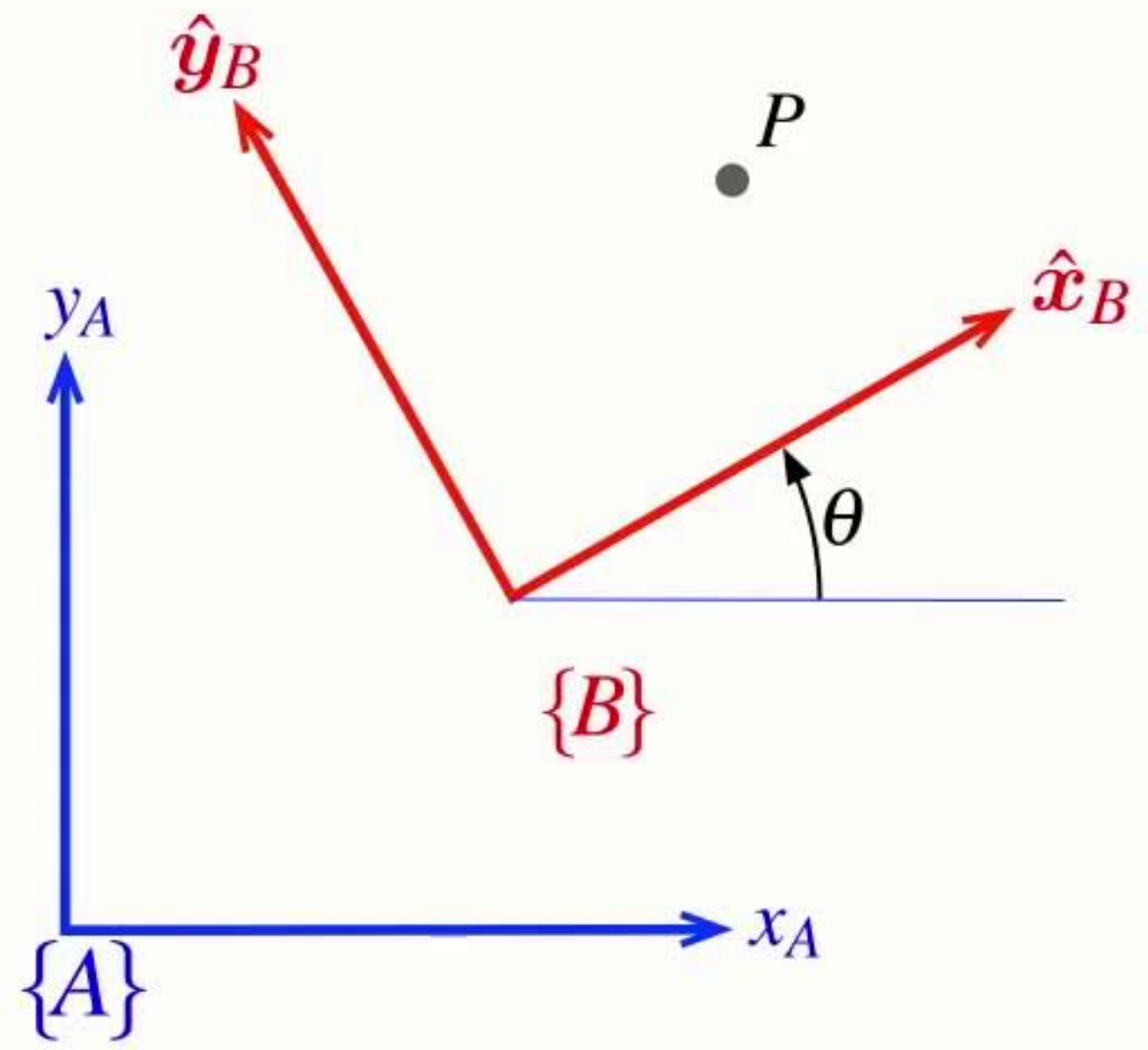
```
1 clear variables
2 close all
3 clc
4 clear livescript
```

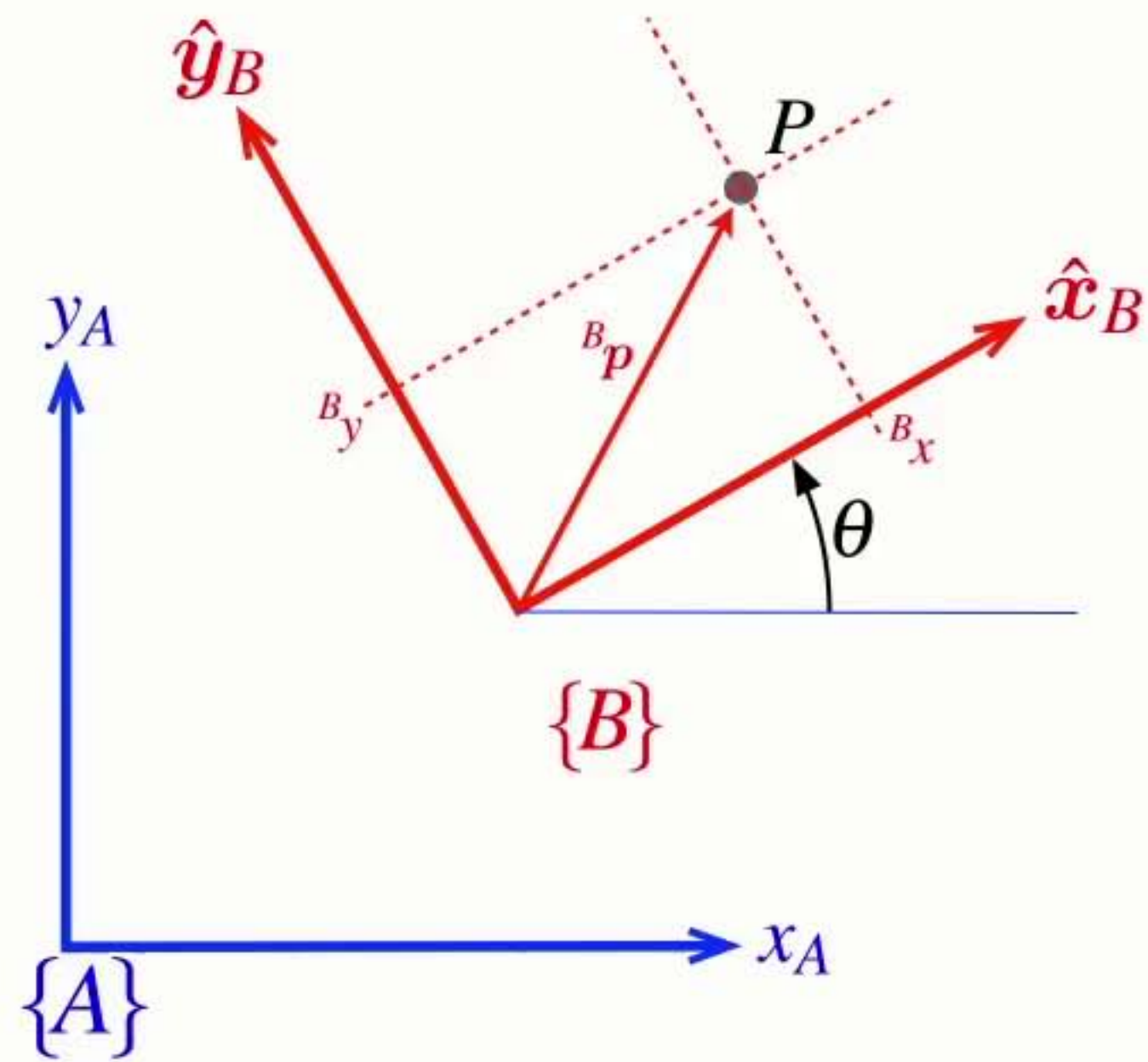
2.6 2D rotations

```
5 % Create a rotation matrix
6 R = rot2(0)
7
8 % Rotate by 0.2 radians
9 R=rot2(0.2)
10
11 % Rotate by 30 degrees
12 R=rot2(30,'deg')
13
14 % Rotation matrices are
15 % 1) orthonormal: columns are orthogonal to each other and unit length
16 % 2) full rank: independent/invertible
17 % 3) symmetrical: the inverse is equal to the transpose
18 c1=R(:,1)
19 c2=R(:,2)
20
21 norm(c1),norm(c2)
22 dot(c1,c2)
23
24 % Full rank
25 det(R)
```

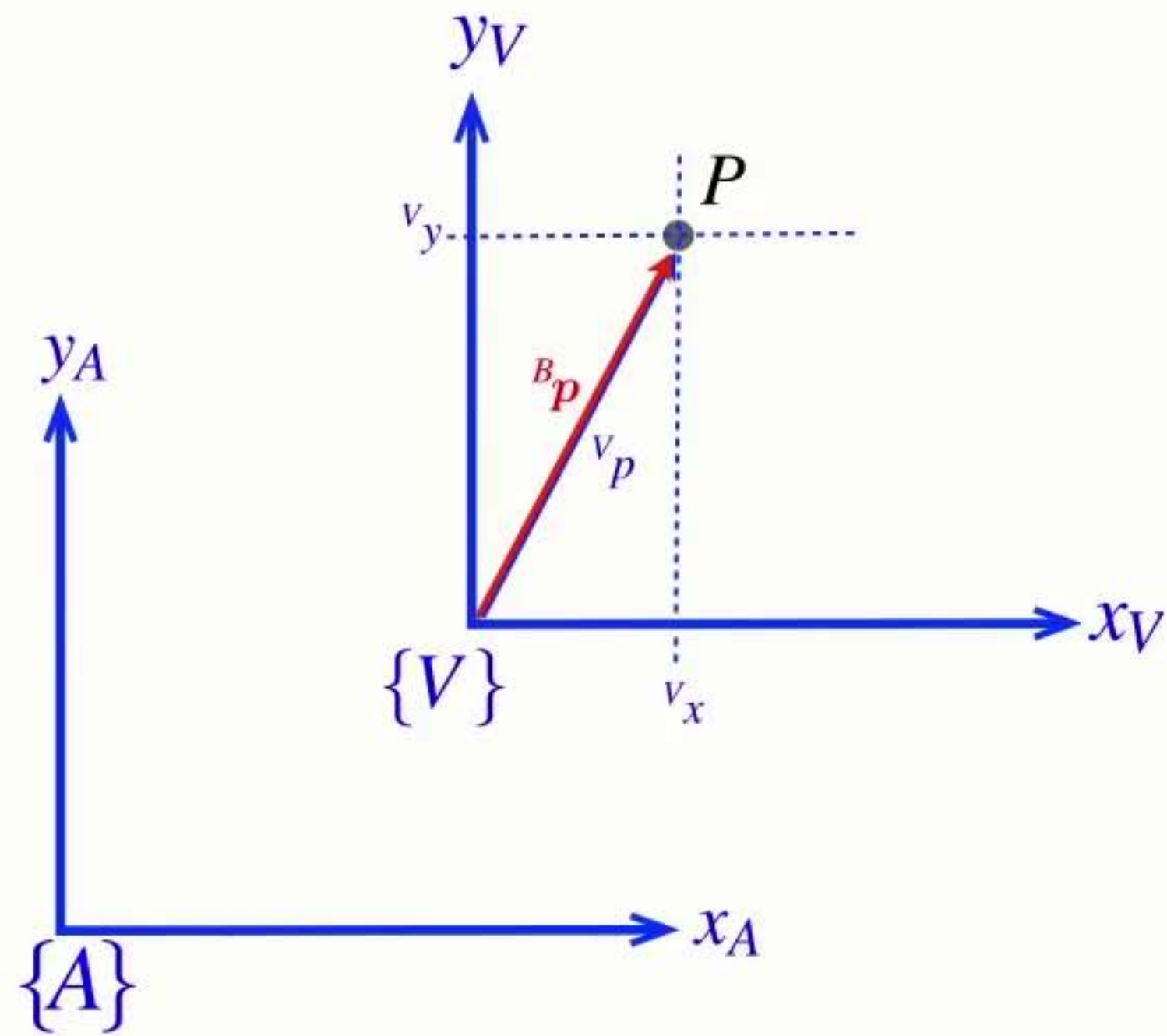
Section 7

Describing rotation and translation

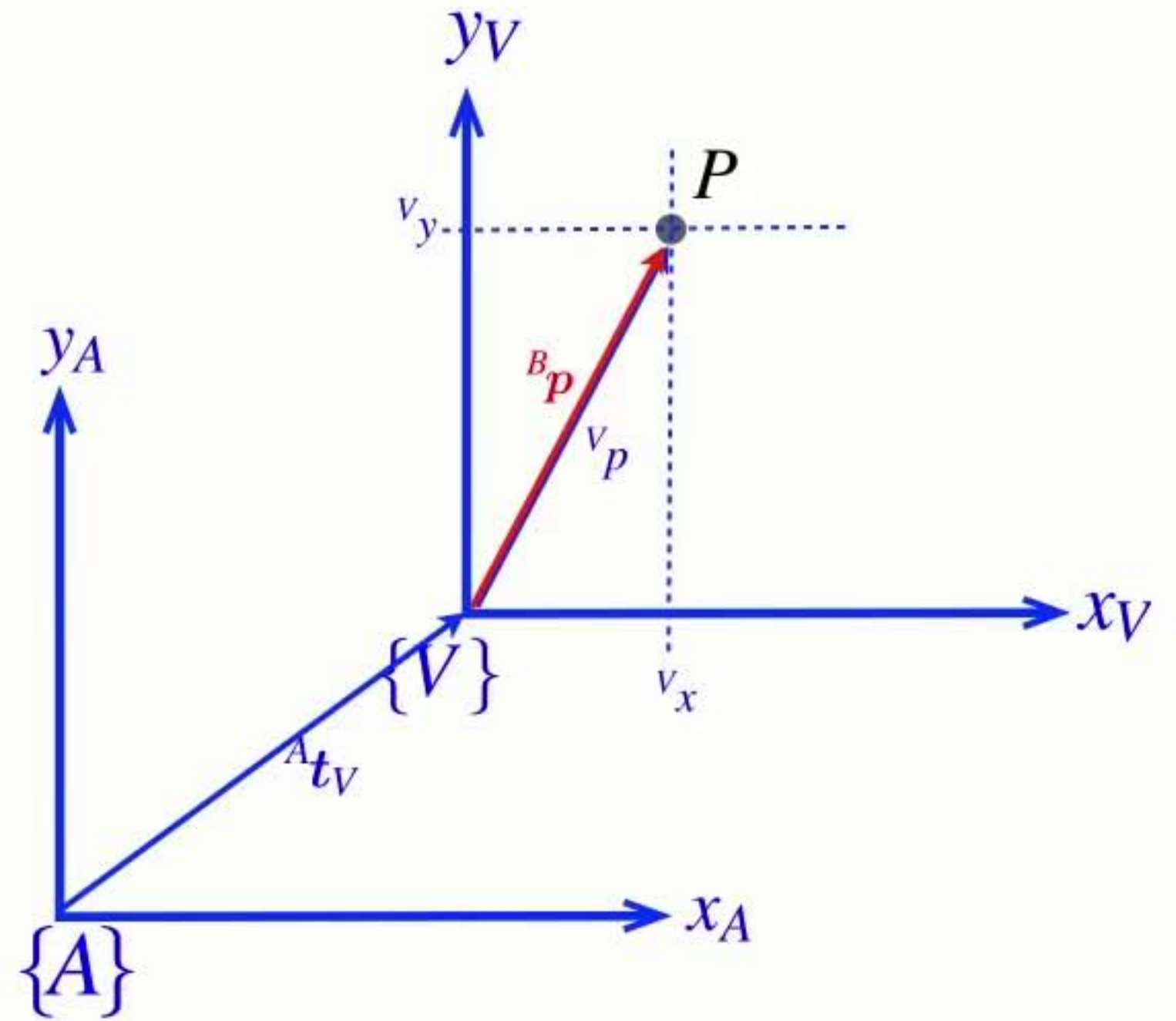




$${}^V\mathbf{p} = {}^V\mathbf{R}_B {}^B\mathbf{p}$$



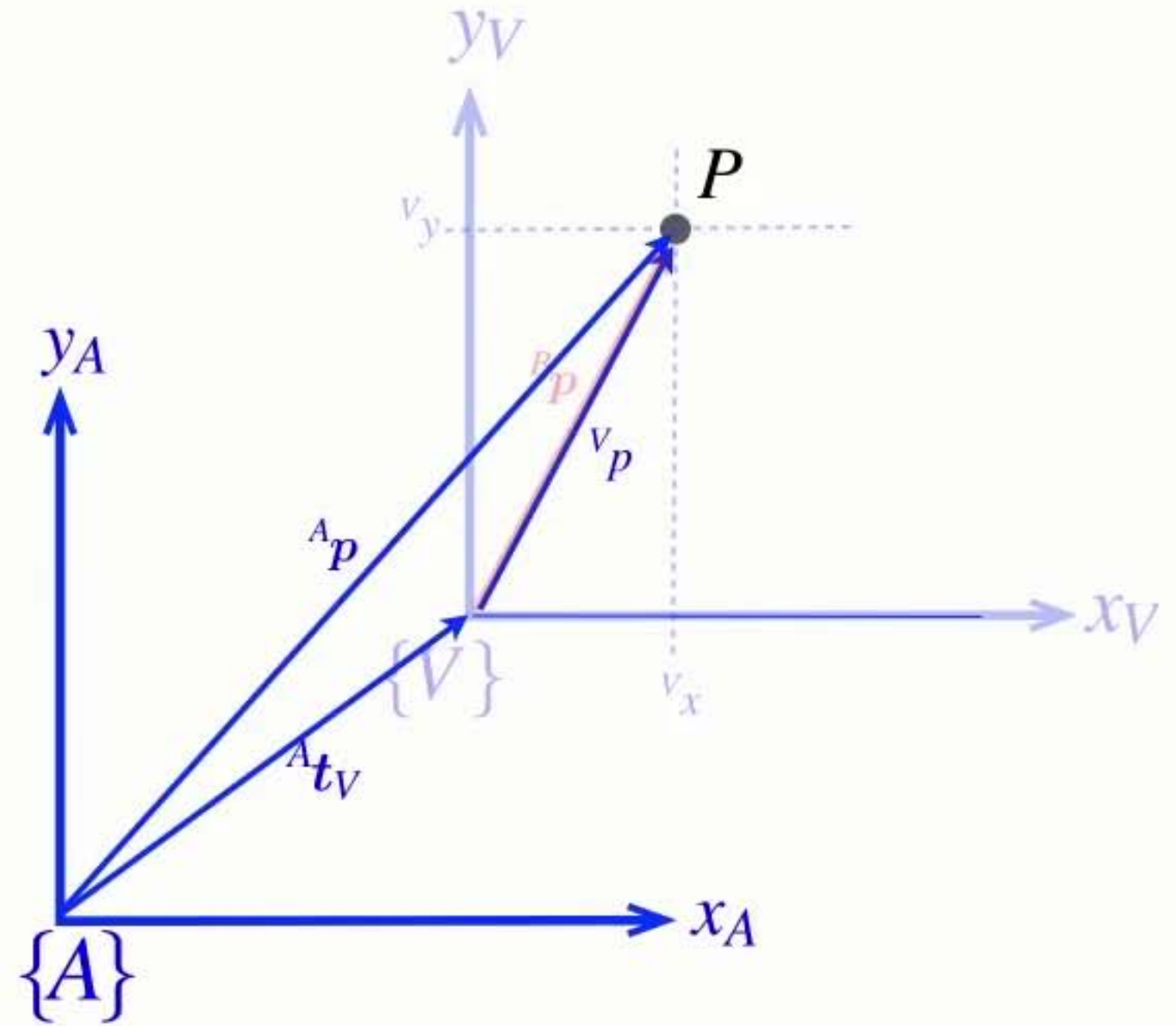
$${}^V\mathbf{p} = {}^V\mathbf{R}_B {}^B\mathbf{p}$$



$${}^V\mathbf{p} = {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{p}$$

$$\{V\} \parallel \{A\}$$

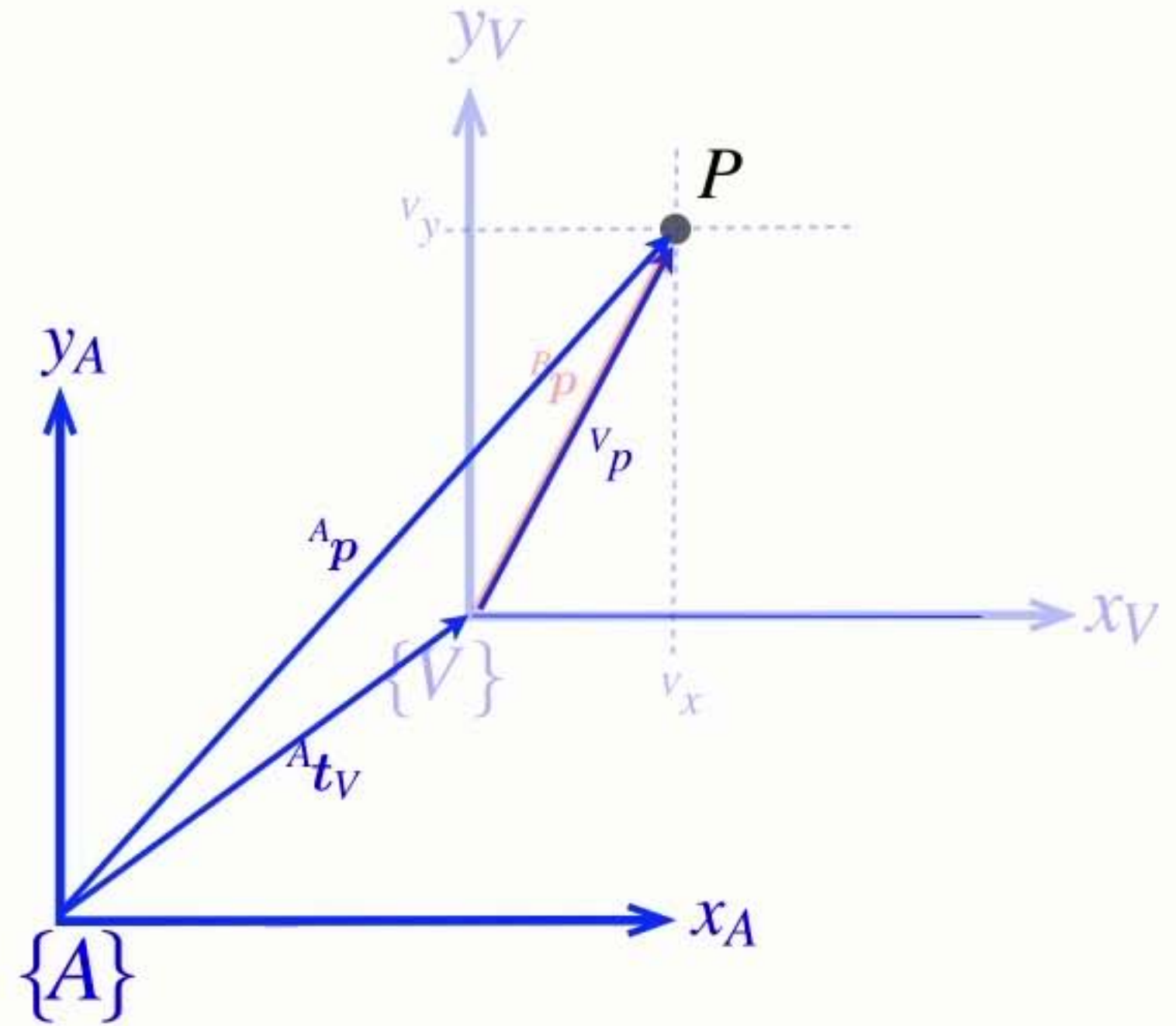


- Only add vectors in the same (or parallel) reference frames

$${}^V\mathbf{p} = {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{p} \quad \{V\} \parallel \{A\}$$

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$



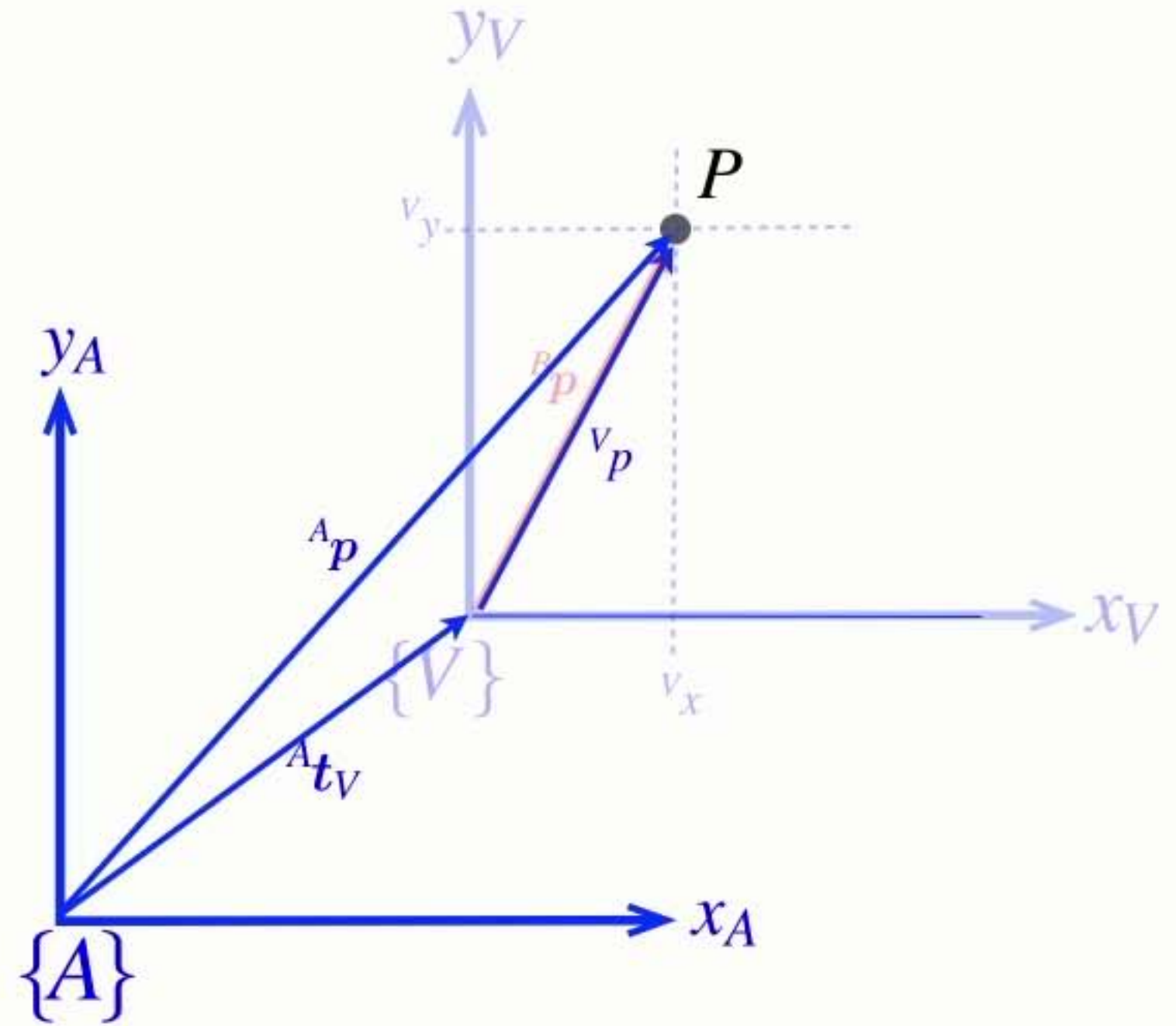
- Only add vectors in the same (or parallel) reference frames

$${}^V\mathbf{p} = {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{p} \quad \{V\} \parallel \{A\}$$

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$

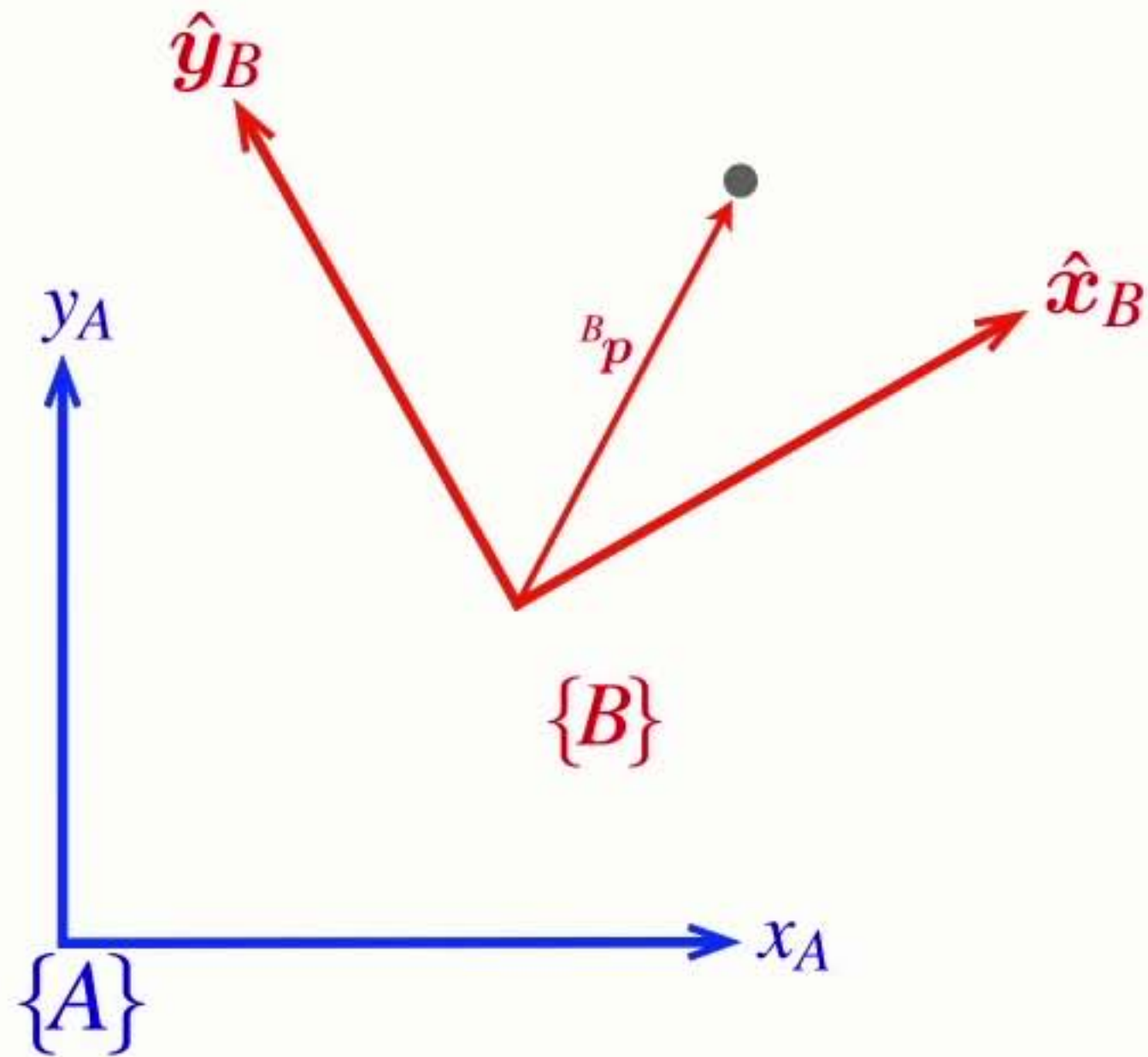
$${}^A\mathbf{p} = {}^A\mathbf{t}_B + {}^A\mathbf{R}_B {}^B\mathbf{p}$$



- Only add vectors in the same (or parallel) reference frames

Homogeneous **form**

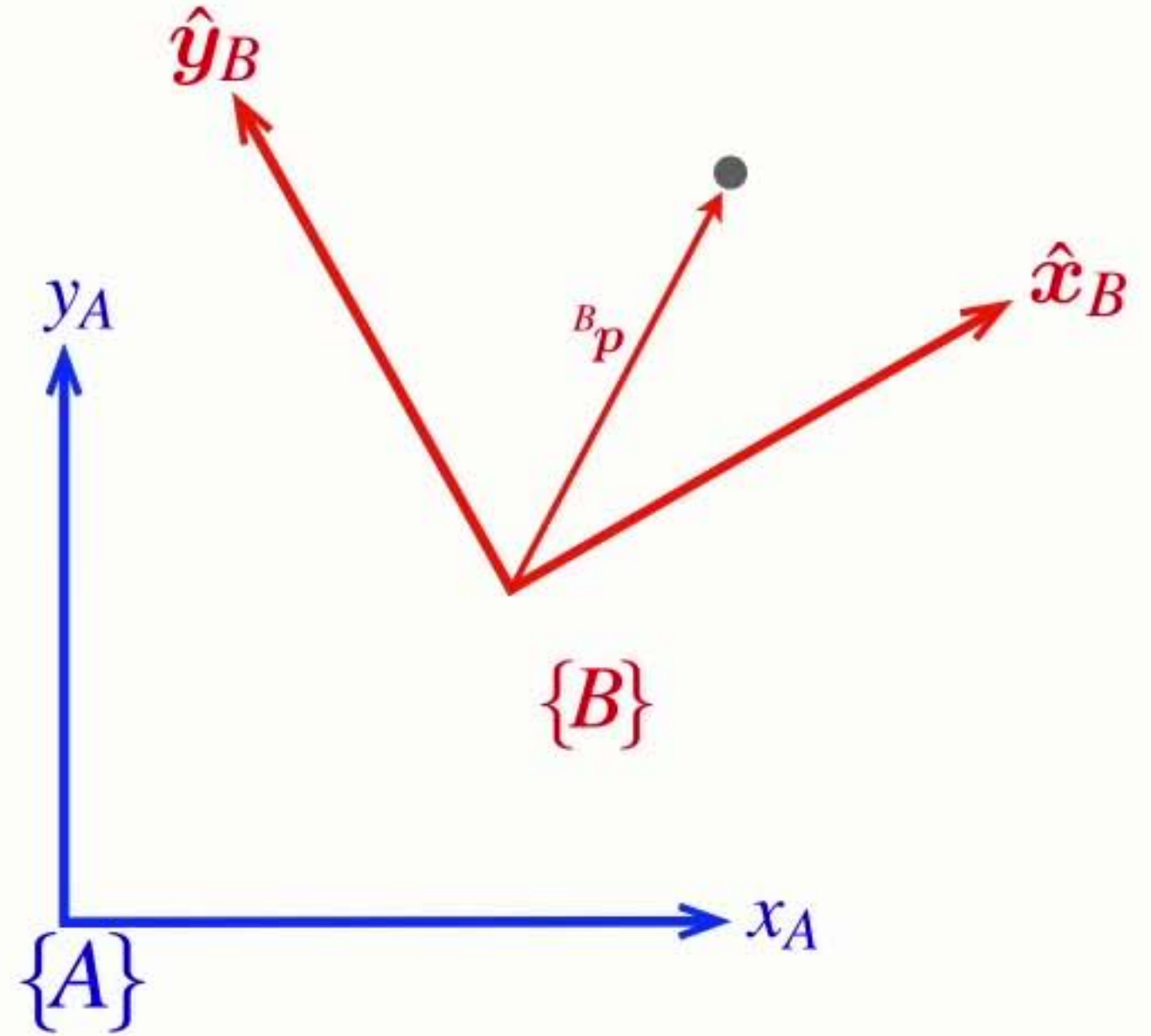
$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$



Homogeneous **form**

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \end{pmatrix} + \begin{pmatrix} {}^Ax_B \\ {}^Ay_B \end{pmatrix}$$

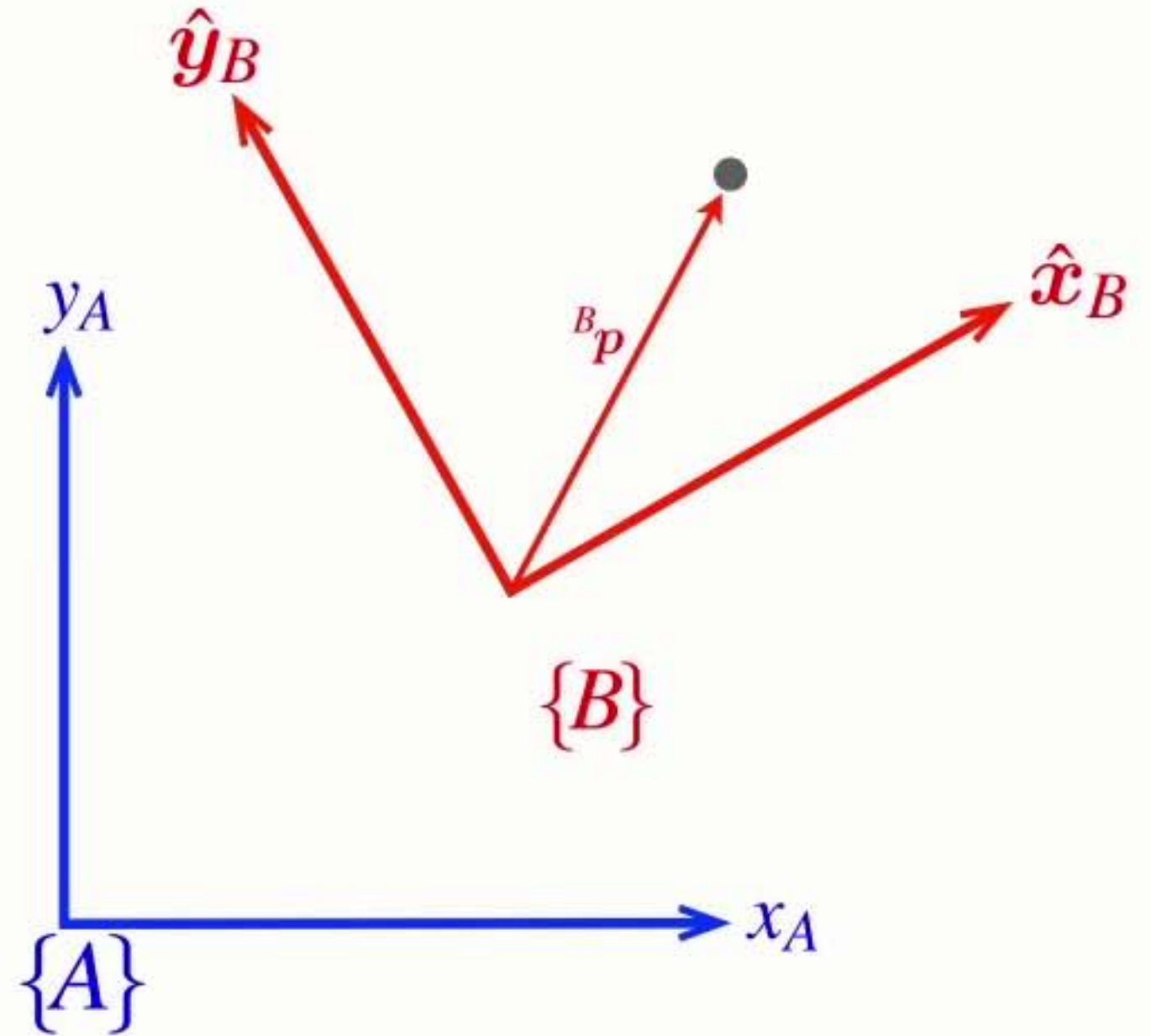


Homogeneous **form**

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \end{pmatrix} + \begin{pmatrix} {}^Ax_B \\ {}^Ay_B \end{pmatrix}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^Ax_B \\ \sin \theta & \cos \theta & {}^Ay_B \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \\ 1 \end{pmatrix}$$



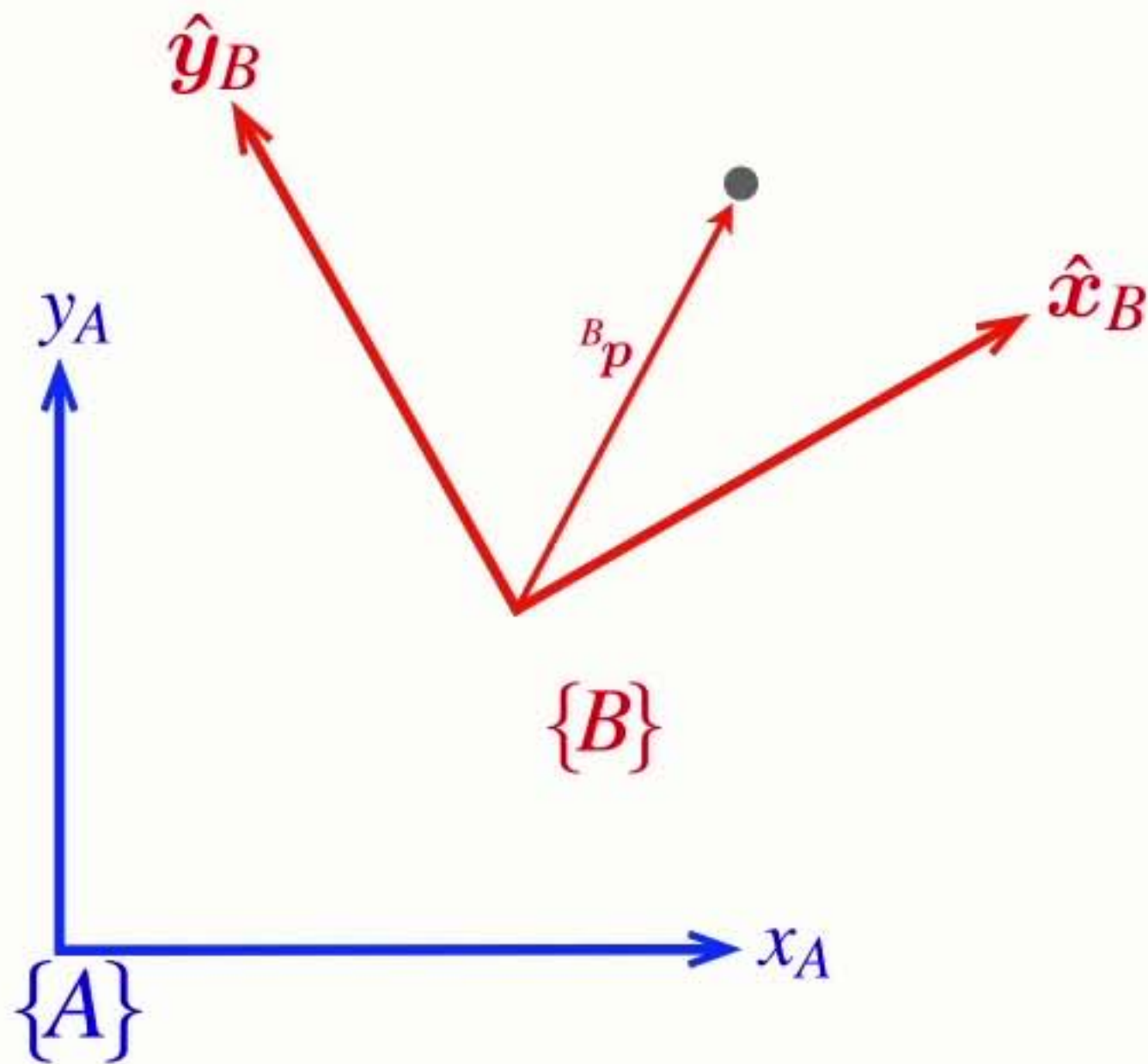
Homogeneous **form**

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \end{pmatrix} + \begin{pmatrix} {}^Ax_B \\ {}^Ay_B \end{pmatrix}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^Ax_B \\ \sin \theta & \cos \theta & {}^Ay_B \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \\ 1 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^Ax_B \\ \sin \theta & \cos \theta & {}^Ay_B \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \\ 1 \end{pmatrix}$$



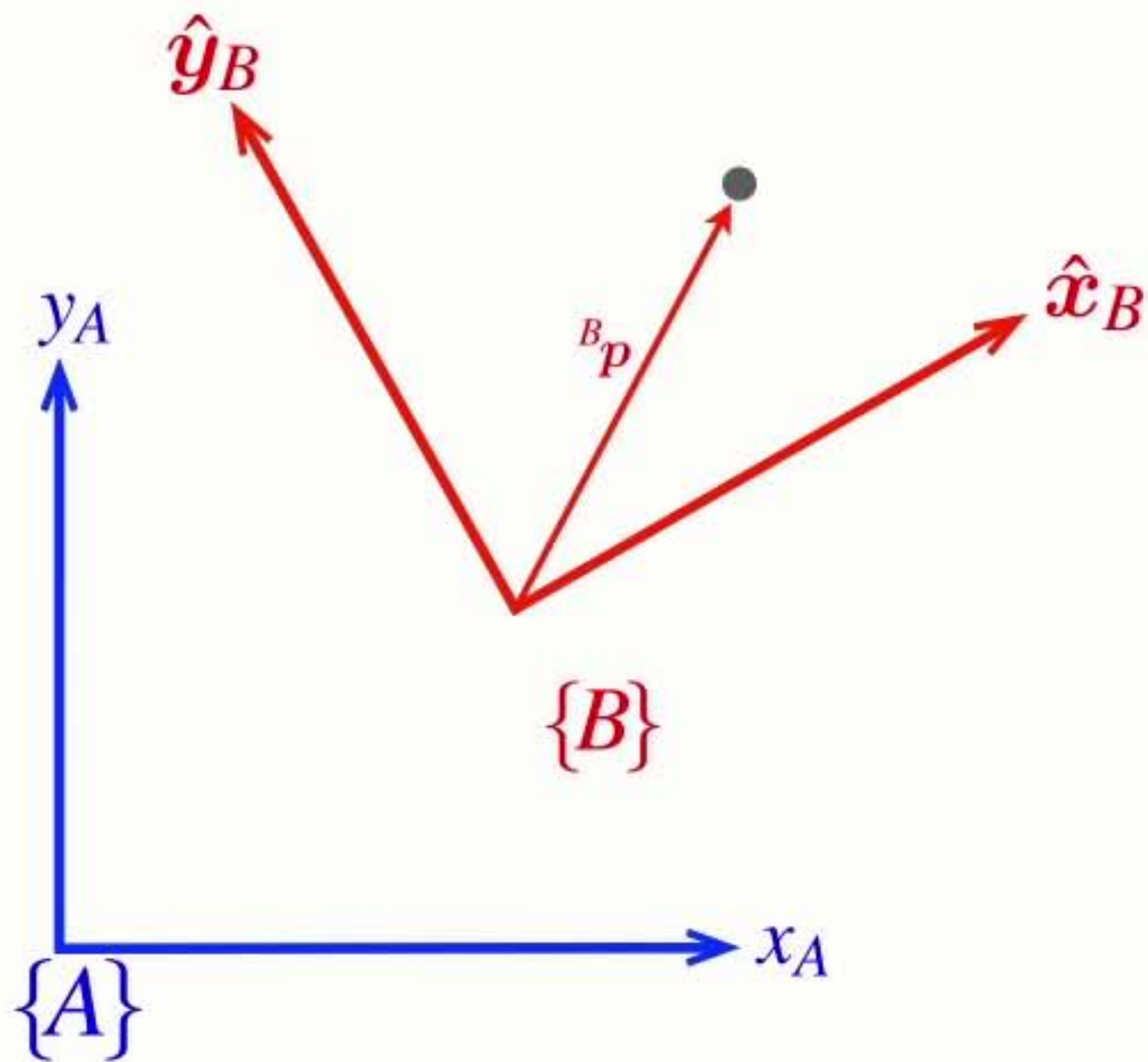
Homogeneous **form**

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \end{pmatrix} + \begin{pmatrix} {}^Ax_B \\ {}^Ay_B \end{pmatrix}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^Ax_B \\ \sin \theta & \cos \theta & {}^Ay_B \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \\ 1 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^Ax_B \\ \sin \theta & \cos \theta & {}^Ay_B \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \\ 1 \end{pmatrix}$$



$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$

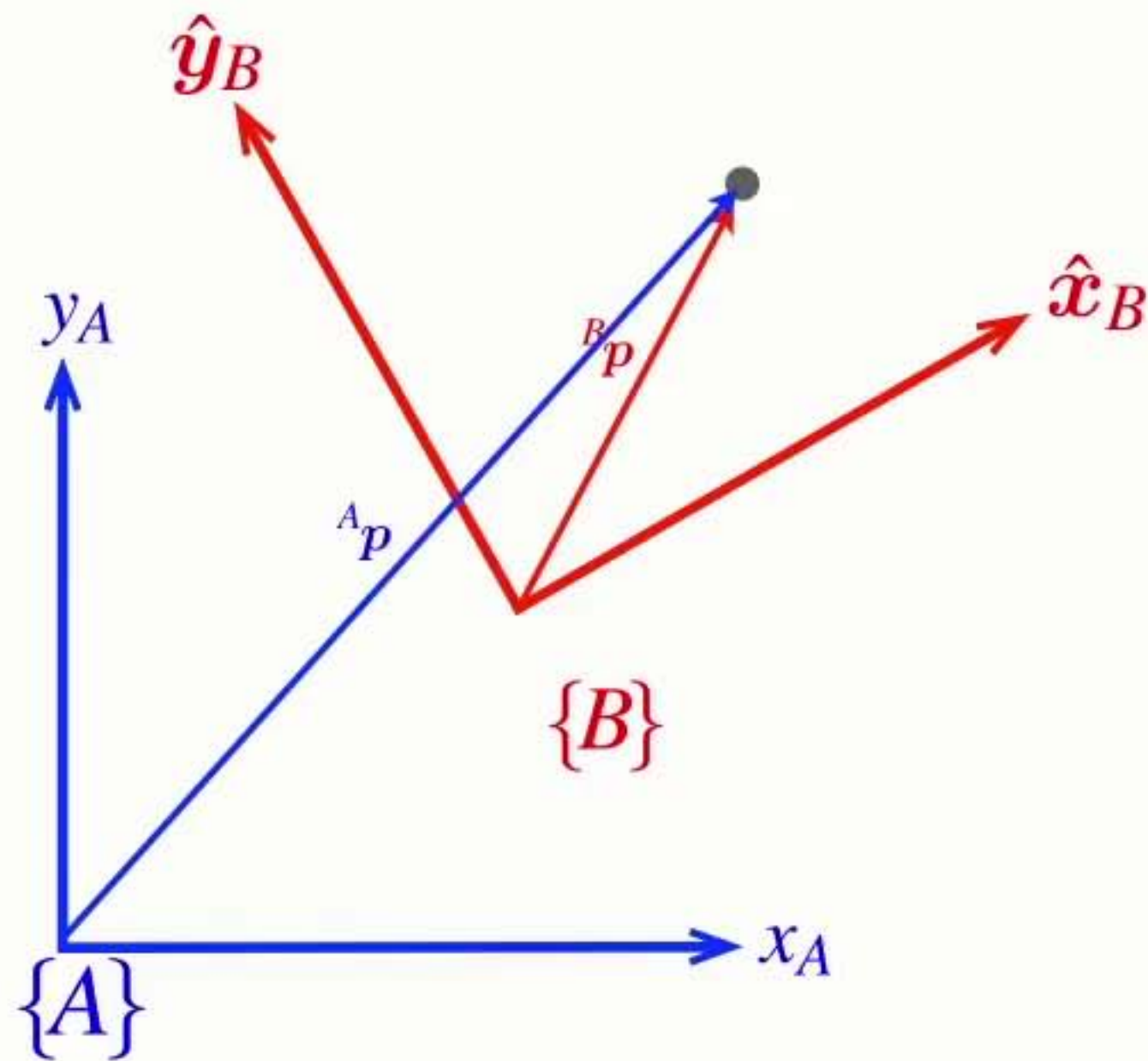
Homogeneous **form**

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \end{pmatrix} + \begin{pmatrix} {}^Ax_B \\ {}^Ay_B \end{pmatrix}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^Ax_B \\ \sin \theta & \cos \theta & {}^Ay_B \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} {}^Ax \\ {}^Ay \\ 1 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^Ax_B \\ \sin \theta & \cos \theta & {}^Ay_B \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} {}^Bx \\ {}^By \\ 1 \end{pmatrix}$$



$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$

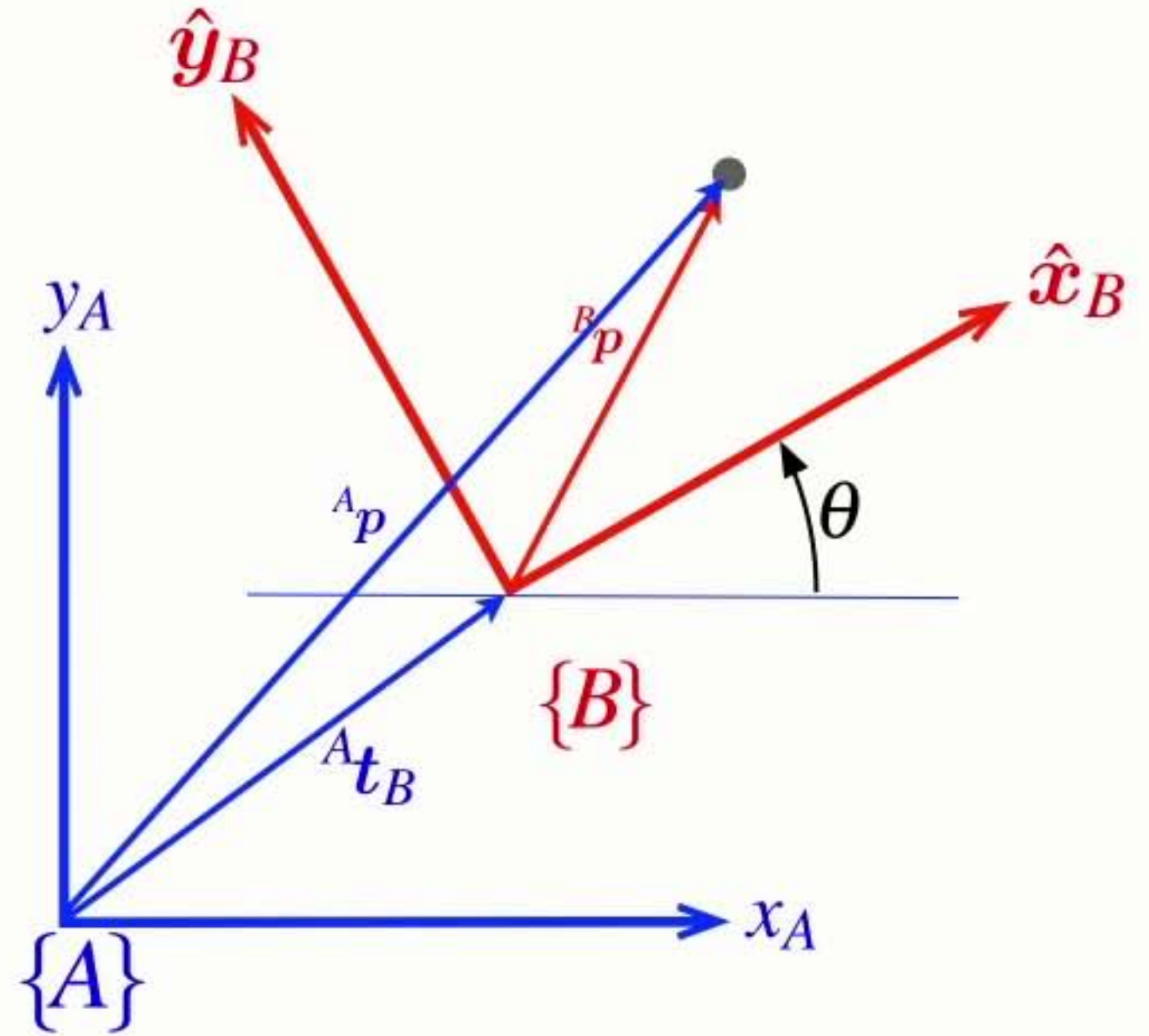
Homogeneous **form**

$${}^A\mathbf{p} = {}^A\mathbf{t}_V + {}^V\mathbf{R}_B {}^B\mathbf{p}$$

$$\begin{pmatrix} {}^A x \\ {}^A y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} {}^B x \\ {}^B y \end{pmatrix} + \begin{pmatrix} {}^A x_B \\ {}^A y_B \end{pmatrix}$$

$$\begin{pmatrix} {}^A x \\ {}^A y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^A x_B \\ \sin \theta & \cos \theta & {}^A y_B \end{pmatrix} \begin{pmatrix} {}^B x \\ {}^B y \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} {}^A x \\ {}^A y \\ 1 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & {}^A x_B \\ \sin \theta & \cos \theta & {}^A y_B \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} {}^B x \\ {}^B y \\ 1 \end{pmatrix}$$



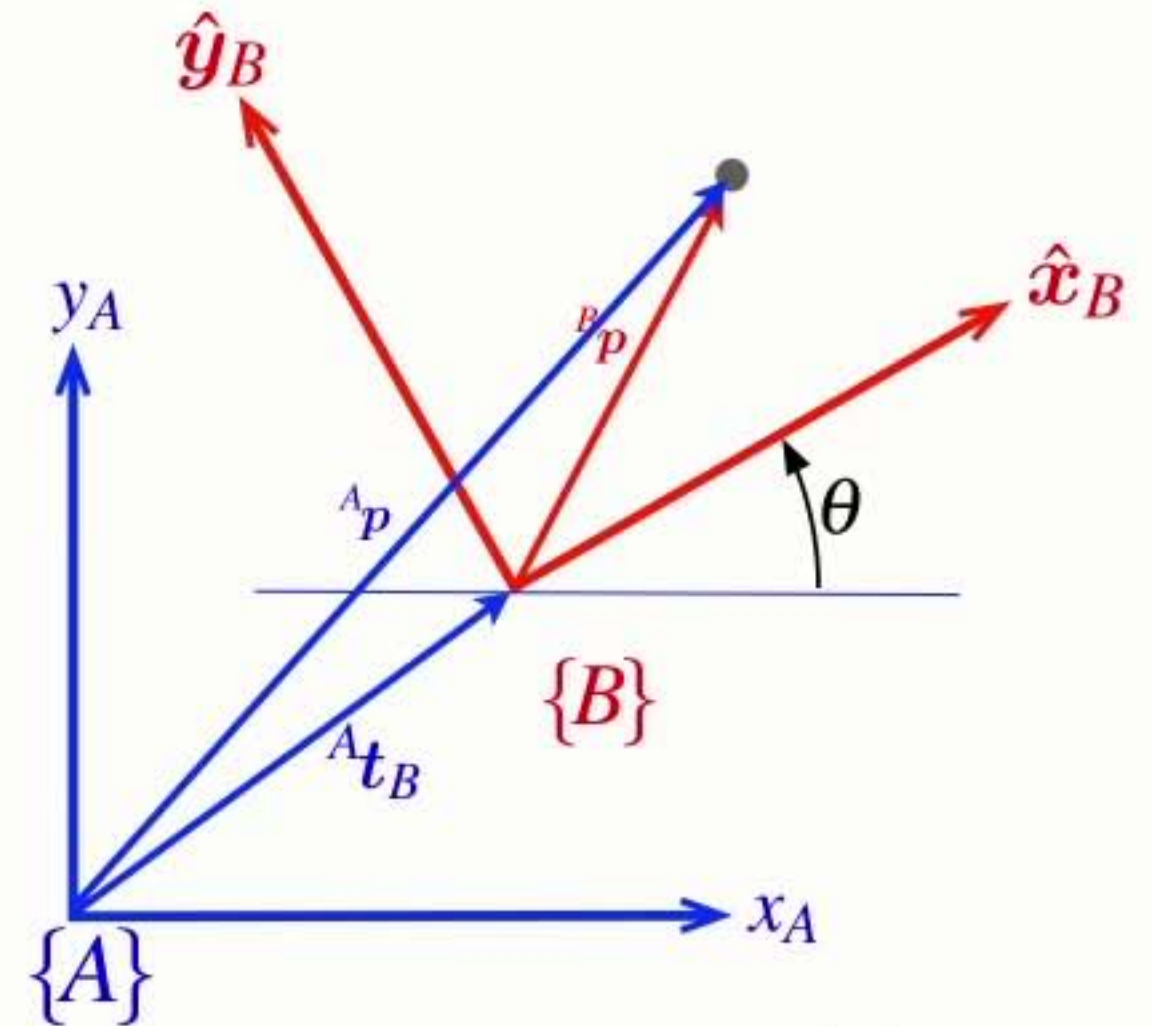
$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$

Properties of the **homogeneous transform**

homogeneous vector

$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$

homogeneous vector



$$\mathbf{p} = \begin{pmatrix} a \\ b \end{pmatrix} \quad \tilde{\mathbf{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

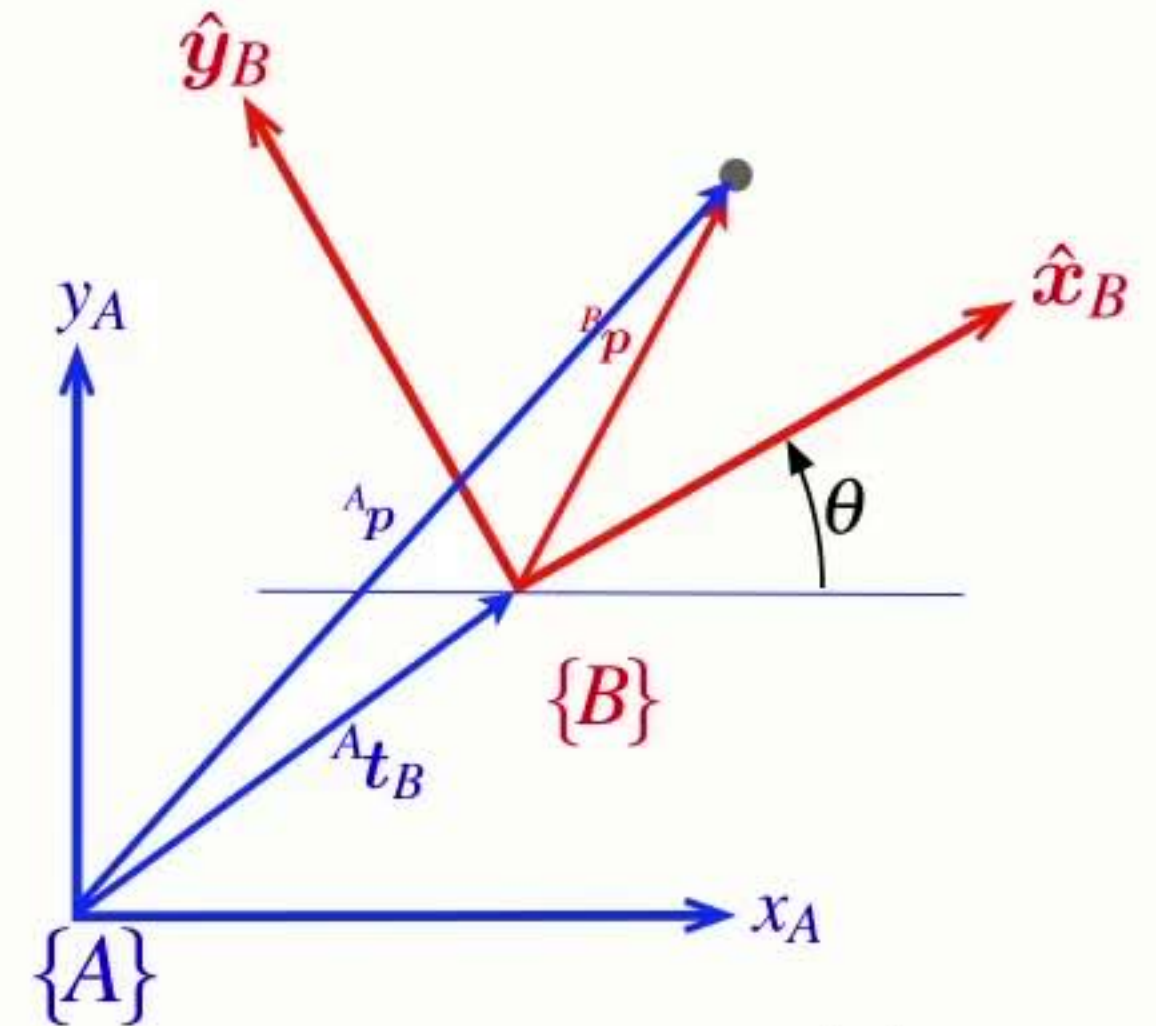
Properties of the **homogeneous transform**

homogeneous vector

homogeneous transform

homogeneous vector

$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$



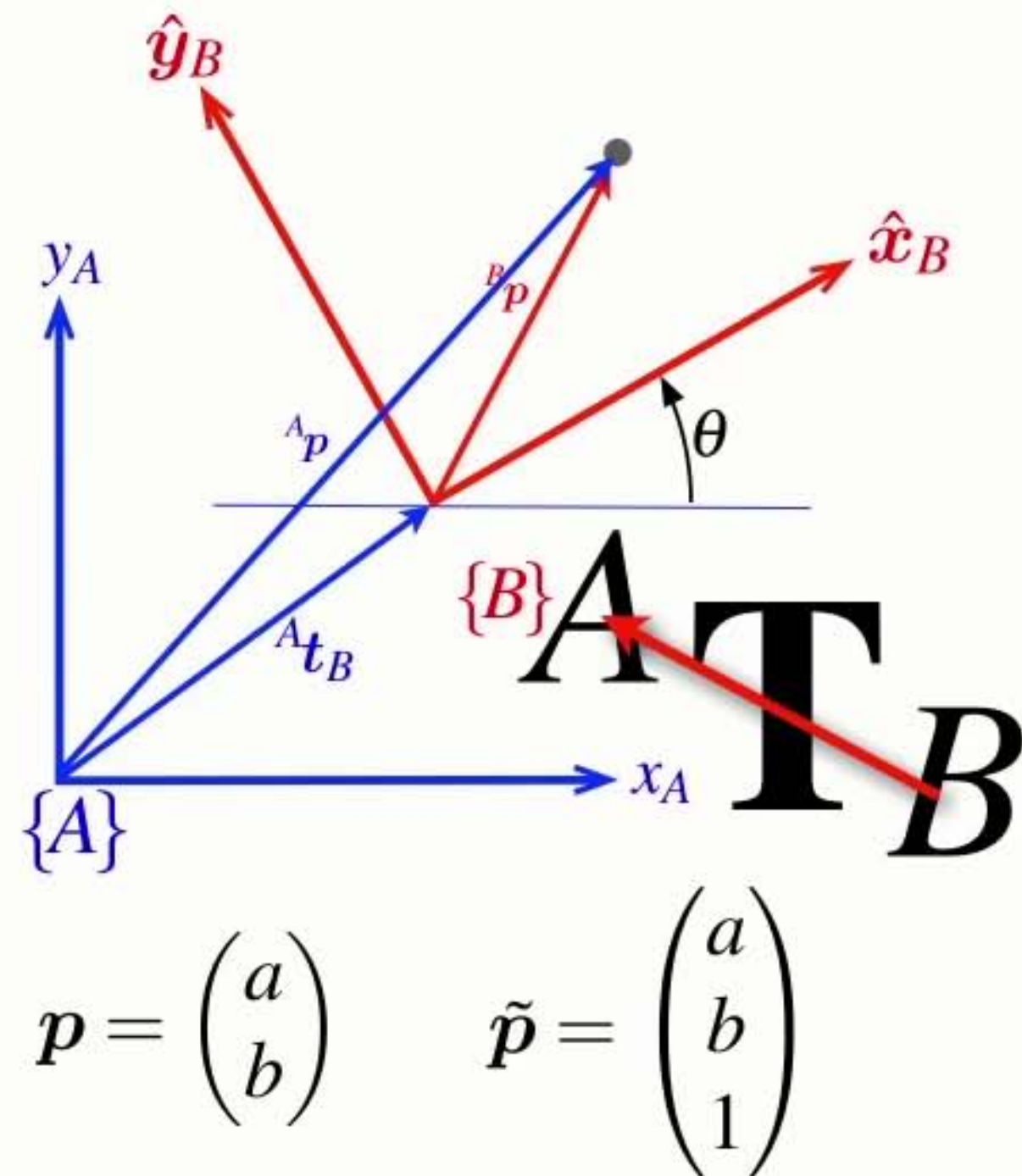
$$\mathbf{p} = \begin{pmatrix} a \\ b \end{pmatrix} \quad \tilde{\mathbf{p}} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

Properties of the **homogeneous transform**

homogeneous vector homogeneous transform homogeneous vector

$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$

$${}^A\tilde{\mathbf{p}} = {}^A\mathbf{T}_B {}^B\tilde{\mathbf{p}}$$



Properties of the **homogeneous transform**

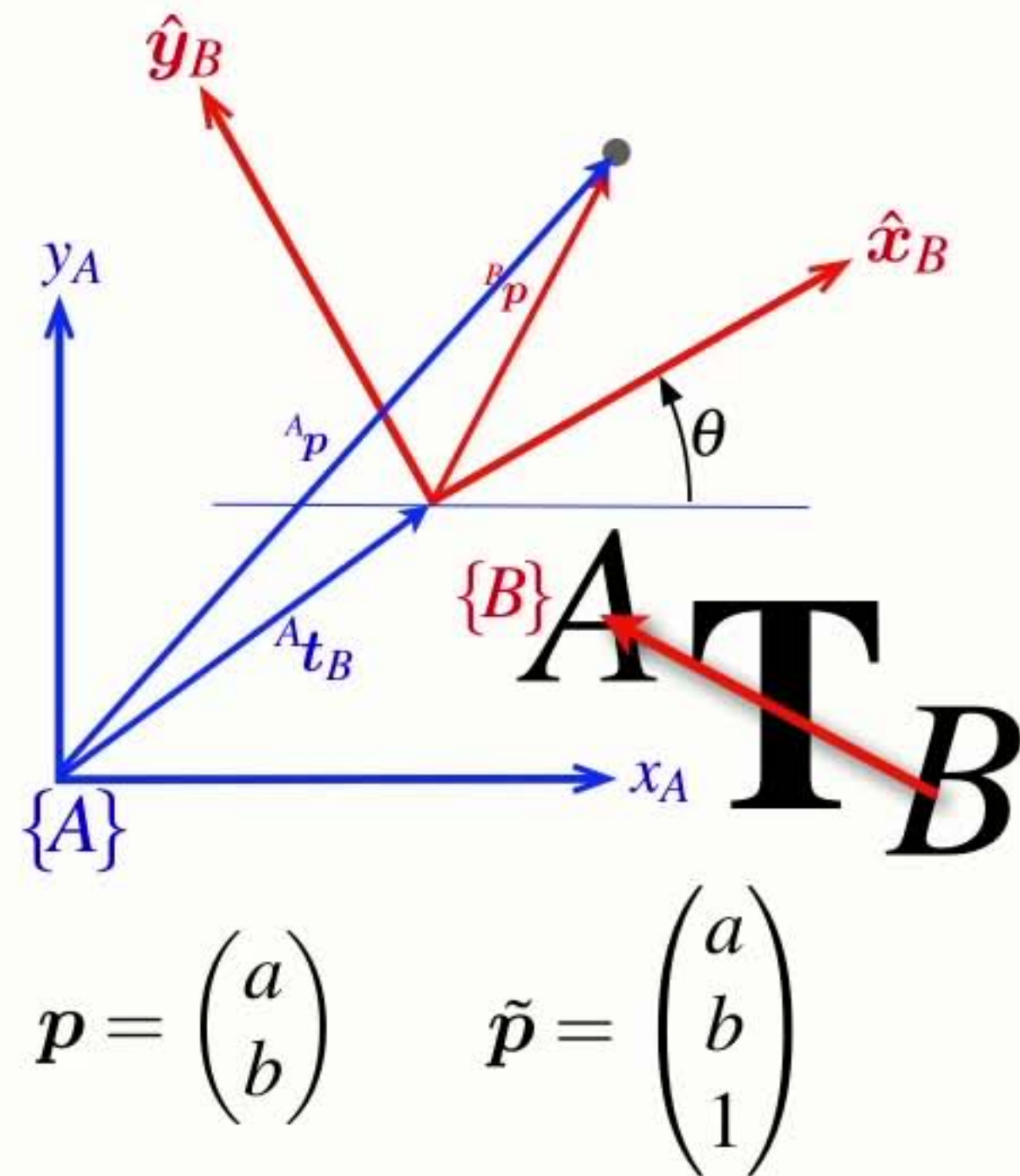
homogeneous vector

homogeneous transform

homogeneous vector

$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$

$${}^A\tilde{\mathbf{p}} = {}^A\mathbf{T}_B {}^B\tilde{\mathbf{p}} \quad {}^A\mathbf{T}_B = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix}$$



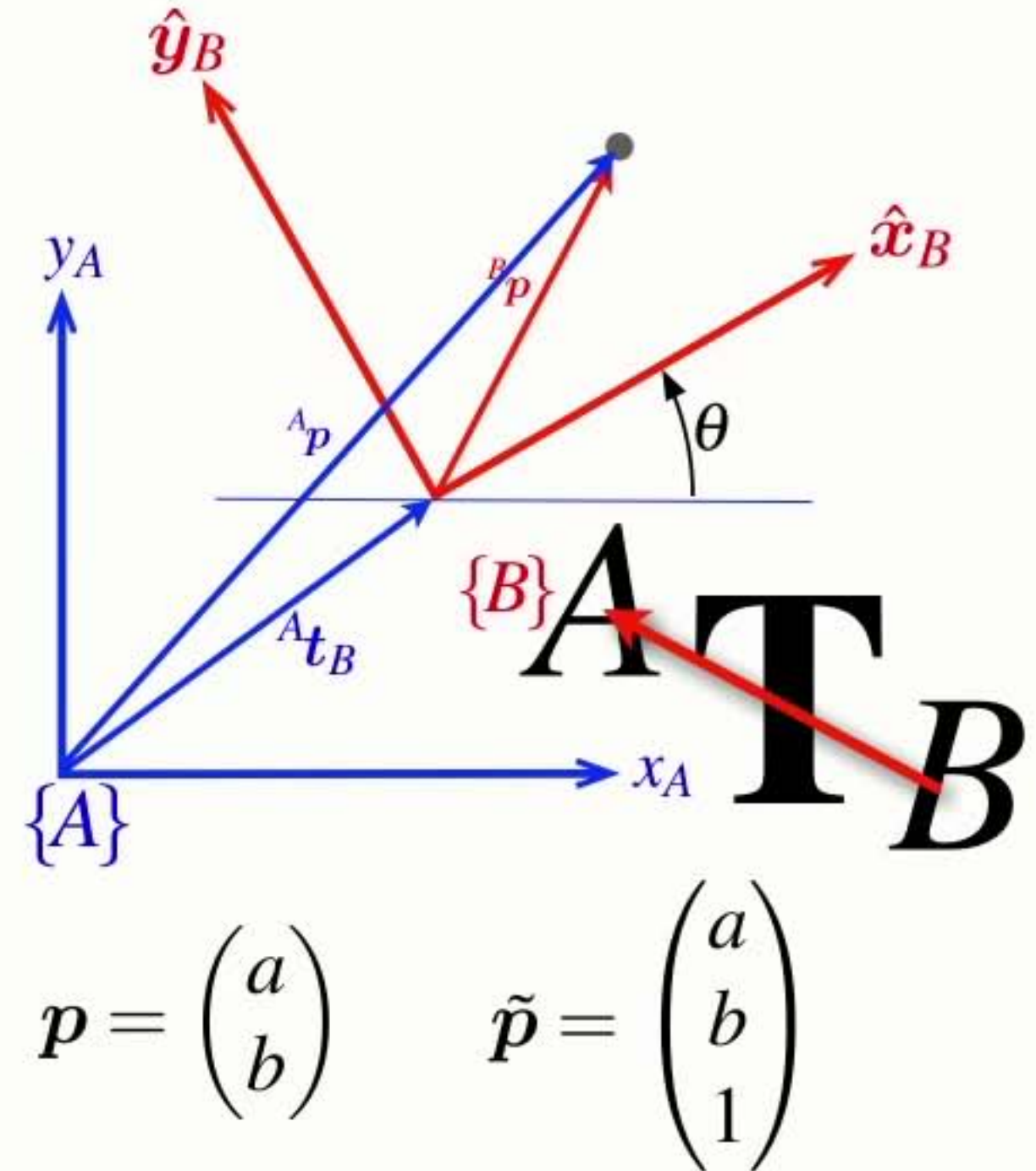
Properties of the **homogeneous transform**

homogeneous vector homogeneous transform homogeneous vector

$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$

$${}^A\tilde{\mathbf{p}} = {}^A\mathbf{T}_B {}^B\tilde{\mathbf{p}} \quad {}^A\mathbf{T}_B = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix}$$

- Describes a relative pose as a 3x3 matrix



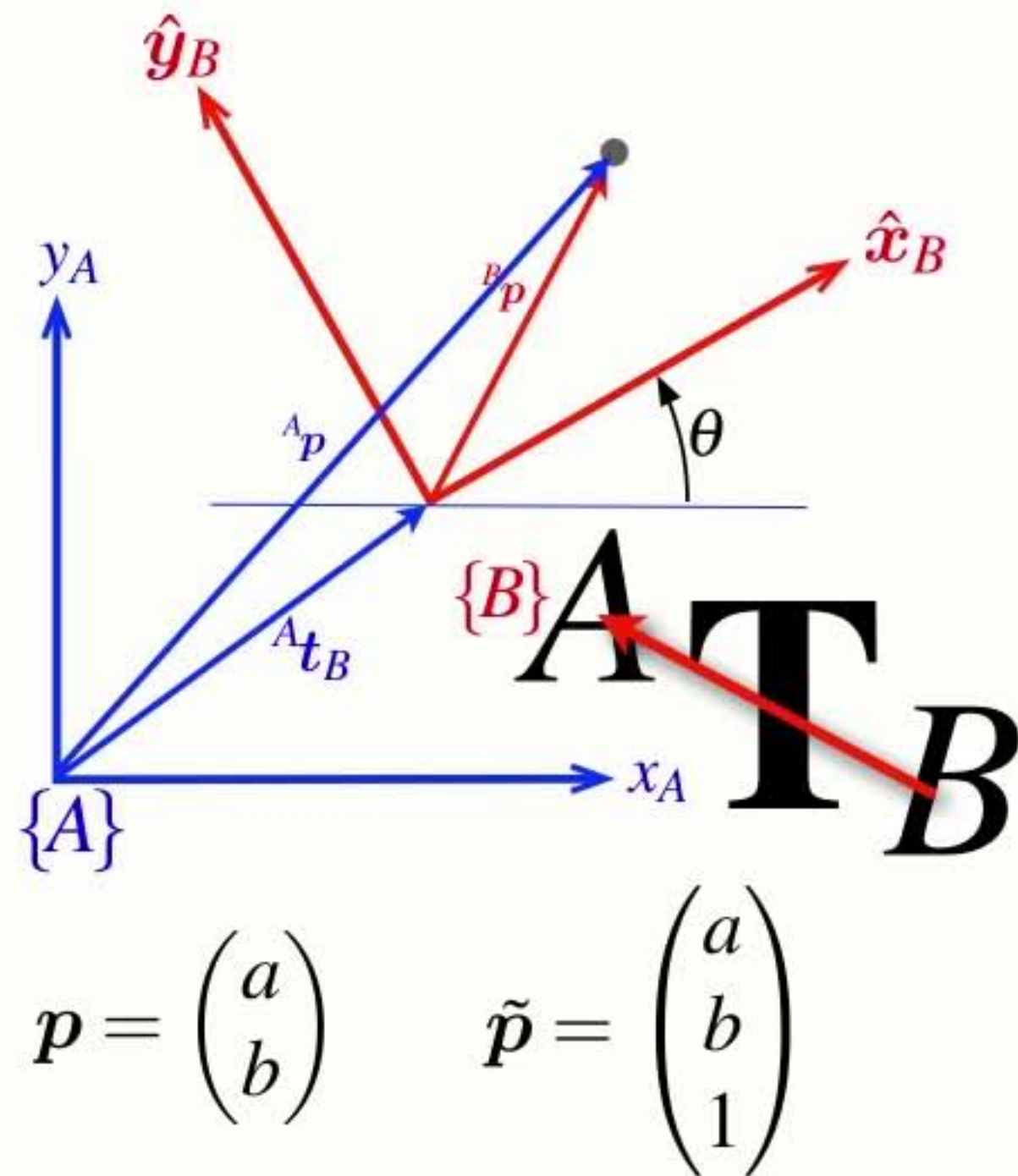
Properties of the **homogeneous transform**

homogeneous vector homogeneous transform homogeneous vector

$${}^A\tilde{\mathbf{p}} = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix} {}^B\tilde{\mathbf{p}}$$

$${}^A\tilde{\mathbf{p}} = {}^A\mathbf{T}_B {}^B\tilde{\mathbf{p}} \quad {}^A\mathbf{T}_B = \begin{pmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ 0,0 & 1 \end{pmatrix}$$

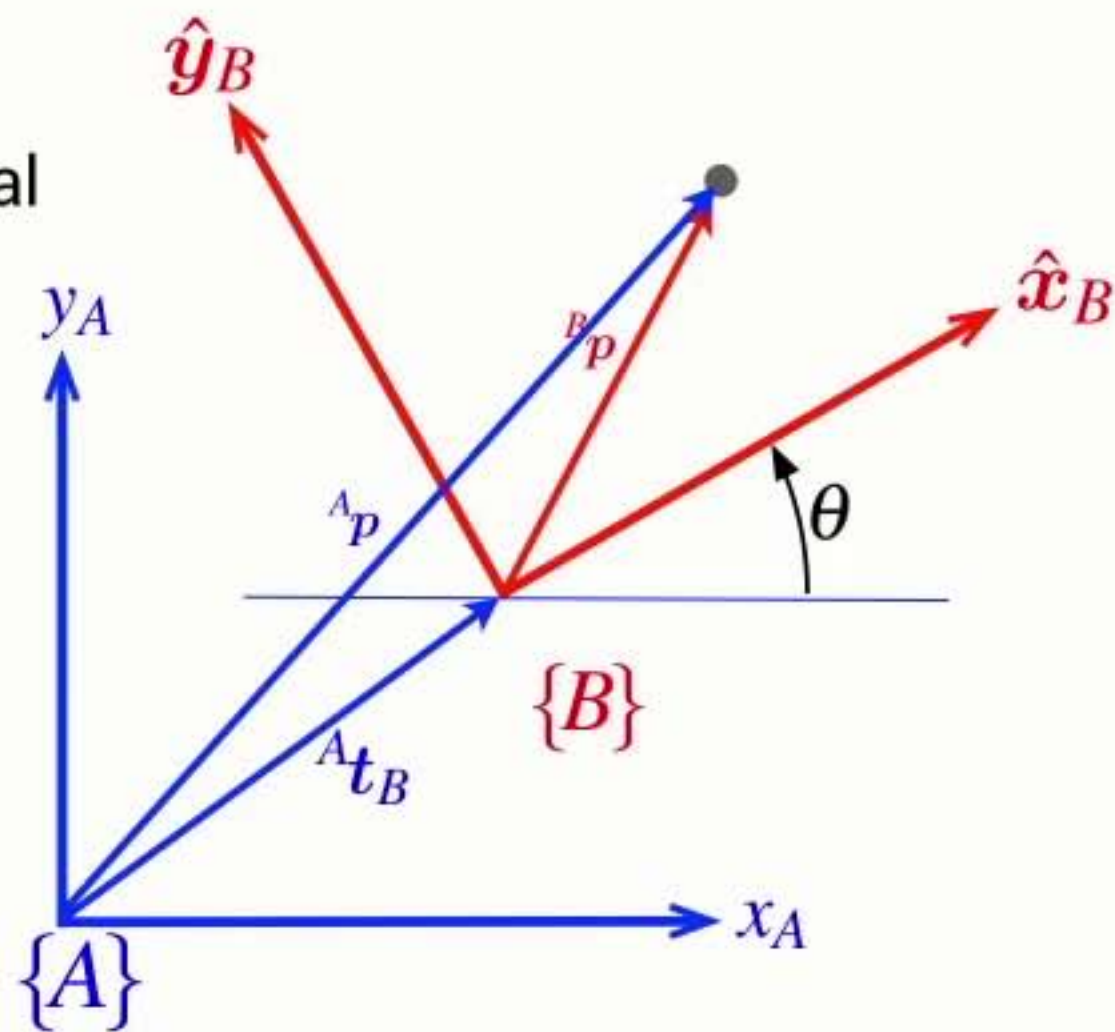
- Describes a relative pose as a 3x3 matrix
- Homogeneous transforms belong to the special Euclidean group of dimension 2 $\mathbf{T} \in SE(2)$



From abstract to real

- Finally we can make these abstract symbols and concepts real

→ Pose is a matrix ${}^A\xi_B \sim {}^A\mathbf{T}_B = \begin{pmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$



$$p = \begin{pmatrix} a \\ b \end{pmatrix} \quad \tilde{p} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

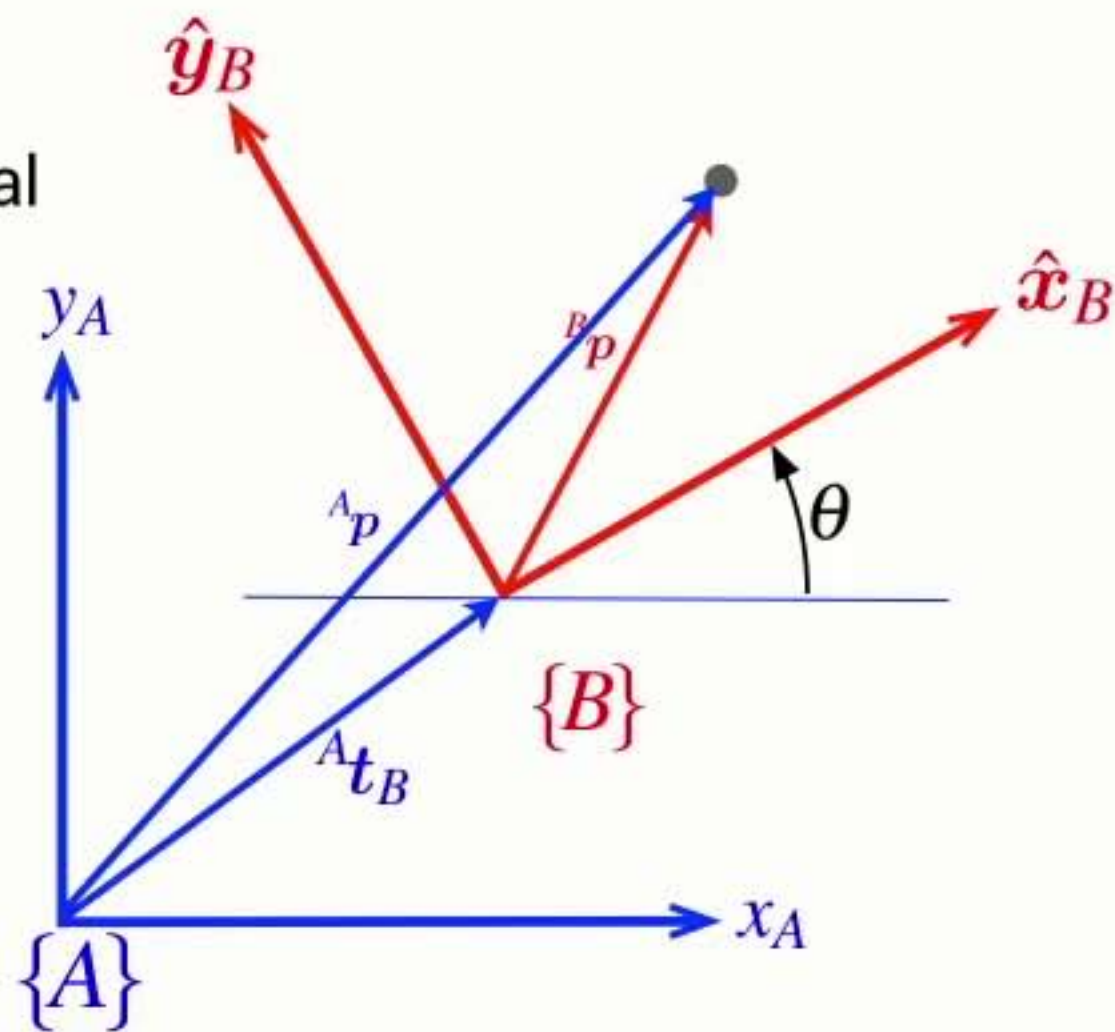
From abstract to real

- Finally we can make these abstract symbols and concepts real

→ Pose is a matrix ${}^A\xi_B \sim {}^A\mathbf{T}_B = \begin{pmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$

→ Compounding (composition) is a matrix-matrix product

$${}^X\xi_Y \oplus {}^Y\xi_Z \mapsto {}^X\mathbf{T}_Y {}^Y\mathbf{T}_Z$$



$$p = \begin{pmatrix} a \\ b \end{pmatrix} \quad \tilde{p} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

From abstract to real

- Finally we can make these abstract symbols and concepts real

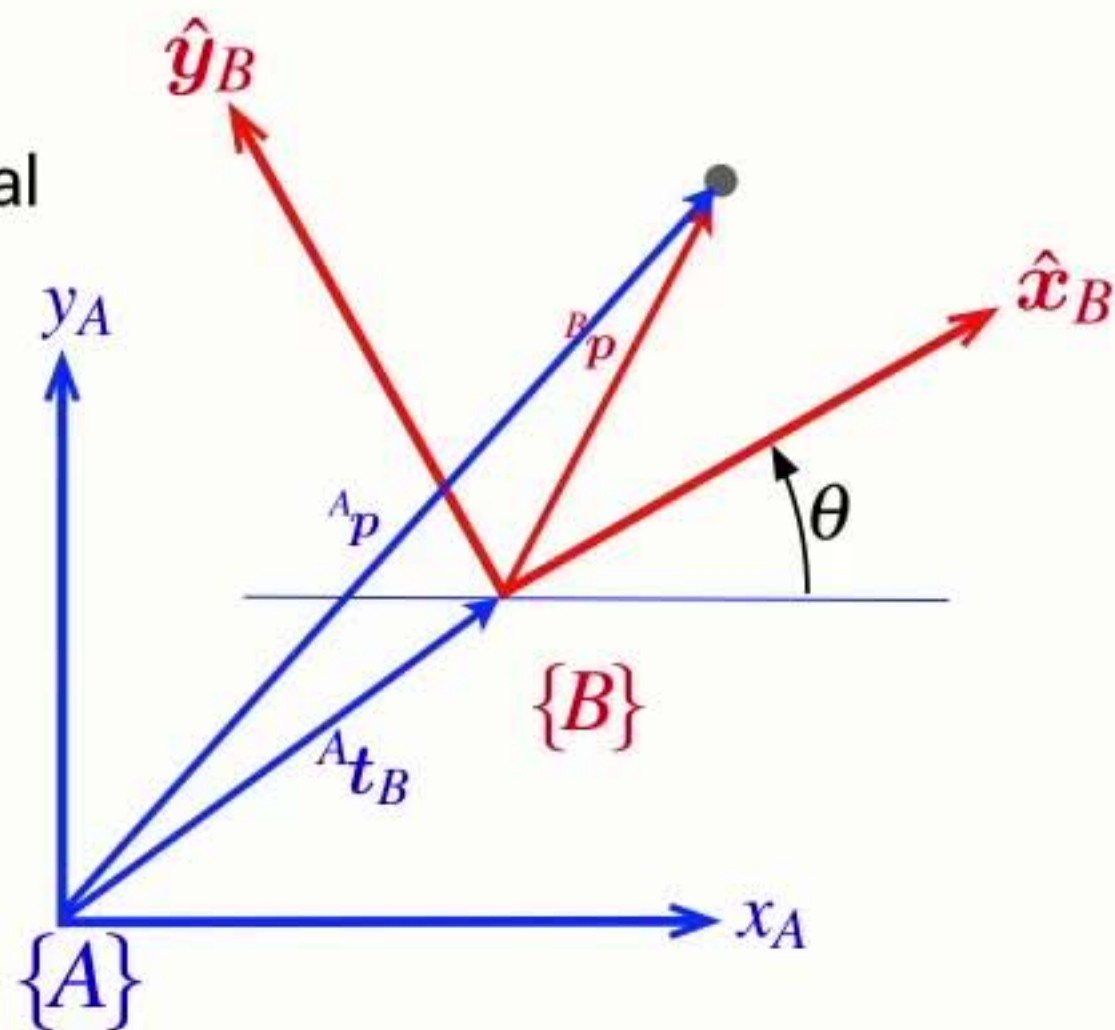
→ Pose is a matrix ${}^A\xi_B \sim {}^A\mathbf{T}_B = \begin{pmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$

→ Compounding (composition) is a matrix-matrix product

$${}^X\xi_Y \oplus {}^Y\xi_Z \mapsto {}^X\mathbf{T}_Y {}^Y\mathbf{T}_Z$$

→ Negation is a matrix inverse

$$\ominus \xi \mapsto \mathbf{T}^{-1}$$



$$p = \begin{pmatrix} a \\ b \end{pmatrix} \quad \tilde{p} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

From abstract to real

- Finally we can make these abstract symbols and concepts real

→ Pose is a matrix ${}^A\xi_B \sim {}^A\mathbf{T}_B = \begin{pmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$

→ Compounding (composition) is a matrix-matrix product

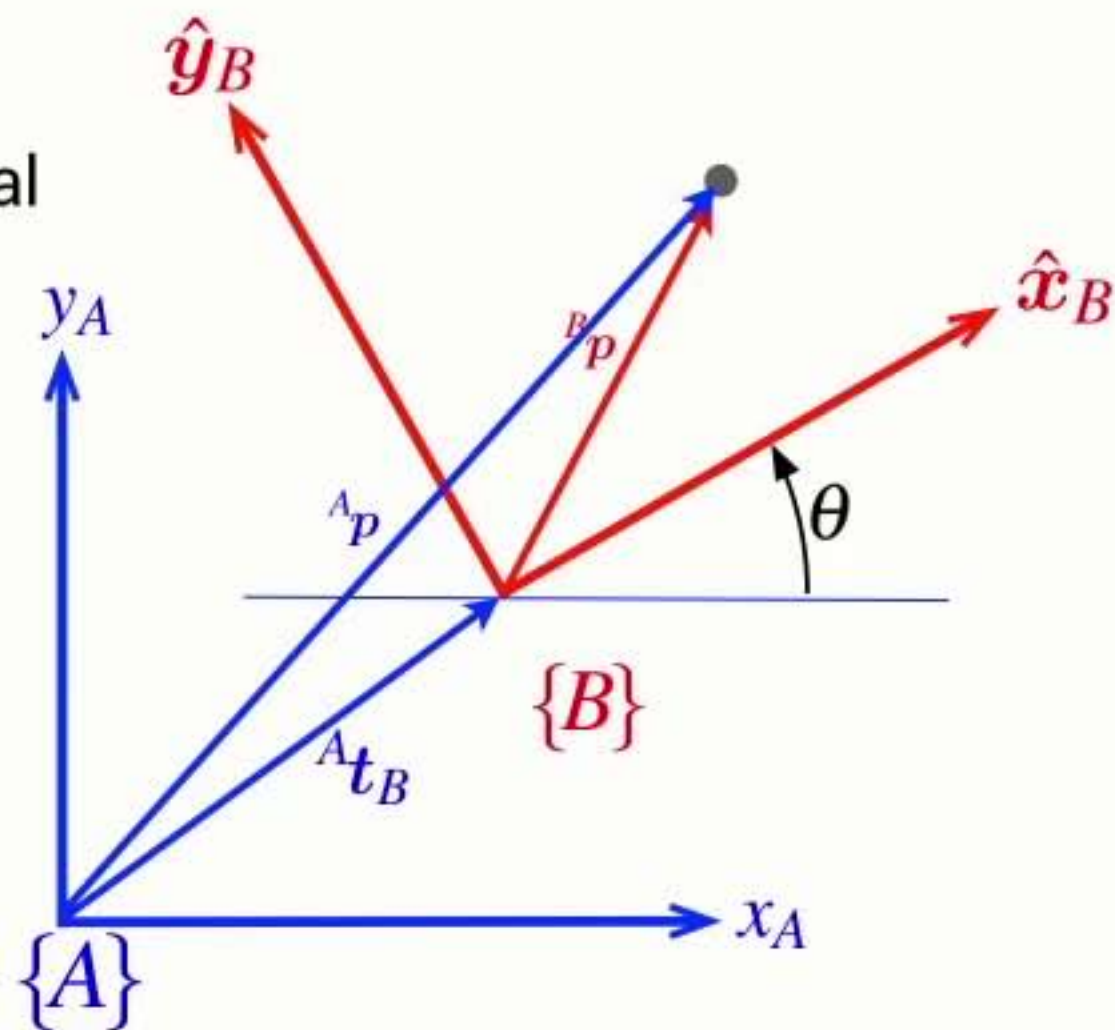
$${}^X\xi_Y \oplus {}^Y\xi_Z \mapsto {}^X\mathbf{T}_Y {}^Y\mathbf{T}_Z$$

→ Negation is a matrix inverse

$$\ominus \xi \mapsto \mathbf{T}^{-1}$$

→ Vector transformation is a matrix-vector product

$$\xi \cdot p \mapsto \mathbf{T}\tilde{p}$$



$$p = \begin{pmatrix} a \\ b \end{pmatrix} \quad \tilde{p} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$

Command Window

```
>> transl2(1,2)
```

```
ans =
```

```
    1    0    1  
    0    1    2  
    0    0    1
```

```
fx >>
```

Workspace

Name ▲	Value	Class
ans	[1,0,1;0,1,2;0,0,1]	double

Command Window

```
>> transl2(1,2)
```

```
ans =
```

```

    1    0    1
    0    1    2
    0    0    1

```

```
>> rot2(30, 'deg')
```

```
ans =
```

```

    0.8660   -0.5000
    0.5000    0.8660

```

fx >>

Workspace

Name	Value	Class
ans	[0.8660,-0.5000;0.5000,...	double

Current Folder

Command Window

```

>> transl2(1,2)

ans =

    1    0    1
    0    1    2
    0    0    1

>> rot2(30, 'deg')

ans =

    0.8660   -0.5000
    0.5000    0.8660

>> trot2(30, 'deg')

ans =

    0.8660   -0.5000    0
    0.5000    0.8660    0
         0         0    1.0000
    
```

fx >>

Workspace

Name ▲	Value	Class
ans	[0.8660,-0.5000,0;0.500...	double

Current Folder

Command Window

```
>> transl2(1,2)

ans =

    1    0    1
    0    1    2
    0    0    1

>> rot2(30, 'deg')

ans =

    0.8660   -0.5000
    0.5000    0.8660

>> trot2(30, 'deg')

ans =

    0.8660   -0.5000    0
    0.5000    0.8660    0
         0         0    1.0000

>> transl2(1,2) * trot2(30, 'deg')

ans =

    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000
```

fx >>

Workspace

Name	Value	Class
ans	[0.8660,-0.5000,1;0.500...	double

```
>> transl2(1,2)
```

```
ans =
```

```
    1    0    1
    0    1    2
    0    0    1
```

```
>> rot2(30, 'deg')
```

```
ans =
```

```
    0.8660   -0.5000
    0.5000    0.8660
```

```
>> trot2(30, 'deg')
```

```
ans =
```

```
    0.8660   -0.5000         0
    0.5000    0.8660         0
         0         0    1.0000
```

```
>> transl2(1,2) * trot2(30, 'deg')
```

```
ans =
```

```
    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000
```

```
>> se2(1, 2, 30, 'deg')
```

```
ans =
```

```
    0.8660   -0.5000    1.0000
    0.5000    0.8660    2.0000
         0         0    1.0000
```

```
fx >>
```

Workspace

Name ▲	Value	Class
ans	[0.8660,-0.5000,1;0.500...	double

Command Window

```
0.8660 -0.5000 1.0000
0.5000 0.8660 2.0000
0 0 1.0000
```

```
>> se2(1, 2, 30, 'deg')
```

```
ans =
```

```
0.8660 -0.5000 1.0000
0.5000 0.8660 2.0000
0 0 1.0000
```

```
>> axis([0 5 0 5])
```

```
>> axis square
```

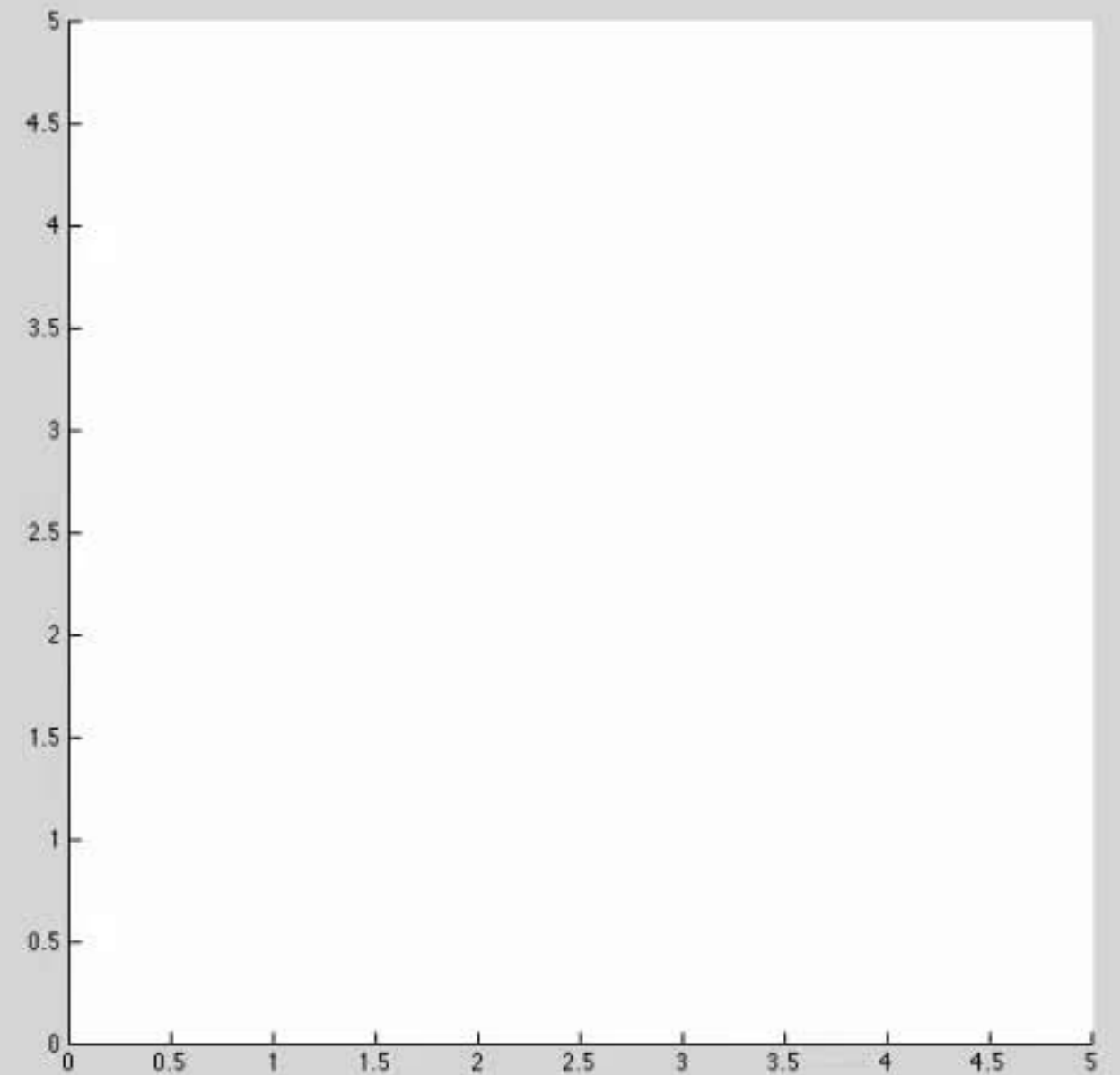
```
fx >>
```

Workspace

Name	Value	Class
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

```
0.8660 -0.5000 1.0000
0.5000 0.8660 2.0000
0 0 1.0000
```

```
>> axis([0 5 0 5])
>> axis square
>> hold on
>> T1 = se2(1, 2, 30, 'deg')
```

T1 =

```
0.8660 -0.5000 1.0000
0.5000 0.8660 2.0000
0 0 1.0000
```

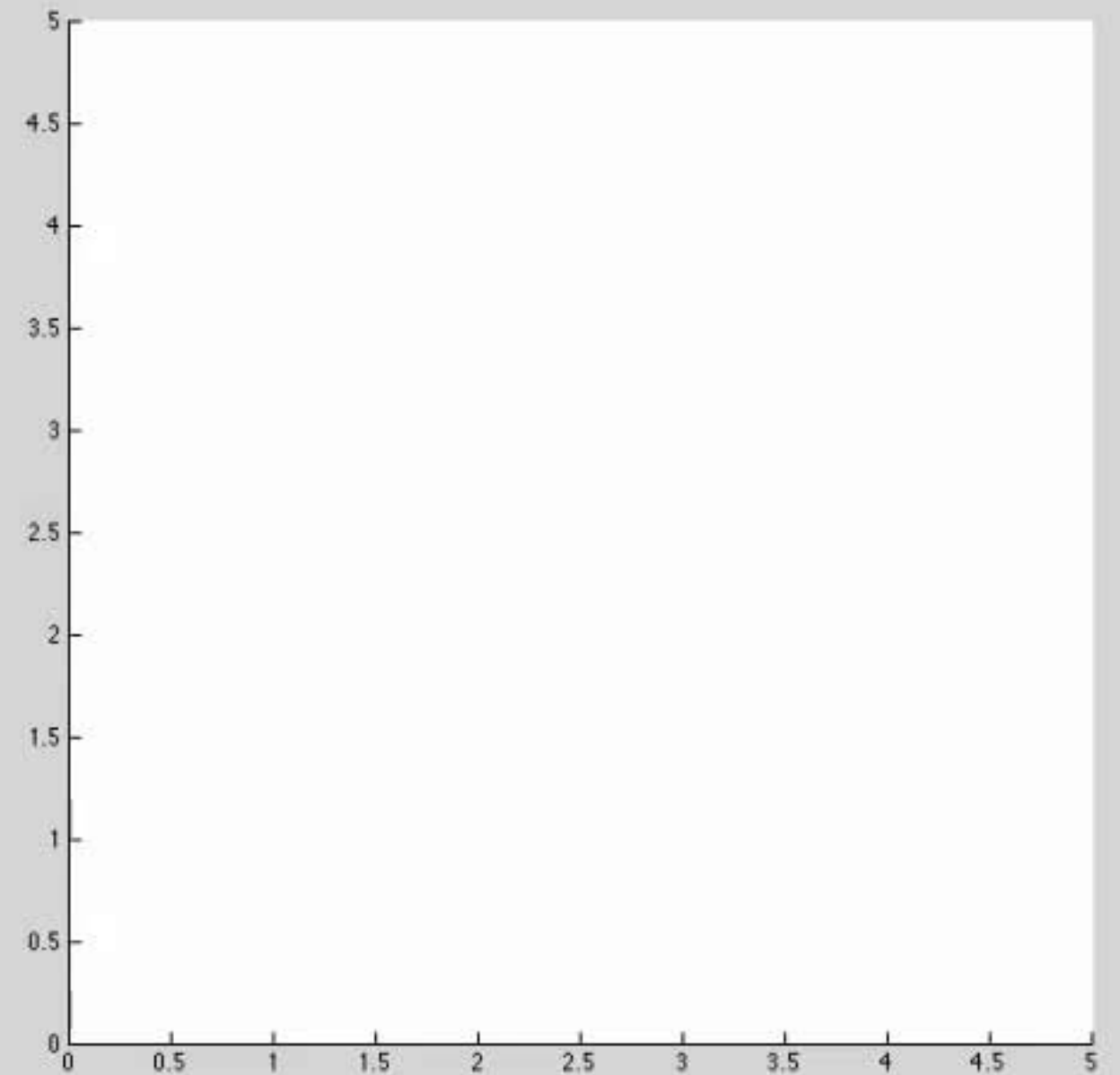
fx >> |

Workspace

Name ▲	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

```
0.5000    0.8660    2.0000
0          0        1.0000
```

```
>> axis([0 5 0 5])
>> axis square
>> hold on
>> T1 = se2(1, 2, 30, 'deg')
```

T1 =

```
0.8660   -0.5000    1.0000
0.5000    0.8660    2.0000
0          0        1.0000
```

```
>> trplot2(T1, 'frame', '1', 'color', 'b')
```

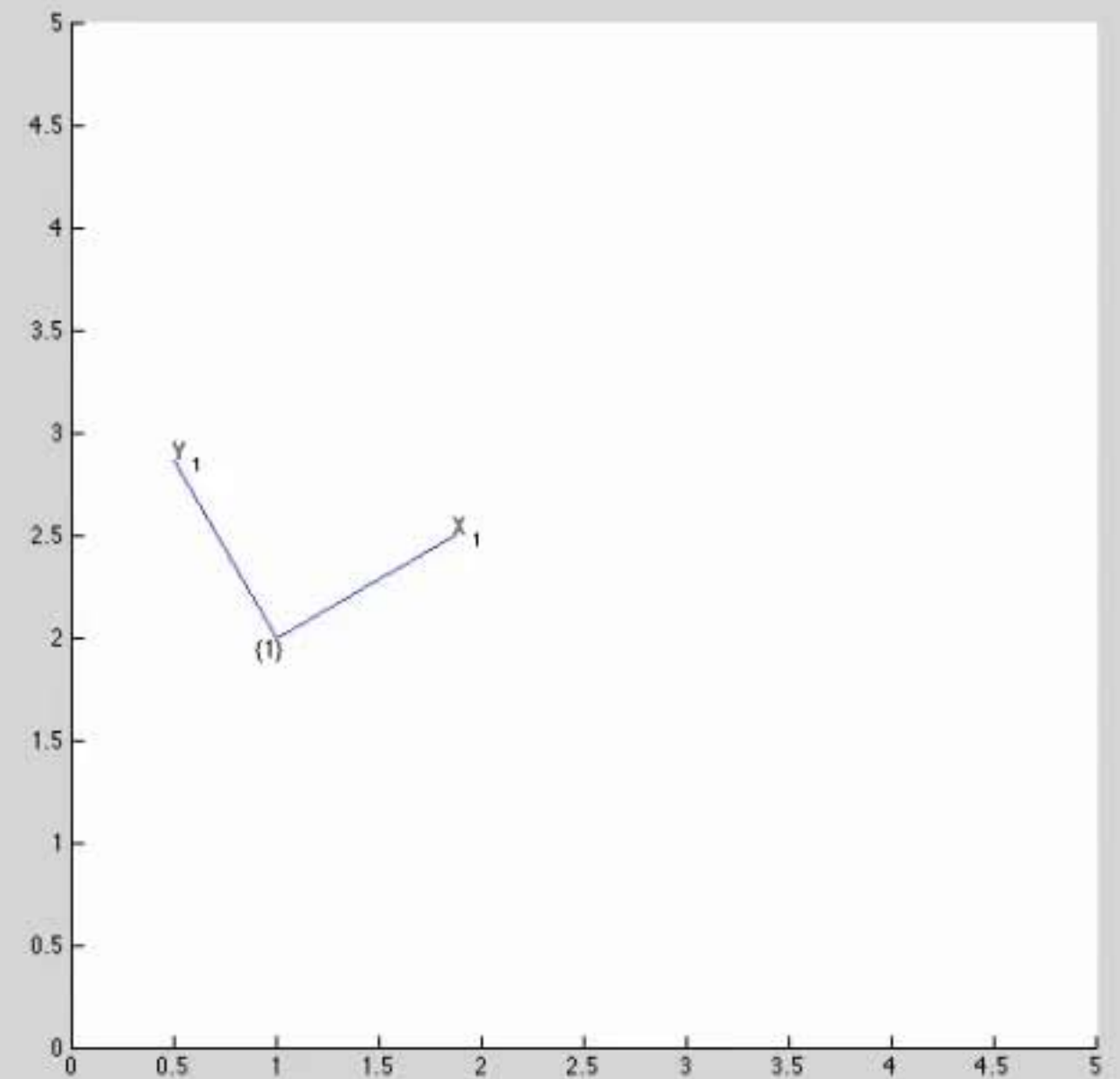
fx >> |

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

T1 =

```
0.8660    -0.5000    1.0000
0.5000     0.8660    2.0000
0          0         1.0000
```

```
>> trplot2(T1, 'frame', '1', 'color', 'b')
```

```
>> T2 = se2(2, 1, 0)
```

T2 =

```
1    0    2
0    1    1
0    0    1
```

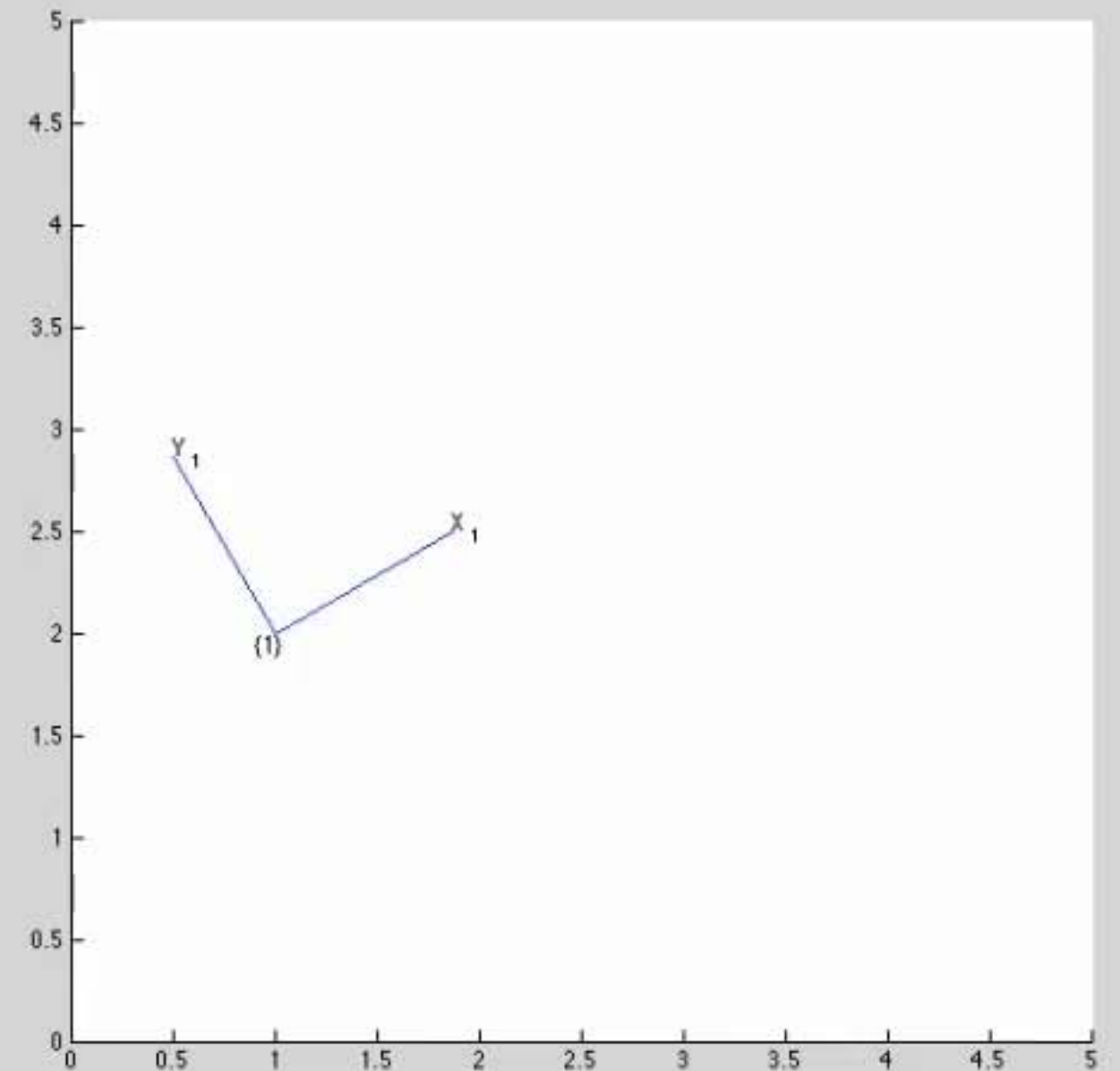
fx >>

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

```
0.8660   -0.5000   1.0000
0.5000    0.8660   2.0000
0         0       1.0000
```

```
>> trplot2(T1, 'frame', '1', 'color', 'b')
>> T2 = se2(2, 1, 0)
```

T2 =

```
1    0    2
0    1    1
0    0    1
```

```
>> trplot2(T2, 'frame', '2', 'color', 'r')
```

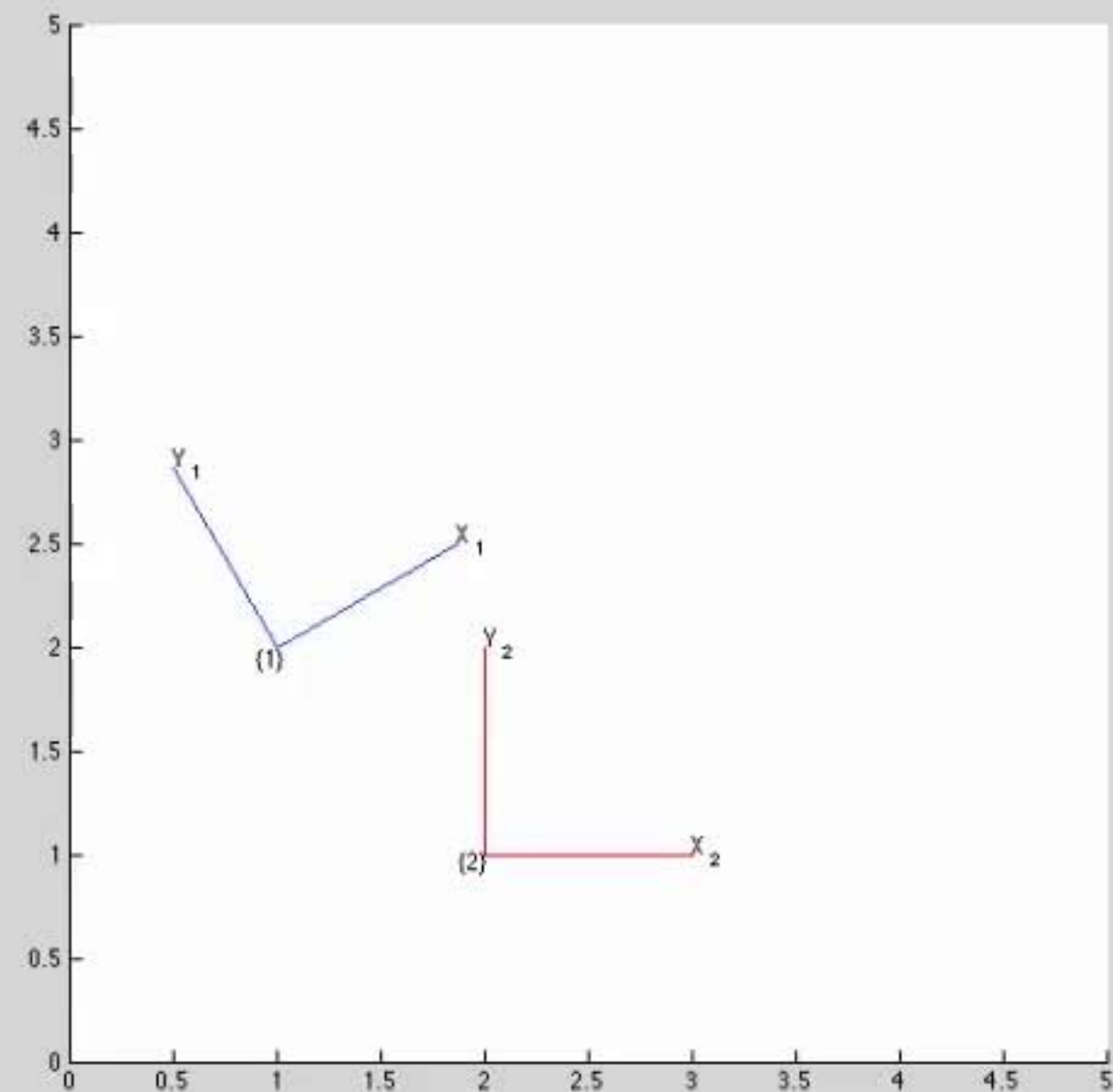
fx >> |

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

T2 =

```

1    0    2
0    1    1
0    0    1

```

```
>> trplot2(T2, 'frame', '2', 'color', 'r')
```

```
>> T3 = T1 * T2
```

T3 =

```

0.8660   -0.5000   2.2321
0.5000    0.8660   3.8660
0         0        1.0000

```

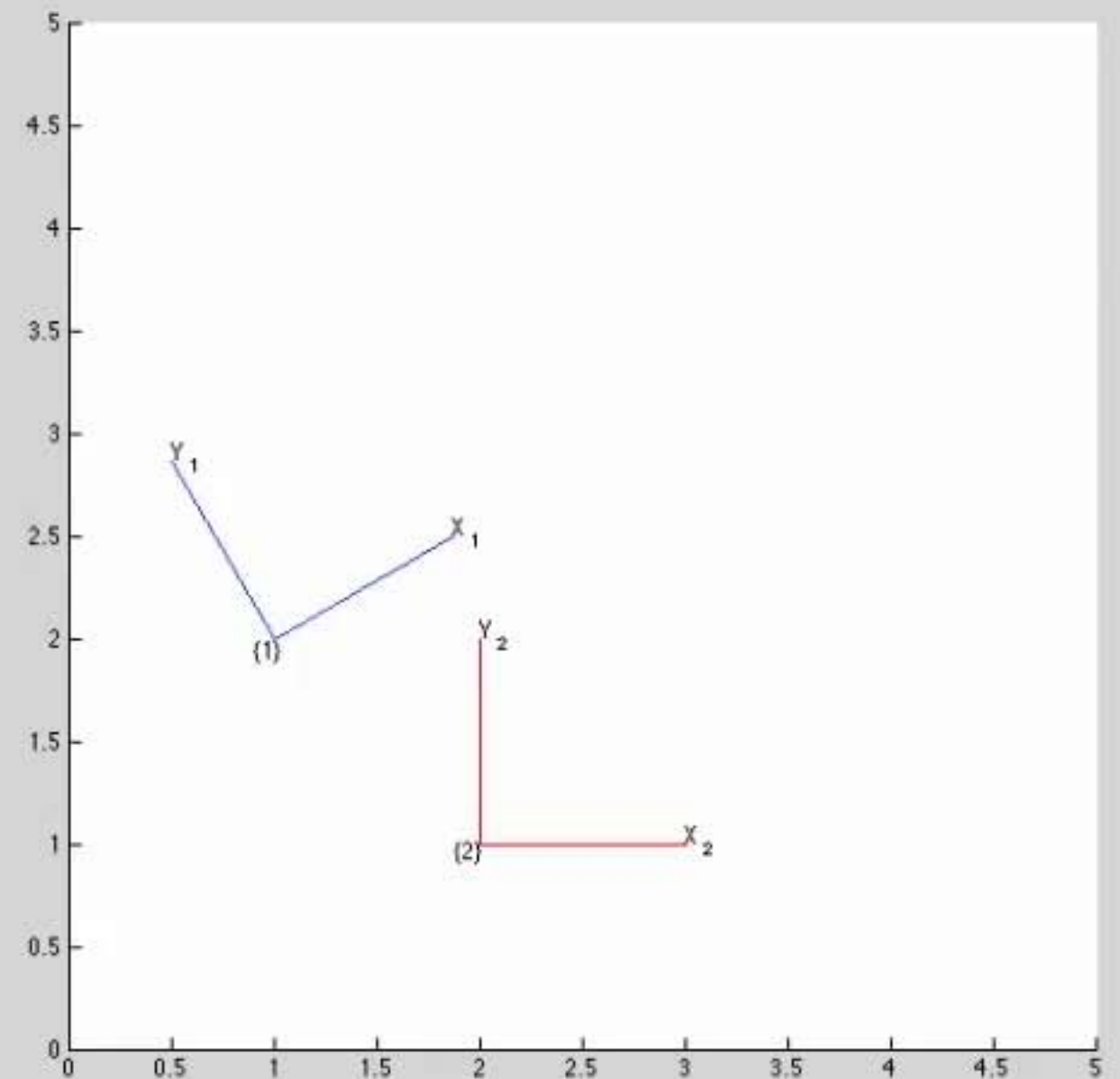
fx >> |

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

```

1    0    2
0    1    1
0    0    1

```

```

>> trplot2(T2, 'frame', '2', 'color', 'r')
>> T3 = T1 * T2

```

T3 =

```

0.8660   -0.5000   2.2321
0.5000    0.8660   3.8660
0         0        1.0000

```

```

>> trplot2(T3, 'frame', '3', 'color', 'g')
fx >> |

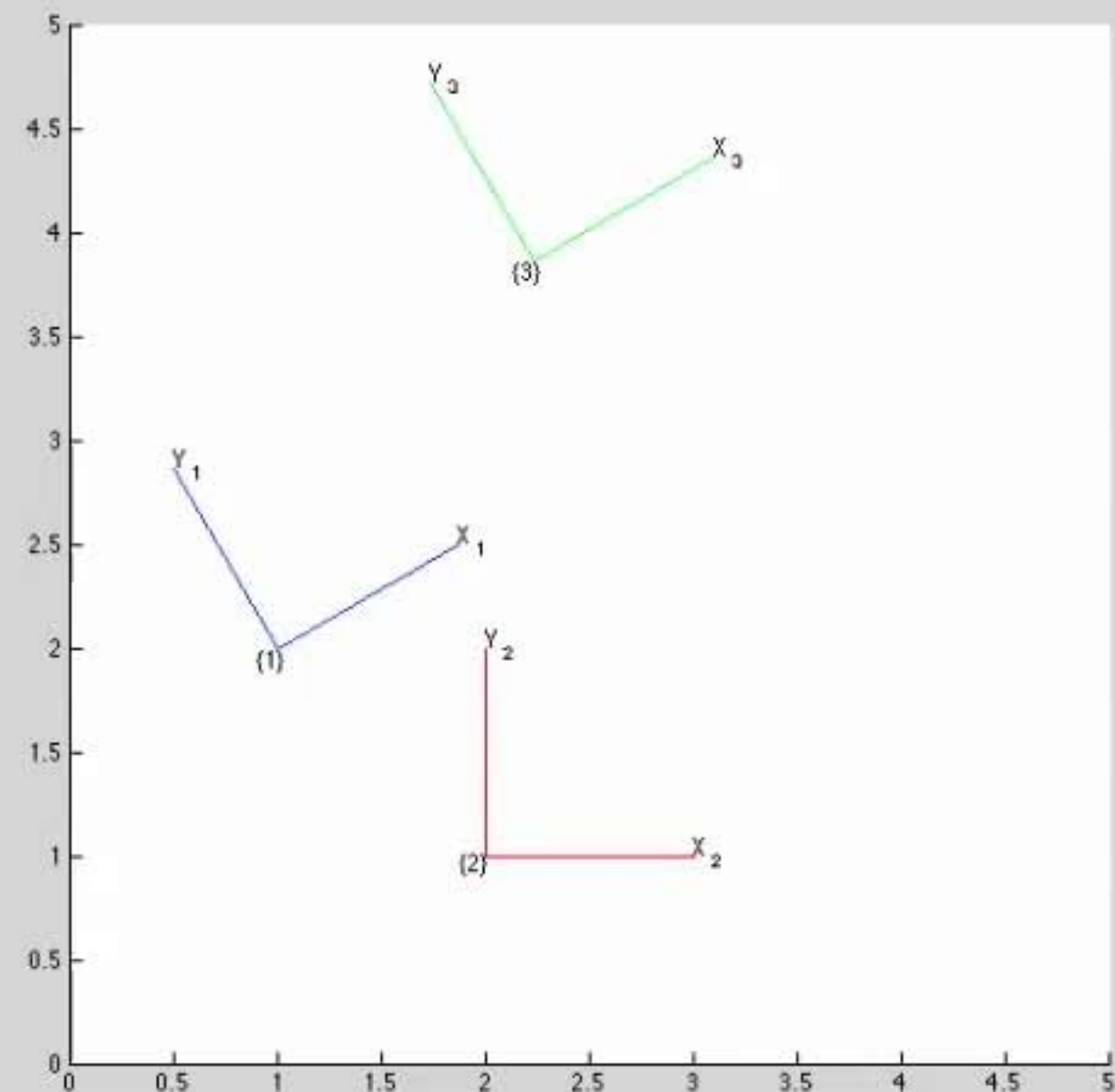
```

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

```
0.8660   -0.5000   2.2321
0.5000    0.8660   3.8660
0         0        1.0000
```

```
>> trplot2(T3, 'frame', '3', 'color', 'g')
>> T4 = T2 * T1
```

T4 =

```
0.8660   -0.5000   3.0000
0.5000    0.8660   3.0000
0         0        1.0000
```

```
>> trplot2(T4, 'frame', '4', 'color', 'c')
```

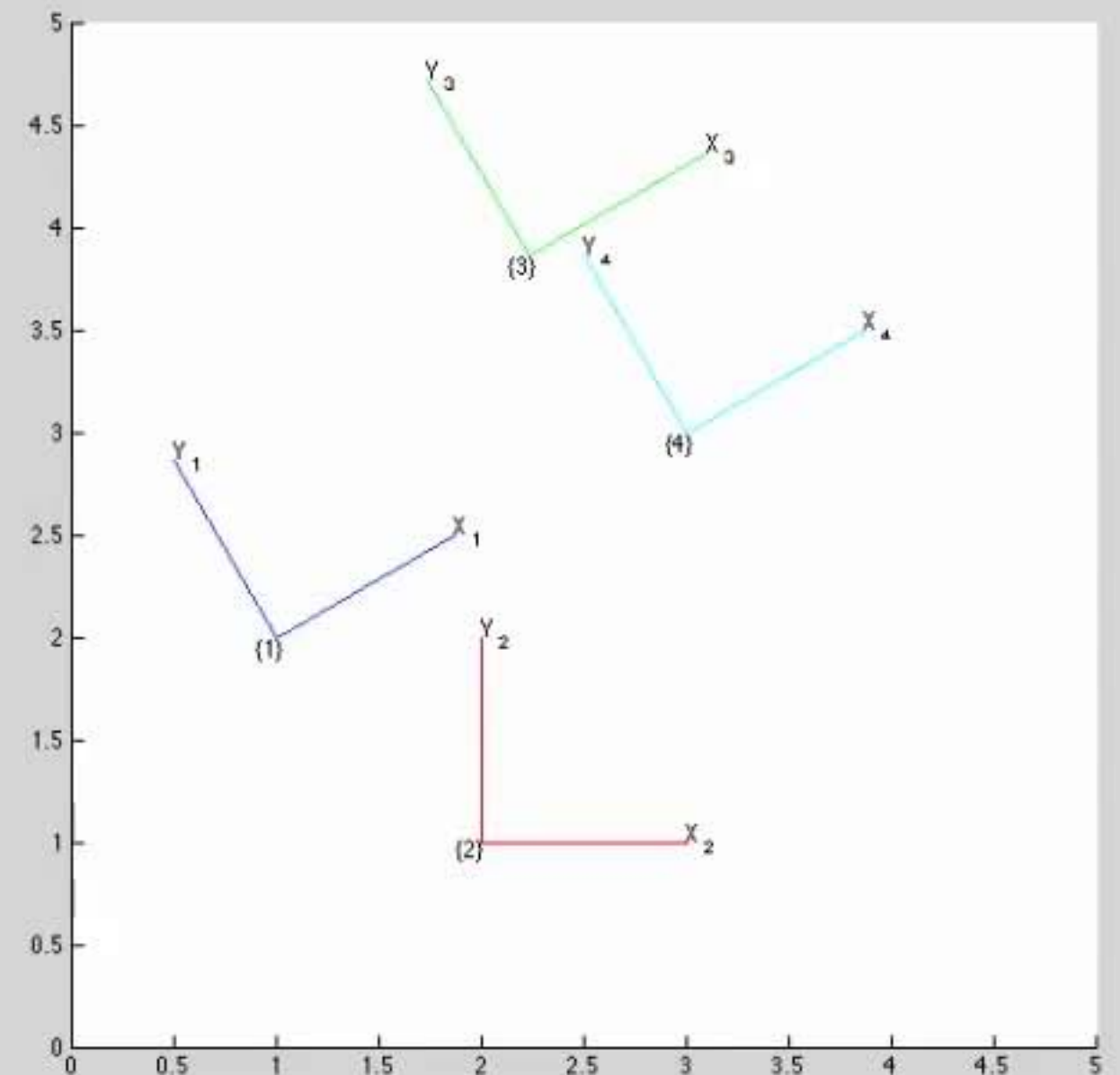
fx >> |

Workspace

Name	Value	Class
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
T4	[0.8660,-0.5000,3;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

T4 =

```
0.8660    -0.5000    3.0000
0.5000     0.8660    3.0000
0          0         1.0000
```

```
>> trplot2(T4, 'frame', '4', 'color', 'c')
```

```
>> P = [ 3 2]'
```

P =

```
3
2
```

```
>> plot_point(P, '*')
```

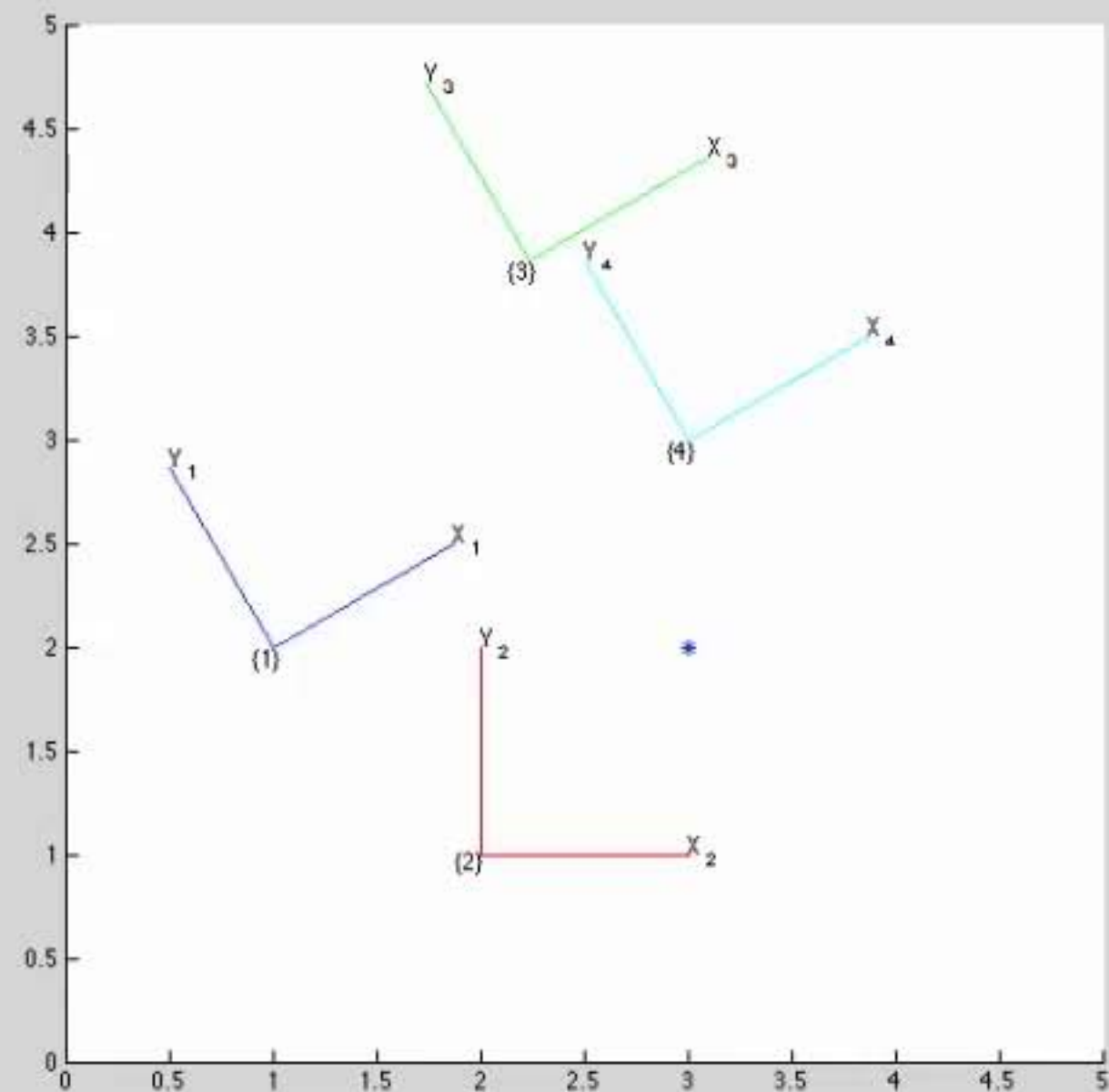
fx >>

Workspace

Name	Value	Class
P	[3;2]	double
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
T4	[0.8660,-0.5000,3;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

Figure 1



Command Window

```
P =  
  
    3  
    2  
  
>> plot_point(P, '*')  
>> P1 = inv(T1) * [P ; 1]
```

```
P1 =  
  
    1.7321  
   -1.0000  
    1.0000
```

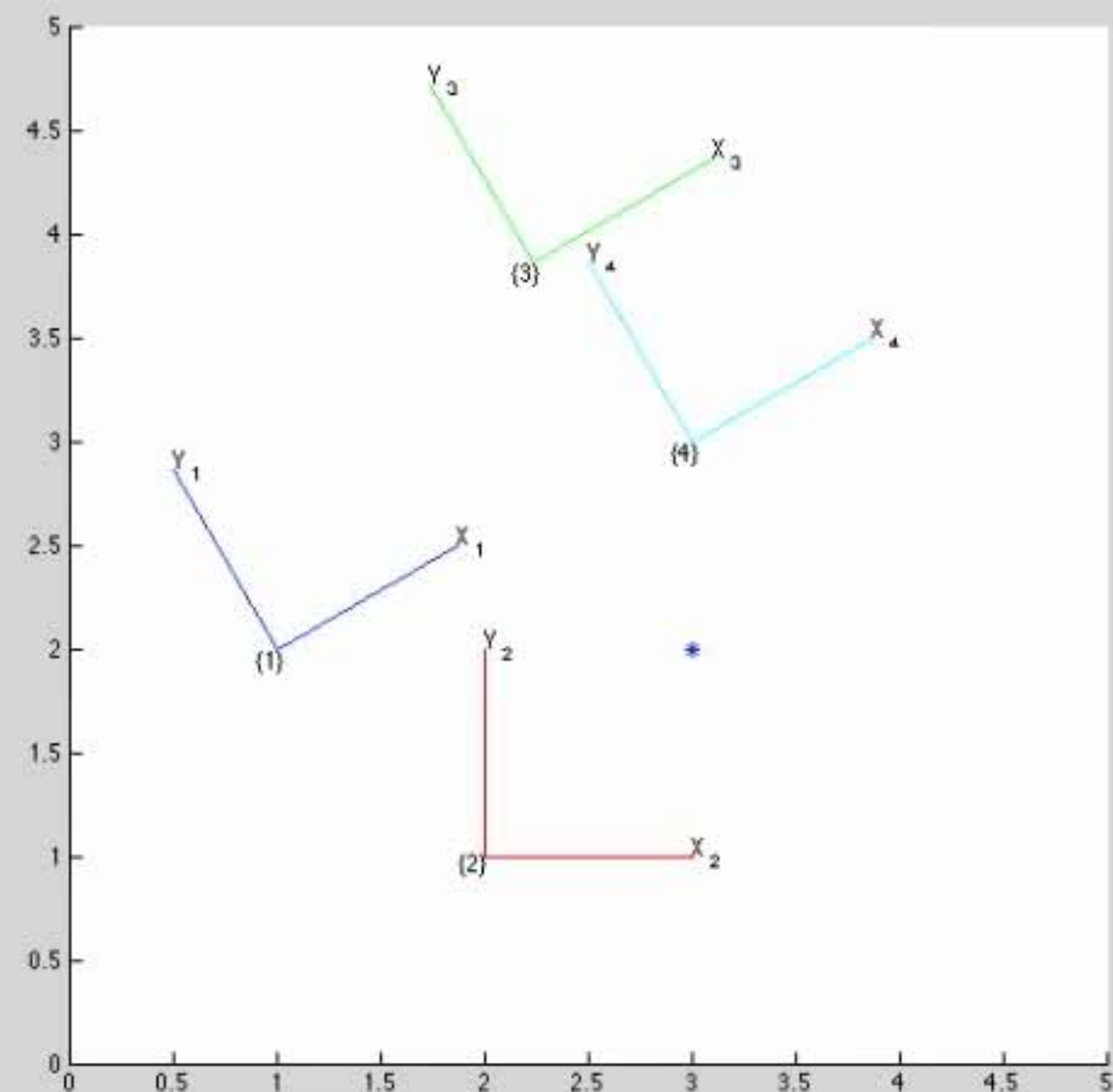
fx >> |

Workspace

Name	Value	Class
P	[3;2]	double
P1	[1.7321;-1.0000;1]	double
T1	[0.8660,-0.5000,1;0.500...	double
T2	[1,0,2;0,1,1;0,0,1]	double
T3	[0.8660,-0.5000,2.2321;...	double
T4	[0.8660,-0.5000,3;0.500...	double
ans	[0.8660,-0.5000,1;0.500...	double

Figures

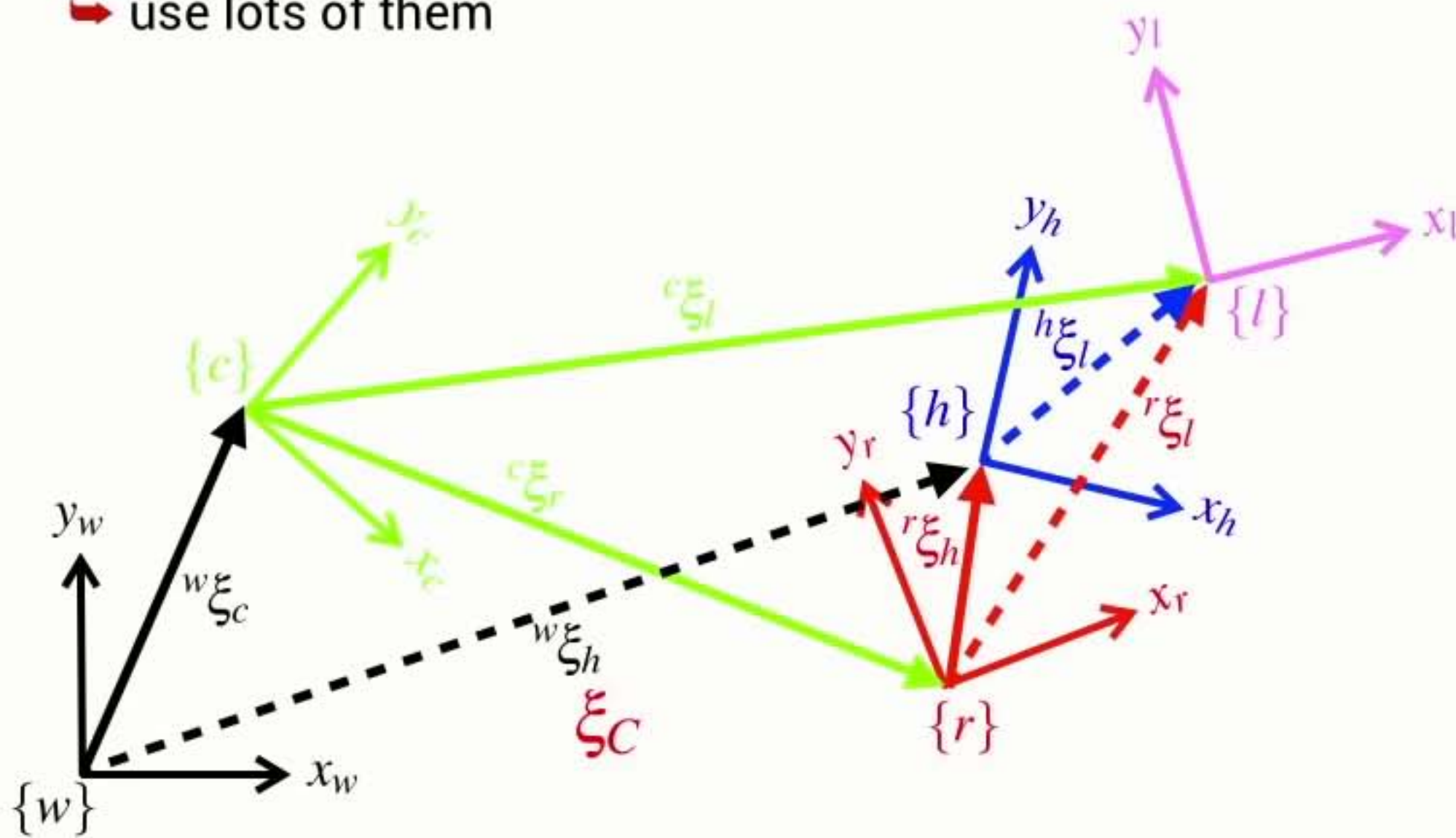
Figure 1



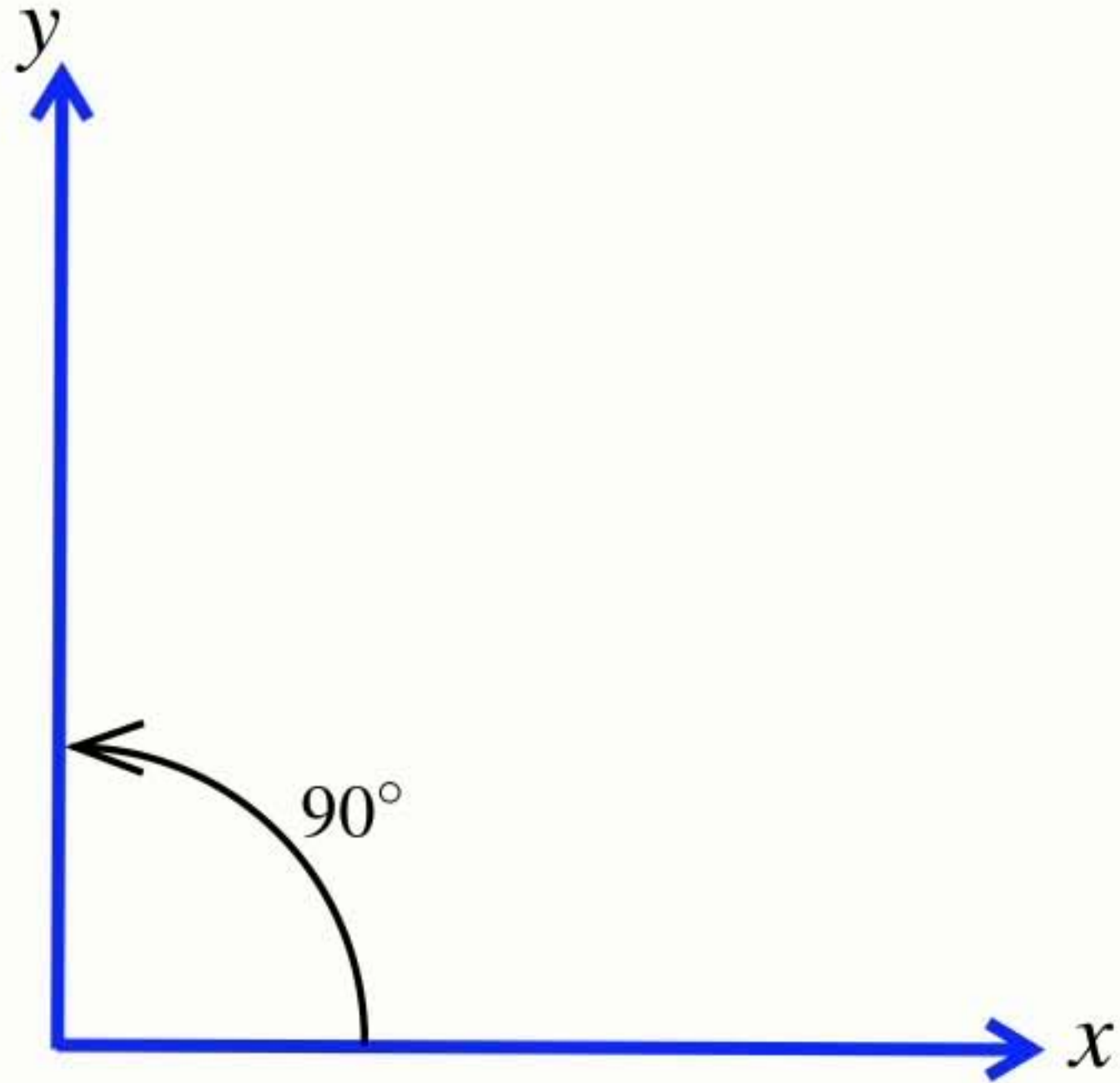
Section8 Summary

Coordinate frames

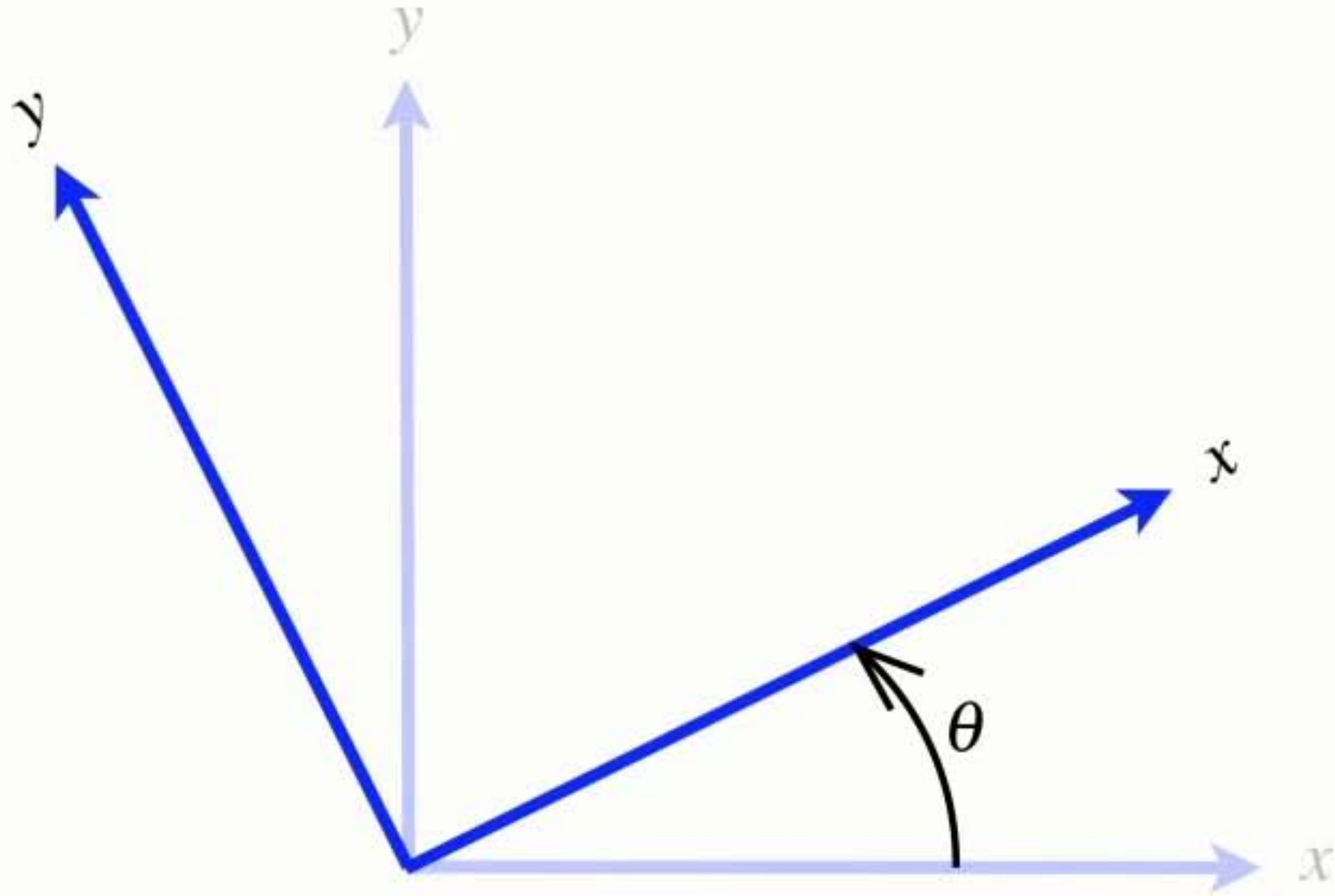
- Coordinate frames are your friend
 - ➡ they describe the pose of objects in the world
 - ➡ use lots of them



Right-handed **coordinate frame**



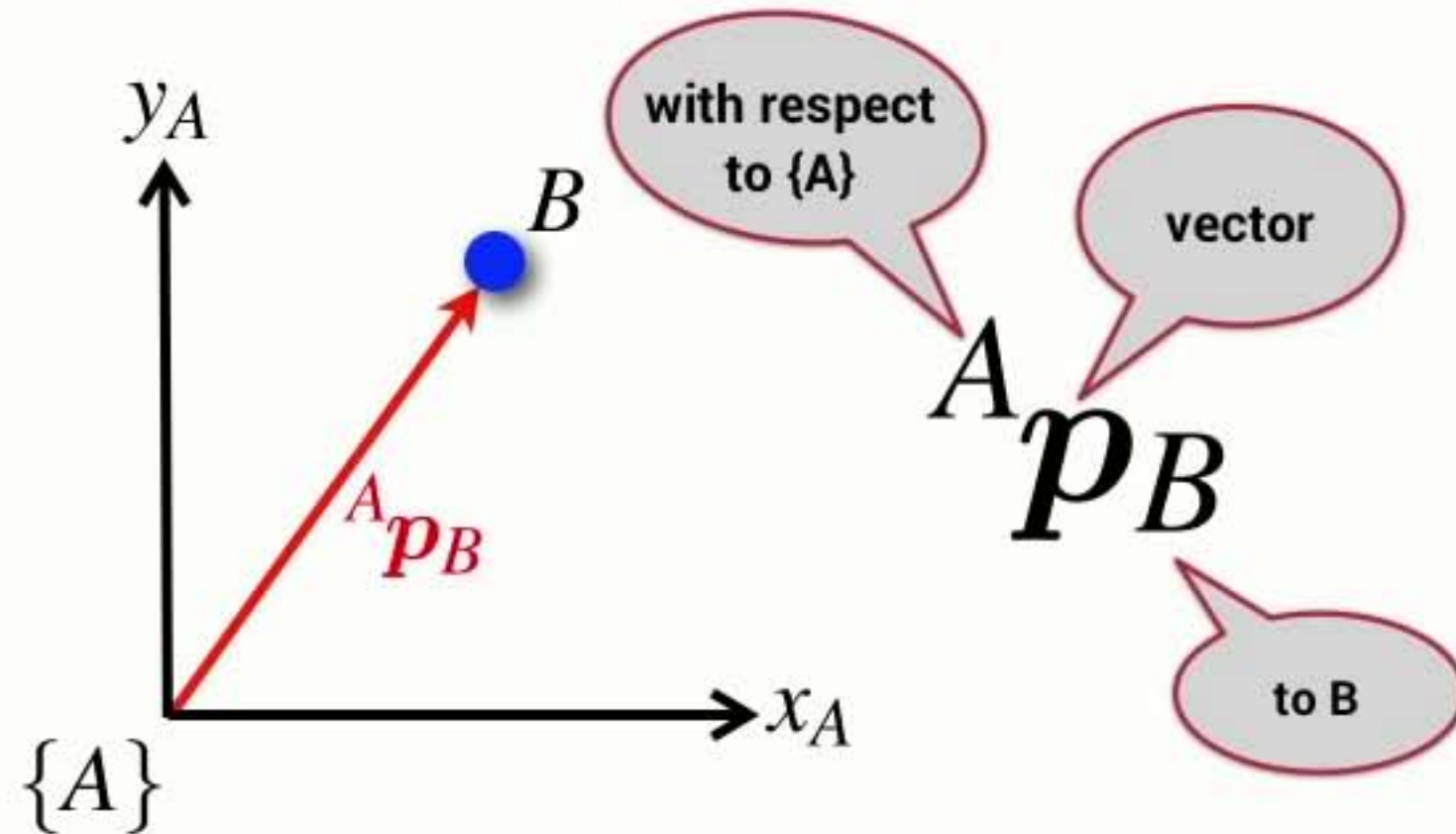
Coordinate frame **rotation convention**



- Angles increase positively in the anti-clockwise direction

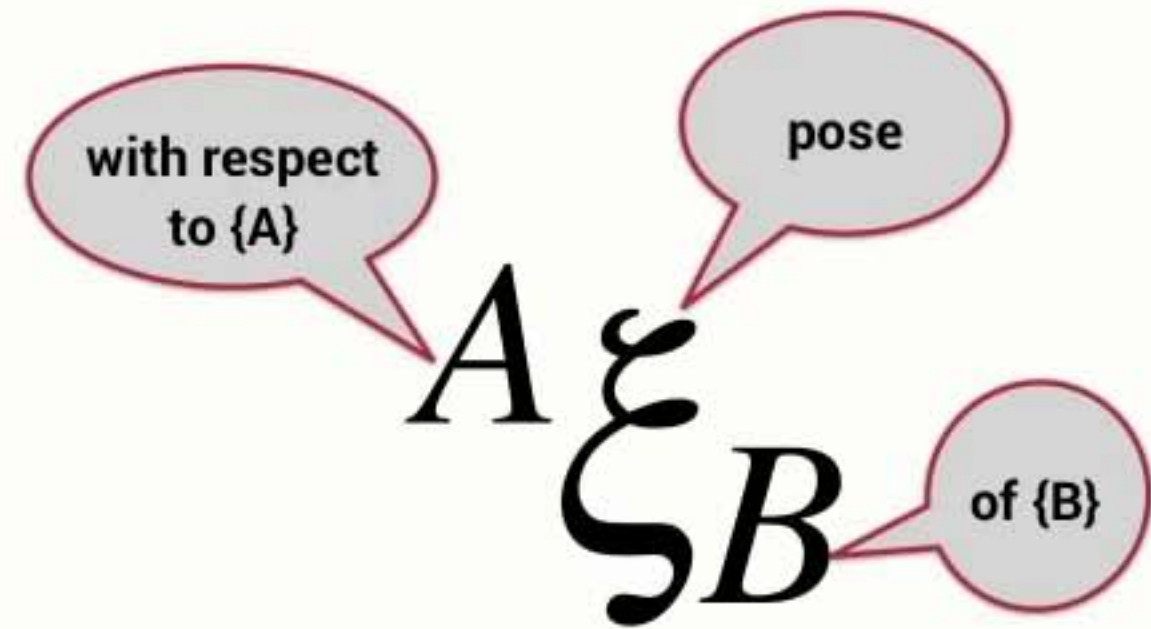
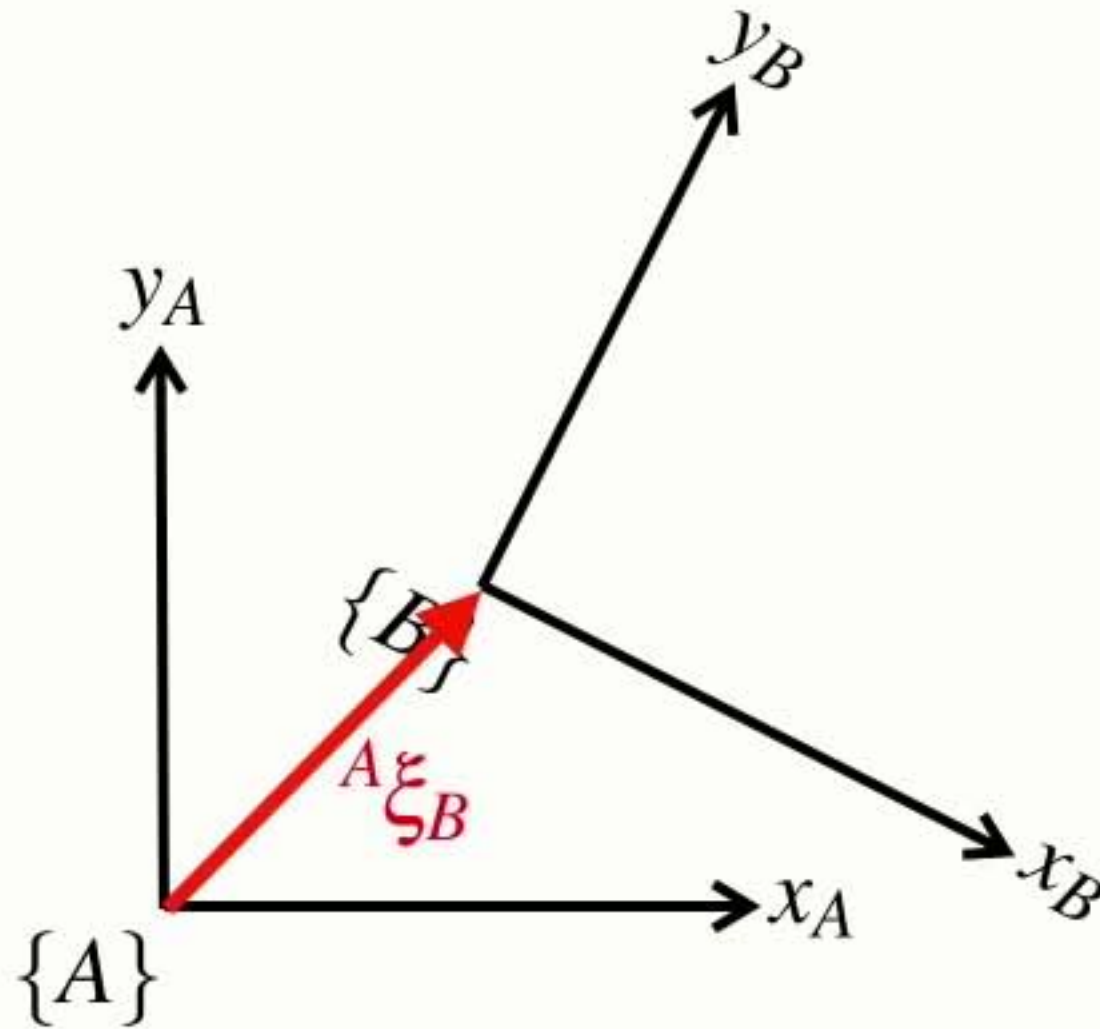
Points & vectors

- A point is described by a vector with respect to a coordinate frame

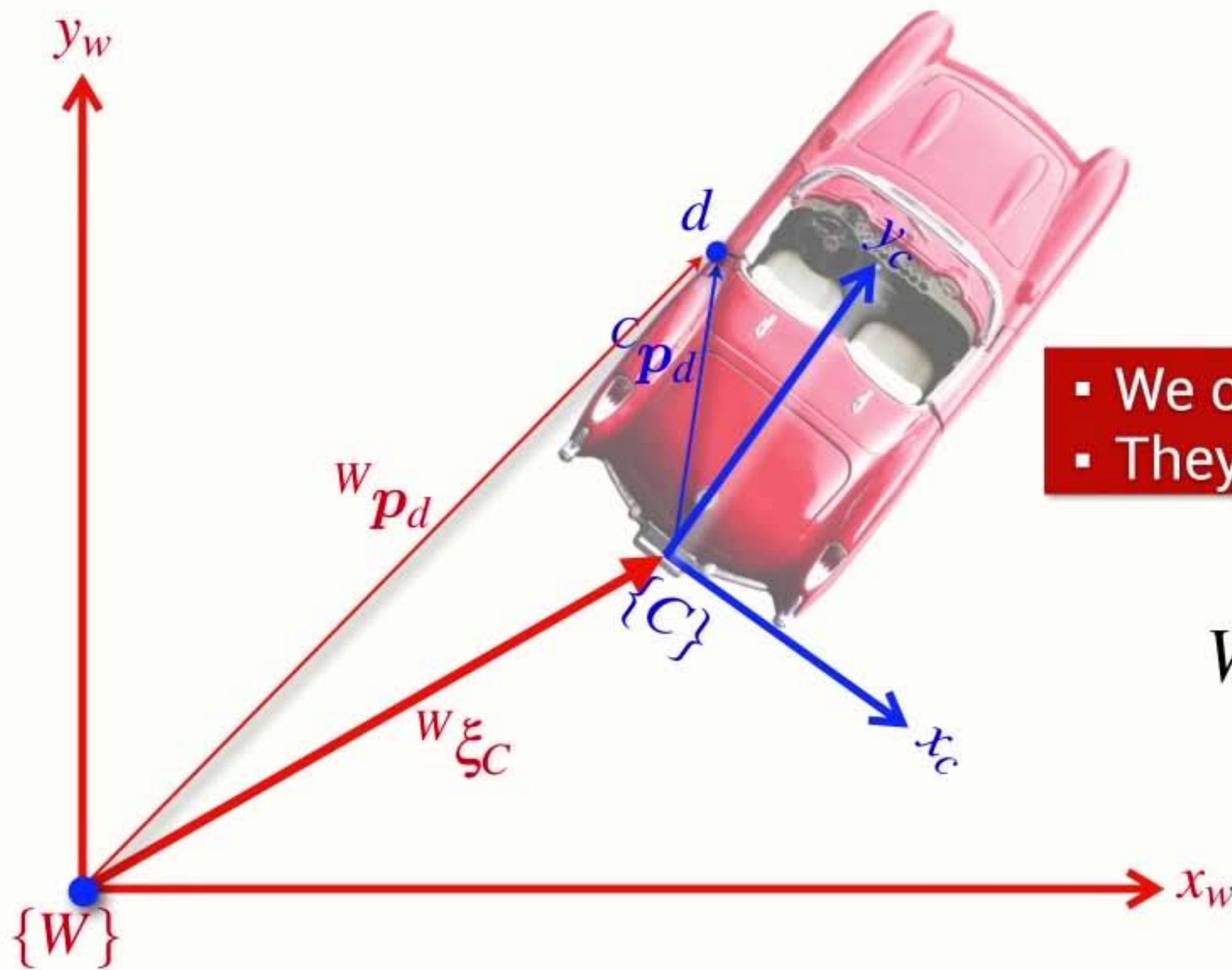


Pose

- The pose of one coordinate frame with respect to another is a relative pose
 - ➔ it has three parameters (x, y, θ)
 - ➔ we can consider it as a simple motion comprising translation and rotation



pronounced **ksi**

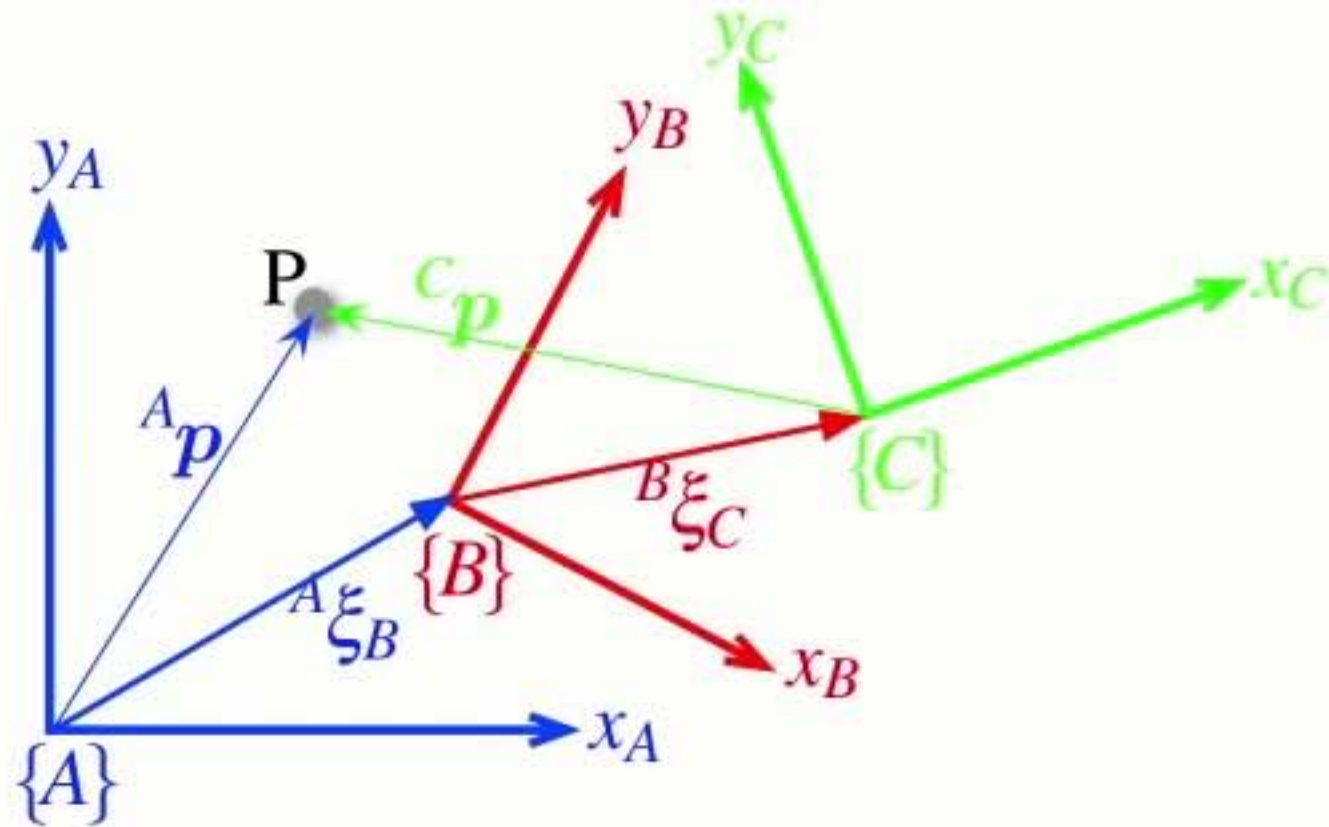


- We can't add **Poses** and **Vectors**
- They are different things

$${}^W p_d = {}^W \xi_C \cdot {}^C p_d$$

transform
vector from one frame
to another

Composition of **poses**



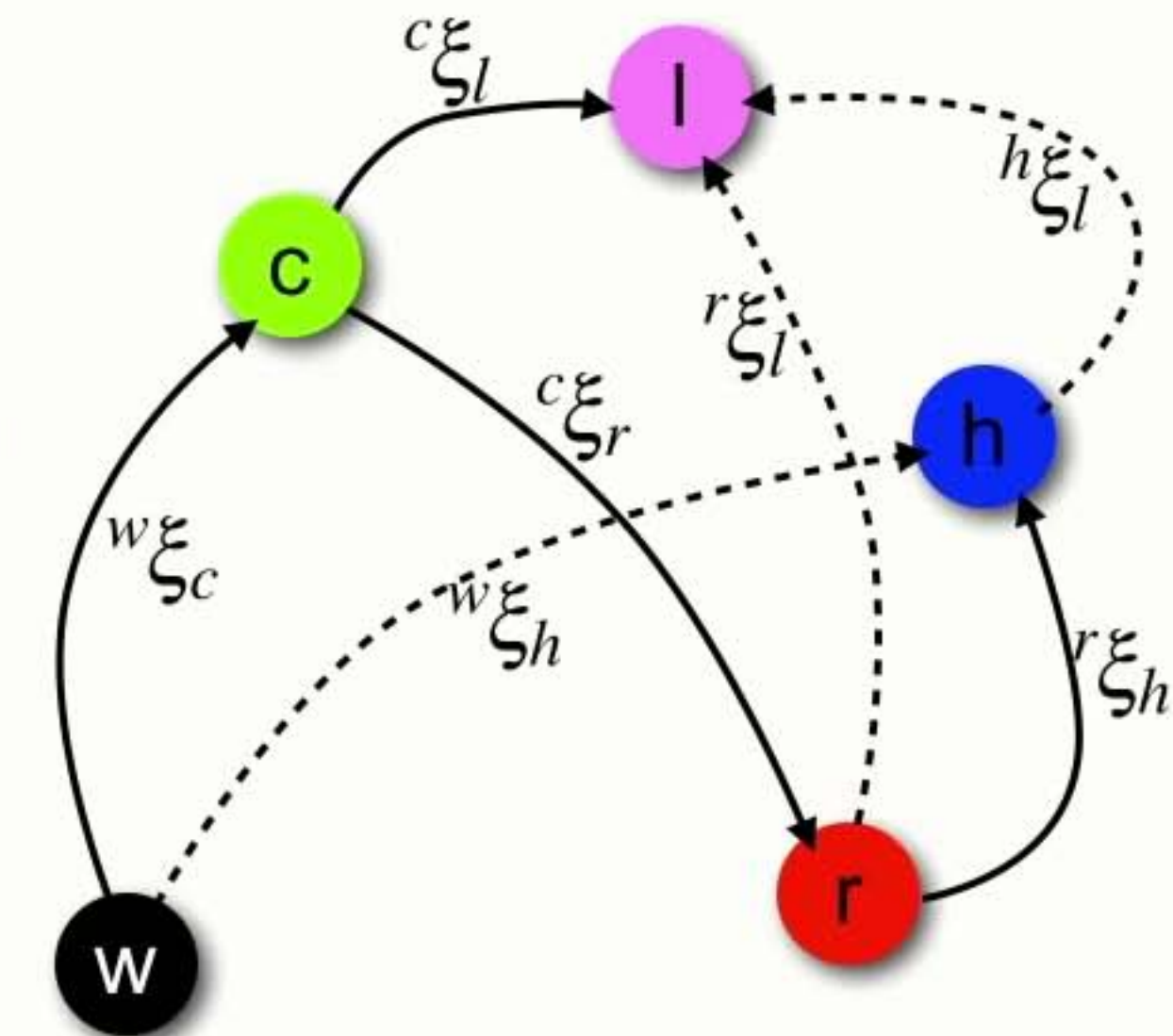
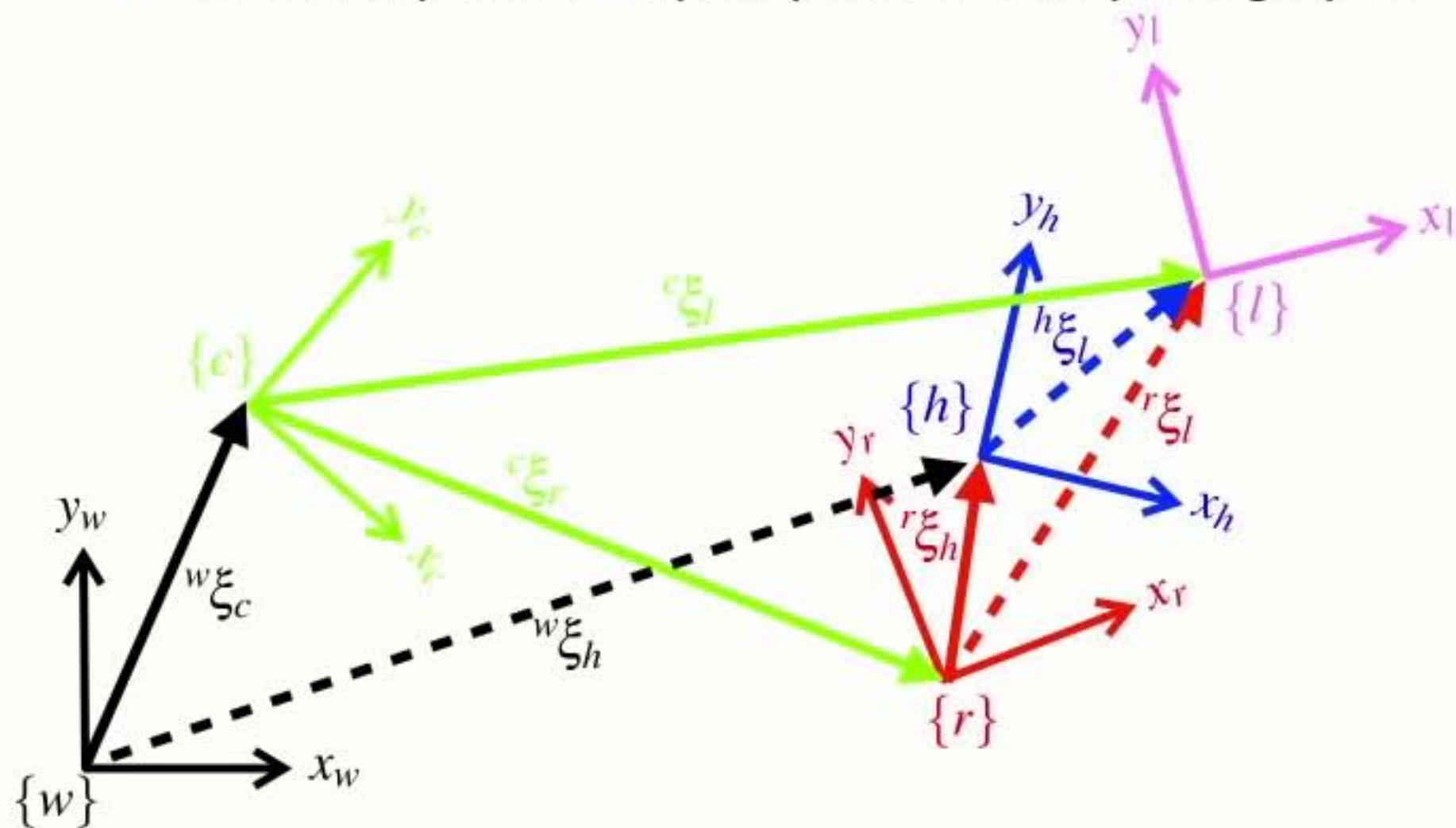
$${}^A \xi_C = {}^A \xi_B \oplus {}^B \xi_C$$

composition

- Relative poses can be **compounded** or **composed**
- We can extend this approach indefinitely...

Pose graphs

- We can express complex problems as pose graphs



$$w\xi_h = w\xi_c \oplus c\xi_r \oplus r\xi_h$$

$$h\xi_l = \ominus r\xi_h \ominus c\xi_r \oplus c\xi_l$$

Spatial algebra

- We can use a spatial algebra to solve for spatial relationships

$${}^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z$$

$${}^Xp = {}^X\xi_Y \bullet {}^Yp$$

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1$$

$$\xi \ominus \xi = 0$$

$$\xi \oplus 0 = \xi$$

$$\ominus \xi \oplus \xi = 0$$

$$\ominus {}^X\xi_Y = {}^Y\xi_X$$

$$\xi \ominus 0 = \xi$$

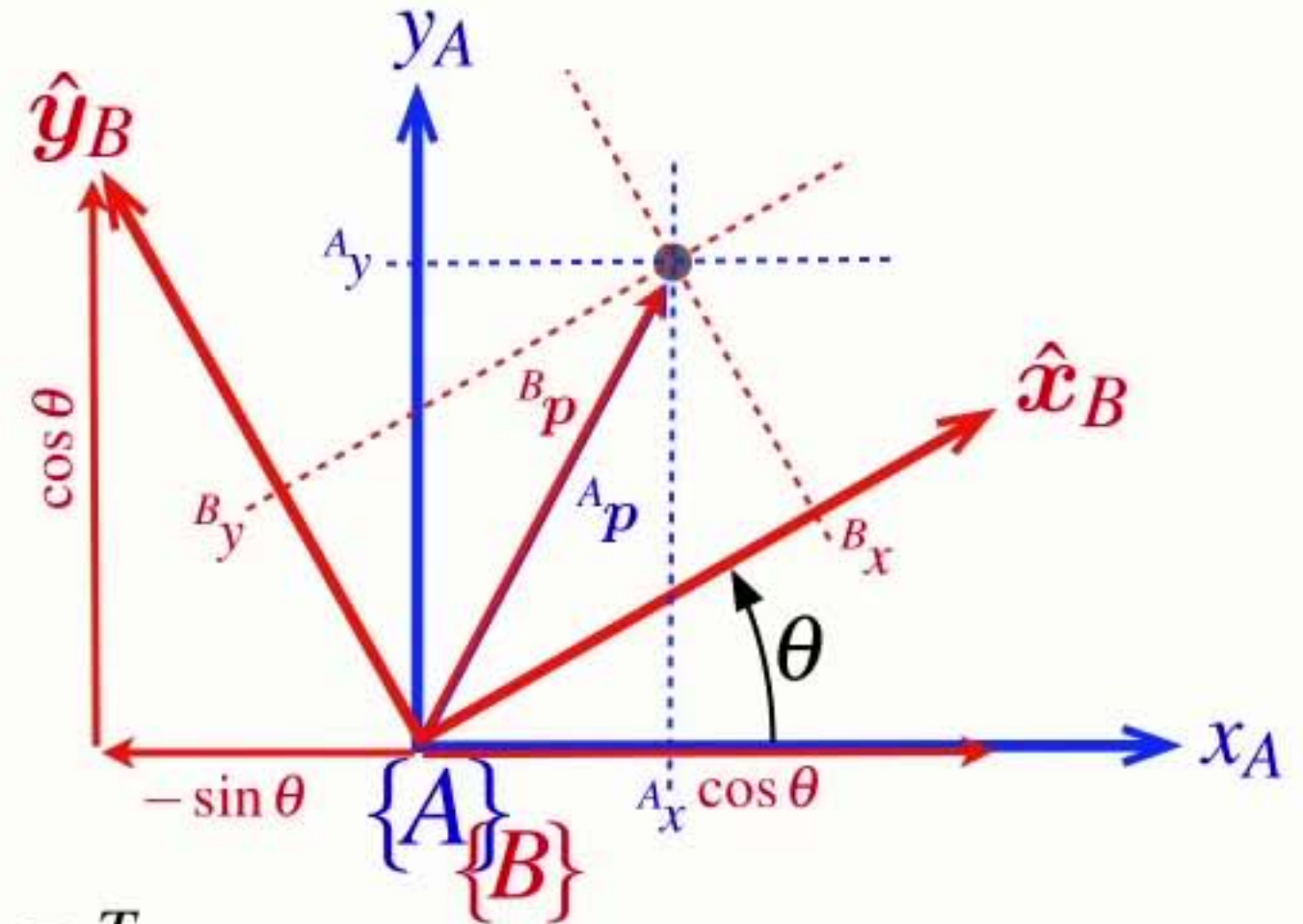
Rotation matrix

- We can rotate a vector from one frame to another

$${}^A\mathbf{R}_B = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$${}^B\mathbf{R}_A = {}^A\mathbf{R}_B^{-1}$$

- Rotates vectors from frame {B} to frame {A}
- Is a function of the rotation angle from frame {A} to frame {B}
- An orthogonal (orthonormal) matrix
 - ➔ The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^T$
 - ➔ The determinant is +1
- Rotation matrices belong to the Special Orthogonal group of dimension 2 $\mathbf{R} \in SO(2)$



$${}^A\mathbf{R}_B$$

Homogeneous transformation

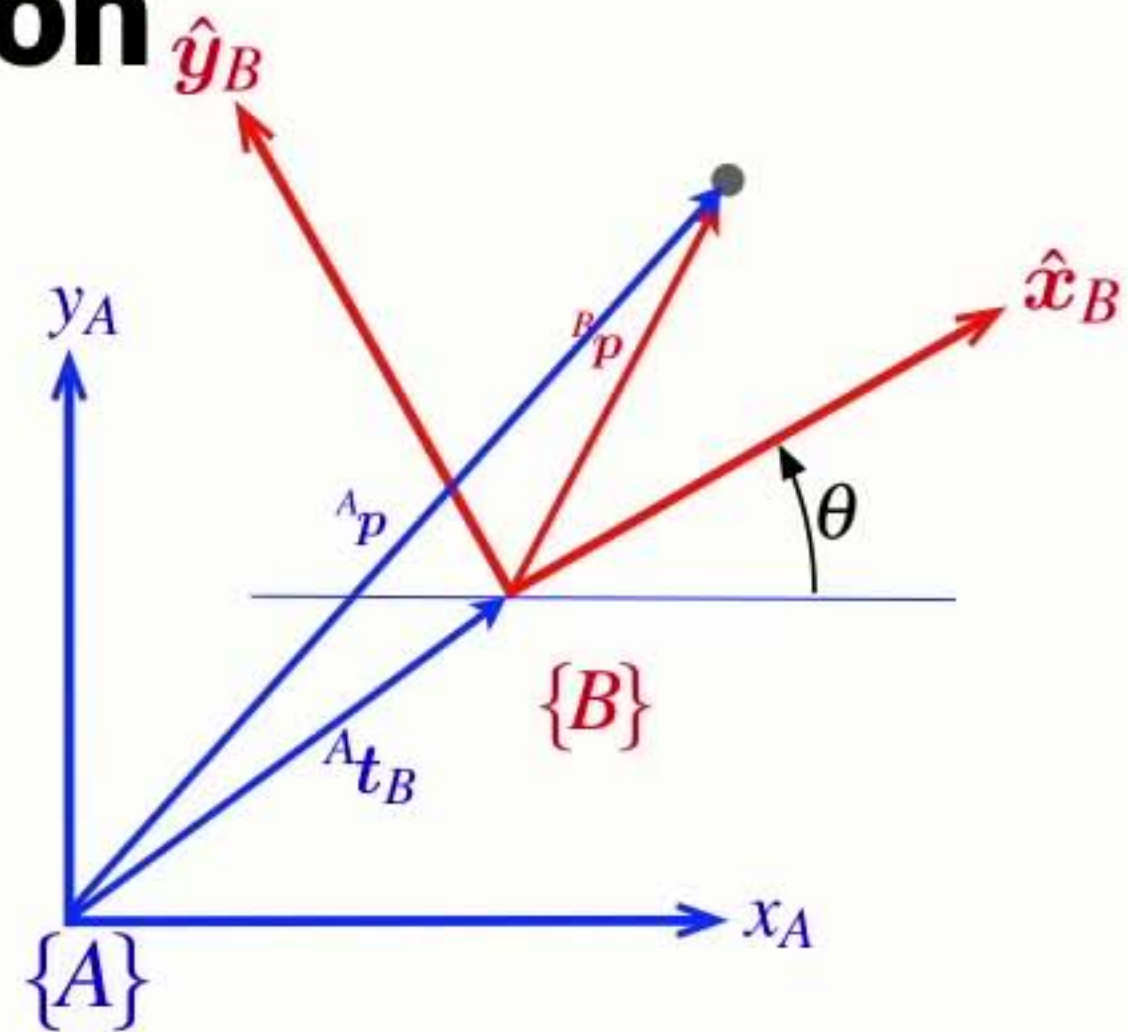
- In concrete terms we can represent pose by 3x3 homogeneous transformation matrix $\mathbf{T} \in SE(2)$

$${}^A\xi_B \sim {}^A\mathbf{T}_B = \begin{pmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$$

$${}^X\xi_Y \oplus {}^Y\xi_Z \mapsto {}^X\mathbf{T}_Y {}^Y\mathbf{T}_Z$$

$$\ominus \xi \mapsto \mathbf{T}^{-1}$$

$$\xi \cdot p \mapsto \mathbf{T}\tilde{p}$$



$${}^A\mathbf{T}_B \quad p = \begin{pmatrix} a \\ b \end{pmatrix} \quad \tilde{p} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$$